

Abstracts from the Fourth Chinese Congress on Gerontology and Health Industry

P1

The Orothogeriatrics Model: Testing the Effectiveness of an Integrated Model of Care in Older Patients with Femur Fracture

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OBJECTIVES: At the end of this session the participants will be able to:

- 1 Understand the principles of the orthogeriatrics model of care
- 2 Improve their clinical practice by learning the most important aspects of treating older persons after a hip fracture
- 3 Treat the pre- and post-operative periods in hip fracture patients, prevent the most common complications and develop interventions to prevent further falls and fractures in this population

BACKGROUND: Older adults with hip fracture have a 5- to 8-fold increased risk for all-cause mortality and much higher risk of institutionalization. Therefore, standardized and evidence-based interventions are highly needed. In this study, we aimed to assess the effectiveness of an integrated model of orthogeriatrics care.

METHODS: Two geriatricians run the program with the assistance of a multidisciplinary team. Assessment includes surgical risk, physical examination, nutrition, cognition, and blood tests. A comprehensive intervention plan was designed following the guidelines from the NSW Agency of Clinical Innovation (http://www.aci.health.nsw.gov.au/_data/assets/pdf_file/0013/153400/aci_orthogeriatrics_clinical_practice_guide.pdf). Two populations of neck of femur (NOF) fracture patients admitted prior (2006) and after (Jan. 2011-June 2012) the implementation of the program were compared.

RESULTS: Two groups of NOF fracture patients were compared: $n = 270$ in 2006, and $n = 150$ in 2011–12. Mean age (83 ± 7) and gender (73% female) were similar in both groups. The orthogeriatrics group showed a significantly shorter delay in surgery (from 59.7% to 21.6%, $P < .01$), lower in-hospital mortality (5.4% vs 1.9%, $P < .05$) and lower rate of di novo admission to nursing homes (24% vs 17%, $P < .05$), with a higher number of patients receiving rehabilitation at discharge in the orthogeriatrics group (50% vs 61.5%, $P < .05$).

CONCLUSION: Our data demonstrates the effectiveness of an evidence-based standardized orthogeriatrics program run by geriatricians together with a multidisciplinary team.

P2

Sirtinol Suppresses Human Cytomegalovirus (hCMV) Infection and hCMV-induced Cellular Senescence and Production of Reactive Oxygen Species in Human Fibroblasts

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BACKGROUND: Human cytomegalovirus (hCMV) causes birth defects in newborns and severe infections in immunocompromised patients. hCMV also contributes to immunosenescence. Anti-hCMV treatments are limited with significant drawbacks. We aimed to investigate effects of sirtinol, a sirtuin inhibitor, on CMV-induced cellular senescence.

METHOD: Human fibroblast WI-38 cells at population doubling (PD) of 30 was infected by hCMV Towne strain at multiplicity of infection (MOI) of 0.01. Sirtinol or resveratrol was added 2 h before hCMV infection. Viral replication was measured using qPCR. Senescent phenotype was determined by senescence-associated β galactosidase (SA- β -Gal) activity. Expression of hCMV and senescence proteins and levels of reactive oxygen species (ROS) production were assessed by Western blot and flow cytometry, respectively.

RESULTS: Sirtinol suppressed hCMV infection in a dose dependent fashion with over 95% reduction of hCMV replication at day post infection (dpi) 7 and showed a potent inhibition on expression of viral immediate early (IE) and early (E) proteins at dpi 3, at 20 μ M. Similarly, sirtinol suppressed or prevented hCMV-induced cellular senescence as demonstrated by significant reduction of SA- β -Gal activity, expression of pRb and Rb, p16 and p53, as well as level of ROS production. Of note, the inhibitory role of sirtinol on hCMV infection and hCMV induced cellular senescence was comparable to that of resveratrol, a sirtuin activator known to suppress hCMV replication.

CONCLUSIONS: Our findings suggest the potential of sirtinol as a novel anti-CMV therapeutic agent and provide insights into molecular pathways for CMV control, and cellular senescence and aging.

P3

Dignity-Conserving End-of-Life (EoL) Care Model for Nursing Homes in Chinese Societies

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BACKGROUND: The provision of end-of-life (EoL) care in long-term-care settings remains largely underdeveloped in most Chinese societies, and nursing home residents often fail to obtain good care as they approach death. This presentation systematically describes the development and implementation mechanisms of a novel Dignity-Conserving EoL Care model that has been successfully adopted by three nursing homes in Hong Kong, and presents preliminary evidence of its effectiveness on enhancing dignity and quality of life (QoL) of terminally-ill residents.

DESIGN AND METHODS: Nine terminally-ill nursing home residents completed the McGill Quality of Life Questionnaire and the Nursing Facilities Quality of Life Questionnaire at baseline and 6-months post EoL program enrollment. Wilcoxon signed rank test was used to detect significance changes in each QoL domains across time.

RESULTS: Although significant deterioration was recorded for Physical QoL, significant improvement was observed for Social QoL. Moreover, a clear trend towards significant improvements was identified for the QoL domains of Individuality and Relationships.

IMPLICATIONS: A holistic and compassionate caring environment, together with the core principles of family-centered care, inter-agency and interdisciplinary teamwork, as well as cultural-specific psycho-socio-spiritual support, are all essential elements for optimizing quality of life and promoting death with dignity for nursing home residents facing mortality. This study provides a useful framework to facilitate the future development of EoL care in long-term-care settings in the Chinese context.

P4

Lifelong Learning and Higher Education: Opportunities and Outcomes

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Given the growing levels of education among elders in many countries, there is increased interest in how lifelong learning can enhance the well-being of older adults and offer opportunities for them to contribute to society. This presentation addresses a conceptual model for lifelong learning programs, provides two examples, and offers a context for lifelong learning research and policy agendas globally. The 4A model (Activity, Affiliation, Awareness and Attitude) by Corley emerged from research on successful aging and formed the basis of the CSULA Lifelong Learning Program Corley co-founded in 2004. In the last 10 years a Creative Longevity and Wisdom (CLW) Initiative at FGU has also offered doctoral students and community members engagement in academic and participatory programming, focused on adult learners. A doctoral concentration in CLW emerged from this initiative and reflects the 4A model. This model will be situated in relation to examples in North American, European and Asian countries as more national lifelong learning initiatives emerge. Addressing cultural considerations, diversity and disparities is a key in ensuring access to lifelong learning. Higher education initiatives across institutions such as EncoreU will be examined as they create incentives for promoting age-friendly universities and colleges. The opportunity for building a global research and policy agenda, including an examination of costs and outcomes of offering lifelong learning for older adults and society at large, will be discussed.

P5

Neighborhood Context and Recurrent Falls among Older Adults Aging in Place

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OBJECTIVE: Neighborhood context contributes both risk and protective factors for falls in community settings. Despite the high risk of institutionalization following a fall, the relationship between fall events and physical/social neighborhood environment not well understood. Social and environmental context has been linked to a variety of health outcomes among older adults, as represented in the International Classification of Functioning, Disability and Health (WHO, 2001).

METHODS: Using the WHO (2001) model, we examined the relationship between neighborhood context (neighborhood disorder and neighborhood social cohesion, each measured 0–7) and self-reported fall events among community-dwelling adults aged 65+ in the Health & Retirement Study. We first regressed baseline and longitudinal (2006–2012) falls separately on perceived disorder and cohesion measures (Mendes de Leon et al., 2009). Second, we regressed baseline and longitudinal falls (2006–2012) separately for specific neighborhood contextual factors.

RESULTS: Participants with higher perceptions of neighborhood cohesion had decreased risk of experiencing recurrent falls, both at baseline (RRR: 0.93) and longitudinally (RRR: 0.94), controlling for model covariates. Each unit increase in perceived social disorder was associated with a higher risk of experiencing recurrent falls at baseline (RRR: 1.10) and longitudinally (RRR: 1.07). Not all measures consistently predicted fall events.

CONCLUSIONS: These results have implications for clinical practice (i.e., geriatric social workers, nurses, and geriatricians assessing risk of falls for a patient) and policy (i.e. prioritize investment in certain neighborhood contextual factors). Identifying—then preventing—falls in one's community improve opportunities for aging in place. Implications for policy and practice are discussed.

P6

The Potential of Technology Supports for Healthy Aging: From Apps to Robots

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In industrialized societies a shift is occurring in the medical model toward self-management of health. Adults 65 and older are most impacted by this shift. There is tremendous potential for technology to improve health and quality of life for older adults. However, technology must be designed with older adults' needs, limitations, and preferences in mind. We must understand facilitators and barriers to technology adoption. Advanced technology design guidelines are needed to support and motivate healthy behaviors and self-management. Our work is targeted towards advancing psychological science and translating that knowledge to facilitate independence, health, and well-being. This presentation will describe our research on technology design for older adults ranging from apps (e.g. supporting medication adherence or pain management) to robotics (e.g. personal robots for the home, telepresence robots to support social connectedness, companion robots that reduce loneliness). The overarching goal is to ensure the psychological science foundation necessary for effective technology integration for supporting health-care in the lives of older adults. Our research advances deployment of technologies in the home, supports translation of fundamental scientific knowledge for design of technology tools, and guides design of emerging technologies. Technology should be empowering, enabling, efficacious, unobtrusive, and seamlessly integrated into everyday activities. Technology has the potential to improve the lives of older adults; older adults are willing to embrace those technologies that are of benefit to them; and psychological science can provide the fundamental knowledge necessary to advance these objectives.

P7

Analysis of the Clinical Risk Factors of No-reflow in Elderly Patients with ST-segment Elevation in Acute Myocardial Infarction Undergoing Percutaneous Coronary Intervention

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BACKGROUND: No-reflow during percutaneous coronary intervention (PCI) could damage the micro-vessels, and cause adverse outcome of PCI. This study aimed to investigate the clinical risk factors of no-reflow during PCI in the elderly patients with ST-segment elevation in acute myocardial infarction (STEMI).

METHODS: Sixty-eight elderly patients with STEMI successfully treated with PCI were divided into normal reflow group (54 patients) and no-reflow group (14 patients). The related risk factors of no-reflow were investigated by comparing the clinical factors between the two groups.

RESULTS: The incidence rate of no-reflow was 20.6% in the elderly patients with STEAMI undergoing emergency PCI. The differences between the no-reflow and the normal groups in systolic blood pressure (101.3 ± 20.5 vs 122.4 ± 17.6 mmHg), type 2 diabetes (71.4% vs 37%), reperfusion time (11.8 ± 2.4 vs 6.6 ± 0.5 h), and balloon expansion times (3.85 ± 1.55 vs 2.2 ± 0.88) were statistically significant ($P < .05$). It was identified by multivariate logistic regression that low systolic blood pressure (SBP < 100 mmHg [95% confidence interval (CI) 1.006–1.032, $P = .02$], type 2 diabetes (95% CI 1.984–3.245, $P = .005$), balloon expansion times (95% CI 1.545–3.345, $P = .002$) and reperfusion time (95% CI 1.545–3.345, $P = .03$) were risk factors for no-reflow in the elderly patients with STEAMI undergoing emergency PCI.

CONCLUSIONS: No-reflow in the elderly patients with STEAMI undergoing emergency PCI is associated with four clinical risk factors, SBP < 100 mmHg, type 2 diabetes, balloon expansion times and reperfusion time.

P8

Clinical Analysis of High Sensitive C-reactive Protein, Atrial Structural Remodeling and Heart Rate Variability in Patients with Persistent Atrial Fibrillation

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BACKGROUND: Atrial structural remodeling is the core of the development of atrial fibrillation. The levels of inflammatory factors are increased in patients with atrial fibrillation. This study aimed to explore the relationships of high sensitive C-reactive protein (Hs-CRP) and left atrium remodeling with heart rate variability (HRV) in patients with persistent atrial fibrillation.

METHODS: Thirty-five cases of persistent atrial fibrillation patients were enrolled as atrial fibrillation group and 35 healthy cases as control group. All cases received the detection of HRV index with 24 h of dynamic electrocardiogram, Hs-CRP concentration and the diameter of left atrium (LAD).

RESULTS: The HRV parameters such as SDNN (84.43 ± 2.86 vs 124.78 ± 3.05 ms), SDANN (80.91 ± 2.03 vs 107.28 ± 5.89 ms), RMSSD (23.65 ± 1.90 vs 31.09 ± 1.22 ms), and PNN50 ($4.33 \pm 0.33\%$ vs $8.74 \pm 0.21\%$) in atrial fibrillation group was significantly lower than that in control group ($P < .01$), while the levels of LAD (43.65 ± 5.75 vs 29.34 ± 4.23 mm) and Hs-CRP (7.20 ± 1.52 vs 2.02 ± 0.23 mg/L) were significantly higher ($P < .05$ or $P < .01$). SDNN, SDANN, RMSSD, and PNN50 values were lower in LAD > 40 mm subgroup than those in LAD ≤ 40 mm subgroup (81.07 ± 2.26 ms, 79.65 ± 0.93 ms, 20.70 ± 1.07 ms, $3.86 \pm 0.15\%$ vs 85.03 ± 2.04 ms, 83.04 ± 1.91 ms, 23.92 ± 0.48 ms, $4.43 \pm 0.25\%$), so as Hs-CRP levels (7.64 ± 0.12 vs 6.94 ± 0.43 mg/L, $P < .05$).

CONCLUSIONS: Altered atrial structure, decreased HRV index, and increased Hs-CRP levels were observed in the patients with persistent atrial fibrillation, suggesting dysregulation of the autonomic nerve system and heightened inflammation in this patient population.

P9

Clinical Observation of Gamma-knife Stereotactic Radiotherapy in 50 Elderly Patients with Advanced Lung Cancer

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BACKGROUND: The first-line therapy for advanced lung cancer in the elderly patients is chemotherapy. This study aimed to analyze the effects and prognostic factors of gamma-knife stereotactic radiotherapy on advanced lung cancer in the elderly patients.

METHODS: From January 2005 to November 2011, 50 patients aged ≥ 70 years with advanced lung cancer were only treated with gamma-knife stereotactic fraction radiotherapy and none previously had radiotherapy, chemotherapy and target therapy. A total dose of (42.30 ± 4.25) Gyr was prescribed to the 40–80% isodose line and in 5–14 fraction of 3.5–8 Gyr each for primary tumor. A total dose of (35.32 ± 5.17) Gyr were prescribed to the 40–80% isodose line and in 6–13 fraction of 3.5–5 Gyr each for metastatic tumor, one time per day, five times per week. Blood count, blood chemistry, tumor markers and imaging were tested every 1–3 months after the treatment. Follow-up was carried out for 2–54 months and ended in December 2012.

RESULTS: The response rate was 86%, with 17 cases of complete remission and 26 cases of partial remission. Overall survival time was 2–54 months with a median survival time of 12 months and an average survival time of 19.76 months. 1-, 2-, 3-, 4-year survival rate was 50%, 28%, 13% and 8% respectively. According to RTOG standard radiotherapy reaction, 5 (10%) patients presented with grade I and 3 (6%) patients with grade II hematology adverse effects. 5 (10%) patients presented with grade I and 4 (8%) patients with grade II gastrointestinal adverse effects. 12 (24%) patients presented with grade I and 3(6%) patients with grade III radiation pneumonitis.

CONCLUSIONS: Gamma-knife treatment as an optional therapy was effective for elderly patients with advanced lung cancer with mild toxicity, and could prolong patients' survival time and improve their quality of life.

P10

Significance of Endoscopic Mucosal Surface Features in Diagnosing Colorectal Polyps

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BACKGROUND: Colorectal polyps were common in the elderly. This study aimed to investigate whether the morphological features of mucosal surface observed by high definition colonoscopy without optical magnification could help predicting the histological type of colorectal polyps.

METHODS: A total of 142 cases with 330 polypoid lesions were inspected by high-definition colonoscopy. The mucosal pit type (pit pattern, PP) was evaluated by close observation and the mucosal vascular morphology (capillary pattern, CP) was observed by narrow-band imaging (NBI). Predicted histological diagnoses were compared with pathological diagnoses to determine sensitivity, specificity and accuracy.

RESULTS: The sensitivity, specificity, positive predictive value, negative predictive value, accuracy, negative likelihood ratio and positive likelihood ratio of PP combining CP in predicting the histological type of colorectal polyps were 88.2%, 83.6%, 95.5%, 64.4%, 87.3%, 5.4, and 0.1, respectively.

CONCLUSIONS: The morphological features of mucosal surface observed by close observation combined with NBI by high-definition colonoscopy could help predicting the histological type of colorectal polyps and determining further treatment and follow-up monitoring.

P11

Effect of Rosuvastatin on Inflammatory Cytokines in Patients with Acute Coronary Syndrome

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BACKGROUND: Inflammatory cytokines play an important role in promoting plaque rupture in patients with acute coronary syndrome (ACS). This study aimed to investigate the clinical effect of rosuvastatin on inflammatory cytokines in patients with ACS.

METHODS: Eighty-seven cases of ACS were divided into rosuvastatin group (44 cases) and atorvastatin group (43 cases), and the rosuvastatin group was divided into 20 mg group and 10 mg group with 22 cases in each. Another 35 healthy cases from medical examination center were enrolled as control group. The plasma levels of inflammatory cytokines were measured, and the patch shape and echogenicity, size, number and thickness were evaluated. The level of intima-media thickness (IMT) was measured.

RESULTS: The levels of TNF- α , IL-6, MMP-9 and hs-CRP in 20 mg group (66.78 ± 45.28 ng/L, 14.97 ± 4.79 pg/ml, 121.27 ± 41.72 μ g/L, 4.15 ± 1.76 mg/L) and 10 mg group (70.87 ± 46.38 ng/L, 15.73 ± 5.78 pg/ml, 126.76 ± 54.38 μ g/L, 5.37 ± 2.31 mg/L) were significantly lower than those in the atorvastatin group (251.41 ± 58.56 ng/L, 22.26 ± 6.83 pg/ml, 176.17 ± 51.93 μ g/L, 5.74 ± 2.60 mg/L, $P < .05$). The patch thickness, IMT were more significantly improved in 20 mg group (1.98 ± 0.48 mm, 1.28 ± 0.48 mm) and 10 mg group (2.23 ± 0.65 mm, 1.34 ± 0.53 mm) than those in atorvastatin group (2.52 ± 0.69 mm, 1.70 ± 0.57 mm, $P < .05$).

CONCLUSIONS: High doses of rosuvastatin could decrease plasma levels of inflammatory cytokines and stabilize atherosclerotic plaque, likely leading reduced incidence of serious heart vascular disease and improved prognosis of patients with ACS.

P12

Analysis of FEV1/FVC < 70% as a Criterion for Chronic Obstructive Pulmonary Disease in the Elderly

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BACKGROUND: A fixed value of FEV1/FVC used for diagnosis of chronic obstructive pulmonary disease (COPD) in all people might result in under diagnosis in young people and over diagnosis in old people. This study aimed to investigate the diagnostic criteria of COPD using FEV1/FVC, by describing lung function in a population aged 45 years or over.

METHODS: 508 normal subjects who came to the clinic of the First Affiliated Hospital of China Medical University for physical examination from December 2011 to December 2012 were divided into three groups according to their age: 45–59 years, 60–69 years, and ≥ 70 years. Lung function was evaluated by spirometers (Jaeger, Germany).

RESULTS: The mean values of FEV1/FVC of the three groups were 80.7%, 79.3%, 76.5%, respectively. The incidence rate of COPD defined as FEV1/FVC < 70% was 1.7%, 2.2%, 11% in 45–59 years group, 60–69 years group, and ≥ 70 years group, respectively ($P < .01$). When COPD was defined as FEV1/FVC < 65%, the incidence rate of COPD in ≥ 70 years group decreased to 4.7%.

CONCLUSIONS: Adjustment of the GOLD criteria for diagnosing COPD is necessary and FEV1/FVC < 65% in people older than 70 years would be a better choice.

P13

Blood Uric Acid Levels in Patients with Multiple System Atrophy-cerebellar Type

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BACKGROUND: Uric acid as the end-product of purine metabolism in human body shows the effects of anti-oxidative stress. This study aimed to investigate the blood uric acid level in patients with multiple system atrophy-cerebellar type (MSA-C) and to delineate the relationship between blood uric acid level and the course of MSA.

METHODS: Twenty-one patients with MSA-C in neurological ward of our hospital from June 2011 to April 2013 were enrolled as observation group. Twenty-two healthy people were enrolled as control group in the same period. Fasting blood samples were collected and blood uric acid levels were evaluated. The relationship of blood uric acid level with the course of MSA-C was analyzed.

RESULTS: Blood uric acid levels (279.19 ± 71.48 mmol/L) of patients with MSA-C were significantly lower than those of healthy people (340.96 ± 58.02 mmol/L) ($P < .05$). Blood uric acid levels were not significantly related to the course of MSA-C ($r = 0.041$, $P = .859$).

CONCLUSIONS: MSA-C patients might be influenced by oxidative stress and had lower blood uric acid levels. However, blood uric acid levels had no relations with the course of MSA-C.

P14

Clinical Efficacy of Gemcitabine Plus Cisplatin in Comparison with Gemcitabine Alone in the Elderly Patients with Advanced Non-small Cell Lung Cancer

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BACKGROUND: Most elderly patients with non-small cell lung cancer (NSCLC) were in advanced stage, and could not undergo operation. This study aimed to analyze the efficacy and toxicity of gemcitabine plus cisplatin in comparison with gemcitabine alone in patients aged 70 years or over with advanced NSCLC.

METHODS: Data from 83 patients with histologically confirmed NSCLC, staged II, III or IV, aged 70 years or over, and with an Eastern Cooperative Oncology Group Performance status 0 or 1 were retrospectively analyzed. Forty of them received gemcitabine, and forty-three received gemcitabine plus cisplatin. The primary endpoint was progression free survival (PFS).

RESULTS: The average age of the patients in gemcitabine alone group was 74.75 ± 1.72 years, compared with 75.72 ± 1.29 years in gemcitabine plus cisplatin (GC) group. Twenty-nine (72.5%) patients in gemcitabine alone group and 28 (65.1%) in GC group had at least one comorbidity. The mean number of treatment cycles completed was 4.65 and 4.02 respectively in gemcitabine alone group and GC group. Overall response rate (ORR), disease control rate (DCR) and median progression-free survival (PFS) were 25%, 70% and 4.45 months in gemcitabine alone group, compared with 39.5%, 79.1% and 6.60 months respectively in GC group. A statistically significant difference in efficacy was observed in favor of GC group, though the rate of adverse events staged III or IV was also higher in GC group than that in gemcitabine alone group (data not shown).

CONCLUSIONS: Gemcitabine plus cisplatin may be considered as an effective option with an acceptable toxicity in chemotherapy for advanced NSCLC in patients aged 70 years or over with good performance status.

P15

Clinical Efficacy of Gastrodin and Donepezil on Vascular Dementia

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BACKGROUND: It is important to improve cognitive function for patients with vascular dementia. This study aimed to observe the therapeutic effect of gastrodin and donepezil on vascular dementia.

METHODS: Forty-eight elderly patients were randomly divided into two groups (the donepezil group vs the donepezil plus gastrodin group). The mini-mental state examination (MMSE), activity of daily living (ADL), Hachinski ischemic score and the EEG expression were evaluated and the adverse effect was observed. The treatment lasted for 12 months.

RESULTS: The cognitive function, the ability of activity of daily living, and the EEG expression were improved in the donepezil group after treatment (MMSE: 18.58 ± 4.49 vs 16.13 ± 4.53 ; ADL: 41.29 ± 9.42 vs 43.54 ± 9.18 ; α wave: 33.54 ± 5.10 vs 25.90 ± 4.93 ; θ wave: 32.29 ± 7.44 vs 37.14 ± 7.48 ; δ wave: 18.68 ± 4.57 vs 21.91 ± 5.80), also improved in the donepezil plus gastrodin group (MMSE: 21.29 ± 3.99 vs 16.96 ± 4.20 ; ADL: 35.54 ± 8.83 vs 42.33 ± 9.30 ; α wave: 36.60 ± 4.87 vs 26.20 ± 4.68 ; θ wave: 27.35 ± 7.54 vs 35.40 ± 7.57 ; δ wave: 16.03 ± 3.77 vs 22.82 ± 5.94). The donepezil plus gastrodin group had even better therapeutic effect compared with the donepezil group, which was inflected in higher score in MMSE (21.29 ± 3.99 vs 18.58 ± 4.49), lower score in ADL (35.54 ± 8.83 vs 41.29 ± 9.42), and higher α band relative power value (36.60 ± 4.87 vs 33.54 ± 5.10) and lower θ (27.35 ± 7.54 vs 32.29 ± 7.44) and δ band relative power value (16.03 ± 3.77 vs 18.68 ± 4.57) in quantitative EEG.

CONCLUSIONS: Donepezil plus gastrodin could effectively improve the life quality and cerebral function of the elderly patients with vascular dementia.

P16

Relationships between Ambulatory Pulse Pressure and Serum Levels of Uric Acid and High Sensitivity C-reactive Protein in Elderly Patients with Essential Hypertension

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BACKGROUND: High sensitivity C-reactive protein (hs-CRP) might be a risk factor atherosclerosis. Serum uric acid (UA) is an important risk factor for cardiovascular disease. This study aimed to investigate the relationships between ambulatory pulse pressure (APP) and serum levels of UA and hs-CRP in elderly patients with essential hypertension (EH).

METHODS: According to the level of APP, 112 elderly patients with EH were divided into 2 groups: PP1 group ($30 \text{ mmHg} \leq \text{APP} < 60 \text{ mmHg}$, $n = 52$) and PP2 group ($\text{APP} \geq 60 \text{ mmHg}$, $n = 60$). The levels of UA, hs-CRP and related clinical biochemical index were evaluated and compared.

RESULTS: The 24 h MSBP in PP2 group was higher than that in PP1 group (135.52 ± 9.22 vs $125.83 \pm 10.63 \text{ mmHg}$, $P < .05$). The 24 h MDBP in PP2 group was lower than that in PP1 group (65.77 ± 6.66 vs $72.61 \pm 8.64 \text{ mmHg}$, $P < .05$). The levels of UA, hs-CRP in PP2 group were higher than those in PP1 group (371.63 ± 86.85 vs $311.25 \pm 74.36 \mu\text{mol/L}$, $P < .05$; 4.19 ± 1.85 vs $2.23 \pm 1.53 \text{ mg/L}$, $P < .01$). Pearson linear correlation showed that APP level in EH patients was significantly related to UA ($r = 0.206$, $P < .05$) and hs-CRP ($r = 0.427$, $P < .01$).

CONCLUSIONS: UA and hs-CRP in elderly patients with EH are closely correlated with APP, and may play an important role in the pathophysiological process of the increased APP.

P17

Efficacy of Noninvasive Positive Pressure Ventilation in the Patients with Acute Respiratory Failure Due to COPD Complicated with Multi-Pneumatocele

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BACKGROUND: Noninvasive positive pressure ventilation (NIPPV) has been widely used in the treatment of chronic obstructive pulmonary disease (COPD) complicated with acute respiratory failure. This study aimed to evaluate the efficacy of NIPPV in the treatment of acute respiratory failure due to COPD complicated with multi-pneumatocele.

METHODS: Fifty-seven patients in respiratory intensive care unit were enrolled in the study. They were divided into two groups, the treatment group ($n = 30$) and the control group ($n = 27$). While patients in the control group received usual care, patients in the treatment group received NIPPV treatment. IPAP began from $8-12 \text{ cmH}_2\text{O}$ to $16-20 \text{ cmH}_2\text{O}$, and EPAP was $2-4 \text{ cmH}_2\text{O}$, with an oxygen flow of $2-5 \text{ L/min}$. The level of SaO_2 maintained above 90%. The clinical symptoms and blood gas analysis were evaluated 2 h, 24 h, 2 days, 5 days after the therapy.

RESULTS: Clinical symptoms and blood gas parameters were improved after the treatment. PH was changed from 7.31 ± 0.12 to 7.36 ± 0.17 ($P > .05$), PaCO_2 was decreased from $69.17 \pm 11.09 \text{ mmHg}$ to $38.11 \pm 11.12 \text{ mmHg}$ ($P < .05$), and PaO_2 was increased from $50.08 \pm 12.12 \text{ mmHg}$ to $70.11 \pm 14.15 \text{ mmHg}$ ($P < .05$). Efficacy was 93.3% and 66.6% for the treatment group and control group, respectively ($P < .05$). The rate of endotracheal intubation (6.7% vs 29.2%) and length of stay (8.4 ± 3.5 vs 12.5 ± 4.3 days) were improved in the treatment group compared to those in the control group ($P < .05$), and there were no pneumothorax in either group.

CONCLUSIONS: NIPPV is an effective method for acute respiratory failure due to COPD complicated with multi-pneumatocele with minimal risk for pneumothorax.

P18

Analysis of Risk Factors in Elderly Patients with Hospital-acquired Pneumonia

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BACKGROUND: Hospital-acquired pneumonia (HAP) is one of the main factors for hospital-related mortality among elderly patients. This study aimed to investigate the risk factors of HAP in the elderly patients and to provide reference for clinical prevention and treatment.

METHODS: A case-control study was conducted, and 60 elderly hospitalized patients were divided into research group and control group with 30 cases in each according to whether they were complicated with HAP. This retrospective analysis was conducted to describe the characteristics of risk factors in the two groups. Univariate analysis and multivariate logistic regression were used for statistics analysis.

RESULTS: Compared with the control group, the ratios of patients with underlying disease, indwelling feeding tube, application of acid-suppressing drugs, hypoalbuminemia, arteriovenous catheterization, were higher in research group ($P < .05$). The results of multivariate unconditional logistic regression analysis showed that the independent risk factors for HAP in the elderly patients were cerebrovascular disease (95% CI 1.938–358.356, $P = .013$), application of acid-suppressing drugs (95% CI 1.402–11.038, $P = .032$), indwelling feeding tube (95% CI 2.245–26.303, $P = .002$) and hypoalbuminemia (95% CI 3.454–10.232, $P = .003$).

CONCLUSIONS: Elderly patients with HAP have more risk factors. Taking comprehensive precaution measures positively to reduce the risk factors is the key to decrease the incidence rate of HAP.

P19

Effect of Pelvic Floor Rehabilitation on Stress Urinary Incontinence of Elderly Female

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BACKGROUND: Stress urinary incontinence (SUI) is a common disease in the middle aged and elderly women. This study aimed to observe the effect of pelvic floor rehabilitation on SUI in the elderly female.

METHODS: Eighty-six female cases with SUI were randomly allocated into training group (n = 43) and control group (n = 43). The training group attended a pelvic floor rehabilitation course, and the control group received the routine education only. The effect was assessed after the treatment.

RESULTS: In the training group, the symptoms of all patients were improved. The symptom improvement rates in patients with mild or moderate SUI were 95.65% and 81.25%. Their pelvic floor muscle strength was 3.29 ± 0.98 after treatment, which was higher than that of pre-treatment (2.56 ± 0.42). The average 1HPTV was 5.93 ± 2.05 g before treatment and 4.16 ± 1.84 g after treatment, and there was significant difference ($P < .05$). But I-QOL showed no significant difference between the two groups after treatment ($P > .05$).

CONCLUSIONS: Pelvic floor rehabilitation course is effective in the improvement of the pelvic floor muscle function status. It is a safe and effective method in the treatment of female stress urinary incontinence.

P20

Clinical Application of Endobronchial Ultrasound-guided Transbronchial Needle Aspiration in the Diagnosis of Mediastinal Lesions and Mediastinal Lymphadenopathy

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BACKGROUND: How to confirm the pathological characteristics of mediastinal lesions is one of the difficult problems for thoracic surgeons. This study aimed to assess the efficacy and safety of endobronchial ultrasound-guided transbronchial needle aspiration (EBUS-TBNA) in qualitative diagnosis of mediastinal lesions or mediastinal lymphadenopathy.

METHODS: From April 2012 to December 2013, 203 patients with mediastinal lesions or mediastinal lymphadenopathy (minimum diameter >1 cm) who were diagnosed by chest computed tomography (CT) or PET-CT were included in this study. EBUS-TBNA was performed in all cases. All the patients with negative EBUS-TBNA were confirmed by mediastinoscopy, thoracoscopy or open thoracotomy.

RESULTS: All 203 patients underwent EBUS-TBNA. The results showed metastatic lymphnodes involvement in 180 patients, including 54 cases of small-cell lung cancer, 119 cases of non-small-cell lung cancer, 1 case of esophageal carcinoma, 1 case of cardiac cancer, 2 cases of nasopharyngeal cancer and 3 cases of lymphoma.

Twenty-three patients showed negative EBUS-TBNA, and 5 patients were diagnosed as metastatic node. The sensitivity, specificity, and accuracy of EBUS-TBNA in defining malignant lymph nodes were 96.8%, 100%, 97%, respectively. No patients were presented with complications.

CONCLUSIONS: EBUS-TBNA is a safe and effective method for evaluating mediastinal lesions.

P21

Relationship between Adiposity and the Prevalence of Hypertension in Adults

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BACKGROUND: Hypertension has been an important public health problem threatening people's health. This study aimed to investigate the relationship between the parameters of adiposity and the prevalence of hypertension in adults.

METHODS: The data of "Jiangsu Provincial Surveillance on Chronic Diseases and Risk Factors" of 2010 were used to evaluate prevalence of newly-confirmed of hypertension among different populations. The relationships of adiposity and blood pressure parameters with the prevalence of hypertension were evaluated by bivariate correlation analysis and unconditional logistic regression analysis.

RESULTS: The prevalence of new hypertension was 35.3%, which was higher in male than that in female. The prevalence showed an obvious rising trend with the increase of age. The results of bivariate correlation analysis showed that body mass index (BMI), waist circumference, waist-height ratio and percentage of body fat had positive correlation with both systolic and diastolic blood pressure. The results of unconditional logistic regression analysis showed that obesity ($\chi^2 = 129.67$, $P < .05$), central obesity ($\chi^2 = 157.05$, $P < .05$), BMI ($\chi^2 = 176.21$, $P < .05$), waist circumference ($\chi^2 = 187.73$, $P < .05$), waist-height ratio ($\chi^2 = 221.49$, $P < .05$) and percentage of body fat ($\chi^2 = 155.22$, $P < .05$), were significantly associated with hypertension. 1-SD increase of BMI, waist circumference, waist-height ratio and percentage body fat increased the risk of hypertension by 59% (OR = 1.59, 95% CI 1.50–1.70), 60% (OR = 1.61, 95% CI 1.50–1.71), 68% (OR = 1.68, 95% CI 1.57–1.79) and 149% (OR = 2.49, 95% CI 2.21–2.80).

CONCLUSIONS: BMI, waist circumference, waist-height ratio and percentage of body fat showed great influences on the prevalence of hypertension. The risk of hypertension increased with age and degree of obesity.

P22

Correlation between Serum Levels of Oxidized Low Density Lipoprotein and Carotid Artery Intima-media Thickness in the Elderly

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BACKGROUND: The increase of carotid artery intima media thickness (CA-IMT) is one of the early features of atherosclerosis (AS). Low-density lipoprotein (LDL) is an important risk factor of AS. This study aimed to determine whether serum level of oxidized LDL (ox-LDL) was associated with CA-IMT and further to study the role of serum ox-LDL in carotid artery stenosis.

METHODS: 217 elderly in-patients who received the detection of carotid artery color Doppler ultrasound during the period from October 2013 to February 2014 were enrolled. 113 cases with

CA-IMT <1.5 mm were divided into normal group and 104 cases with CA-IMT ranging 0.9–1.5 mm were divided into the thickening group. The clinical data were recorded, and levels of blood lipid and serum ox-LDL were compared between the two groups and Logistic regression correlation analysis was carried out.

RESULTS: Serum level of ox-LDL in the thickening group was significantly higher than that in the normal group (1.04 ± 0.37 vs 0.86 ± 0.47 mmol/L, $P < .01$). The multiple Logistic regression indicated that serum ox-LDL level was the most important factor to CA-IMT (OR = 2.911, $P < .01$). The serum level of ox-LDL was significantly correlated with CA-IMT ($r = 0.762$, $P < .01$).

CONCLUSIONS: Serum levels of ox-LDL of the elderly may play an important role in the prediction and diagnosis of CA-IMT.

P23

Differences in Immune Function between Elderly and Young Patients with Pulmonary Tuberculosis

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BACKGROUND: Cellular immunity plays an important role in resisting mycobacterium infection. Aging might influence the immunologic function. This study aimed to investigate the differences of clinical characteristics in cells and humeral immunity between elderly and young patients with pulmonary tuberculosis, and to explore the clinical significance of the related immune parameters.

METHODS: The immune function of 175 elderly and 151 young patients with pulmonary tuberculosis as well as 50 healthy persons were analyzed and compared. Flow cytometry FCM) was used to detect cell ratio of CD3 + T cells, CD4 + T cells, CD8 + T cells, NK cells and B cells, and the level of immunoglobulin was detected by immune nephelometry.

RESULTS: The levels of CD3 + T cells (58.55 ± 11.45 vs 73.61 ± 7.46 , 71.21 ± 5.13) and CD4 + T cells (31.4 ± 10.54 vs 41.84 ± 7.29 , 34.76 ± 4.74) were lower in elderly group than those in young and healthy groups ($P < .01$). The levels of CD4 + T cells and CD8 + T cells (29.23 ± 7.20 vs 24.50 ± 5.13) were significantly higher in young group than those in the healthy group ($P < .01$). NK cells were the highest in elderly group (24.13 ± 8.24), followed by healthy group (16.57 ± 8.33) and young group (11.68 ± 5.36), and these differences among the three groups were of statistical significance ($P < .01$). Serum IgA of elderly group (3.03 ± 1.64 g/L) and young group (2.47 ± 0.97 g/L) were higher than that of healthy group (2.00 ± 0.57 , $P < .01$).

CONCLUSIONS: Young patients with tuberculosis had significantly higher CD4 + T cells and CD8 + T cells than young healthy controls. Cell immune function of elderly patients with tuberculosis is significantly lower than healthy young control, and the main immune cells are NK cells. There are no significant differences in the humoral immune function between young and elderly patients with tuberculosis.

P24

Effects of Impaired Glucose Metabolism on Carotid Atherosclerosis in Patients with Essential Hypertension

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BACKGROUND: Patients with hypertension usually suffer from impaired glucose metabolism (IGM). This study aimed to investigate the effects of IGM on pathologic process of carotid atherosclerosis (AS) in patients with essential hypertension (EH).

METHODS: Forty-six EH patients (EH group) and 36 EH patients combined with type 2 diabetes mellitus (T2DM) were

enrolled in this study. Metabolic parameters such as body mass index (BMI), blood glucose, blood lipidogram, serum uric acid (UA) and fibrinogen (Fib), serum homocysteine (Hcy), and the carotid atherosclerotic plaques were detected in simple EH group and EH combined T2DM group. Relationship of Hcy with carotid intima-media thickness (CIMT), parameters of blood glucose were analyzed in EH combined with T2DM group.

RESULTS: BMI (22.79 ± 2.01 vs 24.33 ± 3.13 , $P < .05$), FBG (5.34 ± 0.51 vs 6.54 ± 1.13 mmol/L, $P < .01$), PBG (7.38 ± 0.50 vs 9.20 ± 1.53 mmol/L, $P < .01$), HbA1c (5.63 ± 0.42 vs 7.06 ± 0.90 , $P < .01$), TG (1.50 ± 0.55 vs 1.83 ± 0.63 , $P < .05$), LDL-C (2.28 ± 0.60 vs 2.84 ± 0.73 , $P < .01$), Apo B (0.69 ± 0.20 vs 0.80 ± 0.18 g/L, $P < .05$), serum UA and Fib in EH group were significantly lower than those in EH combined with T2DM group. The incidence rate of AS plaques (80.6% vs 56.5%, $P < .05$), and CIMT levels (1.24 ± 0.28 mm vs 1.04 ± 0.26 mm, $P < .01$) were significantly higher in EH combined with T2DM group than those in EH group. Pearson correlation analysis revealed that CIMT was positively related with the level of serum Hcy ($r = 0.482$, $P = .003$), and the level of serum Hcy was positively correlated with FPG ($r = 0.645$, $P < .01$), HbA1c ($r = 0.652$, $P < .01$), respectively.

CONCLUSIONS: IGM might aggravate vascular dysfunction and accelerate the pathologic process of AS in EH patients by interacting with other cardiovascular disease (CVD) factors synergistically. Therefore, improvement of glucose metabolism could reduce the risk of CVD in those EH patients.

P25

Evaluation of Vitamin D on Prognosis for Patients in Geriatric Intensive Care Unit

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BACKGROUND: Vitamin D deficiency is common among the elderly. This study aimed to investigate the prevalence of vitamin D deficiency in geriatric intensive care unit (ICU) and its relationship with severity of disease and prognosis.

METHODS: Patients ($n = 213$) admitted to geriatric ICU at Xinhua Hospital from June 2012 to December 2013 were included in this historical cohort study. Serum level of 25-hydroxyvitamin D [25-(OH) D] was measured and analyzed. Main biochemistry index, the incidence of multiple organ dysfunction syndrome (MODS), acute physiology and chronic health evaluationII (APACHEII) score and mortality of 30-day kept in hospital were observed and compared among patients with different serum level of 25(OH)D. Potential risk factors for mortality were analyzed by multivariate logistic regression analysis.

RESULTS: Vitamin D deficiency, insufficiency, and sufficiency was identified in 82 (38.50%), 90 (42.25%), and 41 (19.25%) patients respectively. Compared with sufficiency group, APACHE II score (21.65 ± 8.89 vs 17.05 ± 8.03 , $P < .05$), plasma lactic acid (1.89 ± 0.76 mmol/L vs 1.86 ± 1.03 mmol/L, $P < .05$), C reactive protein (101.08 ± 48.23 mg/L vs 92.48 ± 38.56 mg/L, $P < .05$), incidence of MODS and mortality (30.48% vs 9.75%, $P < .05$) were higher in deficiency group ($P < .05$). Multiple logistic regressions suggested that 25 (OH) D deficiency was an independent risk factor for mortality. The level of 25(OH) D was negatively correlated with length of stay in geriatric ICU ($r = -0.18$, $P < .05$).

CONCLUSIONS: Vitamin D deficiency is highly prevalent among patients admitted to geriatric ICU. Vitamin D deficiency is associated with disease severity and may be an independent risk factor for mortality.

P26

Effects of Pulmonary Rehabilitation in Acute Exacerbation of Chronic Obstructive Pulmonary Disease

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BACKGROUND: Acute episodes of chronic obstructive pulmonary disease (COPD) exacerbation cause significant impairment of pulmonary function. This study aimed to investigate the effects of pulmonary rehabilitation on COPD during acute exacerbation of COPD (AECOPD).

METHODS: Ninety AECOPD patients were randomly divided into test group and control group, with 45 cases in each group. The control group received conventional symptomatic treatments, including oxygen, antibiotics, bronchodilators and glucocorticoids, and the test group received pulmonary rehabilitation additionally. Lung function, modified British Medical Research Council (mMRC), 6 min walk distance (6MWD) before and after the treatment were observed and compared between the two groups.

RESULTS: There was statistical difference in lung function index, mMRC and 6MWD after treatment in the two groups. Compared with control group, the degree of dyspnea (1.8 ± 0.6 vs 2.4 ± 0.7) and 6MWD (410.8 ± 34.4 m vs 396.0 ± 20.4 m) were significantly improved in test group ($P < .05$).

CONCLUSIONS: Conventional pulmonary rehabilitation could increase the exercise tolerance, and improve the symptoms in AECOPD patients.

P27

Effects on the Function of the Tricuspid Valve Caused by Cardiac Implantable Electronic Devices

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BACKGROUND: Tricuspid regurgitation (TR) is a common valvular lesion. The incidence of TR may be increasing in frequency coincident with the use of implanted cardiac devices, such as implantable cardioverter-defibrillators (ICDs) and permanent pacemakers (PPMs). However, the role of lead-placement duration in the development of severe tricuspid regurgitation is unclear.

METHODS: A total of 410 patients, who were hospitalized in East Hospital Affiliated to Tongji University from January 2010 to January 2014, were enrolled and implanted permanent pacemakers or cardiac defibrillators or cardiac resynchronization during the period. Patients whose clinical records and their echocardiography follow up data were complete were included in this retrospective analysis. They were divided into normal and abnormal group based on the degree of tricuspid valve regurgitation in their preoperative echocardiography examination. The clinical and echocardiographic data between the groups were compared and analyzed for correlations.

RESULTS: Among the 420 enrolled patients, 316 patients were normal in preoperative tricuspid valve function and the other 104 patients were abnormal in preoperative tricuspid valve function. Follow-up period (from the first operation of implantation) was 36.06 ± 38.51 months. The result of tricuspid valve regurgitation area/right atrial area (TR/RA) from echocardiographic measurement showed that the value in postoperative measurement was higher than that in preoperative measurement ($P < .001$). In comparison with the ratio of TR/RA among the groups which were divided by the degree of TR, the new onset cases of tricuspid valve regurgitation in preoperative normal group were only 41 (12.97%) while it was significant in the postoperative groups ($P < .05$). Within the patients with normal preoperative tricuspid valve func-

tion, by comparing their preoperative ratio of TR/RA with the postoperative one after their operations of CRT, DDD, VVI, ICD it was found that TR/RA ratio was decreased in CRT subgroup with no significant difference in statistics ($P = .887$) while it was increased in DDD and VVI subgroup ($P < .001$). In comparison of the ratio between DDD and VVI subgroup, it was found that there was no significant statistical difference ($P = .478$). The incidence of TR regurgitation in the patients who were implanted cardiac implantable electronic device increased ($P < .05$) after 12 months of their operations while the difference was not significant ($P > .05$) within 12 months after operations. The postoperative degree of TR was positively correlated with the preoperative degree of TR, and the time span of device's implantation. The more serious the preoperative degree of TR, the longer the time span of device implantation was, and the larger the right atrium, the higher NT-proBNP as well as the higher the degree of postoperative tricuspid regurgitation was ($P < .05$).

CONCLUSIONS: Cardiac implantable electronic devices can cause various degree of postoperative tricuspid valve regurgitation regardless of whether the patient has the condition of preoperative tricuspid valve regurgitation or not. The degree of regurgitation is positively correlated with preoperative tricuspid regurgitation, right atrial area, NT-proBNP levels and the time span of device's implantation but it has nothing to do with the type of the implanted devices (DDD, VVI and ICD).

P28

Evaluation of Recent Prognosis of Elderly Patients with Posterior Circulation Transient Ischemic Attack through Vertebral Artery Dominance

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BACKGROUND: Posterior circulation ischemia refers to a variety of causes of vertebral basilar artery system ischemic lesions, including the posterior circulation transient ischemic attack (TIA) and cerebral infarction. TIA is an important risk factor of ischemic stroke. Therefore, timely accurate evaluation is very important. This study aimed to study the influence of vertebral artery dominance on recent prognosis of elderly patients with posterior circulation TIA.

METHODS: 108 cases of elderly patients with posterior circulation transient ischemic attack were selected with 70 patients with vertebral artery advantage and 38 patients without vertebral artery advantage. The cases were divided into low risk group ($n = 31$), medium-risk group ($n = 65$) and high-risk group ($n = 12$) according to ABCD2 score for cerebral infarction risk stratification. The Vertebral artery MRA, Clinical data ABCD2 score and the incidence of cerebral infarction within 7 days were collected and their differences were compared.

RESULTS: There were 31 patients (28.7%), who developed into cerebral infarction in 108 patients with transient ischemic attack. Incidence of cerebral infarction in medium-risk group was higher than that in the low risk group (30.8% vs 9.7%, $P < .05$), and it was higher in the high-risk group than in the low risk group (66.7% vs 9.7%, $P < .01$) and the medium-risk group (66.7% vs 30.8%, $P < .05$). The incidence of cerebral infarction in group with the vertebral artery dominance was significantly higher than that of the non-vertebral artery dominance group (35.7% vs 15.8%, $P < .05$). If ABCD2 score was ≤ 4 points, there was no significant difference in incidence of cerebral infarction between the groups

with vertebral artery dominance and non-vertebral artery dominance (9.1% vs 11.1%, $P > .05$). If ABCD2 score was $\geq$ four points, the incidence of cerebral infarction in vertebral artery dominant group was significantly higher than that of the non-vertebral artery dominant group (47.9% vs 17.2%, $P < .05$).

CONCLUSIONS: The vertebral artery dominance is one of the risk factors for cerebral infarction for patients with posterior circulation TIA. With ABCD2 score in the medium-risk and high-risk patients, the risk of vertebral artery advantage is particularly evident.

P29

The Relationship between Serum Levels of Insulin-like Growth Factor-1 and Cystatin C and Diabetic Nephropathy in Elderly Patients with Type 2 Diabetes

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Supported by: Natural Science Foundation of Shanghai (13ZR1414700)

BACKGROUND: Diabetic nephropathy (diabetic nephropathy, DN) is a common chronic complications of diabetes, and it has become a major clinical cause of chronic renal failure. Insulin-like growth factor-1 (IGF-1) is a multifunctional regulator of cell proliferation, which is closely related to the occurrence and development of DN. This study aimed to investigate the relationship between changes of serum insulin-like growth factor-1 (IGF-1) and cystatin C levels in patients with diabetic nephropathy.

METHODS: According to the urine albumin excretion rate (UAER) level, the patients were divided into simple type 2 diabetes mellitus (T2DM) group, early diabetic nephropathy (DN) group, the clinical stage of DN group, and another 60 healthy persons as control group. Serum IGF-1 and cystatin C as well as urine albumin were measured using electrochemiluminescence immunoassay.

RESULTS: IGF-1 levels of each group increased significantly ($P < .05$) compared with control group; the IGF-1 levels in the early DN group, clinical DN group is significantly higher than simple T2DM group ($P < .05$); compared with the serum cystatin C levels in the control group and the simple T2DM group, it increased significantly in early and clinical stage of DN group, the difference was statistically significant ($P < .05$).

CONCLUSION: There was a significant correlation between the occurrence and development of the levels of IGF-1, serum cystatin C and diabetic nephropathy in type 2 diabetics, and it is a sensitive indicator with early implications.

P30

Association of Androgen, Androgen Receptor and AR Gene (CAG)n Repeat Polymorphism with Obesity in Older Males

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OBJECTIVE: This study aimed to investigate androgen, androgen receptor level and AR gene CAG length polymorphism associated with obesity in older males.

METHODS: A total of 323 males aged from 60–92 years old were randomly selected from Wanshoulu area, Beijing. Luteinizing hormone (LH), follicle-stimulating hormone (FSH), estradiol (E_2) and total testosterone (TT) were measured by chemiluminescence technique. Free testosterone (FT), sex hormone binding globulin (SHBG) and dehydroepiandrosterone sulphate (DHEA-s) were

measured by ELISA. Mean fluorescent channel (MFC) of androgen receptor in peripheral lymphocyte was analyzed by flow cytometry. DNA sequencing of the CAG repeat from the N-terminal domain of the first exon of the AR gene were analyzed by applied Biosystems 3730xl DNA Analyzer.

RESULTS: With increase of waist circumference, DBP and E_2/T were increased gradually, and SHBG were decreased gradually, $P < .01$. TT was descended, and E_2 was ascended in the heavy obesity (waist circumference ≥ 97 cm), $P < .05$. MFC had no correlation with BMI, WHR and waist circumference. CAG repeats were positively associated with BMI ($P = .044$), but its association was no longer significant after adjusting for age ($P = .067$). Multiple linear regression analysis showed that E_2 was positively related to waist circumference, and TT was negatively correlated with waist circumference.

CONCLUSION: TT and SHBG were decreased, and E_2 and E_2/T were increased in obesity. E_2 positively and TT negatively were related to waist circumference. MFC and CAG length polymorphism had no correlation with body fat distribution.

P31

Relationship between Carotid Remodeling and Plaque Stability by Magnetic Resonance Imaging

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BACKGROUND: This paper aimed to study the relationship between carotid remodeling index (CRI) and plaque tissue characteristics and clinical factors by high-resolution MRI.

METHODS: Forty-six consecutive hypertensive patients with carotid plaque ≥ 1.5 mm (74 carotid plaque, MaxWT = 3.64 ± 1.33 mm) were selected, controlling for ischemic symptoms, treatment with renin-angiotensin system inhibitors and other clinical factors associated with vascular remodeling. Carotid plaque composition (intraplaque hemorrhage, lipid-rich necrotic core and calcification), fibrous cap status and carotid artery morphological measurements (normalized wall index and carotid remodeling index) were analyzed by blinded review.

RESULTS: Table 1 summarizes effects and lack of for major demographic and some clinical factors on CRI. CRI was significantly higher in symptomatic patients compared with asymptomatic patients (OR = 2.12, $P < .05$), and it was significantly lower in receiving renin-angiotensin system inhibitors patients ($\beta = -0.60$, $P = .001$). Normalized wall index was positively correlated with CRI ($r = 0.47$, $P < .001$). Compared with intact-thick fibrous caps, the higher CRI increased the probability of the rupture of fibrous caps (OR=4.63, $P < .05$).

CONCLUSIONS: Carotid remodeling is correlated with plaque stability in the hypertension patients. Monitoring CRI by magnetic resonance imaging may be able to warn the rupture of fibrous caps and the risk of ischemic stroke.

Table 1: Logistic regression analysis of the effect of clinical and imaging factors on the CRI.

Effect Factors	R value	P value
Age	-0.088	0.453
BMI	-0.036	0.730
Stability of hypertension	0.178	0.090
Course of hypertension	0.069	0.945
Taking ACEI/ARB	-0.342	0.001
Normalized Wall Index	3.577	0.001

Figure: The difference of CRI in the different status of fibrous cap.

P32

Association between the Variability of Self-measured Blood Pressure at Home and Cold Pressor Test in Elderly Patients with Mild or Moderate Essential Hypertension

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BACKGROUND: Variability in blood pressure (BP) is independently associated with target organ damage. Self-monitoring of blood pressure by patients at home can provide information on day-to-day variability of BP under relatively well-controlled conditions. However, the association of variability in self-measured BP at home and autonomic nerve function is unknown.

METHODS: From May 2012 to August 2013, 259 elderly patients with mild or moderate essential hypertension were recruited from residential communities in Shandong Province, China. Self-measured BP at home and cold pressor test were completed for all study patients. Parameters of variability in self-measured BP at home were calculated from measurements of the last six consecutive days. Function of autonomic nerve was assessed using cold pressor test. Participants were divided into positive response group (n = 116) and the negative response group (n = 143) according to the results of cold pressor test.

RESULTS: The results showed that in the positive group, morning diastolic (D) BP, standard deviation (SD) of systolic (S) BP, SD of DBP, maximum SBP, maximum DBP, difference between morning and evening SBP and DBP, difference between maximum and minimum SBP, difference between maximum and minimum DBP and mean heart rate were significantly higher than those in the negative response group (all $P < .05$). Evening SBP and evening DBP were lower in the positive response group than those in the negative response group (both $P < .05$). Correlation and regression analyses showed that the increments of 0s and 60s in SBP were significantly influential factors for the SD of SBP and DBP, difference between morning and evening SBP and DBP, difference between maximum and minimum SBP, and difference between maximum and minimum DBP (all $P < .05$).

CONCLUSION: There was a significant association between the variability in self-measured blood pressure at home and the response to cold pressor test in elderly patients with mild or moderate essential hypertension. Higher response to cold pressor test was associated with greater variability in blood pressure.

P33

A Cross-sectional Study of Cognitive Performance in Multiple System Atrophy-Parkinsonian Variant and Parkinson's Disease

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BACKGROUND: The clinical manifestations of multiple system atrophy-Parkinsonian variant (MSA-P) largely overlap with that of Parkinson's disease (PD). Cognitive impairment is well documented in PD, while dementia is a required feature in current consensus diagnostic criteria for MSA. Nevertheless, increasing evidence indicates that cognitive impairment is more common than is recognized, affecting single or multiple cognitive domains and varying from mild cognitive impairment to severe dementia. This study aimed to examine the difference of the cognitive function and clinical characteristics in MSA-P and PD by using systemic assessing scales for cognition in this study.

METHODS: Nineteen patients with MSA-P and 36 patients with PD were consecutively selected. Mini Mental Status Examination (MMSE) and Montreal cognitive assessment (MoCA) scale were used to assess the general cognition. Digital span, similarities, block design and picture arrangement from Wechsler Adult Intelligence Scale-Revision (WAIS-R) as well as visual recognition, visual reproduction and logical memory from Wechsler Memory Scale - Revision (WMS-R) were used to evaluate the cognitive performances in executive function, visuospatial ability, attention and memory.

RESULTS: The activities of daily living (ADL) score and Unified Parkinson's Disease Rating Scale III (UPDRS-III) score were significantly higher in MSA-P group compared to PD group ($P < .05$). The MMSE and MoCA performances were similar between MSA-P and PD patients. On the verbal scale score from WAIS-R and total WMS-R score, patients with MSA-P showed lower scores ($P < .05$). As to different cognitive domains, patients with MSA-P had worse performances in block design, picture arrangement and logical memory ($P < .05$).

CONCLUSIONS: Patients with MSA-P had a faster clinical progression and worse life quality than patients with PD. The general cognitive impairments in MSA-P were similar to those in PD, but impairments in visuospatial function, executive function and memory were more severe in patients with MSA-P.

P34

The Effect of Clinical Intervention for Glucose Excursion on Vascular Endothelial Vasodilatory Function in the Elderly Type 2 Diabetic Patients

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BACKGROUND: The incidence of myocardial infarction and other heart disease is high in the elderly type 2 diabetes mellitus. This study aimed to investigate whether vascular endothelial fluctuation is correlated with glucose, and to demonstrate the therapeutic effects under low-sugar diet for the elderly type 2 diabetic patients.

METHODS: The elderly type 2 diabetic patients were divided into three groups, control group (65–83 years old, n = 21), low-sugar diet group (73–82 years old, n = 20) and insulin therapy group (70–83 years old, n = 25). Blood glucose levels were measured by continuous glucose monitoring, and the indexes of vascular endothelial function was measured using standard laboratory kits.

RESULTS: 8-iso-PGF 2 α , TG, endothelin, FPG, SIB, MAGE and MODD were significantly lower in low-sugar diet group than in control group ($P < .05$). 8-iso-PGF 2 α and endothelin were positively correlated with HbA1c, MAGE, SIB and LAGE in low-sugar diet group and insulin therapy group ($P < .05$). Multiple linear stepwise regression analysis revealed that HbA1c, LAGE and MAGE were significantly correlated with endothelin ($P < .05$), and MODD were significantly associated with 8-iso-PGF 2 α ($P < .05$).

CONCLUSIONS: Although low-sugar diet had no obvious advantage in therapeutic effects with insulin treatment, it could be considered as a practical method to prevent the risk of cardiovascular disease in the elderly type 2 diabetic patients after recurrence. In addition, the results also indicate that blood glucose levels play a significant role in vascular endothelial function in type 2 diabetes.

P35

Correlation of Ambulatory Blood Pressure Rhythm and Morning Blood Pressure Surge in Elderly Hypertensive Patients with Renal Dysfunction

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BACKGROUND: Both abnormal circadian rhythm of blood pressure and morning blood pressure surge are important risk factors for target-organ damage. Our goal was to investigate the correlation of ambulatory blood pressure rhythm and morning blood pressure surge in elderly hypertensive patients with renal dysfunction.

METHODS: One hundred thirty eight hypertensive patients were enrolled from July 2012 to August 2013. Twenty-four-hour ambulatory blood pressure monitoring (ABPM) was detected in all patients. Then, patients were divided into dipper rhythm group and non-dipper rhythm group according to the results of ABPM. Similarly, patients were also divided into morning surge group and non-morning surge group. Plasma creatinine and urine microalbumin were measured in all patients. Glomerular filtration rate was evaluated using CKD-EPI formula.

RESULTS: Compared to non-dipper rhythm group, urine microalbumin was significantly lower [(22.67 ± 10.81) mg/L vs (38.92 ± 16.34) mg/L, $P < .01$], and glomerular filtration rate was markedly higher [(105.93 ± 16.47) ml/min·1.73 m² vs (90.61 ± 10.73) ml/min·1.73 m², $P < .01$] in dipper rhythm group. Compared to non-morning surge group, urine microalbumin was significantly higher [(37.16 ± 16.57) mg/L vs (23.58 ± 10.25) mg/L, $P < .01$], and glomerular filtration rate was markedly lower [(92.38 ± 11.23) ml/min·1.73 m² vs (103.76 ± 14.34) ml/min·1.73 m², $P < .01$] in morning surge group. Non-dipper rhythm and morning surge of blood pressure were independent risk factors for urine microalbumin and glomerular filtration rate, respectively. There was a synergistic effect of non-dipper rhythm and morning surge of blood pressure on urine microalbumin and glomerular filtration rate.

CONCLUSION: Twenty-four-hour ambulatory blood pressure rhythm and morning blood pressure surge were independent risk factors for renal dysfunction, respectively, and there was a synergistic effect on renal dysfunction in elderly hypertensive patients.

P36

Changes and Clinical Significance of Serum Hepcidin – 25 and Iron Metabolism Index in Acute Ischemic Stroke

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BACKGROUND: Dysregulated iron metabolism has been suspected to be linked to atherosclerosis and to cardiovascular disease (CVD). For the purpose of clarifying the relationship between dysregulated iron metabolism and acute ischemic stroke, this study aimed to evaluate change of serum hepcidin-25 and iron metabolism index in acute ischemic stroke patients and analyze their relationship with the severity and early prognosis of the disease.

METHODS: A total of 178 patients presenting with acute ischemic stroke (AIS) were recruited in the case group. Based on the National Institutes of Health Stroke Scale (NIHSS) score, the patients in the case group were divided into 57 cases of light group, 62 cases of medium group and 59 cases of heavy group according to their severity of neurological deficit. The levels of serum hepcidin-25 and ferritin were tested within 24 h, 3, 7, 14 and 28 days after stroke onset. And 150 physically healthy subjects were selected as controls.

RESULTS: The difference of the levels of serum hepcidin-25 and ferritin within 24 h, 3, 7, 14 and 28 days after stroke onset in AIS group was significant ($P < .01$). The levels of serum hepcidin – 25 and ferritin in different stages were higher in AIS group than in control group (all $P < .05$). The difference of the levels of serum hepcidin-25 and ferritin among the light, medium and heavy group was significant ($P < .001$). The levels of serum ferritin and IL-6 were positively correlated with the level of serum hepcidin-25 ($r = 0.992, 0.809, P < .001$).

CONCLUSIONS: The changes of serum hepcidin-25 and ferritin may serve important bioindicators to assess the severity and prognosis of AIS. These results suggest that iron metabolism may be involved in the whole process of cerebral infarction.

P37

Correlation between Cerebral Microbleeds and Circulating Endothelial Microparticles in Lacunar Stroke Patients

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BACKGROUND: The incidence of cerebral microbleeds (CMB) in elderly persons without cerebrovascular disease was 5% and the incidence of CMB in patients with ischemic stroke was 17%. CMB is a cerebral small vessel disease with bleeding tendency. However, the relationship of CMB with circulating endothelial microparticles (EMP) remains unclear.

METHODS: A cohort of 84 patients after 3 months of first-ever lacunar stroke onset were enrolled. The patients were divided into two groups: 33 cases of CMB group and 51 cases of without CMB group according to the results of MRI imaging study detection. The clinical baseline data from all patients were collected. The plasma levels of CD31+/CD42-EMP were detected by flow cytometry.

RESULTS: Serum contents of EMP were remarkably higher in CMB group than in without CMB group and the difference was significant [(451.93 ± 94.46) and (310.93 ± 55.47), $P = .000$]. There was positive correlation between the serum level of EMP and the severity of CMB ($r = 0.817, P = .0001$). Binary logistic regression analysis showed that after adjusting for age, hypertension and fibrinogen, elevated level of EMP was still an independent risk factor that affected the incidence of CMB.

CONCLUSIONS: Elevated level of EMP was an independent risk factor that affected the incidence of CMB. Therefore, the detection of serum contents of EMP could be used as a tool to predict the presence of CMB.

P38

Candesartan Effects on Insulin Sensitivity of the Skeletal Muscle in Spontaneously Hypertensive Rats

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BACKGROUND: Candesartan is one of the angiotensin II receptor blockers (ARBs) known to reduce blood pressure through direct inhibition of the binding between the angiotensin II and their receptors on vascular smooth muscle cells. In recent years, many clinical studies have found that ARBs can reduce the incidence of diabetes. It is hypothesized that candesartan could improve the body's insulin sensitivity by blocking the angiotensin II disruptive effect on skeletal muscle cells. This study was designed to investigate the effects and mechanism of candesartan on insulin resistance of the skeletal muscle in spontaneously hypertensive rats (SHR).

METHODS: Thirty male SHR (12-week-old) were randomly divided into model group, high dose group and low dose group. Ten male WKY (12-week-old) rats served as control group. All the rats were fed with fructose, while the high and low dose group rats received candesartan consecutive treatment with appropriate dose (0.8 mg/kg/d for high dose group and 0.4 mg/kg/d for low dose group). For the first 8 weeks, the serum samples were collected from all the four-group rats. Fasting insulin (FINs) and angiotensin II (AngII) levels were measured using standard assays. At the same time, the gene and protein level of protein kinase B (Akt) in the skeletal muscles were detected by RT-PCR and Western Blot method.

RESULTS: The AngII, FINs and insulin resistance index (HOMA-IR) in model group rats were significantly higher than those in the control (FINs: $13.3 \pm 2.8 \mu\text{U/ml}$ vs $9.0 \pm 2.9 \mu\text{U/ml}$, $P < .01$; HOMA-IR: 3.3 ± 0.8 vs 2.3 ± 0.9 , $P < .05$), and their Akt mRNA and protein level in skeletal muscle were significantly lower than those in the control (mRNA: 0.58 $\pm$ 0.12-fold vs 0.95 $\pm$ 0.06-fold, $P < .01$; protein: 0.09 $\pm$ 0.02-fold vs 0.14 $\pm$ 0.04-fold, $P < .01$). With the treatment of candesartan, the FINs and HOMA-IR were significantly lower in high dose candesartan group than in the model group (FINs: $9.7 \pm 2.6 \mu\text{U/ml}$ vs $13.3 \pm 2.8 \mu\text{U/ml}$, $P < .01$; HOMA-IR: 1.9 ± 0.6 vs 3.3 ± 0.8 , $P < .05$), while their Akt mRNA and protein level in skeletal muscle were significantly higher than those in the model group rats (mRNA: 0.90 $\pm$ 0.25-fold vs 0.58 $\pm$ 0.12-fold, $P < .01$; protein: 0.13 $\pm$ 0.03-fold vs 0.09 $\pm$ 0.02-fold, $P < .01$).

CONCLUSION: AngII can cause the insulin resistance by reducing the regulation of Akt mRNA and protein expression in the skeletal muscle, while candesartan can improve skeletal muscle insulin resistance by inhibiting the effect of AngII.

P39

Prevention of cerebrovascular events by high-dose Folic Acid Supplement in Elderly Female Patients with Hypertensive Emergency and Hyperhomocysteinemia

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BACKGROUND: Hyperhomocysteinemia (Hcy) has been associated with an increased risk of cardiovascular and cerebrovascular events (CCE) in Chinese population, especially elderly female

with hypertensive emergency (HE). This study was to make the assessment of the dosage that folic acid supplementary could better reduce the events of CCE for older women with HE.

METHODS: Elderly females ($y \geq 65$ year, $n = 319$) with HE and homocysteine (Hcy) ($>10 \mu\text{mol/L}$) were selected for this study from June 2006 to June 2009. Medical information was gathered, and all the patients were told to have regular medication, to have well salt restriction ($\leq 5 \text{ g/d}$) and to take vitamin B₁₂ 500 $\mu\text{g/d}$. They were randomly divided into two groups: low dose folic acid (LDFA) (0.4 mg/d) group ($n = 136$) and high dose folic acid (HDFA) (0.8 mg/d) group ($n = 144$). All patients were followed for 5 years monitoring for ischemic stroke admission events (ISAE) orthostatic hypotension admission events as measures of CCE.

RESULTS: After 5 years of follow-up, ISAE in the supplemental HDFA group were significantly lower than those in the LDFA group [41.0% vs 54.4%, $P = .02$]. By comparison of the hospitalization of orthostatic hypotension events, HDFA group was significantly lower than the incidence of LDFA group [45.8% vs 68.4%, $P = .00$]. By logistic regression analysis, the danger of index value (OR) of HDFA group was 0.765, namely the HDFA group can reduce ISAE 23.5% than LDFA group ($P = .021$, $P < .05$). By Kaplan-Meier, HDFA group was better than the LDFA group in reducing hospitalized rate of cerebrovascular disease ($P = .002$, $P < .05$) and acute coronary syndrome ($P = .115$, $P > .05$).

CONCLUSION: Keeping a stable blood pressure, salt restriction and taking VitaminB₁₂, the supplementary folic acid (0.8 mg/d) could further prevent CCE in older women with HE and Hcy than folic acid (0.4 mg/d).

P40

Changes of Brachial Artery Endothelial Function and Carotid Intima-media Thickness in Elderly Patients with Pre-diabetes

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BACKGROUND: Pre-diabetes is considered to be a risk factor for cardiovascular disease. Since the population with pre-diabetes is large and the stage is reversible, it is significant to study their large vascular lesions. However, little is known about brachial artery endothelial function and carotid intima-media thickness (IMT) of elderly people with pre-diabetes.

METHODS: Subjects (60–79 years old, $n = 143$) were divided into four groups by Oral Glucose Tolerance Test (OGTT): the group of normal glucose tolerance (NGT, 18 male, 14 female, age: 68 ± 4), the group of pure impaired fasting glycaemia (IFG, 20 male, 15 female, age: 69 ± 3), the group of pure impaired glucose tolerance (IGT, 22 male, 18 female, age: 70 ± 4), and the group of impaired glucose regulation, with combined IFG and IGT (IGR, 21 male, 15 female, age: 69 ± 5). TC, TG, LDL-C, HDL-C, FPG, FINS, HbA1c, and cystatin C (Cys C) were measured after fasting overnight. 2hPG and 2hINS were measured after 2 h of OGTT. Ultrasound evaluation was conducted on brachial artery endothelial function and carotid IMT. Multiple stepwise regression analysis was used to assess the influence factors of EDD and IMT.

RESULTS: Carotid IMT in the group of NGT was lower [(0.83 $\pm$ 0.13) mm vs (0.98 $\pm$ 0.15) mm, (0.99 $\pm$ 0.14) mm, (0.99 $\pm$ 0.16) mm, $P = .025$, 0.023, 0.023], and brachial artery endothelium-dependent dilation (EDD) was higher [(7.46 $\pm$ 0.96)% vs (6.45 $\pm$ 0.92)%, (5.60 $\pm$ 0.95)%, (4.85 $\pm$ 0.86)%, $P = .037$, 0.009, 0.005] than those of the three remaining groups. EDD gradually were reduced in the group of IFG and the group of IGT as well as the group of IGR ($P < .05$), but IMT had no obvious change ($P > .05$). Stepwise regression analysis showed that EDD was negatively related to FPG, 2hPG, 2hINS, HbA1c and Cys C ($r = -0.56$, -0.68 , -0.57 , -0.59 , -0.64 , $P < .05$). IMT was

positively related to TC, TG and LDL-C ($r = 0.62, 0.59, 0.65$, $P < .05$).

CONCLUSION: Gradual decrease of brachial artery EDD and increase of carotid IMT were observed in elderly patients with pre-diabetes.

P41

Blood Coagulation and its Influencing Factors in Patients with Diabetes Mellitus

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BACKGROUND: Diabetes mellitus (DM) can induce imbalanced blood coagulation system. The aim of this study was to investigate the difference of coagulation state between diabetic patients and healthy individuals and to find out the possible factors associated with abnormal coagulation state in diabetic patients.

METHODS: Two hundred and fifty-one DM patients served as a DM group and were further divided into great vessel lesion group ($n = 163$) and non-great vessel lesion group ($n = 88$). Eighty subjects undergoing physical examination served as a control group. Their thrombin time (TT), activated partial thromboplastin time (APTT) and prothrombin time (PT) were recorded and their serum levels of fibrinogen (Fib), fasting blood-glucose (FBG), glycosylated hemoglobin A1c (HbA1c), total cholesterol (TC), low density lipoprotein (LDL-C), high density lipoprotein (HDL-C) was also measured.

RESULTS: The TT, APTT, and PT were significantly shorter and the serum Fib levels were significantly higher in DM group than in control group (16.28 ± 1.44 s vs 17.80 ± 0.97 s, 35.57 ± 3.73 s vs 37.46 ± 4.17 s, 12.60 ± 1.08 s vs 13.84 ± 0.60 s, 3.21 ± 0.87 g/L vs 2.91 ± 0.55 g/L, $P = .01$, $P = .00$, $P = .04$, $P = .04$, respectively). The serum levels of HbA1c, TC and LDL-C were the major influencing factors for blood coagulation in DM patients. Great vessel lesion was related with the elevated serum Fib level.

CONCLUSIONS: Abnormal blood coagulation was closely related with the serum levels of HbA1c, TC, and LDL-C in DM patients.

P42

Clinical Study of Symptomatic Carotid Artery Stenosis and Concomitant Intracranial Aneurysms

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Supported by: Natural Science Foundation of China (81371296)

BACKGROUND: This study aimed to investigate the clinical features, incidence and clinical management of symptomatic carotid artery stenosis and concomitant intracranial aneurysms.

METHODS: We performed a retrospective review of all patients with symptomatic carotid artery stenosis who underwent digital subtraction angiography (DSA) at our institution between 2012 and 2013. Eighteen patients were found to have un-ruptured intracranial aneurysms in 287 patients with a stenosis $\geq 50\%$. They were divided into aneurysm intervention group ($n = 13$) and non-intervention group ($n = 5$). All patients were followed up by phone or at outpatient services for 3–12 months, and they had imaging results.

RESULTS: The incidence of intracranial aneurysm in symptomatic carotid artery stenosis patients was 6.3%. 17 patients had risk factors such as hypertension, diabetes mellitus or smoking history. Patients in the aneurysm intervention group were followed up without stroke or death, but 1 patient had aneurysms recurrence. No patient in the non-intervention group had aneurysms rupture in

follow-up period. 1 patient with transient ischemic attack was improved after antiplatelet therapy.

CONCLUSIONS: The incidence of intracranial aneurysm in symptomatic carotid artery stenosis patients was higher than that of normal population. Patients with symptomatic carotid artery stenosis and concomitant intracranial aneurysms can get superior effectiveness by individual treatment.

P43

Relationship of Polymorphisms in CXCL16 Genes and its Serum Level with Atherosclerosis Ischemic Stroke

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BACKGROUND: Experimental and clinical observations suggest that CXCL16, a recently discovered transmembrane chemokine combining functions of a chemokine and a scavenger receptor, could be an important player in atherosclerosis, but the relationship of its gene polymorphism with atherosclerosis ischemic stroke has not been extensively studied. Our aim was to study the relationship of CXCL16 gene polymorphism and its serum level with atherosclerosis ischemic stroke and its risk factors.

METHODS: One hundred and twenty atherosclerosis ischemic stroke patients served as a patient group and 75 individuals undergoing physical examination served as a control group in this study. The distribution of CXCL16 gene A181V polymorphism in 2 groups was analyzed by PCR-RFLP. The serum CXCL16 level was measured by ELISA.

RESULTS: Patient group had significantly higher frequency of arachidonic acid (AA) genotype in CXCL16 gene A181V and A allele than the control group (35.8% vs 20.0%, $\chi^2 = 8.702$, respectively, $P = .013$; 59.6% vs 44.0%, $\chi^2 = 9.013$, respectively, $P = .003$). After adjustment of age, smoking, blood glucose, blood pressure by multivariate logistic regression analysis, the incidence of ischemic stroke was higher in those with AA genotype of CXCL16 gene A181V ($OR = 3.734$; 95% CI 1.291–10.805; $P = .015$). However, presence of A181V genotype was not associated with the severity of ischemic stroke ($P = .733$). Serum CXCL16 levels were significantly higher in patient group than those in control group [(0.25 ± 0.20) ng/L vs (0.09 ± 0.15) ng/L, respectively, $P = .000$]. They are also higher in person with AA genotype of CXCL16 gene A181V than those in those with GG genotype of CXCL16 gene A181V in patient group and control group [(0.28 ± 0.21) ng/L vs (0.12 ± 0.11) ng/L, respectively, $P = .001$; (0.16 ± 0.18) ng/L vs (0.03 ± 0.02) ng/L, respectively, $P = .025$]. In addition, CXCL16 genotype was significantly associated with higher serum CXCL16 levels ($r = 0.359$; $P = .000$).

CONCLUSIONS: The AA genotype of CXCL16 gene A181V could up-regulate CXCL16 expression and convey significant risks for atherosclerosis ischemic stroke.

P44

The Clinical Research of Plasma Platelet Activity in Patients with H Type Hypertension

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OBJECTIVE: This study aimed to evaluate plasma platelet activity in patients with H type hypertension.

METHODS: One hundred and sixty-nine patients with essential hypertension were randomly classified into the H type hypertension group ($n = 86$) and the control group ($n = 83$). Mean

values of 24 h ambulatory blood pressure and concentrations of fasting serum TXB2, GMP-140, CD62p, platelet aggregation rate and Homocystein (Hcy) were measured in two groups.

RESULTS: There were no significant differences in the mean value of ambulatory blood pressure between the two groups ($P > .05$). But the differences in concentrations of TXB2, GMP-140, CD62p, platelet aggregation rate and Hcy between the two groups showed statistical significance ($P < .05$ or $P < .01$).

CONCLUSION: The platelet activity was increased in patients with H type hypertension, and resulted in happening of cardiac disease and cerebrovascular disease. Positive measures should be taken to control it.

P45

Research on the Relationship between Inflammatory Reaction of Carotid Atherosclerotic Plaque and Serum Bilirubin

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BACKGROUND: There has been a lot of evidence showing that the main cause of cardiovascular event is the instability of plaque. Inflammatory reaction is one of the major causes of its instability. With the development of molecular imaging techniques, and through ¹⁸F-FDG positron emission tomographic imaging (PET-CT) we can apply noninvasive techniques to quantitatively monitor the degree of inflammatory reaction in carotid plaques. Serum bilirubin has been considered as a harmful metabolite. However, recent studies show that decrease of serum bilirubin was associated with increased incident of cardiovascular events. In vitro, bilirubin is proved to be a strong antioxidant and anti-inflammatory factor. So there is the possibility of bilirubin that reduces the risk of cardiovascular events by inhibiting the inflammatory response in the plaques, however, there is no in vivo supporting evidence. Our objective was to investigate the relationship between the degree of inflammatory reaction of carotid atherosclerotic plaque and serum bilirubin levels.

METHODS: Thirty patients with intimal-medial thickness of carotid artery ≥ 2 mm (detected by ultrasound) were included. All the patients underwent PET-CT to measure the target-to-background ratio (TBR) to quantify the degree of plaque inflammation. Correlations between TBR and serum bilirubin and between TBR and cardiovascular risk factors were analyzed by Pearson's correlation analysis and multiple linear regression analysis.

RESULTS: Pearson correlation analysis showed that TBR was significantly correlated with BMI ($r = 0.38$, $P = .038$), ApoB ($r = 0.392$, $P = .032$), ApoB/apolipoprotein A1 ($r = -0.534$, $P = .002$), and direct serum bilirubin ($r = -0.535$, $P = .002$). After adjusting these risk factors, multiple linear regression analysis showed that direct bilirubin was the only variable that was correlated independently with TBR ($\beta = -0.257$, $P = .002$).

CONCLUSIONS: Our results indicate that serum direct bilirubin was independently associated with the degree of inflammatory reaction of carotid atherosclerotic plaque. The association between decreased level of serum direct bilirubin and increased incident of cardiovascular events may be due to independent correlation between serum direct bilirubin levels and the degree of inflammatory reaction of atherosclerotic plaques.

P46

Effect of Different Permanent Pacemakers on Cardiac Function in Aged Patients

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BACKGROUND: More and more permanent pacemakers have been used in different patients for cardiac dysfunctions. Many studies evaluated the efficacy and safety of pacemaker, few evaluated the long-term use of pacemakers in aged patients. In this study, our objective was to assess the long-term (≥ 2 years) effect of different permanent pacemakers on cardiac function in aged patients.

METHODS: Fifty-three older patients with pacemakers were divided into single-chamber pacemaker group ($n = 28$) and double-chamber pacemaker group ($n = 25$). Thirty aged patients without obvious cardiac disease served as a control group. Their left ventricular end-diastolic volume (LVEDV), left ventricular end systolic volume (LVESV) and left ventricular ejection fraction (LVEF) were recorded and their quality of life was assessed according to the SF-36 scale.

RESULTS: At the end of follow-up, LVEF was significantly lower in the single-chamber pacemaker group than itself before implantation, double-chamber pacemaker group and the control group ($53.84 \pm 8.47\%$ vs $59.56 \pm 7.88\%$, $59.45 \pm 8.15\%$, $64.11 \pm 5.64\%$, respectively, $P < .05$). The LVESV was significantly higher in the single-chamber pacemaker group than itself before implantation, the double-chamber pacemaker group and the control group (47.73 ± 13.33 ml vs 39.27 ± 9.79 ml, 38.88 ± 12.15 ml, 34.50 ± 9.95 ml, respectively, $P < .05$). As to the LVEDV, the changes among the groups didn't reach statistical significance ($P > .05$). After pacemaker implantation in both groups, several index values in SF-36 showed increasing trends comparing with their individual baselines, although no statistically significant difference was detected ($P > .05$).

CONCLUSIONS: Permanent single-chamber pacemaker can reduce the cardiac systolic function in aged patients. The effect of double-chamber pacemaker is better than that of single-chamber pacemaker on cardiac function in aged patients.

P47

Analysis of the Association between Smoking and Middle Cerebral Arterial Stenosis in Males with Ischemic Stroke

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Supported by: Guangdong Provincial Science and Technology Foundation, China (No. 2012B031800410)

BACKGROUND: Middle cerebral artery stenosis is an important cause of ischemic stroke. In our clinical work, we often come across middle cerebral artery stenosis in young male patients who have only one risk factor of smoking. This study was designed to investigate the relationship between smoking and middle cerebral arterial stenosis as well as the onset age of ischemic stroke.

METHODS: A total of 257 males were diagnosed as ischemic stroke, and they were consecutively admitted in Guangdong General Hospital, and were recruited and evaluated by cerebral digital subtraction angiography. According to their smoking status, they were divided into smoker group ($n = 140$) and non-smoker group ($n = 117$). The prevalence of middle cerebral arterial stenosis (M1) and the onset age of ischemic stroke were evaluated, respectively.

RESULTS: The prevalence of middle cerebral arterial stenosis in smoker group was significantly higher than that in non-smoker

group (49.3% vs 24.8%, $P < .01$). Moreover, the onset of ischemic stroke was earlier in smoker group [(60.13 ± 10.52) years old vs (65.26 ± 11.77) years old, $P < .01$].

CONCLUSION: Smoking is probably an important risk factor for middle cerebral arterial stenosis (M1) and can result in earlier onset of ischemic stroke in males.

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A Comparative Analysis of Clinical and MRI Features of Pontine Infarction

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OBJECTIVE: This paper aimed to summarize the clinical and MRI features of pontine infarction and to study the mechanism underlying different types of pontine infarctions.

METHODS: Forty five pontine infarction patients who underwent cerebral MRI and MRA were included in this study. Their pontine infarction was classified into 5 types according to its location on MRI. The relation of different types of pontine infarction with basilar artery sclerosis or stenosis was analyzed.

RESULTS: Hypertension, diabetes mellitus, and carotid arteriosclerosis were the independent risk factors for pontine infarction ($P = .029$, $P = .002$, $P = .000$). Paramedian artery infarction (type A) and watershed area infarction (type D) were detected in 19 patients (42.2%) and 16 patients (35.6%), respectively. The main clinical manifestations of the patients were hemiplegic paralysis (73.3%), dysarthria (55.6%) and hemihypalgesia (28.9%). No pontine damage was observed in 53.3% of the patients. The incidence of basilar artery sclerosis or stenosis was significantly higher in patients with type D pontine infarction than in those with type A pontine infarction ($P = .004$).

CONCLUSION: Paramedian artery infarction and watershed area infarction are the main types of pontine infarction, which must be diagnosed according to their MRI. Type D pontine infarction usually suggests pons basilar artery sclerosis or stenosis while type A pontine infarction is mainly caused by small artery occlusion.

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Predictive Factors and Protective Factors after Primary Percutaneous Coronary Intervention in Patients with No-reflow

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BACKGROUND: According to reports, incidence of no-reflow was higher than 15% when patients with acute myocardial infarction (AMI) experienced percutaneous coronary intervention (PCI). No-reflow can reduce the clinic benefit of PCI greatly. The method of diagnosing those critical patients with no-reflow and early intervention will lower the incidence of no-reflow.

METHODS: A total of 258 AMI patients undergoing primary PCI were enrolled and divided into no-reflow group and reflow group for the study. Levels of troponin I (cTnI), high-sensitivity C-reactive protein (hs-CRP), von Willebrand factor (vWF), brain natriuretic peptide (BNP), and use of the loading-dose atorvastatin in both groups were compared. Risk factors and protective factors of no-reflow phenomenon were analyzed using multivariate logistic regression analysis.

RESULTS: Levels of hs-CRP, vWF, BNP were significantly higher in no-reflow group than in reflow group [(31.6 ± 4.9 vs

23.3 ± 4.0) mg/L, (738 ± 94.7 vs 635 ± 100.2) μg/L, and (409.9 ± 122.8 vs 271.5 ± 99.4) ng/L, $P < .01$]. The use of loading-dose atorvastatin (10.5% vs 40.9%) also showed significant difference statistically ($P < .05$). Multivariate logistic regression analysis showed that hs-CRP, vWF, BNP were the risk factors of no-reflow, and loading-dose atorvastatin usage was the protective factor against this phenomenon.

Table 1. Comparison of clinical data between the two groups

Item	No-reflow group (38 cases)	Reflow group (220cases)	P
Age	61.72±4.81	60.58±7.45	0.292
male [n (%)]	23 (60.5)	138 (62.7)	0.870
Smoking [n (%)]	21 (55.3)	130 (59.1)	0.822
Hypertension [n (%)]	19 (50.0)	100 (45.5)	0.775
Diabetes [n (%)]	16 (42.1)	70 (31.8)	0.412
Drug treatment [n (%)]			
Aspirin	38 (100)	220 (100)	1
Clopidogrel	38 (100)	220 (100)	1
ACEI or ARB	25 (65.8)	156 (71.0)	0.769
Beta blockers	17 (44.7)	142 (64.5)	0.218
Nitrate	32 (84.2)	190 (86.4)	0.789
Infarct artery [n (%)]			
LAD	21 (55.3)	122 (55.4)	0.726
LCX	6 (15.8)	38 (17.2)	0.900
RCA	11 (28.9)	60 (27.3)	0.781
stent diameter (mm)	3.18±0.31	3.03±0.28	0.124
Stent length (mm)	29.78±3.64	29.23±4.49	0.677
Balloon pressure (atm)	12.56±1.65	12.27±1.67	0.595

Table 2. Comparison of laboratory test results and loading dose of atorvastatin application between two groups

Item	No reflow group (38 cases)	Reflow group (220cases)	P
cTnI (μg/L)	8.19±1.17	3.05±1.03	0.000
hs-CRP (mg/L)	31.6±4.9	23.3±4.0	0.000
vWF (μg/L)	738±94.7	635±100.2	0.002
BNP (ng/L)	409.9±122.8	271.5±99.4	0.000

Table 3 Multivariate logistic regression analysis

Item	β	Standard error	OR	95% CI	P
hs-CRP	0.366	0.108	1.442	1.166~1.783	0.001
vWF	0.012	0.005	1.012	1.003~1.022	0.009
BNP	0.010	0.003	1.010	1.004~1.017	0.002
Loading dose of atorvastatin	-1.712	0.866	0.181	0.033~0.986	0.048

CONCLUSIONS: hs-CRP, vWF, BNP levels were predictors of no-reflow phenomenon. Loading-dose atorvastatin usage can reduce no-reflow phenomenon.

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The Value of "Nine Points Method" in the Prediction of Primary Cerebral Hemorrhage Patients with Hematoma Enlargement and Adverse Clinical Outcomes

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BACKGROUND: If patients present symptoms of hematoma enlargement, they will have a poor outcome. If we can predict hematoma enlargement, and give more intervention, it might reduce the mortality rate and patients' poor outcome. Professor H. Bart Brouwers presented that CTA spot sign and "nine points" scoring methods could predict hematoma enlargement. But its clinical application is unclear.

METHODS: Seventy nine patients with spontaneous cerebral hemorrhage were selected from our hospital. Based on the scores obtained using this "nine points method", 55 patients were in the 0–3 group and 24 cases in the 4–9 group. The imaging methods included original image for identifying "spot sign" of CTA as well as baseline CT scan for hematoma volume calculation. The baseline data in the two groups, NIHSS score, hematoma volume, the number of surgical operation intervention rate, and mortality rate were compared in order to confirm the significance of this method. The "nine points method" was also compared with the "spot sign" to predict hematoma expansion in PPV, NPV, sensitivity, and specificity.

RESULTS: The average age of 55 patients in 0–3 group was 62.7 ± 16.2 years, and the average age of 24 patients in 4–9 group was 60.4 ± 18.4 years. Compared with 0–3 group, point syndrome positive rate and hematoma enlargement, 30 days NIHSS score, surgical intervention, 90 day mortality in 4–9 group was significantly increased. The difference was statistically significant ($P = .022, 0.000, 0.000, 0.014, 0.000, 0.003$) respectively. The "nine points method" was used to predict hematoma expansion. The positive predictive value (PPV) was 50%, negative predictive value (NPV) 96.4%, sensitivity 85.7%, and specificity 81.5% respectively. A total of 71 patients underwent CTA examination, 12 people with positive "sign" accounted for 16.9%, and 10 people had hematoma enlargement. There were 59 people with negative "little sign", and 4 had hematoma enlargement. The positive predictive value of "syndrome" (PPV) was 83.3%, and the negative predictive value (NPV) was 93.2%. Sensitivity was 71.4%, and specificity was 96.5%. Comparing the simple use of "spot sign" and "nine point method", PPV decreased and NPV increased

2 groups of test results

Group	0 ~ 3 group (55 cases)	4 ~ 9 group (24 cases)	P
No. of spot-positive patients (%)	2(3.6%)	10(41.7%)	0.000
Hematoma Expansion (%)	2(3.6%)	12(50%)	0.000
30-d NIHSS	5.28 ± 2.43	11.13 ± 3.67	0.014
Surgical evacuation(%)	2(3.6%)	14(58.3%)	0.000
3-mo mortality(%)	1(1.8%)	5(20.8%)	0.003

CONCLUSION: "Nine points method" could effectively predict hematoma enlargement and poor outcome. The clinical practice should be combined with the "nine point method" for the treatment in patients with cerebral hemorrhage.

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A Preliminary Study of Prediction of Cog-12 to the Deterioration of Cognition in the Elderly

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BACKGROUND: Due in part to the lack of brief measures, Alzheimer disease (AD) and other dementias were under diagnosed in the community. There had been some studies confirming the value of Cog-12 in cognitive function screen, therefore, the value of this scale was validated to predict the deterioration of cognitive in the elderly. As Cog-12 was composed of two parts, scores

of each part of the scale and total score were analyzed in this study.

METHODS: At the beginning, 56 individuals with normal cognition or mild cognitive impairment (MCI) patients were enrolled in this averagely two-year study. The Cog-12 scale, the Montreal cognitive assessment (MoCA) and the Mini Mental Status Examination (MMSE) were administered to all the patients. The cognitive function was evaluated at the baseline and after averagely two years of follow-up. According to the cognitive function at baseline, normality or MCI, the individuals were divided into two groups to analyze respectively.

RESULTS: There were 17 MCI patients converting to Alzheimer disease patients (AD group), while 10 MCI patients did not convert (MM group). 18 individuals maintained normal cognitive function (NC group) after following up, while 11 patients converted to MCI (CM group). The MoCA and MMSE scores were significantly higher whereas the Cog-12-I score was significantly lower in NC group than that in CM group ($P < .05$). The MoCA score was significantly higher whereas the total Cog-12-I and Cog-12 scores were significantly lower in MM group than that in AD group ($P < .05$). The area under ROC curve showed that the sensitivity of Cog-12 score in predicting MCI was 81.4%, when its cutoff value was 4.5. (See the Table)

CONCLUSIONS: Cog-12 scale was an effective tool for assessing the intelligence, mental state and behavior of patients with cognitive impairment.

	NC (n=18)	CM (n=11)	P	MM (n=10)	AD (n=17)	P
Baseline						
Cog-12-I	2.2 ± 1.3	3.7 ± 2.0	0.017	5.7 ± 3.9	11.1 ± 7.0	0.037
Cog-12-II	0.5 ± 0.9	0.2 ± 0.4	0.214	1.3 ± 1.4	2.1 ± 2.2	0.305
Cog-12	2.7 ± 1.5	3.9 ± 2.1	0.072	7.0 ± 4.7	13.8 ± 8.5	0.030
MoCA	27.1 ± 1.4	25.6 ± 1.8	0.013	22.0 ± 2.8	18.4 ± 4.0	0.024
MMSE	29.7 ± 0.8	28.3 ± 1.3	0.011	25.4 ± 1.9	25.2 ± 2.9	0.805
After 2 years						
Cog-12-I	2.5 ± 1.6	6.7 ± 3.9	0.003	8.2 ± 7.1	17.6 ± 7.4	0.007
Cog-12-II	0.5 ± 0.9	0.4 ± 0.5	0.783	2.1 ± 3.2	3.6 ± 3.4	0.310
Cog-12	3.1 ± 1.9	7.1 ± 4.2	0.042	10.3 ± 9.6	21.2 ± 10.3	0.021
MoCA	27.5 ± 1.4	22.3 ± 3.0	0.000	22.1 ± 3.7	13.1 ± 6.4	0.002
MMSE	29.1 ± 1.2	26.8 ± 3.2	0.025	26.5 ± 3.0	18.2 ± 6.3	0.003

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Early Prediction of Cerebral Infarction Risk after TIA with ABCD² Score and hs-CRP

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BACKGROUND: This study aimed to explore the predictive value of ABCD² score combined with high sensitive C-reactive protein (hs-CRP) for early cerebral infarction risk in transient ischemic attack (TIA) patients.

METHODS: TIA patients ($n = 192$) were divided into low risk group ($n = 113$) and moderate to high risk group ($n = 79$) according to the ABCD² scores. Clinical data and laboratory test results were collected. The 7d and 90d incidence of cerebral infarction after the presence of TIA was observed. The patients were then divided into hs-CRP ≤ 3.0 mg/L group and hs-CRP > 3.0 mg/L group according to their serum hs-CRP levels, and we observed the 7 days and 90 days incidence of cerebral infarction. Receiver operating characteristic curve (ROC) were used to calculate the area under the curve (AUC) and to evaluate the performance of ABCD² scores group, hs-CRP group and ABCD² scores group combined with hs-CRP group in predicting cerebral infarction.

RESULTS: Age ≥ 60 years (63/79 vs 61/113), a history of hypertension (66/79 vs 50/113), diabetes (44/79 vs 33/113) and hyperlipidemia (29/79 vs 13/113) were more common in the moderate to high risk group compared to those in the low risk group and the difference was statistically significant ($P < .05$). The serum hs-CRP levels were higher in the moderate to high risk group compared to those of the low risk group and the difference was statistically significant (3.61 ± 4.83 vs 1.63 ± 2.92 mg/L, $P < .05$). Cerebral infarction occurred 0 and 24 (30.4%) cases respectively in the low risk group and moderate to high risk group within 7 days after TIA ($P < .001$), and 5 (4.4%) and 33 (41.8%) cases within 90 days ($P < .001$). The patients were 149 cases in hs-CRP ≤ 3.0 mg/L group and 43 cases in hs-CRP > 3.0 mg/L group. Cerebral infarction occurred 7 cases (4.7%) in the hs-CRP ≤ 3.0 mg/L group and 17 cases (39.5%) in the hs-CRP > 3.0 mg/L group within 7 days after TIA ($P < .001$), and 11 (7.4%) and 27 (62.8%) cases within 90 days ($P < .001$) respectively. For predicting 7d cerebral infarction risk, the AUC of ABCD2 scores group, hs-CRP group and ABCD2 scores group combined with hs-CRP group were 0.836, 0.777 and 0.928, for 90 days risk, the AUC were 0.785, 0.803 and 0.903 respectively, all $P < .05$.

CONCLUSIONS: ABCD² scores predicted the early cerebral infarction risk, and the additional use of hs-CRP levels improved the accuracy of the prediction.

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Clinical Value of Plaque Thickness Evaluated by CT Angiography among Moderate to Severe Carotid Stenosis Patients

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OBJECTIVE: This study aimed to investigate the relationship between different thickness of plaque composition and cerebrovascular events (cerebral infarction or TIA) on CTA images among moderate to serve carotid stenosis patients.

METHODS: This is a retrospective analysis of 120 neck CTA scanning patients. The patients were divided into symptomatic group (77) and asymptomatic group (43) according to the cerebrovascular events. Extracranial carotid stenosis $\geq 50\%$ was defined as moderate to severe stenosis. The thickness of main components and the CT value in soft plaque were measured. Logistic regression analysis was used to study the relevance with cerebrovascular events.

RESULTS: Thickness of soft plaque in cerebrovascular event group was significantly greater than that of asymptomatic group [(3.29 ± 1.00) mm vs (2.51 ± 0.80) mm], and thickness of hard plaque in cerebrovascular event group was smaller than that of asymptomatic group [(1.44 ± 0.97) mm vs (1.94 ± 0.75) mm, $P < .01$]. According to logistic regression analysis, soft plaque thickness was closely related to cardiovascular events (OR = 4.882, 95% CI 1.914–12.448, $P = .001$).

CONCLUSION: Soft plaque thickness was significantly associated with cardiovascular risks, and CTA source image provides an easy and reliable clinical evaluation method to evaluate the incidence of cerebrovascular accident of moderate to severe carotid artery stenosis.

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Correlation of OSAS with Blood Pressure, Diabetes Mellitus and Peripheral Atherosclerosis in the Elderly Patients

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BACKGROUND: Obstructive sleep apnea syndrome (OSAS) is common in the elderly patients. Whether there are correlations between OSAS and peripheral atherosclerosis, hypertension, and diabetes mellitus (DM) in the elderly patients is still unknown. This study aimed to investigate these relationships.

METHODS: One hundred and thirteen elderly OSAS patients who had been admitted to our hospital from January 2010 to October 2014 served as an experimental group and the patients were further divided into mild OSAS group (65–86 years old, mean age 73.6 ± 9.0 years, $n = 32$), moderate OSAS group (60–81 years old, mean age 71.5 ± 7.3 years, $n = 33$) and severe OSAS group (63–83 years old, mean age 71.7 ± 8.5 years, $n = 48$) according to their AHI. Forty age-, sex-, and smoking-matched subjects served as a control group (62–84 years old, mean age 69.6 ± 7.9 years, $n = 40$). The patients underwent femoral artery ultrasonography. Their 24 h blood pressure and serum HbA1c level were measured.

RESULTS: The incidence of T2DM and hypertension was significantly higher in mild, moderate and severe OSAS groups than that in control group (28.1%, 33.3%, 50.0% vs 0%, $P < .05$). The incidence of T2DM was significantly higher in severe OSAS group than that in mild OSAS group (50.0% vs 28.1%, $P < .05$). The BMI and the incidence of hyperlipidemia and fatty liver were significantly higher in severe OSAS group than that in control group (27.7 ± 3.1 vs 24.4 ± 2.5 , 68.8% vs 40.0%, 56.3% vs 27.5%, all $P < .05$). Systolic blood pressures (SBP mmHg) were significantly higher in moderate and severe OSAS groups than that in control group in the 24 h mean SBP (134.7 ± 11.3 , 134.8 ± 11.5 vs 124.6 ± 10.9 , $P < .05$), daylight SBP (135.4 ± 14.1 , 135.6 ± 13.6 vs 126.9 ± 11.7 , $P < .05$), night SBP (133.8 ± 10.2 , 134.5 ± 11.0 vs 122.8 ± 11.6 , $P < .01$), vigil SBP (141.2 ± 15.9 , 143.5 ± 18.0 vs 125.2 ± 11.9 , $P < .01$) and diastolic BP (79.7 ± 7.2 , 79.9 ± 6.7 vs 74.4 ± 8.4 , $P < .05$). Multivariate logistic regression analysis showed that unstable plaques in carotid and femoral arteries were positively correlated to AHI ($B = 0.135$, $P = .017$, $B = 0.118$, $P = .029$).

CONCLUSIONS: Hypertension, T2DM, hyperlipidemia and unstable plaques in carotid and femoral arteries are more likely to occur in severe OSAS patients.

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Relationship between Nocturnal Pulse Pressure Index and Serum 25-hydroxyvitamin D3 Level in Elderly Men with Hypertension

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BACKGROUND: Primary hypertension is an important risk factor of cardiovascular and cerebrovascular disease. With the application of ambulatory blood pressure monitoring, it was found that hypertensive patients with abnormal circadian rhythm, pulse pressure or pulse pressure index had a high risk of occurring cardiovascular and cerebrovascular disease. This study aimed to investigate the relationship between serum 25-hydroxyvitamin D3 level and nocturnal pulse pressure index in elderly men with hypertension.

METHODS: Four hundred and seventy-one cases in our department were selected. With blood pressure monitoring, they

were divided into two groups according to the level of nocturnal pulse pressure index (PPI), low level group (PPI <0.46) and high level group (PPI ≥0.46). Their body height, body weight, BMI and serum levels of fasting blood glucose, blood lipid and HbA1C were measured. Their serum 25-hydroxyvitamin D3 level was measured by ELISA. The difference of above parameters between the two groups and the factors affecting nocturnal pulse pressure index were analyzed.

RESULTS: Serum 25-hydroxyvitamin D3 of the group with low nocturnal pulse pressure index was higher than that of the group with high nocturnal pulse pressure index [(38.47 ± 20.20) nmol/L vs (34.18 ± 12.78) nmol/L, respectively, $P = .006$], and there was significant difference in grading of 25-hydroxyvitamin D3 level ($P < .05$). Multivariate logistic regression analysis revealed that nocturnal pulse pressure index was independently related with serum 25-hydroxyvitamin D3 and age.

CONCLUSIONS: Among elderly men with hypertension, level of nocturnal pulse pressure index was significantly and inversely relative with 25-hydroxyvitamin D3 and low 25-hydroxyvitamin D3 level may be an independent risk factor for high level of nocturnal pulse pressure index in elderly men with hypertension.

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Utility of Ultrasonography in Diagnosis and Treatment of Acute Pulmonary Embolism in the Elderly

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OBJECTIVE: This study aimed to evaluate the utility of ultrasonography in diagnosis and treatment of acute pulmonary embolism (APE) in the elderly.

METHODS: Twenty-seven elderly APE patients (71–96 years of age, mean age 79.7 ± 9.3 years old) were included in this study. They underwent echocardiography, lower extremity deep vein ultrasonography, distal oxygen saturation test, blood pressure and plasma D-dimer level measurement before the onset and after treatment of APE.

RESULTS: Systolic and diastolic blood pressure and distal oxygen saturation level were significantly lower, while the serum D-dimer level was significantly higher at the onset of APE than before the onset of APE in elderly APE patients ($P < .05$). Blood pressure and distal oxygen saturation level returned to normal, while the serum D-dimer level decreased significantly after treatment ($P < .05$) Table 1. The right atrium and ventricle were significantly larger, the pulmonary aorta and inferior vena cava were significantly wider, the reflex speed in tricuspid valve was significantly rapider, and the systolic pressure of pulmonary artery was significantly higher after treatment than before treatment, (Table 2). The incidence of deep vein embolism in lower extremity was 77.8% detected by ultrasonography in elder APE patients.

CONCLUSION: Echocardiography and lower extremity deep vein ultrasonography in combination with other clinical tests were necessary for the early diagnosis and treatment of APE in the elderly.

Table 1 The partial clinical details in elder patients comparability in before APE, underway APE and after treatment APE ($\bar{x} \pm s$)

	before APE (n=27)	underway APE (n=27)	after treatment APE (n=23)
blood pressure (mmHg)			
systolic blood pressure	141±21	103±16 ^a	131±22 ^a
diastolic blood pressure	82±14	61±12 ^a	78±16
distal hemooxygen saturation (%)	96±4.92	80±6.71 ^a	93±5.42
serum D-dimer (mg/l)	2.12±0.78	11.07±1.89 ^a	0.49±0.13 ^a

annotate: compared with before APE, a: $P < .05$; 1mmHg=0.133kPa

Table 2 The details of echocardiography in elder patients before APE, underway APE

and after treatment APE ($\bar{x} \pm s$)

	RASD (mm)	RVDD (mm)	IVC (mm)	MPA (mm)	TRV (m/s)	TRPG (mmHg)	SPAP (mmHg)
Before onset(27cases)	35.6±4.1	34.2±3.7	16.6±2.3	22.6±2.4	2.92±0.78	34.87±3.01	39.13±4.6
At the onset(27cases)	44.3±4.6 ^a	42.1±3.5 ^a	24.1±2.6 ^a	27.7±2.8 ^a	4.16±1.89 ^a	69.58±6.45 ^a	72.24±9.8
After relief of disease(23cases)	37.0±3.9	36.3±3.1	17.4±2.1	24.4±2.5	3.21±1.13	40.34±4.21	45.76±4.2

annotate: compared with before APE, a: $P < .05$

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Role of Stromal Interacting Molecule 1 in Forming Atherosclerotic Plaques in ApoE Gene Knock-out mice

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BACKGROUND: Atherosclerosis is characterized by a complex multifactorial pathophysiology, which involves lipid disorder, oxidative stress, and vascular wall remodeling. Intracellular calcium concentration change is closely related to proliferation and migration of vascular smooth muscle cell. This study was to investigate the association between stromal interacting molecule 1 (STIM1) and formation of atherosclerotic plaques by observing the changes of STIM1 in apoE gene knock-out mice.

METHODS: Forty-five male apoE gene knock-out mice aged 7–8 weeks served as a model group and 45 age-matched wild type C57BL/6J mice served as a control group in this study. The animals were killed 20, 27 and 33 weeks after they were fed with hyperlipidic diet. Aortic tissue samples taken from the animals were stained with HE and Masson to calculate the ratio between plaque area and luminal area. The serum lipid level was measured with oil red O staining. Expression of STIM1 in smooth muscle cells was detected with immunohistochemical staining. Expression of STIM1 in the formation of vulnerable plaques was detected by Western blot.

RESULTS: The expression levels of STIM1 was significantly higher in model group than that in control group (0.2352 ± 0.0165 vs 0.1435 ± 0.0262 , 0.2985 ± 0.0241 vs 0.1226 ± 0.0270 , 0.3552 ± 0.0340 vs 0.1385 ± 0.0156 , $P < .01$) and increased with the age. The ratio of positive STIM1 expression area was positively related with the adjusted plaque area ($r = 0.754$, $P < .01$) and negatively related with the adjusted collagen area ($r = -0.783$, $P < .01$).

CONCLUSIONS: STIM1 was involved in the formation of atherosclerotic placelously related with their formation and progress.

P58**Dynamic Changes of High-sensitivity C-reactive Protein Levels in Patients with Atherosclerotic Cerebral Infarction at High Altitude**

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BACKGROUND: High altitude affects inflammatory response in patients with stroke. This study aimed to investigate the plasma dynamic level of high-sensitivity C-reaction protein (hs-CRP) in patients with acute atherosclerotic cerebral infarction at high altitude.

METHODS: Fifty-eight native and 55 Han nationality patients with atherosclerotic cerebral infarction were selected as case 1 and case 2 group respectively from affiliated hospital of Qinghai University (about 2,200 m height) and Hainan Tibetan Autonomous Prefecture Hospital (about 3,000 m height). At the same period, another 60 native and Han nationality volunteers at the Health Center were selected respectively as control 1 and control 2 group. Nephelometry was used to determine the concentration of plasma hs-CRP, and neurologic functional default scoring (CNFDS) was scored at different time within 1 month and then compared and analyzed.

RESULTS: Plasma hs-CRP dynamic levels in case 1 and 2 group were higher than those in the control 1 and 2 group [$(4.46 \pm 1.95, 8.87 \pm 2.01, 5.45 \pm 1.13, 3.84 \pm 0.82)$ mg/L vs (1.89 ± 0.92) mg/L and $(6.68 \pm 1.43, 11.14 \pm 2.12, 7.49 \pm 1.21, 6.33 \pm 1.12)$ mg/L vs (2.02 ± 0.96) mg/L], the differences had statistical significance ($P < .05$). The plasma dynamic level of hs-CRP in case 2 group was higher than that in case 1 group, and the difference had statistical significance ($P < .05$). The plasma level of hs-CRP in severe group of CNFDS were higher than that in medium group, and medium group of CNFDS were higher than that of mild group, and the difference had statistical significance ($P < .05$).

CONCLUSIONS: Inflammatory response may play a role in acute atherosclerotic cerebral infarction and secondary brain damage at high altitude. The plasma of hs-CRP concentration increases and secondary brain damage maybe aggravate in the hypoxic environment.

P59**Associations between AMPK Concentration and Metabolic Syndrome in the Elderly of Tianjin**

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BACKGROUND: Imbalance of energy metabolism is a major cause for metabolic syndrome. AMP-activated protein kinase (AMPK) is a major component of a protein kinase cascade that plays a pivotal role in the regulation of intracellular energy. By testing of the AMPK concentration in peripheral blood of the elderly, it is hoped to find out the associations between the AMPK concentration and metabolic syndrome (MS).

METHODS: A total of 595 elderly patients were enrolled and divided into experimental group (MS group) and control group (non-MS group) according to the diagnostic standards of MS. The AMPK concentration was tested with venous blood sample of each patient in both groups after 8 h fasting. Patients in experimental group with hypertension, hyperglycemia or hyperlipidemia were given corresponding therapy according to the guideline and all

patients in MS group received health lifestyle education. One year later, the test of AMPK concentration was repeated for every patient in order to observe the change of AMPK concentration before and after medical intervention.

RESULTS: The BMI, FPG, systolic pressure, diastolic pressure and TG were higher in MS group than those in non-MS group. HDL-C and the AMPK concentration in MS group was lower than that in non-MS group ($P < .01$). After one year of intervention in MS group, the FPG [$(6.36.3 \pm 1.1)$ mmol/L vs (6.8 ± 1.2) mmol/L, $P = .000$], systolic pressure [(136.3 ± 12.3) mmHg vs (145.3 ± 13.6) mmHg, $P = .003$], diastolic pressure [(80.3 ± 8.6) mmHg vs (89.9 ± 9.1) mmHg, $P = .000$], and TG [(2.2 ± 0.4) mmol/L vs (2.5 ± 0.3) mmol/L, $P = .008$] decreased, while the HDL-C [(1.0 ± 0.3) mmol/L vs (0.8 ± 0.1) mmol/L, $P = .010$] and AMPK concentration [(4.65 ± 0.22) nmol/L vs (4.07 ± 0.24) nmol/L, $P = .006$] increased than before.

CONCLUSIONS: The elderly with MS has a lower AMPK concentration, with the improvement of blood pressure, blood glucose and lipid level after relative therapy. AMPK concentration in peripheral blood increases. So AMPK is associated with MS and reflects the condition of MS.

P60**Influence of Overweight and Obesity on Coagulation and MPV in Elderly Patients with Essential Hypertension**

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BACKGROUND: Hypertension is a well-recognized risk factor for coronary artery disease, atherosclerosis, and cerebrovascular disease. Some comorbidities in hypertension, such as overweight and obesity, are increasingly recognized in the elderly, predisposing them to cardiovascular disease. Previous studies have related overweight and obesity with their high risks of cardiovascular events, but the mechanisms are fully understood. As thrombosis is an important part of cardiovascular disease, obesity may lead to cardiovascular events by influencing the occurrence of thrombus. This study aimed to analyze the relationships between overweight, obesity, blood coagulation indexes, and mean platelet volume, and to investigate overweight, obesity and their influences on blood coagulation, MPV in elderly patients with essential hypertension.

METHODS: According to the body mass index (BMI), a total of 204 elderly essential hypertension patients were divided into control group ($n = 79, 18.5 \text{ kg/m}^2 \leq \text{BMI} < 23.9 \text{ kg/m}^2$), overweight group ($n = 67, 24.0 \text{ kg/m}^2 \leq \text{BMI} < 28 \text{ kg/m}^2$) and obese group ($n = 58, \text{BMI} \geq 28.0 \text{ kg/m}^2$). We measured and compared plasma Fibrinogen (Fib), plasma D-dimer, antithrombin-III, platelet distribution width, MPV, etc. Fib, D-dimer, MPV were made correlation analysis with BMI.

RESULTS: Compared with the control group, levels of Fib of the overweight and obese group were significantly higher ($2.8 \pm 0.70 \text{ g/L}$ vs $3.1 \pm 0.77 \text{ g/L}$ vs $3.4 \pm 0.8 \text{ g/L}$, $P < .05$). Compared with the control group, the level of D-dimer of the overweight and obese group were significantly higher ($165.0 \pm 31.6 \mu\text{g/L}$ vs $178.9 \pm 49.9 \mu\text{g/L}$ vs $194.9 \pm 43.9 \mu\text{g/L}$, $P < .05$). Compared with the control group, levels of MPV of the overweight and obese group were significantly higher ($9.9 \pm 0.6 \text{ fl}$ vs $10.1 \pm 0.7 \text{ fl}$ vs $10.4 \pm 1.0 \text{ fl}$, $P < .05$). Pearson correlation analysis showed that Fib level was positive correlated with BMI significantly ($r = 0.393$, $P < .01$). D-dimer level was positively correlated with BMI significantly ($r = 0.252$, $P < .01$). MPV level was positively correlated with BMI significantly ($r = 0.275$, $P < .01$).

CONCLUSION: Obesity and overweight were related with impaired blood coagulation and increased platelet activation in elderly patients with essential hypertension.

P61

Effects of Rosuvastatin on the Carotid Vulnerable Plaque in the Elderly Patients

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BACKGROUND: The detection rate of carotid artery plaque in aging population increased every year, especially vulnerable plaque. The role of statins in atherosclerosis has been confirmed by many research institutes in recent years. This study aimed to observe the clinical effects of rosuvastatin on carotid vulnerable plaque in the elderly patients.

METHODS: Elderly patients with carotid vulnerable plaque ($n = 103$) were randomly divided into three groups: small dose rosuvastatin group ($n = 38$, taking rosuvastatin 5 mg qd); conventional dose rosuvastatin group ($n = 37$, taking rosuvastatin 10 mg qd); control group ($n = 28$, only treating with lifestyle intervention). Blood levels of blood lipid, high-sensitivity C-reactive protein (hs-CRP), neck intima-media thickness (IMT), area of atherosclerotic clots and plaque Crouse score were assessed and compared among the three groups before and after six months treatment, and the incidence of adverse reactions and ischemia stroke were observed after six months treatment.

RESULTS: All above parameters had no significant changes after six months lifestyle intervention in control group (data not shown). The total cholesterol (TC), low density lipoprotein cholesterol (LDL-C), hs-CRP were significantly lower in small dose group and conventional dose group ($t = 2.259, 2.191, 2.221, 5.565, 5.680, 14.43, P < .05, P < .01$). There were marked differences between the two groups ($t = 2.847, 3.306, 10.324, P < .01$), and IMT, plaque area, Crouse score, were significantly reduced in small dose group and conventional dose group ($t = 2.259, 2.191, 2.221, 5.565, 5.680, 14.43, P < .05, P < .01$). There were obvious difference between the two groups ($t = 2.847, 3.306, 10.324, P < .01$). The incidence of ischemic stroke was significantly decreased in conventional dose group compared with that of small dose group and control group (8.10% vs 15.78% vs 25.00%), and there were no serious adverse reactions in two drug groups.

CONCLUSIONS: Our results indicate that rosuvastatin had strong anti-atherosclerosis function in the elderly patients with carotid vulnerable plaque, and could reverse and stabilize the vulnerable plaque and reduce ischemia stroke incidence rate. Higher dose at 10 mg/d was superior to the lower 5 mg/d. No significant adverse reactions were observed.

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Risk Factors of Ischemic Stroke in Elderly Patients with Intracranial Atherosclerotic Stenosis: A Retrospective Study of a Large Single Center Patient Cohort

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Supported by: The Local Ethics Committee of the Third Affiliated Hospital of Sun Yat-sen University

BACKGROUND: Mounting evidence indicates that intracranial atherosclerotic stenosis (ICAS) is one of the most important risk factors of ischemic stroke in Asia, especially in China, and is related to recurrent stroke, morbidity and mortality. However, there is less scientific data regarding the relationship between the

risk factors and ICAS related stroke in a South Chinese population. This study aimed to investigate the risk factors in ischemic stroke elderly patients with intracranial atherosclerotic stenosis (ICAS).

METHODS: A retrospective study of 3,563 patients was conducted from June 2012 to June 2014. Of all the patients, 235 consecutive elderly patients in first-ever stroke with ICAS were recruited in our study. Clinical data were collected, which included demographics, and the presence of risk factors [hypertension, diabetes mellitus, heart disease, hyperlipidemia, peripheral vascular disease, current smoking, and alcohol assumption]. Metabolic syndrome (MetS), magnetic resonance imaging (MRI) with MRA (within the next 3 post-stroke days), carotid duplex ultrasonography, 24 h electrocardiography (Holter), electrocardiography, and other routine admission laboratory tests (within 24 h of ischemic stroke) were performed.

RESULTS: The rate of first-ever stroke with ICAS was 6.6%. Among the 235 enrolled patients, the young (45–64 years) patients were 99 (42.13%), the young-old (65–74 years) patients were 84 (35.74%), and old-old (75–99 years) were 52 (22.13%). Especially, the old patients were 136 (57.87%). Statistical analysis showed a significant difference in drinking history, history of smoking, hyperlipidemia, and biological indicators Hcy, HbA1c, Lp (a), TG/HDL-C, TC / HDL-C among these groups ($P < .05$). The young-old (65–74 years) patients had the higher TC/HDL-C ratio ($P < .01$), levels of plasma Hcy ($P < .05$) and the lower TG/ HDL-C ratio ($P < .05$), levels of plasma Hcy ($P < .05$), the drinking history ($P < .05$) compared with those of the young (45–64 years) patients. Furthermore, the old-old (75–99 years) patients had also the higher TC/ HDL-C ratio ($P < .01$), levels of plasma Hcy ($P < .05$) and the lower the drinking history, history of smoking, levels of plasma Lp (a) ($P < .05$), compared with those of the young (45–64 years) patients. Compared with the young-old (65–74 years) patients, the old-old (75–99 years) patients had significantly higher TG/ HDL-C ratio ($P < .01$), and lower levels of plasma Lp (a) ($P < .05$).

CONCLUSION: Healthy lifestyle (cessation of smoking and drinking) had the potential to reduce or delay the occurrence of ischemic stroke with ICAS. Hcy and disorders of lipid metabolism are major risk factors for ischemic stroke with ICAS in the elderly patients.

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Correlation between Coronary Artery Stenosis Degree and 20S Proteasome in Elderly Patients with Metabolic Syndrome

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BACKGROUND: Atherosclerosis is defined as a chronic inflammation of the arterial wall and is the leading cause of cardiovascular disease. Metabolic syndrome is one of the most important risk factors for atherosclerosis. 20S proteasome is the active core of ubiquitin-proteasome system. The objective of the present study was to investigate the relationship between coronary artery stenosis degree and 20S proteasome in elderly patients with metabolic syndrome.

METHODS: Two hundred and four elderly patients with metabolic syndrome, who underwent coronary angiography from August 2011 to April 2014, were classified into coronary heart

disease group (CHD group) and free of coronary heart disease group (control group). Coronary artery stenosis degree, concentration and activity of 20S proteasome, and concentration of tumor necrosis factor (TNF)- α were also evaluated.

RESULTS: Concentration and activity of 20S proteasome and concentration of TNF- α in CHD group were significantly higher than those in control group (for concentration of 20S proteasome, 166.24 ± 82.96 ng/L vs 96.64 ± 54.15 ng/L; for activity of 20S proteasome, 80.59 ± 37.32 pnmol/s/mg vs 41.56 ± 22.56 pnmol/s/mg, and for TNF- α , 71.55 ± 19.81 ng/L vs 58.73 ± 21.26 ng/L, $P < .001$, respectively). Results of Pearson and multiple linear regression analysis showed that Gensini grade of coronary artery and concentration of TNF- α were positively correlated with concentration (r values were 0.484 and 0.352, all $P < .001$, respectively) and activity (r values were 0.417 and 0.396, all $P < .001$, respectively) of 20s proteasome. Gensini grade of coronary artery and concentration of TNF- α were important impact factors of concentration and activity of 20s proteasome.

CONCLUSION: Coronary artery stenosis degree were closely related to concentration and activity of 20s proteasome in elderly patients with metabolic syndrome.

P64

Distribution of Cerebral Infarction Caused by Occlusion or Stenosis of Perforating Artery: a Diffusion-weighted Imaging Study
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BACKGROUND: Cerebral infarction caused by occlusion or stenosis of perforating artery accounts for nearly 30% of cerebral infarction, distribution of which was important to the prognosis of patients. This study aimed to investigate the distribution of cerebral infarction caused by occlusion or stenosis of perforating artery.

METHODS: Patients with acute cerebral infarction caused by occlusion or stenosis of perforating artery were retrospectively evaluated. Diffusion weighted imaging was applied to determine the localization of cerebral infarction. Distribution of cerebral infarctions was summarized.

RESULTS: Three hundred and six-nine patients were included in this study, 261 of whom were male and 108 were female. The mean age was 62.5 ± 12.9 years. Infarctions were located in the corona radiata or centrum semiovale in 23 patients (6.2%), basal ganglia in 129 patients (35.0%), internal capsule in 59 patients (16.0%), unilateral thalamus in 45 patients (12.2%), pons in 78 patients (21.1%), midbrain in 11 patients (3.0%), medulla in 18 patients (4.9%), and bilateral paramedian thalamus simultaneously in 6 patients (6.2%).

CONCLUSIONS: Cerebral infarction caused by occlusion or stenosis of perforating artery is most commonly located in basal ganglia, which is followed by pons. Relatively rare sites include midbrain, medulla and bilateral paramedian thalamus.

P65

Aspirin Resistance and its Risk Factors in Female Ischemic Stroke Patients

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BACKGROUND: This study aimed to analyze the risk factors for aspirin resistance (AR) by observing its incidence in female ischemic stroke patients.

METHODS: One hundred and sixteen female ischemic stroke patients were divided into AR group ($n = 47$) and aspirin sensitive group ($n = 69$), according to the AA inhibitory rate. All patients were given 100 mg aspirin tablet in the morning for 10 days. The inhibition rate of arachidonic acid (AA) on platelet aggregation and the serum glycosylated serum protein level were then measured. The general clinical data were compared. The patients were further divided into glycosylated serum protein <236 μ mol/L group ($n = 67$) and glycosylated serum protein ≥ 236 μ mol/L group ($n = 49$). The incidence of AR in the two groups was compared. The risk factors for AR were analyzed by logistic binary regression analysis.

RESULTS: The body weight was significantly higher in AR group than in aspirin sensitive group (66.9 ± 10.2 kg vs 63.5 ± 7.9 kg, $P < .05$). The incidence of AR was significantly higher in glycosylated serum protein ≥ 236 μ mol/L group than in glycosylated serum protein <236 μ mol/L group (57.1% vs 28.4%, $P < .01$). Multivariate logistic binary regression analysis showed that glycemic control was the risk factor for AR (OR = 3.368, 95% CI 1.550–7.320, $P = .002$).

CONCLUSIONS: The incidence of AR is rather high in female ischemic stroke patients. Glycemic control and body weight have influence on the incidence of AR.

P66

Calibrated Automated Thrombography for Monitoring Coagulation Function in Patients with Intracerebral Haemorrhage

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BACKGROUND: Intracerebral hemorrhage (ICH) is a common type of stroke. In patients with ICH, coagulation, anticoagulation and fibrinolysis are disrupted, which triggers a series of reactions to restore balance in the body. With increase in bleeding and exacerbation of ICH, prothrombin time, activated partial thromboplastin time and thrombin time all increase, with conventional coagulation tests (e.g. prothrombin time and activated partial thromboplastin time) in evaluation of the initial stage of clot formation, which only reflects a small part of the coagulation process, as $>95\%$ of thrombin is formed after the initial stage. Calibrated automated thrombography is the only test for evaluating the entire coagulation process. In the present study, the course of coagulation function in patients with ICH was investigated using calibrated automated thrombography.

METHODS: Patients admitted to hospital with ICH (confirmed within 18 h of symptom onset) were enrolled. Patients' history and blood samples were obtained within 6 h of admission; further blood samples were collected on days 4, 8 and 15 (or on discharge between days 9–15, grouped with day 15 data). Blood samples were also collected from age- and sex-matched healthy controls. All samples underwent calibrated automated thrombography.

RESULTS: At admission, thrombin lag time and time to peak was longer, and endogenous thrombin potential and peak height were lower, in patients with ICH ($n = 20$) than in healthy controls ($n = 29$). Lag time in patients with ICH gradually decreased, but it remained significantly longer than in controls until day 8 [(6.56 ± 4.44) min vs (3.26 ± 0.77) min, $P < .05$]. Time to peak also gradually decreased, but remained longer in patients than in controls by day 15 [(9.62 ± 4.87) min vs (5.39 ± 1.06) min, $P < .05$]. Endogenous thrombin potential gradually increased in

patients, but remained lower than in controls on day 15 [(1570.44 ± 400.94) nM min vs (2445.60 ± 529.51) nM min, $P < .05$]. Peak height gradually increased in patients, but remained lower than in controls on day 15 [(210.92 ± 93.47) nmol/L vs (412.12 ± 56.25) nmol/L, $P < .05$].

CONCLUSIONS: Patients with ICH had poorer coagulation function than healthy individuals, but this function gradually recovered during hospitalization.

P67

Utility of for Assessing Motor Symptoms, Olfactory Test and Transcranial Ultrasonography in Diagnosis of Parkinson's Disease

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OBJECTIVE: This study aimed to investigate the utility of assessing motor symptoms, olfactory test and transcranial ultrasonography in diagnosis of Parkinson's disease (PD).

METHODS: Fifty PD patients and fifty age- and gender-matched non-PD individuals as control group were recruited. Unified Parkinson's Disease Rating ScaleIII (UPDRS III), T&T olfactometry and transcranial ultrasonography were applied respectively to assess motor asymmetry, olfactory function and substantia nigra (SN) echogenicity. Hyposmia and SN hyperechogenicity (SN+)'s influencing factors were analyzed, and three assessments and the combined's sensitivity, specificity and positive predictive value for diagnosis of PD were calculated respectively.

RESULTS: Compared with the control group, motor asymmetry, hyposmia and SN+ in PD group have statistical significance ($P < .05$, $P < .01$). Sensitivity, specificity and positive predictive value for assessment of motor symptoms, olfactory test and transcranial ultrasonography for diagnosis of PD are as follows: 80.0%, 78.0%, 78.4%, 58.0%, 74.0%, 69.0%, 90.5%, 81.3%, and 80.9%. Results of combined three assessments present 38.1%, 97.9%, 94.1%. Sensitivity of combined is lower than that of assessment of motor symptoms and transcranial ultrasonography, but specificity is significantly higher than that of three assessments alone.

CONCLUSIONS: Assessment of motor symptoms, olfactory test and transcranial ultrasonography may be useful methods for the diagnosis of Parkinson's disease, and the combination of the three might improve the diagnostic specificity.

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Association between Lacunar Infarction and Serum Bilirubin Level in Aged Male Patients

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BACKGROUND: Lacunar infarction is a common ischemic cerebral small vessel disease in aged people, and it is believed that it is associated with increased risk of dementia, stroke and morbidity, but little is known the exact pathogenesis and the clinical significance of lacunar infarction (LI). This study aimed to investigate the association of LI with serum bilirubin level in male elderly patients.

METHODS: Ninety-eight cases of male LI patients were selected in this study. LI was detected on MRI, and the patients were divided into single (36 cases) and multiple LI (62 cases) groups according to the number of lesions. Medical histories were recorded for all the patients, including chronic bronchitis, chronic obstructive pulmonary disease (COPD), coronary disease, stroke,

transient ischemic attack (TIA), age, the history of smoking and drinking. Laboratory test was made including serum bilirubin, homocysteine, total cholesterol, triglycerides, and low density lipoprotein cholesterol and so on. The serum bilirubin level and other clinical information between the two groups were compared. According to the total bilirubin, all the LI patients were divided into high total bilirubin (10.6–21.0 $\mu\text{mol/L}$) group ($n = 35$) and low total bilirubin (0–10.5 $\mu\text{mol/L}$) group ($n = 63$), and the number of lesions and other clinical information between the two groups were compared. Data were analyzed using SPSS, version 17.0. Continuous data were presented as mean ± SD. Student t test was used for the group comparisons. Wilcoxon rank sum test was used for Non-normal distribution data. The chi-square test was used for nominal data. The risk factors for LI were calculated by using multivariate logistic regression. All tests were considered significant if $P < .05$.

RESULTS: For serum levels of indirect reacting bilirubin, direct reacting bilirubin and total bilirubin, there were no significant differences between the single and multiple groups ($P > .05$). Number of lesions was 5.3 vs 4.7 ($P = .50$) according to total bilirubin groups, and there was no significant difference between the two groups ($P > .05$). Multivariate logistic regression for LI shows that there were no risk factors after adjustment for other factors.

CONCLUSIONS: There was no association of lacunar infarction with serum bilirubin level in male elderly patients. No risk factors were identified for LI in male elderly patients in this study.

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Association of Cystatin C and Risk of Essential Hypertension in Elder Patients with Diabetes

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BACKGROUND: Serum cystatin C is an independent risk factor for essential hypertension, and meanwhile, hypertension is a common diabetic vascular complication in elder patients. It was hypothesized that serum cystatin C would be an independent risk factor for essential hypertension in elder patients with diabetes.

METHODS: Elderly diabetic patients ($n = 538$) were enrolled from January 2013 to August 2014 in our hospital. Patients were divided into two groups: non-hypertension group (191 cases) and hypertension group (347 cases) according to their blood pressure. General clinical and laboratory data were collected, including age, gender, body mass index (BMI), blood pressure, blood lipid, serum cystatin C, serum creatinine, high sensitive C reactive protein, N terminal B type natriuretic peptide, glycosylated hemoglobin and fasting blood glucose.

RESULTS: Compared with the non-hypertension group, age, smoking, body mass index, systolic pressure, and diastolic pressure were significantly increased in the hypertension group patients. According to the blood pressure levels, patients were divided into 4 groups: control group, level 1, level 2 and level 3 patients with essential hypertension. Compared with the control group, age, BMI, systolic blood pressure, diastolic blood pressure, Cystatin C, hs-CRP, N terminal B type natriuretic peptide levels in level 3 hypertension patients were significantly increased, and compared with the control group, the systolic blood pressure, diastolic blood pressure levels were significantly increased in level 1 and level 2 hypertension patients. Compared with cystatin C levels in level 1 hypertension patients, those were significantly increased ($P < .05$).

The relationship between risk factors and the incidence of hypertension was determined by spearman's assay. The results of Spearman correlation analysis suggested that hypertension was associated with age, smoking history, BMI, SBP, DBP, hs-CRP, creatinine and cystatin C. Multivariate logistic regression analysis demonstrated that age, smoking history, BMI, creatinine, cystatin C and N terminal B type natriuretic peptide levels were the independent risk factors of essential hypertension in elderly patients with diabetes. The relationship between risk factors and Cystatin C level was also determined by spearman's assay, which suggested that the risk factors of plasma cystatin C level in elderly patients with diabetes mainly include age, gender, smoking history, systolic blood pressure, TC, NT-proBNP, LDL-C, HDL-C, hs-CRP and serum creatinine level.

CONCLUSIONS: With the increase of serum cystatin C level in elderly diabetic patients, the incidence of hypertension is on a rising trend, which may suggest that cystatin C could be an independent risk factor for essential hypertension in elderly diabetic patients.

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Dyslipidemia and its Risk Factors in Elderly Residents in Renqiu, Hebei, China: Screening Analysis

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Supported by: The Health Bureau of Renqiu City; Science and Technology Department of Hebei Province

BACKGROUND: A number of studies reported that there was a significant correlation between dyslipidemias and age, and that the prevalence of dyslipidemia increased gradually with age. Much more attention was paid to dyslipidemia or its characteristics of 18 years old adults, and dyslipidemia and its risk factors for the 60- to 70-year-old elderly has been rarely reported. This study aimed to investigate and to analyze the prevalence and risk factors of dyslipidemia for the elderly aged 60–70.

METHODS: A total of 5010 permanent residents (male 2163 and female 2847) aged 60–70 years (mean 64 (62–67) years) from Renqiu County were included in this study. They underwent face to face investigation with health questionnaires, anthropometric measurements, and laboratory testing. Risk factors of dyslipidemia and specific individual dyslipidemia were analyzed by multivariate logistic regression analysis.

RESULTS: In the surveyed population, the incidence of total dyslipidemia, high TC, high TG, low HDL-C and high LDL-C were 53.3%, 31.1%, 35.4%, 7.4%, and 19.9% respectively in the investigated residents. The incidence of total dyslipidemia, high TC, high LDL-C, high TG were significantly higher in female than males (62.9% as 40.6%, 39.5% as 20.0%, 43.1% as 25.2%, 24.1% as 12.2%, $P < .05$). Multivariate logistic regression analysis showed that overweight (OR = 1.407, 95% CI 1.211–1.635), abdominal obesity (OR = 1.382, 95% CI 1.191–1.604), smoking (OR = 1.181, 95% CI 1.036–1.347), fasting glucose (OR = 1.165, 95% CI 1.111–1.221), insulin levels (OR = 1.026, 95% CI 1.017–1.036), systolic pressure (OR = 1.006 95 CI 1.004–1.009) and uric acid (OR = 1.003, 95% CI 1.002–1.004) were independent risk factors for dyslipidemias.

CONCLUSIONS: The incidence of dyslipidemia is high in the 60–70 year old Renqiu residents. Public education and regular check-up are necessary in order to effectively control dyslipidemia, and ultimately to achieve effective prevention and control of cardio-cerebrovascular events.

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Role of Lipoprotein-associated Phospholipase A₂ and hs-CRP in Risk Stratification of Patients with Transient Ischemic Attack

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BACKGROUND: Transient ischemic attack (TIA) and thrombotic stroke have identical etiologies. Patients who experience TIA are at risk of stroke. However, most of them do not receive timely evaluation and treatment in China. The ABCD2 score is a common clinical tool to predict patients with TIA at high risk for early stroke. Although the ABCD2 score has been recommended for use in several national guidelines including China, improvements in predictive accuracy remain necessary. The measurement of biomarkers indicating the presence of coagulation activation might offer additional risk assessment beyond that of clinical factors alone. The candidate markers include lipoprotein-associated phospholipase A₂ (Lp-PLA₂) and hypersensitive C-reactive protein (hs-CRP). The aim of this paper was to determine whether measurement of D-dimer (DD) would improve the risk stratification after TIA.

METHODS: One hundred and thirty-two TIA patients admitted to our hospital were divided into positive event group ($n = 34$) and negative event group ($n = 98$). The patients were scored according to the ABCD2 scores and further were divided into low risk group with 0–3 points ($n = 46$), moderate risk group with 4–5 points ($n = 70$) and high risk group with 6–7 points ($n = 16$). Their serum Lp-PLA₂ activity and hs-CRP were measured. Their primary outcomes were analyzed by composite endpoint analysis. The incidence of ischemic stroke within 90 days was calculated in patients with their conducting artery stenosis $\geq 50\%$ or cardiogenic embolism needing anticoagulation treatment.

RESULTS: Serum Lp-PLA₂ activity was significantly higher in positive event group than negative event group and in patients with their conducting artery stenosis $\geq 50\%$ than in those with no conducting artery stenosis ($P < .05$). Composite endpoint analysis showed that 4–5 points, 6–7 points, and Lp-PLA₂ activity $> 33 \mu\text{mol (L.min)}$ were the risk factors for composite end events in TIA patients (RR = 2.18, 95% CI 1.26–3.61, $P = .008$; RR = 3.43, 95% CI 1.91–6.39, $P = .001$; RR = 2.73, 95% CI 1.04–7.06, $P = .042$).

CONCLUSION: The combination of Lp-PLA₂ activity $> 33 \mu\text{mol (L.min)}$ and the ABCD2 score can predict ischemic stroke and death within 90 days in TIA.

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Quality Control of Single Disease in Acute Ischemic Stroke Patients

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BACKGROUND: In order to strengthen the medical quality management, standardize the clinical behavior, the Ministry of Health of China implemented quality control of single disease. Acute ischemic stroke (AIS) was included in the catalogs of quality control of single disease in 2009. This study aimed to find out if there was any improvement after the policy was implemented in our hospital.

METHODS: Seventy-six AIS patients who underwent quality control of single disease served as a post-quality control group and

67 AIS patients who did not undergo quality control of single disease served as a pre-quality control group. Their antithrombotic therapy within 48 h, lipid-regulating therapy, rehabilitation therapy, hospital cost, medication ratio, length of hospital stay, and neurological defect score were compared.

RESULTS: Lipid-regulating therapy ratio was significantly higher while the total hospital cost and medication ratio were significantly lower. The length of hospital stay was significantly shorter in post-quality control group than in pre-quality control group (90.8% vs 52.2%, $P < .01$; 16659.90 ± 9775.62 RMB yuan vs 19957.17 ± 8712.93 RMB yuan, 48.94% ± 10.46% vs 52.45% ± 9.85%, 14.49 ± 7.76 days vs 17.16 ± 8.23 days, $P < .05$). No significant difference was found in antithrombotic therapy within 48 hours, rehabilitation therapy and neurological defect score between the two groups ($P > .05$)).

CONCLUSIONS: Quality control of single disease can effectively reduce the hospital cost, medication ratio and hospital stay time of AIS patients.

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Changes of Plasma Interleukin-35, Chitinase-3-like-1 protein, Adiponectin in Patients with Acute Coronary Syndrome and its Clinical Significance

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BACKGROUND: Interleukin (IL) -35 is a novel heterodimeric cytokine in the IL-12 family. There are many evidences that demonstrated that IL-35 can efficiently suppress the T cell (such as Th1, Th2) activity and reduce the progression of autoimmune diseases and inflammatory diseases. However, current information about the levels of plasma IL-35 and its potential role in the field of coronary artery disease is still limited. Therefore, in our study, we measured the levels of plasma IL-35, YKL-40 and APN in different clinical presentations and the associations between them in order to determine its independency and predictive power.

METHODS: One-hundred and seventeen consecutive patients with coronary heart disease (CHD) were admitted to the department of cardiology, Second Affiliated Hospital of Xuzhou Medical College and were included in this observational, cross-sectional study between May 2013 and April 2014. They were divided into three groups: acute myocardial infarction (AMI) group (40 cases), unstable coronary artery disease (UAP) group (40 cases), and stable coronary artery disease (SAP) group (37 cases). At the corresponding time period, 30 cases of angiographically normal subjects were included as control group. Besides, all patients with CHD were divided into three groups according to Gensini score: group A (Gensini score < 26), group B (Gensini score between 26 - 54) and group C (Gensini score > 54). The concentration of plasma IL-35, YKL-40 and APN were measured by enzyme linked immunosorbent assay (ELISA).

RESULTS: (1) Concentrations of IL-35 were significantly decreased in SAP group (74.06 ± 21.08 pg/ml, $P < .01$), in UAP group (63.05 ± 17.65 pg/ml, $P < .01$), and in AMI group (52.76 ± 16.85 pg/ml, $P < .01$) compared with control group (92.31 ± 24.42 pg/ml). Furthermore, concentrations of plasma IL-35 in UAP group and AMI group were significantly lower than those in SAP group (respectively $P < .05$ and $P < .01$). The concentrations of YKL-40 in groups with CHD were significantly higher than those in normal group ($P < .01$ respectively). Furthermore, the concentrations of YKL-40 in UAP group and AMI group were higher than those in SAP group ($P < .05$, $P < .01$ respectively). The concentrations of APN in CHD were significantly lower than those in the normal group. Furthermore, those in AMI

group and UAP group were lower than those in SAP group ($P < .01$ respectively). (2) According to the Gensini score, levels of APN were lower in group B and group C than those in group A ($P < .01$, respectively). However, levels of YKL-40 were higher in group B and group C than those in group A ($P < .01$, respectively). There were no significant differences in levels of IL-35 between the three groups (A vs B, A vs C, B vs C, respectively $P = 0.530$, 0.587 , 0.266). (3) There was no correlation between the levels of IL-35 and coronary lesion severity ($r = -0.033$, $P = .726$). Plasma levels of APN were negatively correlated with coronary lesion severity ($r = -0.844$, $P < .01$). The levels of YKL-40 were positively correlated with coronary lesion severity ($r = 0.814$, $P < .01$).

CONCLUSION: The changes in IL-35, YKL-40 and APN proved that they participate in the progression of atherosclerosis, plaque instability and the subsequent onset of acute coronary syndrome. IL-35 was not associated to the Gensini score, suggesting that changes in IL-35 level were associated with the inflammatory status and plaque destabilization of CHD but not the severity of the coronary artery stenosis. While YKL-40 and APN were associated with the Gensini score, to a certain extent, they may reflect the severity of the illness and coronary lesion. Therefore, it may be a useful method for forecasting the acute coronary syndromes by measuring the plasma levels of cytokines.

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Combination of Pulse Wave Velocity and Retinal Artery Lesions to Predict Coronary Artery Disease in Chinese Elderly Patients

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BACKGROUND: Brachial-ankle pulse wave velocity (baPWV) is a simple and non-invasive method of evaluating arterial stiffness and coronary artery disease (CAD). Retinal atherosclerotic changes are associated with the atherosclerotic changes of coronary arteries. The aim of this study was to determine whether the combination of baPWV and retinal artery lesions can be a useful approach to predict CAD in Chinese elderly patients.

METHODS: Coronary angiography, eye fundus examination and baPWV were measured in 209 elderly patients with typical pain or suspected CAD. BaPWV over 1400 cm/s and retinal artery atherosclerosis ≥ Stage 2 were defined as risk factors, respectively.

RESULTS: The ratio of baPWV > 1400 cm/s and retinal artery atherosclerosis ≥ Stage 2 were higher in patients with CAD than those without (58.82% to 13.7%, $P < .001$). In multivariable logistic regression analysis, both baPWV > 1400 cm/s (OR: 7.456, 95% CI 3.254–17.084, $P = .000$) and retinal artery atherosclerosis (OR: 5.376, 95% CI 2.327–12.419, $P = .000$) were associated with the presence of CAD. The combination of the two risk factors remained an independent predictor of the presence of CAD after adjusting for factors such as age, gender, hypertension, hypercholesterolemia, obesity, diabetes, smoking, and typical chest pain (OR: 15.034, 95% CI 5.930–38.119, $P < .001$).

CONCLUSION: Both baPWV over 1400 cm/s and retinal artery atherosclerosis greater than or equal Stage 2 are useful markers to predict the presence of the CAD. The combination of the two non-invasive methods can be a useful approach to predict CAD in Chinese elderly patients.

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The Age Tendency of Serum Lipids Level and Lipid Goal Attainment in the Elderly

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BACKGROUND: As we have less robust evidence to guide the management of dyslipidemia in older patients, especially for those aged 75 years and older, the elderly share the same therapeutic goal with younger adults in our guideline. However, the elderly often have multiple comorbidities and show less tolerance with lipid lowering drugs. Therefore, the aim of this study was to understand the management of dyslipidemia in the elderly by investigating the relationship of serum lipids level and lipid goal attainment with ageing.

METHODS: A retrospective analysis was performed using the database of our laboratory information system and outpatient electrical medical records, in order to retrieve lipid profile, including total cholesterol (TC), low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), triglyceride (TG), and to collect relevant clinical information in a large cohort of outpatient elderly (≥ 65 years old), who visited in the year of 2013. Patients with antineoplastic drugs were excluded, and the rest were classified into various groups according to age to observe the alterations of serum lipids level and lipid goal attainment along with ageing.

RESULTS: A total of 22,847 outpatients elderly (≥ 65 years old) were enrolled in the final analysis. Among these patients, 42.7% (9757) were male and their mean age was 74.4 years, while 57.3% (13090) were female and their mean age was 73.6 years. In addition, 24.6% (5617) were treated with lipid lowering drugs. Serum TC, LDL-C, HDL-C and TG showed declining trends with ageing process in both man and women ($P < .05$), and the down-trend still existed when we ruled out the effect of lipid lowering drugs. The lowest level of TC and LDL-C appeared at the age of 90–94 years old, while TG at the age of 85–89 years old. Among the elderly patients with very high or high cardiovascular risk and without lipid-lowering therapy, the likelihood of attaining LDL-C goal increased with ageing (OR: 1.215; 95% CI 1.169–1.264, for every 5 years increment, $P < .001$), and the likelihood of attaining non-HDL-C goal also concerted with ageing (OR: 1.231; 95% CI 1.188–1.277, for every 5 years increment, $P < .001$).

CONCLUSION: With the increment of age in the elderly, serum lipids level had a declined tendency, and lipid goal attainment rates tended to increase without having lipid-lowering agents. Measures should be taken to personalize lipid-lowering strategy in future guideline recommendations for the management of dyslipidemia in the elderly.

P76

The Association between Plasma Klotho Protein Level and Lacunar Infarction in the Elderly People

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OBJECTIVE: This study aimed to investigate association between plasma Klotho levels and Lacunar infarction in the elderly people.

METHODS: The level of Klotho protein was tested in 88 elderly patients (age ≥ 60 years) with Lacunar infarction as a research group compared with 90 patients without Lacunar infarction as controls by ELISA.

RESULTS: Compared with the control group, Klotho protein level in the elderly with Lacunar infarction group were significantly decreased ($P < .01$). Using multivariate logistic regression analysis, the low plasma Klotho level was identified as independent risk factors of Lacunar infarction (OR 3.325, 95% CI [1.515–7.300], $P = .003$).

CONCLUSIONS: Klotho protein level may be associated with Lacunar infarction in the elderly people. There is an important clinical significance of monitoring plasma Klotho protein level in the prevention, diagnosis and prognosis of Lacunar infarction.

P77

Ability of Ambulatory ECG-based T-wave Alternans to Modify Risk Assessment of Cardiac Events

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BACKGROUND: Exercise-based spectral T-wave alternans (TWA) has been proposed as a noninvasive tool-identifying patients at risk of sudden cardiac death (SCD). The ability of ambulatory electrocardiogram (AECG)-based TWA for risk stratification of cardiac events has not been well established. This study aimed to review data regarding 24-hour AECG-based TWA and to discuss its potential role in risk stratification of fatal cardiac events across a series of patient risk profiles.

METHODS: Prospective clinical trials of the predictive value of AECG-based TWA obtained with daily activity published between January 1990 and October 2014 were retrieved. Major endpoints included composite endpoint of SCD, cardiac mortality, and severe arrhythmic events.

RESULTS: Data were accumulated from 5 studies involving a total of 1,588 patients, including 317 positive and 1,271 negative TWA results. Compared with the negative group, the positive group showed increased rates of SCD (hazard ratio [HR]: 7.49, 95% confidence interval [CI] 2.65 to 21.15), cardiac mortality (HR: 4.75, 95% CI 0.42 to 53.55), and composite endpoint (SCD, cardiac mortality, and severe arrhythmic events, HR: 5.94, 95% CI 1.80 to 19.63). For the 4 studies evaluating TWA measured using the modified moving average method, the HR associated with a positive vs negative TWA result was 9.51 (95% CI 4.99 to 18.11) for the composite endpoint.

CONCLUSIONS: The positive group of AECG-based TWA had a nearly six-fold risk of severe outcomes compared with the negative group. Therefore, AECG-based TWA may provide an accurate means of predicting fatal cardiac events.

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Sleep Quality and Sleep Disturbing Factors of Inpatients in Geriatric Wards in a Chinese General Hospital

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OBJECTIVES: This study aimed to better understand sleep quality and sleep disturbing factors of inpatients and to compare the perception of sleep disturbing factors between nurses and inpatients.

METHODS: We included inpatients ($n = 116$) and licensed nurses ($n = 53$) from October 31, 2012, to March 31, 2013 in geriatric wards in the largest general university-based teaching hospital in central south China. The Pittsburgh Sleep Quality Index and self-designed questionnaires were used to examine the inpatients' sleep quality and to compare perceptions of sleep disturbing factors

between inpatients and nurses. The following statistical methods were used to evaluate the data: chi-square test, logistic regression analysis and Z statistic.

RESULTS: Patients reported poor sleep quality (89.7%) during hospitalization and a reduction in sleep quality and change of sleep habits after hospitalization (31.9%). Age was not an independent factor in inpatients sleep quality. The chi-square and logistic regression analysis showed that the main sleep disturbing factors in inpatients were psychology disturbing factors, and the main individual disturbing factors were worries about illness, discomfort associated with diseases, boredom, and too much sleep in the daytime and uncertainty about the upcoming test results. While the main sleep disturbing factors from nurses' perspective were environmental disturbing factors and the main individual disturbing factors were discomfort associated with diseases, unfamiliar environment, nursing at night, noise from therapeutic equipment and worries about illness. There was a significant difference in the perception of sleep quality and sleep disturbing factors between patients and nurses in the hospital inpatient setting.

CONCLUSION: This study demonstrates that most inpatients reported sleeping problems stemming from multiple sources. Additionally, there was a significant difference in the perception of sleep disturbing factors between nurses and inpatients. Relevance to clinical practice: Our survey shows that most nurses are aware of inpatients' poor sleep quality. However, many patients continue to suffer from poor sleep during hospitalization because of a wide range of sleep disturbing factors. This suggests that nurses need to be better educated and trained to minimize sleep disturbing factors and to improve sleep quality of their patients.

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The Effect of Different Altitude Hypoxia on Chemerin, Leptin and Adiponectin in Elderly Patients with Congestive Heart Failure and its Relationship with Ventricular Remodeling

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OBJECTIVE: More than 140 million people worldwide live more than 2500 meters above sea level. Changes in human physiology, pathology, pathogenesis, and clinical features of the heart and pulmonary circulation are reported during high altitude exposure. Yet, few studies have explored the effect of different altitude hypoxia (2260 m, 3300 m) on Chemerin, Leptin and Adiponectin in elderly patients (≥ 65 years old) with congestive heart failure. The aim of the present study was to explore the significance of the level change of serum Chemerin, leptin and adiponectin in elderly patients with chronic heart failure (CHF) at different altitude, and effects of the level change at different altitude on left ventricular mass index (LVMI) and mean wall stress (MWS).

METHODS: Sixty-two CHF patients from high altitude (3300 m) were studied and 67 CHF from moderate altitude (2260 m) were enrolled as controls. The serum levels of Chemerin, Leptin and Adiponectin were detected by ELISE. Their Left ventricular ejection fraction (LVEF), LVMI and MWS were measured by echocardiography.

RESULTS: Compared with moderate altitude group, the serum levels of Chemerin, Leptin and Adiponectin were increased in high altitude group [(42.87 $\pm$ 7.86) pg/ml vs (26.75 $\pm$ 5.22) pg/ml], (16.74 $\pm$ 2.28) ug/L vs (7.95 $\pm$ 1.53) ug/L and (17.64 $\pm$ 1.99) mg/L vs (7.65 $\pm$ 1.18) mg/L, respectively, all $P < .01$]. Similarly, LVMI and MWS were significantly higher in high altitude group [(189.7 $\pm$ 9.5) vs (96.3 $\pm$ 8.4) g/m² and (476.2 $\pm$ 14.9) vs (304.7 $\pm$ 13.5) dynes $\times 10^3$ /cm², respectively, all $P < .01$]. There was a positive correlation between LVMI, MWS and the levels of Chemerin, Leptin and Adiponectin (all $P < .01$).

CONCLUSION: Serum levels of Chemerin, Leptin and Adiponectin in CHF patients increased with the altitude level. Serum Chemerin, Leptin and Adiponectin may be involved in the pathogenesis and development of chronic heart failure, and they may interact and influence each other and lead to the change of ventricular remodeling in CHF patients at high altitude.

P80

Clinical Analysis for 36 Cases of Primary Hyperparathyroidism

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OBJECTIVE: This study aimed to explore the clinical manifestations, diagnosis, misdiagnosis and treatment of primary hyperparathyroidism (PHPT).

METHODS: The clinical data of 36 patients with PHPT who were admitted to this hospital between Jan. 2009 and Oct. 2014 were reviewed and analyzed retrospectively.

RESULTS: The manifestations of PHPT had various features. Among the 36 patients, the ratio of male to female was 1:1, and the average age was 56 years old. PHPT is often misdiagnosed as other disease, accounting for about 25.0%. Of all the patients, 9 were discovered with joint, 7 with digestive system symptoms, 6 with cervical mass, 5 with urinary calculus, 4 with fatigue symptoms, 2 with osteoporosis and urinary calculus, one with fracture and 2 without any symptoms. Their calcium and parathyroid hormone level were increased by various degrees. The accuracy rates of level diagnosis of Color Doppler ultrasound, CT or ^{99m}Tc-MIBI were improved obviously. 28 patients received surgical treatment and pathology examination confirmed that there were 23 cases of parathyroid adenoma, one case of parathyroid hyperplasia, one case of parathyroid carcinoma, 2 cases of parathyroid adenoma with nodular goiter, and one case of parathyroid adenoma combined with thyroid carcinoma. Serum calcium and parathyroid hormone level declined obviously after operation. Among the other 8 patients who received non-surgical treatment, 7 were alleviated and one died due to high blood calcium crisis.

CONCLUSIONS: Recognition of PHPT should be improved. Measurement of serum calcium should be included in routine physical examinations. The suspected cases should be early detected by serum PTH and parathyroid Color Doppler Ultrasound examination. Early surgical treatment can improve the prognosis and the quality of life in the primary hyperparathyroidism.

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The Effect of Klotho Gene Transfection on the Vitality of Primary Osteoblast

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OBJECTIVE: This study aimed to investigate the effect of Klotho (KL) gene transfection on the vitality of primary osteoblast, and to reveal the effect of KL via fibroblast growth factor-23 (FGF-23) signal path on bone metabolism.

METHODS: The primary bone mesenchymal stem cells (BMSCs) were abstracted from male rats. The morphologies of BMSCs were observed by inverted microscope. After the third generation culture, the specific surface markers of them were identified by flow cytometer. Then BMSCs were cultured using conditioned medium particularly for inducing into osteoblasts for 14-21 days. Alizarin red staining was used to test whether the induction was successful. After successfully induction, the primary osteoblasts

were transfected by adeno associated virus mouse KL gene. The experiment was divided into 5 groups: control group, KL group, KL with BGJ398 (FGFR1 inhibitor) group, adenovirus empty vector group, and adenovirus empty vector with BGJ398 (FGFR1 inhibitor) group. After treatments, the expression of KL in osteoblasts was detected by RT-PCR, inversed fluorescent microscope and ELISA. The expression of osteoblast discharges: osteopontin (OPN) and osteocalcin (OCN) were detected by RT-PCR.

RESULTS: The BMSCs may be seen after replacement of bath solution, namely after 72 h of BMSCs isolation. Confluent cells with long spindle could be seen using the inverted microscope (ckx41, Olympus, Japan) after 6 days of cultivation. When it was cultivated to 13d, cell density could reach 85%. And the BMSCs of three passage showed positive staining against BMSC CD44, CD90, but negative against HSC CD45 and the positive rates were 94.63%, 99.41%, 0.74% respectively. Compared to the control group, adenovirus empty vector group and adenovirus empty vector with BGJ398 (FGFR1 inhibitor) group, the KL group and KL with BGJ398 group which had exogenous rAAV.mKL transfection showed significant expression of KL mRNA by RT-PCR and KL protein by ELISA in osteoblasts ($t = 175.250$, $t = 206.506$, $t = 325.156$; $t = 222.422$, $t = 256.385$, $t = 392.880$; $P < .05$). Compared to the KL with BGJ398 group, the control group, adenovirus empty vector group and adenovirus empty vector with BGJ398, the KL group showed significant expression of OCN mRNA in osteoblasts ($t = 35.910$, $t = 47.131$, $t = 64.771$, $t = 43.871$; $P < .05$) and the KL group also showed significant inhibition expression of OPN mRNA in osteoblasts ($t = 112.710$, $t = 126.180$, $t = 143.501$, $t = 137.161$; $P < .05$). There were no significant differences to be found in the expression of OCN mRNA and OPN mRNA in osteoblasts between the KL with BGJ398 group, the control group, adenovirus empty vector group and adenovirus empty vector with BGJ398 group ($t = 0.120$ – 2.729 , $t = 0.330$ – 2.727 ; $P > .05$). These results showed that KL could improve discharge of OCN and inhibit OPN in osteoblasts when KL-FGF23-FGFR1 pathway was normal.

CONCLUSION: KL can promote the vitality of primary osteoblast probably through KL-FGF23-FGFR1 pathway.

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The Changes and Relationship with Serum Adipokine, Bone Metabolic Biochemical Markers and Osteoporosis in Aged Men

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OBJECTIVE: Adipocytokines have been proposed as new mediators with protective effects of fat mass on the skeleton. The aim of this study was to examine the serum levels of adipokine, bone metabolic biochemical markers in aged men patients with OP, and to investigate the relationship between adipokine, biochemical makers of bone formation and bone resorption and OP in aged men.

METHODS: Lumbar bone mineral density (BMD) values among 183 aged males with ages ranging from 65 to 76 years were measured by dual-energy X-Ray absorptiometry, according to the results of the BMD. The subjects were classified into 3 groups: 49 patients with OP, 74 patients with decrease of bone mass and 60 with normal bone mass. Serum adipokine with leptin and adiponectin (ADP), in addition to serum biochemical markers of bone formation and bone resorption with bone alkaline phosphatase (BAP), osteocalcin (OC) and crosslaps (CTX) were also measured by enzyme linked immunosorbent assay (ELISA).

RESULTS: Levels of ADP, BAP and OC in OP group were lower than those in group with decrease of bone mass and with

normal bone mass [(9.36 ± 2.11) ug/ml vs (15.37 ± 2.76) ug/ml and (23.56 ± 3.98) ug/ml, (7.27 ± 1.84) ng/L vs (11.45 ± 2.86) ng/L and (15.63 ± 3.52) ng/L, (8.76 ± 2.04) ng/mL vs (13.55 ± 3.43) ng/mL and (19.76 ± 4.86) ng/mL, all $P < .05$ respectively], the levels of leptin and CTX in OP group was higher than that in group with decrease of bone mass and with normal bone mass [(15.39 ± 2.08) ug/L vs (11.07 ± 2.16) ug/L and (7.89 ± 1.52) ug/L, (279.45 ± 16.41) ng/L vs (258.99 ± 15.46) ng/L and (236 ± 15.21) ng/L, $P < .05$ and $P > .05$ respectively]. ADP, BAP and OC showed a significant positive correlation ($r = 0.637$ and 0.659 , $P < .05$), and leptin was negatively correlated significantly with BAP, OC ($r = -0.571$ and -0.604 , $P < .05$).

CONCLUSIONS: Incidence of OP in aged men is related to serum adipokine level and serious insufficiency of bone formation, adipokine and biochemical makers of bone formation is an important marker for predicting the incidence of OP in aged men.

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Prognostic Value of Circulating Catestatin Levels for in-hospital Malignant Arrhythmia in Elderly Patients with Acute Myocardial Infarction

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OBJECTIVE: This study aimed to determine whether circulating levels of Catestatin (CST) provide prognostic information independently of conventional risk markers for the development of in-hospital malignant arrhythmia in elderly patients with acute myocardial infarction (AMI).

METHODS: The data of 100 patients (age ≥ 65 years, mean age 68.4 years, 71% male) with AMI were collected from Qinghai People's Hospital between April 2013 and January 2015. The patients were categorized into 2 groups according to their status of whether they had malignant arrhythmia (AMI with malignant arrhythmia, $n = 47$, mean age 69.5 years, 73.2% male; AMI without malignant arrhythmia, $n = 53$, mean age 67.3 years, 69.8% male). Clinical features, therapeutic approaches, coronary angiography results and CST were compared between the groups. Multivariate logistic regression analysis was applied to determine the association between risk factors and in-hospital malignant arrhythmia. Receiver-operator characteristic (ROC) curve was performed to evaluate the power of CST on predicting in-hospital malignant arrhythmia.

RESULTS: Patients with malignant arrhythmia were more likely to be older, to have higher Cystatin-C, blood glucose, hs-CRP, CST value, past history of atrial fibrillation, heart failure, and a lower ejection fraction, and criminals vessels more involved in left anterior descending and three branches (all $P < .05$). After multivariate adjustment, CST remained to be an independent risk factor for the development of in-hospital malignant arrhythmia (OR = 1.112, 95% CI 1.047–1.183, $P < .001$). The area under the ROC curve of CST was 0.736, when CST = 72.17 (pg/ml), and had the best sensitivity (95.8%) and specificity (76.6%).

CONCLUSION: Plasma CST levels were an independent predictor for the development of in-hospital malignant arrhythmia in elderly patients with AMI.

P84

Determination of Pb before and after Cigarettes Smoking

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OBJECTIVE: This study aimed to determine heavy metal lead in three brands of cigarettes through using graphite furnace atomic absorption spectrometry (GFAAS), and to explore the change of Pb content before and after cigarettes smoking.

METHODS: Method of microwave digestion and graphite furnace atomic absorption spectrometry was used to determine the content of heavy metal lead in tobacco and soot in three brands of cigarette.

RESULTS: The results showed that Pb content in Lushan brand cigarette is 5.3750 $\mu\text{g/g}$, and the content in soot is 1.6776 $\mu\text{g/g}$. Pb content in Jinsheng brand cigarette is 3.2250 $\mu\text{g/g}$, and it is 1.3854 $\mu\text{g/g}$ in soot. Pb in Zhonghua brand cigarette is 2.9200 $\mu\text{g/g}$, while it is 0.7538 $\mu\text{g/g}$ in soot. Through comparison, there is marked difference in heavy metal content before and after cigarette smoking.

CONCLUSION: Significant difference of Pb content was identified before and after cigarette smoking, and most Pb may be sucked into lung with smoke, absorbed by human body, and accumulation of Pb would cause harm to smokers' health. This study improved the understanding that smoking does do harm to the human body. At the same time, it provided a scientific basis for protecting human health and strengthened tobacco control.

P85

The Correlation of a High Normal Serum TSH with Blood Pressure, Fasting Plasma Glucose and Dyslipidemia in Chinese Han Population

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OBJECTIVE: This study aimed to explore the association of a high normal serum thyroid-stimulating hormone (TSH) level with blood pressure, fasting plasma glucose (FPG) and dyslipidemia in Chinese Han population.

METHODS: A questionnaire survey of 1510 subjects were conducted and measurement for height, body weight, blood pressure, TSH, FPG, total cholesterol (TC), triglyceride (TG), and high-density lipoprotein cholesterol (HDL-C) was made. According to the TSH level, the subjects were divided into low TSH group (0.30–0.99 mIU/L, n = 351), moderate TSH group (1.00–1.89 mIU/L, n = 703) and high TSH group (1.90–4.80 mIU/L, n = 456). Analysis of variance and linear regression was used for data analysis.

RESULTS: SBP ($P = .009$), DBP ($P = .003$), TG ($P = .028$) and HDL-C ($P = .009$) had significant differences among the low TSH group, moderate TSH group and high TSH group. In the univariate linear regression model, serum TSH within the reference range was negatively associated with SBP ($\beta = -3.743$, $P = .015$) and DBP ($\beta = -3.036$, $P = .006$); positively associated with TG ($\beta = 0.195$, $P = .002$) and HDL-C ($\beta = 0.125$, $P = .016$). However, after adjustment for gender, age, body mass index (BMI), FPG, TC, the correlations were no longer present. In the multiple linear regression model, for males, a significant negative correlation was found between normal range TSH, SBP ($\beta = -7.244$, $P = .021$), FPG ($\beta = -0.407$, $P = .036$).

CONCLUSION: In Han male population, serum TSH level is negatively correlated with SBP and FPG.

P86

Gut Microbiota Composition in the Elderly with Metabolic Diseases

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OBJECTIVE: This study aimed to investigate the gut microbiota composition in healthy elderly vs elderly with metabolic diseases such as hypertension, Type2 DM (T2DM) and Coronary Artery Disease (CAD).

METHODS: A cross-sectional study was conducted among 117 volunteers above 75 years recruited from communities in Zhejiang Province, China. They were divided into healthy, hypertension, hypertension with CAD, and hypertension with T2DM and quantitative PCR was performed to analyze gut microbiota composition.

RESULTS: It is observed that only Bacteroidetes ($P = .023$), Enterococcus ($P = .041$), E.coli ($P = .002$) were significantly higher in hypertension group compared to healthy group. Bifidobacteria ($P = .014$), Butyrate producers such as E.rectale, Atopobium ($P = .005$), Clostridia cluster I & cluster IV (F.praustnizii ($P = .008$), Ruminococcus spp ($P = .035$), and opportunistic bacteria (Bacteroidetes ($P = .015$), Enterococcus ($P = .005$), Streptococcus ($P = .031$) were significantly higher in hypertension group compared to hypertension with CAD group. Butyrate producers (Clostridia cluster I ($P = .038$), E.rectale ($P = .02$), Ruminococcus spp ($P = 0$)) were significantly higher in hypertension group compared to hypertension with DM group. Lactobacillus ($P = .001$), Bifidobacterium ($P = .018$), Bacteroidetes ($P = .028$), Clostridia cluster IV ($P = .004$) were significantly higher in hypertension with T2DM group compared to hypertension with CAD group.

CONCLUSION: Hypertension elderly who have CAD or T2DM have more opportunistic (bad) bacteria composition and less "good" bacteria compared to healthy or hypertension-only elderly. This was an exploratory study and further investigation to confirm these data are required.

P87

Duration of Hypertension is Associated with Cognitive Function: a Cross-sectional Study in Chinese Adults

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BACKGROUND: Hypertension (HTN) is a very prevalent public health problem and as the population ages, cognitive impairment (CI) is also going to be a public health burden. However, the relationship between hypertension duration and cognitive function decline worldwide is unknown. The aim of the study was to investigate the effects of HTN duration on CI in Chinese population.

METHODS: At baseline, 1 386 HTN patients and 293 normotensive (NT) people were enrolled. The HTN patients were further divided into four subgroups (duration of HTN <6 years, 6–10 years, 11–20 years, and >20 years) according to the HTN duration and were screened for cognitive function with neuropsychological tests including mini-mental-state-examination (MMSE) and clock-drawing-test (CDT) in comparison with the NT group.

RESULTS: More HTN patients had CI (45.3%) than NT subjects (30.4%), and increased with HTN duration ($P < .0001$). Compared with the scores of MMSE and CDT in the NT group, the declines were higher in the HTN patients ($P < .0001$), and in the four HTN subgroups, both MMSE and CDT scores fall when the HTN duration increased ($P < .05$). Furthermore, in the HTN population without CI, the trend in alteration of CI was relatively mild but still existed ($P < .05$).

CONCLUSION: People with HTN were more likely to have CI and the possibility and aggravation increased with the prolongation of HTN duration, both in amnesic and nonamnesic function.

P88

Correlation between Subjective Memory Decline and Objectively Assessed Memory Ability among Older People

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OBJECTIVE: The purpose of this study was to explore whether the objectively assessed memory is associated with the self-rated memory, and does the self-rated memory ability differ among ages.

METHODS: Data was obtained from the Evergreen project. A total of 558 people aged between 72 and 92 years were included in the current study. In order to examine the complexity of the relationships between self-rated memory and objective memory ability of the older people, cognitive functioning was assessed by metacognitive self-evaluations and Mini-D cognitive screening test.

RESULTS: Mini-D sum score and self-rated memory decline was significantly correlated ($r = -0.15$, $P < .001$). People with lower Mini-D score had 1.40 times the odds of having self-rated memory decline (95% CI = 0.90–2.18) compared to the individuals with higher Mini-D sum score (≥ 34). No significant age difference for self-rated memory ability was found (OR = 0.99, 95% CI = 0.61–1.43).

CONCLUSION: Objectively assessed memory correlated modestly, but significantly with self-rated memory. Old individuals with lower objectively assessed memory ability are not more likely to experience self-rated memory decline. Self-rated memory decline did not differ with age.

P89

Therapeutic Analysis of 112 Elderly Patients with Malignant Tumor

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OBJECTIVE: This study aimed to analyze the current therapeutic status of elderly patients with malignant tumor, and to explore the effect of different therapies on survival-time and therapeutic-effect for elderly patients

METHODS: Elderly patients with malignant tumor ($n = 112$) were retrospectively analyzed in our department from 2010 to 2012.

RESULTS: Forty-four patients received no treatment. 68 patients were treated with different methods including surgery, radiation, chemotherapy and adjuvant hormonal therapy, and among them 21 patients received surgery and 58 person times received radiation, chemotherapy or adjuvant hormonal therapy. The average Eastern Cooperative Oncology Group Performance Status (ECOG-PS) of patients in the treatment group and palliative group were 1.562, 0.603 and 3.045, respectively. It was much lower in the treatment group than that of the palliative group. Colorectal cancer, prostate cancer and lymphoma were the most common diseases in patients in treatment group. However, in the palliative group, the common diseases were lung cancer, gastric cancer, liver cancer and gallbladder carcinoma, respectively. The two year survival rate of the treatment group and palliative group were 55.9% and 11.3%, meanwhile, the median survival time of the two groups were 48 months and 15 months, respectively.

CONCLUSIONS: ECOG-PS and the type of tumor were the key points for elderly people to choose therapy. Overall survival and median survival time can be improved by active treatments for elderly patients with malignant tumor.

P90

Leptin Levels and the Relationship between Leptin and Diabetes-Related Factors in Alzheimer's Disease

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OBJECTIVE: Leptin and insulin are important regulators for food intake and body weight. Metabolic disorders such as obesity and diabetes are associated with increased risk for the development of Alzheimer's disease (AD). Our study aimed to explore plasma leptin levels and the association between leptin and diabetes-related factors (fasting glucose, fasting insulin and insulin resistance) in AD.

METHODS: Patients with AD (AD group, $n = 62$) and 60 healthy elders (control group) were enrolled in this study. Blood samples were collected for measurement of serum fasting insulin (Fins) by chemiluminescence assay and leptin by enzyme linked immunosorbent assay (ELISA). The two groups were all tested by Mini-Mental State Examination (MMSE) and Clinical Dementia Rating scale (CDR).

RESULTS: Compared to the control group, plasma leptin levels were significantly decreased in AD group ($P < .05$). However, leptin levels were not markedly different among AD patients with different severities of cognitive impairment ($F = 0.15$, $P = .87$). In addition, leptin levels showed an obviously positive correlation with Fins and insulin resistance in the control group ($P = .001$, $P = .002$), while no such results were observed in AD group ($P = .93$, $P = .95$).

CONCLUSION: Plasma leptin levels were decreased in AD, but not altered in these AD patients. There may be a regulating disorder between leptin and insulin/insulin resistance in AD.

P91

Does Kuznets Curve of Longevity Exist? Statistical Modeling and Analysis

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BACKGROUND: In economics, a Kuznets curve graphs the hypothesis that as an economy develops, market forces first increase and then decrease economic inequality. Logically we can assume the similar inverted U curve between the national health level and health inequality level. Since the most important indicator or measurement of health is the average life expectancy, we can put health inverted U curve known as the Kuznets curve of longevity. It implies that with the development of economic and progress of science, the average life span is also increasing, but the differences of life expectancy at birth among people in a country experienced or will experience a first increased and then decreased process.

METHODS: This study uses data of 179 nations in Human Development Report 2011 by United Nations Development Programme (UNDP) and SPSS 16.0 for data processing. And in this report, the inequality-adjusted life expectancy index was created to adjust the life expectancy at birth of a country for inequality in the distribution across the population. Here we use the value of the Inequality-adjusted life expectancy index to represent the national health level, and we use the loss (%) of life expectancy at birth due to Inequality-adjusted to represent health inequality level. The calculation process is divided into two main steps. On the one hand, the classical Kuznets model only discusses the Quadratic relationship, but the possible Linear, Quadratic, Power, Exponential, Cubic, Logarithmic relationships are also discussed here. On the other hand, both the impacts of national health level as

independent variables on health inequality level and health inequality level as independent variables on national health level are discussed here. The Table below lists descriptive statistics of variables.

RESULTS: Below models describe the impacts of national health level on health inequality levels and vice versa.

CONCLUSIONS: It is found that all twelve regression models, including Forward and Inverse, and including Linear, Quadratic, Power, Exponential, Cubic and Logarithmic, are statistical significant. And the R^2 of the twelve models are rather satisfactory, for eleven of them are bigger than 0.8, the rest one are also big than 0.7. But the Health inverse-U shaped patterns of inequality or the Kuznets curve of longevity is falsified in this study, for the graphical representation of the corresponding Quadratic model is actually Regular U type rather than Inverted U Curve. In fact, despite the above tedious figures and calculations, intuitively, these 12 models drawn by the SPSS software are all quite similar to a linear with negative slope of 45 degrees. The most plausible here is that social progress, not only prolongs the life of the people, but also reduces synchronously the differences of life length among people.

Table below lists descriptive statistics of variables

	N	Minimum	Maximum	Mean	Std. Deviation
National Health Level	179	2.21	9.65	6.56	2.13
Health Inequality Level	179	2.92	5.20	1.84	1.34

Independent: National Health Level Dependent: Health Inequality Level	Model Summary					Parameter Estimates			
	R Square	F	df1	df2	Sig.	Constant	b1	b2	b3
Linear	.965	4.932E3	1	177	.000	58.954	-61.865		
Logarithmic	.968	5.338E3	1	177	.000	2.122	-33.191		
Quadratic	.975	3.378E3	2	176	.000	68.722	-99.611	31.542	
Cubic	.975	2.242E3	3	175	.000	66.998	-89.192	12.902	10.254
Power	.820	808.607	1	177	.000	5.612	-1.826		
Exponential	.917	1.947E3	1	177	.000	145.709	-3.602		

Independent: Health Inequality Level Dependent: National Health Level	Model Summary					Parameter Estimates			
	R Square	F	df1	df2	Sig.	Constant	b1	b2	b3
Linear	.965	4.932E3	1	177	.000	.943	-.016		
Logarithmic	.917	1.947E3	1	177	.000	1.322	-.254		
Quadratic	.974	3.348E3	2	176	.000	.987	-.021	.000	
Cubic	.974	2.219E3	3	175	.000	.987	-.022	.000	-2.613E-8
Power	.820	808.607	1	177	.000	1.988	-.449		
Exponential	.968	5.338E3	1	177	.000	1.047	-.029		

P92

Relationships between the Prothrombotic State Markers and Leukoaraiosis in Aged Patients with Metabolic Syndrome

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OBJECTIVE: This study aimed to investigate the relationship between prothrombotic state markers and leukoaraiosis in aged patients with metabolic syndrome.

METHODS: A total of 350 metabolic syndrome patients in the affiliated hospital of Zhejiang University who were receiving treatment from January, 2013 to January, 2015. According to the results of head MRI, the patients were divided into LA group and

non-LA group. The severity of LA was graded by its appearance under MRI and the Blennow scale. Then, the plasma levels of fibrinogen (Fib), von willebrand factor (vWF), D-dimer (D-D) and Prothrombin time (PT), thrombin time (TT), and activated partial blood coagulation time (APTT) level of the two groups were measured and compared respectively.

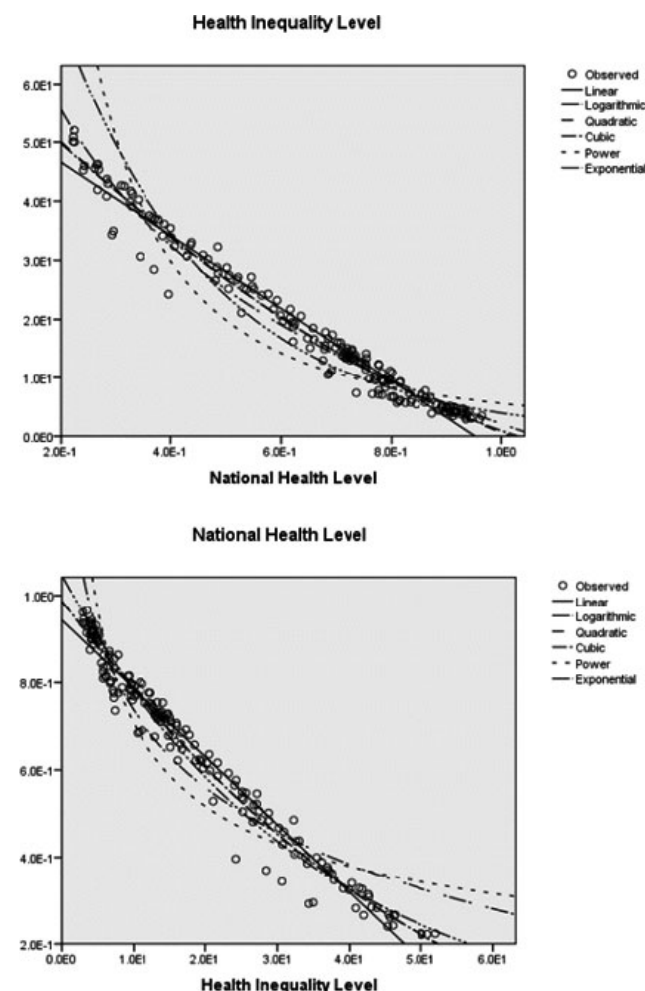
RESULTS: Levels of Fib, D-D, and vWF in LA (n = 203) group were obviously higher than those of non-LA group (n = 147) ($P < .05$). Spearman correlation analysis showed that the stage of leukoaraiosis severity was both negatively correlated with FIB, vWF and D-D ($r = 0.014, 0.089, 0.215; P < .05$). Multivariate liner regression analysis showed that Fib, vWF, HbA1c and the duration of hypertension were the main risk factors for leukoaraiosis in aged patients with metabolic syndrome. (The standard regression coefficient was 2.516, 0.312, 0.206 and 0.167).

CONCLUSION: HbA1c levels and duration of hypertension were positively related to LA. The severity of LA in aged patients with metabolic syndrome can be predicted by Fib and vWF.

P93

Sarco-osteoporosis: Prevalence and Association with Frailty in Chinese Community-dwelling Older Adults

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OBJECTIVES: This study aimed to apply Asian Working Group for Sarcopenia (AWGS) criteria to estimate the prevalence of sarco-osteoporosis and to investigate its relationship with frailty, in a sample of community-dwelling Chinese older people.

METHODS: A total of 316 participants were recruited from Changsha city and its surrounding area, Hunan Province, China. Regression analysis using frailty as the dependent variable for groups of sarcopenia, osteoporosis, sarco-osteoporosis and adjusted for covariates was performed.

RESULTS: The prevalence rate of sarco-osteoporosis was 10.4% in older men and 15.1% in older women. People ≥ 80 years old (OR 4.8; 95% CI, 3.05–10.76; $P = .027$), women (OR 2.6; 95% CI, 1.18–2.76; $P = .036$) and higher level of comorbidity (OR 3.71; 95% CI, 1.61–10.43; $P = .021$) were independently associated with the likelihood of being sarco-osteoporosis. In the frail group, sarco-osteoporosis occurred in 26.3% for men ($n = 5$), 38.5% for women ($n = 10$), and in lower proportion in the prefrail (13.6% of men, $n = 11$; 16.2% of women, $n = 12$) and nonfrail group (1.6% of men, $n = 1$; 1.9% of women, $n = 1$) ($P < .05$, respectively). Furthermore, the likelihood of being frail/prefrail was substantially higher in the presence of sarco-osteoporosis (OR 4.16; 95% CI, 2.17–17.65; $P = .019$ in men; and OR 4.67; 95% CI, 2.42–18.86; $P = .007$ in women).

CONCLUSIONS: Patients with sarco-osteoporosis were more likely older (≥ 80 years), women, frail or prefrail, and with greater burden of comorbidities. Sarco-osteoporosis was more closely associated with frailty/prefrailty than either sarcopenia or osteoporosis alone in community-dwelling Chinese elders.

P94

Time-dependent Increase of Chitinase1 in APP/PS1 Double Transgenic Mice

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BACKGROUND: It is reported that chitinase1 increases in cerebrospinal fluid and peripheral blood of patients with Alzheimer's disease (AD). However, the alteration of chitinase1 in the progress of AD is still unclear. The present study aimed to detect chitinase1 level in different stages of APP/PS1 double transgenic mice.

METHODS: APP/PS1 double transgenic mice at the age of 4 months, 12 months and 22 months were used as experimental models. Cognitive function was detected by Morris water maze test in APP/PS1 double transgenic mice as well as controls. ELISA and the quantitative RT-PCR were used to detect chitinase 1 level in different groups.

RESULTS: The study showed that expression of chitinase1 differed from various time points (4 months, 12 months, and 22 months) of APP/PS1 transgenic mice, and the level of chitinase1 gradually increased in a time-dependent manner. While there were no statistical differences in the chitinase1 level among the wild-type mice in various ages. Moreover, chitinase1 increased significantly in APP/PS1 transgenic mice aged 12 months and 22 months compared with the age matched wild-type group, respectively ($P < .05$). However, no difference of chitinase1 was found between 4-month-old APP/PS1 mice and wild-type mice. Compared with the age matched wild type group, the consequences of qRT-PCR on the increase in chitinase1 was in accordance with ELISA in APP/PS1 transgenic mice. In addition, Morris water maze showed that 4-month-old APP/PS1 mice had normal spatial learning and impaired spatial memory. Both spatial learning and spatial memory in 12 months and 22 months old APP/PS1 mice were declined.

CONCLUSIONS: Time-dependent increase of chitinase1 in APP/PS1 double transgenic mice indicates that the level of chitinase1 is associated with decline of cognition. Chitinase1 might be a biomarker of disease progression in AD.

P95

Study on the Application of DAPK and APC1A Gene Promoter Methylation in Diagnosis of Non-small Cell Lung Cancer

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OBJECTIVE: This study aimed to detect the expression of DAPK and APC1A gene promoter methylation in NSCLC (Non-small cell lung cancer), and then to evaluate the clinical significance and correlation of the expression of DAPK and APC1A in NSCLC.

METHODS: Seventy-eight patients with NSCLC underwent surgical resection at the Tianjin Medical University General Hospital from January to December 2014 and 26 control specimens with normal lung tissue were selected as objects. The expression of DAPK and APC1A were analyzed in 78 NSCLC patients and 26 control specimens were tested by methylation-specific PCR. All statistical analyses were performed with the SPSS 16.0 for Windows software system. The relations of clinical factors and expression of DAPK and APC1A were performed by Pearson Chi-Square test. The correlation of DAPK and APC1A was analyzed by Spearman Correlation Coefficients. The differences were considered significant when $P < .05$, and differences were considered remarkable significant when $P < .01$.

RESULTS: The positive expression rates of DAPK in NSCLC and in normal lung tissue were 73.3% and 10.6%, and the positive expression rates of APC1A in NSCLC and in normal lung tissue were 80.0% and 15.1%, respectively. The differences were significant ($P < .01$). The positive expression of DAPK and APC1A in NSCLC was correlated with age, histological subtype, grading, lymphatic metastasis or TNM stage ($P < .05$). The expression of DAPK was positively correlated with the expression of APC1A in NSCLC ($r = 0.516$, $P < .05$).

CONCLUSION: Expression of DAPK and APC1A was correlated with age, histological subtype, and grading, lymphatic metastasis or TNM stage. DAPK expression level was positively correlated with APC1A in NSCLC. They may closely correlate with progression, invasion and metastasis of lung carcinoma.

P96

Personal and Environmental Correlates of Physical Function in China's Older Adults

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OBJECTIVE: Socio-ecological models emphasize that environmental factors play a role in shaping older adults' moderate to vigorous physical activity (MVPA). However, little research validates this relationship in China. This study aimed to explore the associations of individual and environmental factors with MVPA among Chinese older adults.

METHODS: We used the 2011 China Health and Retirement Longitudinal Survey, a national representative sample of respondents over age 45. MVPA was a binary variable (met guideline or not), using the WHO's recommendations for physical activity. We conducted a multilevel logistic regression using HLM 6.08 to examine if individual factors and environmental/infrastructure factors at the community level affected the MVPA of older adults.

RESULTS: Results showed that at the individual level, single compared to married people and women were less likely to meet MVPA recommendations. Levels of physical mobility and self-rated health were positively associated with MVPA. Health behaviors, such as smoking and drinking, were not associated with MVPA. Interestingly, self-rated standard of living and health before age 15 were negatively associated with MVPA. At the community level, people who lived in plain rather than mountainous areas as well as lived in environments with better infrastructure, such as roads, were more likely to meet activity guidelines. Accessibility to public transportation and shopping were negatively related to MVPA. There was no association between access to sport facility, temperature, and MVPA.

CONCLUSIONS: Strengthening the infrastructure of communities and introducing MVPA programs in accordance with various local conditions may enhance physical activity levels of older Chinese people.

P97

Relationship between Serum Adiponectin and Atherosclerosis in Patients with Hypertension and Obesity

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OBJECTIVE: This study aimed to investigate the relationship between serum adiponectin and atherosclerosis (AS), and to explore potential mechanisms.

METHOD: Subjects were divided into control, simple obesity, simple hypertension and obesity-associated hypertension groups. Hypertension was diagnosed under the guidance of China hypertension diagnostic criteria in 2005, and obesity was diagnosed by body mass index according to $\text{BMI} \geq 25 \text{ kg/m}^2$ (Asia-Pacific standards). A general physical examination and conventional blood glucose, blood lipids and other biochemical tests were performed for all subjects, and serum concentrations of adiponectin were measured by using standard ELISA assay. Common carotid artery intima-media thickness (CCA-IMT) and carotid artery diameter (CADIA) were measured using logic9 color ultrasonography.

RESULTS: BMI, triglyceride (TG), low density lipoprotein-cholesterol (LDL-C) in control group, simple obesity group, simple hypertension group and obesity-associated hypertension group were increased and high density lipoprotein-cholesterol (HDL-C) were decreased across the groups in the order as listed. There is a statistical significance ($P < .05$) in each group. Fasting plasma glucose (FPG), systolic blood pressure (SBP), diastolic blood pressure (DBP) in simple hypertension group and obesity-associated hypertension group was higher than that of control group, and there was a statistical significance ($P < .05$) among the different groups. BMI, SBP, DBP in simple hypertension group and obesity-associated hypertension group were higher than that in simple obesity group, and there was a statistical significance ($P < .05$) among the different groups; HDL-C in simple hypertension group and obesity-associated hypertension group was lower than that of simple obesity group, and there had a statistical significance ($P < .05$). BMI, SBP, DBP, FPG, LDL-C in obesity-associated hypertension group was higher than that in simple hypertension group. HDL-C in obesity-associated hypertension group was lower than that in simple hypertension group, and CCA-IMT in control group, simple obesity group, simple hypertension group and obesity-associated hypertension group were increased gradually. CADIA and serum adiponectin concentrations in four groups were decreased gradually, and there was a statistical significance ($P < .05$). Serum adiponectin concentrations in control group, simple obesity group, simple hypertension group and obesity-associated hypertension group were decreased gradually, and there was a statistical significance ($P < .05$). BMI and CCA-IMT were positively correlated

($r = 0.576$, $P = .000$), both CADIA and serum adiponectin concentrations were negatively correlated with BMI ($r = -0.352$, $P = .000$, $r = 0.284$, $P = .000$), and serum adiponectin concentrations and CCA-IMT were negatively correlated ($r = -0.346$, $P = .000$), and serum adiponectin concentrations and CADIA were positively correlated ($r = 0.222$, $P = .000$).

CONCLUSION: Hypertension and obesity are risk factors of AS. When they exist at the same time, they may play a synergy, which can accelerate the process of atherosclerosis. Serum adiponectin level was negatively correlated to the degree of obesity and atherosclerosis, and its lower levels may accelerate the process of atherosclerosis, and associated with hypertension and obesity.

P98

Correlations between Visfatin and MMP-9 and Insulin Resistance in Patients with Coronary Heart Disease

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OBJECTIVE: This study aimed to explore the relationships of serum levels of visfatin and MMP-9 with insulin resistance (IR) in patients with coronary heart disease (CHD).

METHOD: Eighty patients who had coronary angiography in the first hospital of Jilin university were recruited. According to the level of homeostasis model assessment (HOMA-IR), $\text{HOMA-IR} \geq 2.8$ as the standard of IR, thirty-two patients were classified as CHD with IR, and twenty-eight as CHD without IR. Another twenty healthy individuals were selected as the control group. Height, weight, blood pressure, fasting serum glucose, insulin, triglycerides, total cholesterol, HDL-C, LDL-C, serum concentrations of visfatin and MMP-9 were measured. HOMA-IR and Gensini integral were calculated as an evaluation method of insulin resistance and coronary artery's stenosis degree.

RESULTS: Serum concentrations of visfatin and MMP-9 of the patients in control group, CHD without IR and CHD with IR were increased gradually. And compared with each other, there is a statistical significance ($P < .05$). In the patients with CHD, both IR and Gensini integral were positively correlated with serum concentration of visfatin ($r = 0.5549$, $P < .01$; $r = 0.6231$, $P < .01$). Both IR and Gensini integral were positively correlated with serum concentration of MMP-9 ($r = 0.4211$, $P < .01$; $r = 0.6179$, $P < .01$). There was a positive correlation between visfatin and MMP-9 ($r = 0.6377$, $P < .01$). There was a positive correlation between Gensini integral and IR ($r = 0.5023$, $P < .01$).

CONCLUSIONS: IR is a risk factor of CHD, and serum concentrations of visfatin and MMP-9 have positive associations with CHD and IR. Serum concentrations of visfatin and MMP-9 combined with coronary angiography have synergy in the detection for coronary stenosis degree in the patient with CHD.

P99

Correlation between Unstable Angina Pectoris and Intestinal Flora Imbalance in the Elderly

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OBJECTIVE: This study aimed to analyze the relationship between the Elderly Unstable Angina Pectoris and Intestinal Flora Imbalance in the Elderly.

METHODS: Thirty elderly patients who had unstable angina and 30 control subjects with similar age, gender and BMI in the first hospital of Jilin university were recruited. Fasting blood testing myocardial injury markers, triglycerides, total cholesterol, HDL-C, LDL-C, and homocysteine were taken. Data of electrocardiography

before and after treatment also collected. Feces of participants were collected 0.5 g of wet weight, and then total bacterial DNA was extracted. The primers of *Escherichia coli*, *Bacteroides thetaiotaomicron*, *Lactobacillus* were designed by primer 3.0. We can infer the concentration of different intestinal flora from copy numbers of bacteria in fecal samples by PCR. Statistical analyses were performed by SPSS 17.0.

RESULTS: Unstable angina pectoris group had significantly greater amount of *Escherichia coli* than the normal group ($P < .05$). The normal group had significantly greater amount of *Bacteroides thetaiotaomicron* and *Lactobacillus* than the unstable angina pectoris group ($P < .05$). There was a significant positive correlation ($r = 0.394$, $P = .031$) between the concentration of *Escherichia coli* and the level of HCY in the unstable angina pectoris group. There was a significant negative correlation ($r = -0.524$, $P = .003$) between the *Lactobacillus* and the level of LDL in the unstable angina pectoris group.

CONCLUSIONS: The results of this study indicate that the elderly unstable angina pectoris was characterized by an increased concentration of *Escherichia coli*, while by a decreased concentration of *Bacteroides thetaiotaomicron* and *Lactobacillus*. Intestinal flora imbalance may contribute to the development of unstable angina.

P100

Correlation between Myocardial Nutrients-1 and Left Ventricular Hypertrophy in Hypertension Patients

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OBJECTIVE: This study aimed to investigate the relationship between cardiotrophin-1 (CT-1) changes in hypertension patients and its possible formation mechanism and risk factors.

METHODS: Sixty-four patients with hypertension patients were enrolled and divided into two groups: 31 left ventricular hypertrophy cases and 33 non-left ventricular hypertrophy cases. Thirty healthy individuals matched for the age, gender and weight were enrolled as healthy controls. All subjects were measured for height, weight, blood pressure, fasting blood glucose, blood lipid and chemistry determination of ELISA two groups of Serum CT-1 concentrations were determined using standard ELISA assay. Ultrasonic heartbeat graph in patients with left ventricular end diastolic diameter, end diastolic left ventricular posterior wall thickness, end diastolic interventricular septal thickness were determined according to the formula of left ventricular weight (LVM) and left ventricular mass index (LVMI).

RESULTS: CT-1 levels were significantly higher in the group of hypertension with left ventricular hypertrophy than those in hypertension without left ventricular hypertrophy group; both were significantly higher than those in healthy controls (all $P < .01$).

CONCLUSIONS: Serum CT-1 levels were higher in hypertension patients. CT-1 levels were significantly associated with LVH in these patients.

Table Comparison between CT-1 and LVMI ($\bar{x} \pm s$)

Parameter	Control group (n=30)	The group of hypertensive with left ventricular hypertrophy (n=33)	Hypertension without left ventricular hypertrophy group (n=31)
CT-1(pg/ml)	41.44±9.14	179.6±18.02**	331.28±11.89**ΔΔ
LVMI(g/m ²)	82.9±9.2	104.09±9.59**	154.17±21.25**ΔΔ

** Compared with the control group, $P < .01$;

ΔΔ Compared with hypertension without left ventricular hypertrophy group, $P < .01$

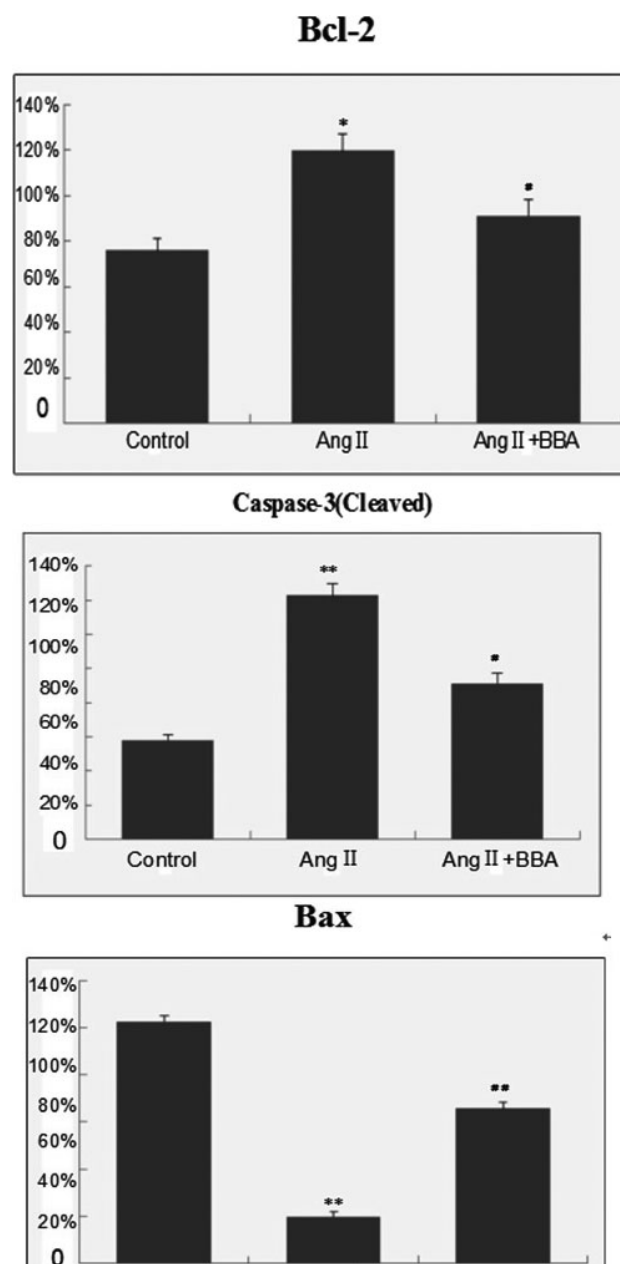
P101

Protective Effects of Blueberry Anthocyanin on Ang II-induced Apoptosis to Human Umbilical Vein Endothelial Cells

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OBJECTIVE: This study aimed to establish HUVEC apoptosis model, using Western blot to detect changes of apoptosis-related protein Bcl-2, Bax and caspase-3 before and after Ang II caused HUVEC apoptosis. The content of intracellular ROS was measured after Ang II reaction, and the impact of blueberry anthocyanin in above circumstances was observed to investigate the protective effect of blueberry anthocyanin in HUVEC.

Figure: Blueberry anthocyanin to HUVEC Bcl-2, Bax, Caspase-3 (Cleaved) expression



METHODS: Human umbilical vein endothelial cells (HUVEC) were chosen, and HUVEC apoptosis model was developed. In vitro cultured human umbilical vein endothelial cells (HUVEC) were divided into control group (Control), Ang II group (Ang II), blueberry anthocyanin Group (BBA), Ang II + blueberry anthocyanin group (Ang II +BBA). MTT assay was used to check out cell activity so as to determine the optimal concentration of Ang II in inducing HUVEC apoptosis and to protect concentration of blueberry anthocyanin. Cell apoptosis were detected by Annexin V-FITC-labeled flow cytometry. Western Blot was used to detect the expression of Bcl-2, Bax and Caspase-3 protein. The intracellular ROS level was detected.

RESULTS: HUVEC apoptosis was significantly increased after Ang II stimulation. Compared to Ang II group, Bcl-2 protein level of the blueberry anthocyanin Group was significantly increased, and the expression of Bax and Caspase-3 protein was decreased, and the intracellular ROS level was on the downward trend (Figure).

CONCLUSIONS: Angv II induced apoptosis in HUVEC cells. Ang II induced HUVEC apoptosis could be secondary to the imbalance expression of Bcl-2 and Bax protein and Caspase-3 activation. Blueberry anthocyanin had inhibitory effect on HUVEC apoptosis.

P102

Purification and Characterization of Plantaricin ZY003, a Novel Bacteriocin Produced by *Lactobacillus Plantarum* ZY003

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OBJECTIVE: The aim of the present study was to purify and characterize a bacteriocin produced by *Lactobacillus plantarum* ZY003 isolated from fresh milk with a broad-spectrum activity against both Gram-positive and Gram-negative bacteria.

METHODS: This bacteriocin was purified by XAD (Xpress-application Developer, XAD) 2, cation exchange chromatograph, gel chromatograph, and HPLC (High Performance Liquid Chromatography, HPLC). The peptide was inactivated by proteinase K, α -chymotrypsin, and trypsin, but was resistant to heat and catalase. Purification of peptide was performed by ammonium sulphate precipitation followed by gel filtration chromatography.

RESULTS: *Lactobacillus plantarum* ZY003 isolated from infant feces produced a novel antimicrobial peptide, named plantaricin ZY003 with thermostability up to 121 °C. This peptide exhibited antibacterial activity against a wide range of bacteria including Gram-positive and Gram-negative bacteria under pH values from 2.0 to 5.0. The molecular mass of the peptide was 2632 Da determined by mass spectrometric analysis. These sequences of this bacteriocin showed no homology with known bacteriocins when compared with those available in the database, which indicates that the plantaricin ZY003 is a novel bacteriocin.

CONCLUSION: These characters of plantaricin ZY003 suggest that this bacteriocin can potentially be useful for pathogen control in food.

P103

Comprehensive Assessment of the Nutritional Status of Elderly Patients in Five Cities in China

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OBJECTIVE: This study aimed to assess nutritional status of elderly Chinese patients and to evaluate the association between nutritional status of elderly patients and their activities of daily living.

METHODS: This was an open and multi-centered investigation in five cities including Beijing, Shanghai, Chengdu, Guangzhou, and Chongqing. All of the participants are 65 years old or elder senior patients. The Mini-Nutritional Assessment was used to investigate the nutritional status of the elderly patients. Activities of Daily Living Scale were used to evaluate their activities of daily living. Data on basic social demographic characteristics such as age, sex, education, and medical insurance were also collected.

RESULTS: A total of 5,973 subjects were enrolled in this study. Proportion of normal nutritional status in Beijing was the highest of 52.8%, and the lowest in Chongqing was 14.3%. The mean rate of malnutrition and nutritional risk for elderly patients was 53.7%. A comparison of the elderly patients' nutritional status showed that it was better in Beijing, Shanghai, and Chengdu than in that in Guangzhou ($P < .05$), and the latter was better than that in Chongqing ($P < .05$). 54.2% of elderly patients with normal nutrition could take care of themselves and 11.7% of them could not. The results suggested that the nutritional status and ability to perform daily activities were positively correlated ($P < .05$).

CONCLUSION: The elderly patients' malnutrition status in these five cities presents a less optimistic outlook and needs a further consideration. In addition, elderly patients who can perform activities of daily living well are more likely to have a better nutritional status.

P104

Chronic Obstruction Pulmonary Disease and Senescence

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OBJECTIVE: Senescence is considered as an important contributing factor in driving chronic obstruction pulmonary disease (COPD). This study aimed to review literature on the relationship between COPD and senescence.

METHODS: Literature retrieval was made in PubMed by using keywords of "COPD, senescence/aging, autophagy, oxidative stress", and analysis of research findings was made.

RESULTS: The pathogenesis of COPD is loosely linked with senescence. Age-related lung structure and function changes increase susceptibility of COPD. In the development of COPD, senescent cell cytokines enlarge inflammatory phenotype including IL-1, IL-6, TNF- α , MMP-9 and others, which accelerate lung aging. Senescence is an important factor of inflammatory reaction. NF- κ B signaling controlling release of inflammatory mediators is a significant SASP trigger. Telomere shortening causes DNA damage response, p21CIP1 expression activating pRB, cells keeping senescence via p16. II alveolar cells, endothelial cells increase in COPD lung. Alveolar cells, pulmonary endothelial cells and fibroblasts of Emphysema have shorter telomeres. Oxidative stress causes DNA damage response and influences telomere length. Oxidation and antioxidation out-of-balance makes apoptosis of lung parenchyma cells increase, and generate lung tissue destruction and airflow obstruction, which eventually develop COPD. There appears to be senile cells with level of autophagy related gene ULK1 increase. Double membrane autophagosomes vesicles prove to be autophagy associating with aging. LC3B-, autophagy-associated proteins and autophagosomes increase expression in human lung specimens from patients with COPD. Aging and anti-aging out-of-balance triggers lung aging, especially cellular senescence regulators like T-box gene and CDKN2A. SIRT1 makes DNA damage and repair via FOXO3, p53, NF- κ B and histones to achieve anti-inflammatory and antioxidant effects. Klotho gene, SMP30 and senescence related genes, DNA repair factors are all involved in the pathological process of COPD.

CONCLUSION: Current literature suggests the role of cellular senescence contributing to COPD. Antioxidants and anti-aging

molecules could be a promising therapeutic interventions for COPD.

P105

Advantages of PET-CT Imaging for Geriatric Patients with Gastric Cancer

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OBJECTIVE: This study aimed to explore the value of PET-CT scan for geriatric patients with gastric cancer.

METHODS: The results of PET-CT scan for 86 geriatric patients with gastric cancer who underwent PET-CT scan in the different courses during treatment were retrospectively summarized. Then, the results of PET-CT scan plus other conventional methods were analyzed to identify the value of PET-CT scan in diagnosis, staging, response evaluation, monitoring recurrence and metastasis following treatment.

RESULTS: For 45 patients who underwent PET-CT scan for post-operation follow up besides conventional modality, PET-CT was performed to be more sensitive, and additional 1 with local recurrence, 2 with lymph node metastasis and 4 with distant metastasis were found compared with conventional screening methods. Among 11 patients, there were 7 responder patients and 4 non-responder patients. Moreover, the difference of SUV max percentage underwent pre and post chemo-radiotherapy PET-CT scan for response evaluation. Of 11 geriatric patients, 6 had clinical PR and 5 had SD, while percentage decrease of SUV max identified changes between the patients with PR and SD is significant. 51 patients underwent PET-CT scan at clinical initial consultation for metastasis prediction. Five in 51 cases were found distant metastasis, including lung in 2, bone in 1, and abdomen wall in 2. Two with additional metastasis was found by PET-CT scan not by conventional methods. PET-CT scan is better than Contrast Enhanced Computed Tomography (CECT) scan in prediction of lymph node metastasis, with sensitivity 38% vs 32%, specificity 95.7% vs 80.1%, accuracy rate 85.2% vs 74.0%, positive prediction rate 65.7% vs 33.4%, and negative prediction rate 85.7% vs 82.3% separately. PET-CT predicts lymph node metastasis more accurately than CECT, with higher area under the ROC curve (0.661 vs 0.543).

CONCLUSION: PET-CT scan was useful in diagnosis of gastric cancer and might give more information on N and M staging in elderly patients.

P106

Evaluation of Health-Related Quality of Life for Senior Chinese with Stable Chronic Obstructive Pulmonary Disease (COPD)

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OBJECTIVE: This study aimed to assess the health-related quality of life for senior Chinese with stable COPD.

METHODS: Patients were recruited from outpatient department of a tertiary hospital in ShaanXi Province, China. Demographic characteristics such as age, gender, weight and height were collected. Spirometry was performed for every patient after salbutamol 400 µg was inhaled. Six-minute walk test was done according to the guidelines that were developed by American Thoracic Society in 2002. Four questionnaires, including COPD Assessment Test (CAT), Clinical COPD Questionnaire-week Version (CCQ), St George's Respiratory Questionnaire for COPD Patients (SGRQ-C) and Chronic Respiratory Questionnaire-Self Administered Stan-

dardized Activities-First Administration Version (CRQ-SAS) were administered to all participants.

RESULTS: Ninety subjects were enrolled between September 2013 and September 2014. Of the 90 participants, 75 (83.3%) were male and 70 (77.8%) were smokers or ex-smokers, the mean age was 67.83 years, and the mean educational history was 8.89 years. The mean body mass index was 23.27 kg/m², and the mean post-bronchodilator FEV1 was predicted to be 51.21%. The average 6-minute walk distance was 424.53 meters. The average score of CAT and CCQ were 20.38 and 23.92, respectively. The mean SGRQ-C score was 38.47. The symptoms, activity and impacts average scores of SGRQ-C were 64.30, 45.11 and 25.67 respectively. Of CRQ-SAS, the average scores of dyspnea, fatigue, emotional function and mastery were 2.92, 2.85, 2.95 and 2.90 respectively. There was a weak correlation between FEV1 and each questionnaire scores. There was a moderate correlation between 6-minute walk distance and SGRQ-C, CCQ and CRQ-SAS. However, there was a weak correlation between CAT and 6-minute walk distance. For 6-minute walk distance, activity score of SGRQ-C, impacts score of SGRQ-C and history of coronary artery disease were independent influencing factors. For BODE index, activity and impacts score of SGRQ-C were independent influencing factors.

CONCLUSIONS: CAT, CCQ, SGRQ-C and CRQ-SAS were suitable to assess health-related quality of life for Chinese senior patients with stable COPD. Among four questionnaires, SGRQ-C appeared to be the best one to evaluate health-related quality of life.

P107

CIK Immunotherapy and Non-small Cell Lung Cancer in Geriatric Patients

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OBJECTIVE: The study aimed to determine the long-term efficacy and effect on cellular immune functions of adjuvant immunotherapy with autologous cytokine-induced killer (CIK) cells for geriatric patients with non-small cell lung cancer.

METHODS: Seventy five geriatric patients with stage II or III non-small cell lung cancer who had undergone pulmonary resection operation were randomly assigned to immunotherapy with CIK or no immunotherapy (control). Patients were followed up for over 5 years.

RESULTS: The 5-year overall survival (OS) and 5-year disease-free survival (DFS) rate for the immunotherapy vs control group were 32.1% vs 25.6% ($P = .085$) and 26.3% vs 11.2% ($P = .041$) respectively. In the immunotherapy group, the mean CD3⁺ level, CD4⁺ level, and CD4⁺/CD8⁺ ratio increased from 53.8%, 27.1%, and 0.9 respectively, at baseline to 67.1%, 35.5%, and 1.5 respectively, 1 week after the first CIK-cell treatment and returned to baseline after 2 months, and these values were significantly higher in immunotherapy patients after 3 courses of immunotherapy than those in controls 6 months after adjuvant chemotherapy.

CONCLUSION: Adjuvant immunotherapy with CIK cells prolongs OS and DFS in geriatric patient with non-small cell lung cancer (stage II or III). This treatment may improve the host's immune function, but multiple cycles are necessary for long-term therapeutic efficacy.

P108

The Assessment of Medicine Treatment for Elderly Patients with Intestinal Infections

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OBJECTIVE: This study aimed to assess the therapeutic effect of intestinal infections via analyzing clinical features and medication in elderly patients.

METHODS: Patients aged 60 and older who were hospitalized for intestinal infection (N = 40) were enrolled into this study. The screening criteria of the case appeared to be sign of intestinal infection in 72 hours after being in hospital, i.e. pathogenic bacteria were detected. Treatments adhered to the principle of selective use of antibiotics and combination of medicine for critical patients. The mild/moderate patients were treated using aminoglycoside antibiotics typically as tobramycin, 3–5 days for a course. The critical patients were treated using the combination of cephalosporin and quinolones, 10 days for a course. In addition, for those patients with senile disease, treatment order was to control infection first and then to give the treatment of primary disease at the same time with strengthening nutrition support. On the other hand, as an auxiliary, bacillus licheniformis was used to adjust the balance of bacteria.

RESULTS: Of 40 patients, recovery 16 (40%), remission 20 (50%), death 4 (10%) with multiple organ failure due to a variety of complications. Also, treatment effect was significantly related to whether there was timely medical care in the community and presence of comorbid chronic diseases. Age or sex did not appear to have a significant impact on patients' outcome.

CONCLUSIONS: Early diagnosis is a key factor for improving treatment effectiveness of intestinal infections in elderly patients.

P109

Utility of Bacillus Licheniformis in the Treatment of Elderly Patients with Chronic Diarrhea

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BACKGROUND: This study aimed to evaluate the efficacy of bacillus licheniformis in the treatment of elderly patients with chronic diarrhea.

METHODS: Patients aged 60 and older with chronic diarrhea that was mainly caused by chronic hepatitis, ulcerative colitis and cancer radiotherapy/chemotherapy induced intestinal flora imbalance (N = 48) were recruited. In the stool routine test (Stool RT) at their first visit, 12 were '+' for both RBC & WBC, 10 were '+' for both WBC & Pus-cell, 10 were '+' for WBC and 16 were '+' for Pus-cell. Oral bacillus licheniformis capsules (viable count 2.5×10^8 CFU per capsule) were given 3 times daily, 2 capsules each time and initial dose was double. A course was 30-days.

RESULTS: Of 48 patients, half of them were given a course of treatment and the others with more stubborn illness were given two courses. All patients' diarrhea, abdominal distension, poor appetite and other symptoms disappeared as well as the tests for RBC, WBC and Pus-cell were wholly changed to '-' in the Stool RT at the end of the treatment. In addition, the main items of Blood RT through the treatment period also showed welcome change that gave average statistical data: $3.31 \pm 0.14 \rightarrow 4.21 \pm 0.21$ for RBC, $87.82 \pm 5.69 \rightarrow 113.60 \pm 8.21$ for HGB, $6.20 \pm 0.71 \rightarrow 7.05 \pm 0.62$ for WBC and $132.71 \pm 27.3 \rightarrow 159.18 \pm 38.1$ for PLT.

CONCLUSIONS: Based on one or two 30-days course, using bacillus licheniformis could effectively treatment the elderly patients with chronic diarrhea and significantly improve the anemia secondary to diarrhea.

P110

Correlation between Cardiovascular Risk Factors and Brachial-ankle Pulse Wave Velocity

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OBJECTIVE: This study aimed to explore the relationship between brachial-ankle pulse wave velocity (baPWV) and cardiovascular risk factors in Fujian coastal population.

METHODS: Cardiovascular risk factors and baPWV were measured for all 3344 objects. Analyses were performed to evaluate the relationships between baPWV and cardiovascular risk factors using statistical methods of correlation, multivariate linear regression, and univariate analysis.

RESULTS: Mean baPWV was 1460.8 ± 367.7 mm/s for all. Correlation analysis showed baPWV was correlated with age, gender, systolic blood pressure (SBP), diastolic blood pressure (DBP), body mass index (BMI), plasma glucose (FPG), triglycerides (TG), total cholesterol (TC), low-density lipoprotein cholesterol (LDL-C), uric acid (UA), glycosylated hemoglobin (HbA1c). Multiple linear regression analysis showed that the risk factors of baPWV was the SBP, age, HbA1c, BMI, TC, TG in turn ($P < .05$). Univariate analysis showed the differences of the abnormal rate of baPWV among the groups of age ($\chi^2 = 44.3$, $P < .01$) and SBP ($\chi^2 = 56.7$, $P < .01$).

CONCLUSION: The major risk factors of baPWV were systolic blood pressure and age, which suggested the importance of controlling blood pressure and cardiovascular risk factors for the prevention of the deterioration of atherosclerosis.

P111

Serum Uric Acid Level and its Correlation with Cardiovascular Risk Factors in Coastal Areas of Fujian Province

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OBJECTIVE: This study aimed to access serum uric acid level and its correlation with cardiovascular risk factors in Fujian province.

METHODS: The enrolled subjects (n = 3349) were selected by multi stage stratified cluster random sampling and variance analysis and binary logistic regression were applied to evaluate cardiovascular risk factors and serum uric acid level.

RESULTS: The morbidity of hyperuricemia was 19.44% in 3349 coastal population in Fujian Province while 18.85% was the standardized age in our country. Combined with hyperuricemia, the morbidity of dyslipidemia, overweight and hypertension accounted for 77.88% (507/651), 48.39% (315/651) and 40.09% (261/651) respectively, which were much higher than those with normal serum uric acid (all $P < .05$). There were positive correlation between serum uric acid level and age, systolic pressure, diastolic pressure, waistline, body mass index, triglyceride, total cholesterol, and low density lipoprotein cholesterol (LDL-C). The correlation between high density lipoprotein cholesterol (HDL-C), estimated glomerular filtration rate (eGFR) and serum uric acid level were negative. Logistic analysis showed that the morbidity of hyperuricemia increased with high triglyceride, high LDL-C, low HDL-C and low eGFR. The odds ratios value and 95% confidence intervals were 2.533 (2.004–3.201), 1.493 (1.161–1.919), 1.133 (1.133–1.675), and 1.521 (1.521–2.208).

CONCLUSIONS: The morbidity of hyperuricemia was 18.85% in coastal population in Fujian Province. People with hyper-

uricemia were always accompanied with many cardiovascular risk factors and metabolic diseases, such as dyslipidemia.

P112

Can Gait Speed Replace Comprehensive Geriatric Assessment in the Elderly?

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OBJECTIVE: Aging is associated with generalized slowing of movement. Gait speed is an indicator of physical performance. The present study aimed to determine whether gait speed could replace the comprehensive geriatric assessment (CGA) and to assess the status of the elderly.

METHODS: A total of 531 participants aged 60 years or over were recruited from inpatients and community population in this study. CGA was performed in a face-to-face examination. The data of general health status, sociodemographic information and neuropsychiatric status were collected. Timed gait speed (m/s) was assessed over 20 meters at a usual pace.

RESULTS: Two hundred and eight-five participants (53.7%) had a low gait speed (gait speed ≤ 0.8 m/s). Low gait speed was associated with age (74.9 ± 7.6 , $P < .001$), higher prevalence of chronic diseases (hypertension (33.3%), coronary heart disease (18.3%), chronic obstructive pulmonary disease (7.3%), cerebrovascular disease (11.9%) and gastrointestinal disease (10.7%)), depression (17.3%), mild cognitive impairment (MCI) (17.9%), higher disability scores (16.0 ± 3.5 , $P < .001$) and length of hospital stay (20 (12,40), $P = .003$). The gait speed was not associated with gender, work type, education, body mass index, liver disease, kidney disease, diabetes, cataract and tumor.

CONCLUSIONS: Gait speed only reflects the status of some chronic diseases, disability, MCI, depression and length of hospital stay of the elderly. Thus, our results do not support that gait speed may replace CGA to assess the health status of the elderly.

P113

Study on 25-OH Vitamin D Level in Older Male Patients with Osteoporosis in Summer at Different Altitude

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OBJECTIVE: Age-related changes and osteoporosis on the serum 25-OH Vitamin D are reported during sea level. Yet, few studies have explored the effect of different altitude hypoxia (2260 m, 3300 m) on 25-OH Vitamin D in senile male patients with osteoporosis. The aim of the present study was to explore the significance of the level change of serum 25-OH Vitamin D in senile male patients with osteoporosis in different altitude, and its relationship with the altitude hypoxia via measuring the serum 25-OH Vitamin D level in senile male patients with osteoporosis.

METHODS: A total of 131 consecutive older male patients with osteoporosis were enrolled in summer (in accordance with the WHO diagnostic criteria for osteoporosis, 1994), and osteoporosis patients were divided into high altitude group (3300 m) ($n = 61$, mean age of 69.4 ± 3.8) and moderate altitude group (2260 m) ($n = 70$, mean age of 70.2 ± 2.5) according to altitude. Chemiluminescence was applied to determine the serum 25-OH Vitamin D. The content of calcium, phosphorus, alkaline phosphatase, and parathyroid hormone were also measured.

RESULTS: Compared with moderate altitude group, the serum 25-OH Vitamin D levels were significantly lower at high altitude group [(19.38 ± 3.14) ng/ml vs (12.97 ± 4.05) ng/ml] $P < .01$. The vitamin D deficiency situation was 11.43% at moderate altitude and 27.88% at high altitude. Significantly high 25-OH Vitamin D deficiency occurred at high altitude compared with that at moderate altitude ($P < .01$). The serum calcium, phosphorus, alkaline phosphatase, and parathyroid hormone levels were within the normal range in two groups.

CONCLUSION: There are different degrees of 25-OH Vitamin D deficiency in male patients with osteoporosis in the same season in different altitude. The serum 25-OH Vitamin D levels at high altitude were lower. The results of this study suggest that residency under hypoxic environment is associated with a decline in Vitamin D and suggests that there is the need of higher vitamin D intake in senile osteoporotic patients at high altitude than that at moderate altitude.

P114

Development of a New Aspiration Pneumonia Risk Assessment Scale in the Elderly

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PURPOSE: The study aimed to develop and validate a brief assessment scale for evaluating the risk of aspiration pneumonia in elderly.

METHODS: The assessment scale was on the basis of standardized swallowing assessment (SSA) and the high-risk factors of aspiration pneumonia in the elderly. The disturbance of consciousness and dysphagia for 2, 4, 6 scores corresponding to mild, moderate and severe condition was defined, according to SSA Scoring criteria. Gastroesophageal reflux, gastric retention, dependent feeding, long-turn bed rest, recumbent feeding, record of using basic anesthesia supine or sedatives, diaphragmatic elevation was scored 1 point as the high risk factor of aspiration pneumonia in the elderly, respectively.

RESULTS: According to this risk assessment scale, a total of 210 elderly patients (Age 82.4 ± 3.1 years) from January 2013 and December 2014 in our hospital were divided into low-risk, mid-risk and high-risk groups 2 corresponding to less than five scores, six to twelve scores and more than thirteen scores, respectively. The incidence of aspiration pneumonia was 5.1% (3/59) in low-risk group, 27.5% (19/69) in mid-risk group and 90.2% (74/82) in high-risk group. The differences between groups were statistically significant ($P < .05$). It suggests that the new established aspiration pneumonia assessment scale has clinical value to evaluate and predict the occurrence of aspiration pneumonia in the older patients.

CONCLUSION: These findings suggest that the new aspiration pneumonia risk assessment scale may have clinical value in predicting aspiration pneumonia in the elderly patients.

P115

The Anti-inflammatory and Anti-osteoarthritis Activities of *Pyrola Calliantha* Extracts

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OBJECTIVE: This study aimed to investigate the anti-inflammatory and anti-osteoarthritis activities of *Pyrola calliantha* H. Andr. (PC) *in vivo* and *in vitro*.

METHODS: Carrageenan-induced hind paw edema and xylene-induced ear edema models were the two acute inflammatory

rat models investigated. Rat knee osteoarthritis (OA) model was also duplicated for synovial fluids (SF) preparation. The levels of pro-inflammatory cytokines, IL-1 β and TNF- α , in SF were determined by ELISA. Histological examination was employed to determine the pathological changes, including apoptosis of chondrocytes. The apoptotic rate of separated cultured chondrocytes was also determined by fluorescent staining. Expression of MMP-13 and Bcl-2/Bax at mRNA and protein levels were evaluated by qRT-PCR and Western-blot, respectively.

RESULTS: PC extracts could significantly decrease the edema induced by carrageenan or xylene, the contents of pro-inflammatory cytokines, TNF α and IL-1 β , in OA SF, and to decrease the apoptotic rate of cultured chondrocytes *in vivo* and *in vitro*, and also to protect joint cartilage *in vivo*. In cultured chondrocytes, after treatment with PC extracts, the genes and proteins expression of MMP-13 and Bax were down regulated, and the expression of Bcl-2 was upregulated.

CONCLUSION: These results indicate that PC extracts, *in vivo* and *in vitro*, were capable of suppressing the inflammatory responses, decreasing apoptosis of chondrocytes, and protecting joint cartilage. PC extracts may have the potential utility in the treatment of OA.

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P116

Application of Oxycodone Hydrochloride Injection or Fentanyl Citrate Injection Combined with Propofol to Elderly Patients during ERCP

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OBJECTIVE: This study aimed to investigate the effectiveness, safety and comfort of propofol combined with oxycodone hydrochloride injection or fentanyl citrate injection in elderly patients during endoscopic retrograde cholangiopancreatography (ERCP).

METHODS: Eighty cases of elderly patients undergoing ERCP were randomly divided into oxycodone combined with propofol group (group O) and fentanyl combined with propofol group (group F), 40 cases in each group. The patients were induced with Oxycodone 0.1 mg/kg in group O, fentanyl 1 μ g/kg in group F, and propofol 2 mg/kg in both groups. Basic vital signs and BIS Value were recorded at following time points: entering the operating room (T1), after induction (T2), entering the endoscope (T3), taking me stone (T4), exiting endoscope (T5), the patient awake (T6) and dose of propofol, duration of surgery, recovery time, 30 min postoperative pain score and adverse effect including glossptosis, twisted body and respiratory depression were observed and recorded.

RESULTS: There was no significant difference in general condition such as weight, age, duration of anesthesia, surgery time, even MAP, HR, RR, et al ($P > .05$), but the maintain dosage of propofol, the incidence of adverse reactions, 30 min postoperative pain score in group F was greater than that of group O ($P < .05$) [(169.21 $\pm$ 16.03) vs (130.30 $\pm$ 13.21), (17.5%) vs (2.5%), (4.2 $\pm$ 0.5) vs (2.0 $\pm$ 0.3), all $P < .05$].

CONCLUSION: Both anesthetic methods can meet the anesthetic requirements for elderly patients during ERCP. Compared with Fantanyl, oxycodone can reduce the dosage of propofol, the incidence of respiratory depression and other adverse effects. Oxycodone could also have some advantages over Fentanyl in visceral pain. So oxycodone combined with propofol may be a better choice for elderly patients undergoing therapeutic ERCP.

P117

Relationship between Serum Adiponectin, Cardiotrophin-1 and Left Ventricular Hypertrophy in Hypertensive Patients

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BACKGROUND: Most studies have showed that left ventricular hypertrophy is an independent risk factor of cardiovascular event. Some hypertensive patients had satisfactorily controlled blood pressures, but left ventricular weight still increased, and circulating and local neuroendocrine hormone may play a more important role in the development of left ventricular hypertrophy.

METHODS: A total of 64 hypertensive patients were enrolled, including 31 cases with left ventricular hypertrophy and 33 cases without left ventricular hypertrophy, while 30 healthy controls were enrolled at the same time. Serum adiponectin (APN) and cardiotrophin-1 (CT-1) were determined with ELISA assay, and left ventricular end diastolic diameter (LVEDD), left ventricular posterior wall thickness (LVPWT) and inter-ventricular septal thickness (IVST) were detected with echocardiography, and left ventricular weight (LVM) and left ventricular mass index (LVMI) were calculated with those results.

RESULTS: Serum APN level of hypertensive patients with left ventricular hypertrophy was 6.78 ± 0.15 μ g/ml, which was significantly lower than that of hypertensive patients without left ventricular hypertrophy (7.87 ± 0.18 μ g/ml, $P < .01$) and healthy controls (11.97 ± 0.13 μ g/ml, $P < .01$). Serum CT-1 levels of hypertensive patients with left ventricular hypertrophy was 331.28 ± 11.89 pg/ml, which was significantly higher than that of hypertensive patients without left ventricular hypertrophy (179.6 ± 18.02 pg/ml, $P < .01$) and healthy controls (41.44 ± 9.14 pg/ml, $P < .01$).

CONCLUSIONS: Serum APN level was lower in hypertensive patients, especially with left ventricular hypertrophy, and serum CT-1 level was higher in hypertensive patients, especially with left ventricular hypertrophy. Serum levels of APN and CT-1 were associated with left ventricular hypertrophy in hypertensive patients.

P118

Relationship between Circadian Rhythm Index of Heart Rate and Dysfunction of Autonomic Nervous System in the Elderly with Common Diseases

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BACKGROUND: Dysfunction of autonomic nervous system is the common process for coronary heart disease (CHD), primary hypertension (EH), diabetes mellitus (DM), and benign prostatic hyperplasia (BPH) which have become common diseases in the elderly. The function of autonomic nervous system can be evaluated by circadian rhythm index of heart rate. The aim of this study was to identify potential links between circadian rhythm index of heart rate and dysfunction of autonomic nervous system in the elderly with common diseases.

METHODS: A total of 144 subjects (aged 75.67 ± 9.10 years) were divided into four groups: A (not suffering from common disease, $n = 11$), B (suffering from one common disease, $n = 36$), C (suffering from two common diseases, $n = 59$), and D (suffering from three or four common diseases, $n = 38$) according to the number of common diseases (CHD, EH, DM, BPH) they were suffering from. Daytime average heart rate, nocturnal mean heart rate, 24-hour average heart rate, and circadian rhythm index of heart rate were determined by 24-hour ambulatory electrocardiogram to evaluate the function of autonomic nervous system in all subjects.

RESULTS: The circadian rhythm index of heart rate was negatively correlated with number of the common diseases that subjects were suffering from ($r = -0.586$, $P < .01$). The nocturnal mean heart rate and 24-hour average heart rate did not show significant difference. Daytime average heart rate and circadian rhythm index of heart rate decreased regularly from group A to D (69.85 ± 7.99 beats/min vs 69.08 ± 9.37 beats/min vs 64.75 ± 8.70 beats/min vs 62.79 ± 9.33 beats/min, $P < .05$; $15.01 \pm 5.28\%$ vs $12.64 \pm 5.90\%$ vs $6.62 \pm 8.00\%$ vs $1.23 \pm 7.89\%$, $P < .05$).

CONCLUSIONS: The elderly have lower circadian rhythm index of heart rate and more serious dysfunction of autonomic nervous system.

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Changes of Heart Rate Variability and Function of Autonomic Nervous System in the Elderly with Cardiovascular Disease Risk Factors

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BACKGROUND: Coronary heart disease (CHD), essential hypertension (EH) and diabetes mellitus (DM) were important risk factors for adverse cardiovascular events and dysfunction of autonomic nervous system. Heart rate variability can evaluate the function of autonomic nervous system. This study aimed to evaluate changes of heart rate variability and dysfunction of autonomic nervous system in the elderly with cardiovascular disease risk factors.

METHODS: A total of 112 subjects (aged 72.63 ± 8.40 years) were enrolled in this study. They were divided into four groups: A (not suffering from diseases, $n = 15$), B (suffering from one disease, $n = 35$), C (suffering from two diseases, $n = 32$), and D (suffering from three diseases, $n = 30$) according to the number of disease (CHD, EH, DM) they were suffering from. SDNN, SDNNI, pNN50, rMSSD, SDANN, triangle index were determined by 24-hour ambulatory electrocardiogram to evaluate the function of autonomic nervous system in all subjects.

RESULTS: SDNNI and pNN50 did not show markedly differential in statistics ($P > .05$). SDNN and triangle index decreased regularly from group A to D (141.64 ± 21.38 ms vs 134.96 ± 36.58 ms vs 123.79 ± 36.06 ms vs 115.59 ± 29.93 ms, $P < .05$; 29.50 ± 4.15 vs 25.30 ± 8.17 vs 22.81 ± 7.40 vs 21.68 ± 9.12 , $P < .05$). SDANN and rMSSD also decreased regularly from group A to D (126.79 ± 16.40 ms vs 106.91 ± 31.41 ms vs 98.69 ± 32.22 ms vs 96.55 ± 37.26 ms, $P < .05$; 80.64 ± 17.71 ms vs 71.55 ± 28.18 ms vs 63.21 ± 33.42 ms vs 56.45 ± 26.59 ; $P < .05$).

CONCLUSIONS: The elderly with more cardiovascular risk factors had lower heart rate variability (SDNN, rMSSD, SDANN, triangle index) and more serious dysfunction of autonomic nervous system.

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Associations of Frequent Nocturnal Micturition and Heart Rate Variability in Elderly Benign Prostatic Hyperplasia Patients

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BACKGROUND: Experimental and clinical data have demonstrated that heart rate is an independent risk factor for cardiovascular disease. This study aimed to evaluate association of frequent nocturnal micturition and heart rate variability in the elderly patients with benign prostatic hyperplasia.

METHODS: From January 2014 to November 2014, a total of 76 elderly subjects (aged 74.58 ± 8.60 years) were divided into benign prostatic hyperplasia (BPH) group ($n = 57$) and the control

group ($n = 19$). The BPH group was divided into three subgroups according to the number of nighttime micturitions: low frequency (LF) group ($n = 18$), medium frequency (MF) group ($n = 18$) and high frequency (HF) group ($n = 21$). 24-hour ambulatory electrocardiogram detection was performed in all the subjects, and the detection indicators included circadian rhythm of heart rate and heart rate variability -SNDD, SDANN and triangular index.

RESULTS: There were no differences in daytime average heart rate, nighttime average heart rate and 24-hour average heart rate between BPH group and the control group ($P > .05$). There were significant difference in circadian rhythm of heart rate, SNDD, SDANN and triangular index between BPH group and the control group (0.98 ± 0.12 vs 0.87 ± 0.07 , 128.60 ± 30.00 ms vs 156.84 ± 42.53 ms, 90.00 ± 41.75 ms vs 136.28 ± 58.42 ms and 22.05 ± 7.33 vs 28.11 ± 10.45 respectively, $P < .05$). All the subjects in the control group were normal in circadian rhythm of heart rate, and in BPH group, the incidence of abnormalities of circadian rhythm of heart rate was significantly higher in HF group than that in MF and LF group (57.14% vs 50.00% vs 5.56% , $P < .05$).

CONCLUSIONS: Frequent nocturnal micturition and heart rate variability were correlated in the elderly patients with benign prostatic hyperplasia.

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Self-management of Oral Anticoagulation in the Elderly with (AF)

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OBJECTIVE: The objective of this study was to evaluate the impact of patient self-management (PSM) program on safety, efficacy, and quality of life about anticoagulation control in elderly patients with atrial fibrillation (AF) requiring chronic warfarin therapy compared with clinic standard treatment in China.

METHODS: This is a prospective cohort study. A total of 274 eligible subjects agreed to participate: 89 subjects who were assigned to PSM group using portable INR monitor for self-testing at home and 185 subjects received standard treatment at an anticoagulation clinic (conventional group). The PSM group participated in instruction courses that aimed to practice OAC management on their own. 12 months follow-up were made. Quality of life (QOL) was assessed using the questionnaire Short-Form 12 Health Survey, version 2 (SF-12v2), Time in therapeutic range (TTR) assessed the quality of the anticoagulation treatment given. An INR range of 2.0 to 3.0 has been established as therapeutic by stroke prevention trials in AF.

RESULTS: Eighty subjects in PSM group and 168 subjects in conventional group completed the study, respectively. The data in two groups shows that the regression equation for measured value of the portable INR monitor (y axis) and traditional central lab measured value (x axis) is $y = 0.927x + 0.177$. Good correlation between the INR measured values of the two groups can be observed ($r = 0.926$, $P < .05$). Meanwhile, its bias can be observed with Bland-Altman diagram and the bias between the Self-testing INR and the conventional method INR in parallel is (-0.173 ± 0.157) and there were no sample bias more than 0.5 INR units. The median frequency of INR measurements in the study period was 7.3 days test INR one time in PSM group compared to 38.7 days in the conventional group ($P = .000$). Medians of the percentage of TTR was 68% vs 57% ($P = .000$). Cumulative risk of the first primary endpoint is performed by using Kaplan-Meier analysis, $P = .131$. Thromboembolic complications occurred in 4 patients in the PSM group vs in 15 patients in the conventional group ($P = .277$). Major bleeding events occurred in 1 patient vs 6 patients in the PSM group and the conventional group, respectively ($P = .534$). At least one minor bleeding occurred in two groups

which was not statistically significant ($P = .465$). There were significant improvements ($P = .038$) of the total physical score between the PSM group (42.3 ± 5.3) and conventional group (40.7 ± 5.8) after 12 months follow-up. Significant improvement was also observed in total mental composite score ($P = .046$) in the PSM group (53.4 ± 6.5) compared to the conventional group (51.5 ± 7.2).

CONCLUSION: Compared to the conventional INR monitoring, PSM appeared to be safe and effective for monitoring anti-coagulation therapy with warfarin in elderly patients with AF.

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Prognostic Value of Pregnancy Associated with Plasma Protein-A in Elderly Patients with Acute Coronary Syndrome

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OBJECTIVE: This study aimed to investigate whether pregnancy associated with plasma protein-A (PAPP-A) is useful for risk assessment in elderly patients with coronary syndrome (ACS).

METHODS: PAPP-A was measured in 455 elderly patients with ACS. These patients were followed for 12 months. A cut point was made according to the ROC curve, and the patients were divided into high level group and low level group, in order to evaluate the association between the PAPP-A and the incidence of the combined cardiovascular events.

RESULTS: The cut point of PAPP-A was 23.5 MIU/L, and the sensitivity and specificity of the incidence of was 66.7% and 75.0% respectively. Multivariate logistic regression analysis demonstrated that there was a correlation between the PAPP-A level and rates of the combined cardiovascular events (OR = 2.832, $P = 0.006$). The Cox proportional hazards regression model showed that PAPP-A was an independent risk factor for prognosis of the combined cardiovascular events (HR = 2.79, $P = .007$).

CONCLUSIONS: PAPP-A was independently associated with recurrent cardiovascular events in elderly patients with ACS. PAPP-A may be useful for risk assessment and monitoring in populations at high risk of cardiovascular events.

P123

Anti-tumor Activity of the Extracts from *Reineckia Carnea* on Human Lung Adenocarcinoma A549 Cells in Vitro and its Apoptosis Mechanisms

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OBJECTIVE: This study aimed to investigate anti-tumor activity of *Reineckia carnea* n-Butanol extracts (RCBE) on the proliferation and apoptosis of human lung adenocarcinoma A549 cells in vitro, and to investigate the mechanism of RCBE inducing A549 cell apoptosis.

METHODS: MTT assay was applied to screen the proliferation inhibitory rate of the extracts of *Reineckia carnea* on A549 cell lines in vitro. Flow cytometry was employed to detect the apoptosis rate, the mitochondrial membrane potential, and the intracellular free Ca^{2+} concentration using annexin-V-FITC/PI double staining, JC-10 staining, and FLUO-8 AM staining, respectively. Caspase-3/7 activity was determined by Luminescence assay.

RESULTS: RCBE exhibited remarkable anti-proliferation activity on A549 cell in a dose- and time-dependent manner, and the IC_{50} was $63.24 \text{ mg} \cdot \text{L}^{-1}$ at 24 hours. Apoptosis was induced by

RCBE and increased as the augmentation of the drug concentration ($P < .05$). By the treatment with RCBE, the mitochondrial membrane potential was demonstrated to decrease ($P < .05$), the intracellular free Ca^{2+} concentration increase ($P < .05$), and the expression of caspase-3/7 up regulate ($P < .05$). Furthermore, pretreated with either intracellular calcium chelator BAPTA-AM or caspases pectrum inhibitors Z-VAD-FMK, the apoptosis induced by RCBE could effectively attenuate ($P < .05$).

CONCLUSION: RCBE exhibited inhibitory effect on proliferation and induced apoptosis of A549 cell lines in vitro. The mechanism of inducing apoptosis may be associated with the decreased mitochondrial membrane potential, increased intracellular calcium ion, and up regulation of the caspase activity.

P124

KAP and its Impact on Oral Health Status of the Elderly in Rural Areas of China

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OBJECTIVE: This paper aimed to investigate the KAP (knowledge, attitude and practice) related to oral health status, to analyze the influential factors, and to provide the prevention knowledge for the countryside in Weifang.

METHODS: A total of 3093 elderly over 60 from the countryside of Qinzhou, Zhucheng and Anqiu areas were enrolled and interviewed by investigators. The data were analyzed with descriptive statistics, single and multi-factor logistic regression analysis.

RESULTS: The awareness of the basic knowledge (11.8%-29.2%), the recognition rate (15.0%-90.0%) and participation rate of the preventive measures (7.4%-60.6%) were very low. The multi-factor logistic regression analysis indicated that the two factors of age (OR = 4.546) and denture (OR = 5.773) were the major risk factors of root caries. The 5 factors such as the frequency of tooth brushing (OR = 0.461), vertical brushing (OR = 0.182), the use of fluoride toothpaste (OR = 0.508), dental calculus (OR = 0.593), and regular oral examination (OR = 0.276) were the protective factors of root caries ($P < .01$).

CONCLUSION: KAP on oral health status was low in the countryside in Weifang. In the future, rural primary medical institutions should play a fundamental role in strengthening oral health care knowledge dissemination.

P125

Relationship between Serum Homocysteine Concentration and Renal Injury in Healthy Elderly Chinese Men

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OBJECTIVE: This study aimed to investigate the relationship between serum homocysteine concentration and renal injury in healthy elderly Chinese men.

METHODS: A cross sectional study was conducted based on the data collected from annual health check in 2014 in our hospital. A total of 349 male subjects, aged over 65, with no history of smoking and alcohol consumption less than 1-2 times/month were enrolled in the study. Subjects with history of severe cardiocerebral vascular disease, diabetes, hypertension, chronic kidney disease, and recently taking nephrotoxicity drugs were excluded. We measured serum homocysteine concentration with direct chemical luminescence method. If serum homocysteine concentration was above $16 \text{ } \mu\text{mol/L}$, it was defined as hyperhomocysteinemia. All subjects were divided into hyperhomocysteinemia group ($n = 121$) and con-

trol group (n = 228). The serum homocysteine concentration, age, height, weight, blood pressure, blood sugar, blood lipid, uric acid, urea, creatinine, glomerular filtration rate and the prevalence of glomerular filtration rate below 60 ml/(min•1.73 m²) were compared between these two groups. The correlation between homocysteine, age, uric acid and glomerular filtration rate was analyzed with Pearson correlation analysis. According to the serum homocysteine level quarterback spacing, all subjects were divided into four groups. The relationship between serum homocysteine concentration and the risk of glomerular filtration rate below 60 ml/(min•1.73 m²) was analyzed in Logistic regression analysis.

RESULTS: The mean serum homocysteine concentration was 15.66 μmol/L in healthy elderly Chinese men. The prevalence of hyperhomocysteinemia in these subjects was 34.67% (121/349), and the prevalence of glomerular filtration rate below 60 ml/(min•1.73 m²) was 36.68% (128/349). There were significant high levels of serum homocysteine concentration, age, uric acid, creatinine and the prevalence of glomerular filtration rate below 60 ml/(min•1.73 m²) (49.59% vs 29.82%, $P < .01$) in hyperhomocysteinemia group than control, while there was a significant low level of glomerular filtration rate (62.22 ± 17.25 ml/(min•1.73 m²) vs 70.81 ± 17.93 ml/(min•1.73 m²), $P < .01$) in hyperhomocysteinemia group. Correlation analysis demonstrated that serum homocysteine level ($r = -0.147$, $P = .006$) and age ($r = -0.661$, $P < .001$) were negative correlated with glomerular filtration rate, but uric acid ($r = -0.074$, $P = .170$) was not correlated with glomerular filtration rate. Logistic analysis showed that higher serum homocystene concentration significantly increased the risk of glomerular filtration rate below 60 ml/(min•1.73 m²). Compared with the first group as reference, the odd ratio of the second to the forth group was 1.882 (95% CI 0.969–3.657), 1.789 (95% CI 0.919–3.484) and 3.995 (95% CI 2.079–7.675). There was a significant high prevalence in the fourth group than those in the first group ($P < .001$).

CONCLUSION: High level of serum homocysteine was a major risk factor for renal injury in elderly men.

P126

Increased ZAP70 in the Older Mice with Dry Skin Pruritus

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BACKGROUND: Dry skin pruritus is common in elderly people. However, the mechanism of dry skin that causes pruritus is still unclear. Recent reports show that immune response may involve in dry skin pruritus via regulation of nerve growth factor (NGF). ZAP70, as a T-cell receptor (TCR), may promote the secretion of NGF. This study aimed to detect the alteration of ZAP70 in a mice model with dry skin pruritus.

METHOD: C57 mice about 22 months old were used as experimental animal. Model with dry skin pruritus was performed as following. Rostral back was treated daily with cutaneous application of acetone /ether (1:1) mixture (AE), water following AE for 5 days. Transepidermal water loss and scratching behavior were evaluated. ZAP70 was examined by immunohistochemistry and western-bolt. Age-matched normal mice were used as controls.

RESULTS: After the treatment with AEW, behavior observation of 22- month mice showed that the number of spontaneous scratching was significantly increased compared to the control. The number of spontaneous scratching AEW group was 58 bouts per hour, and the control group was 16 bouts per hour ($P < .05$). Immunohistochemical analysis revealed that the ZAP70 was significantly increased in the skin in 22- month mice with dry skin pruritus compared to the control. Western-blot analysis of ZAP70 expression was reported as ratio of integrated optical density (IOD) of each band to the IOD of the respective β-actin. The num-

ber of 22- month mice group was 0.09 ± 0.009 , the number of control group was 0.04 ± 0.008 . ZAP70 expression in 22- month mice group was significantly higher than that of control group ($P < .01$) in skin with dry skin pruritus.

CONCLUSION: Our results indicate that increased ZAP70 expression might regulate dry skin pruritus in older mice.

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Changes of Blood IGF-1 before and after Treatment in Patients with Acute Cerebral Infarction and its Molecular Mechanism

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OBJECTIVE: This study aimed to detect insulin-like growth factor-1 (IGF-1), homocysteine (HCY), high sensitive C-reaction protein (hs-CRP), lipoprotein-a (Lp(a)), uric acid (UA) and the lipid of related index (TCH, TG, hdl-c, LDL-C) of level changes, and to discuss their clinical application of the correlation between Butylphthalide capsule and drug analysis.

METHODS: Thirty cases of inpatients with ACI were randomly selected between March 2011 and December 2011 in our hospital. Among them, there were 18 men and 12 women with the age of 47 to 68 years old. The average age was 60 ± 5.8 years. Chemiluminescence immunoassay method was used for the detection of IGF-1, and Japan's Olympus automatic biochemistry analyzer was used to test hs-CRP, HCY, UA, Lp (a), and lipid-related indicators (TCH, and TG, and HDL-C, and LDL-C).

RESULTS: For patients with ACI disease, 72 hours detection of concentration of IGF-1 (239.17 ± 65.3 ng/ml) was lower than that of control group (326.5 ± 72.2 ng/ml), and the difference was statistically significant ($P < .05$); LP (a), HCY and CRP concentrations (481.07 ± 98.2 mg/L, 19.97 ± 5.3 μmol/L, 9.87 ± 2.1 mg/L) was significantly higher than that in control group (190.1 ± 45.8 mg/L, 12.3 ± 3.4 μmol/L, 1.1 ± 0.9 mg/L), with statistically significant differences ($P < .05$, $P < .05$, $P < .01$). After treatment for 13 days, HCY IGF-1, CRP concentrations had restored (278.18 ± 62.5 ng/ml, 14.9 ± 4.2 μmol/L and $3.521.2$ mg/L), and its value of difference was statistically significant ($P < .05$). HCY, CRP had no statistical significance ($P > .05$), and its value of difference had statistical significance ($P < .05$); LP (a) after the treatment (485.14 ± 86.3 mg/L) did not change significantly, and the differences were not statistically significant ($P > .05$). By the comparison of Butylphthalide treating group (N = 10) and routine treatment group (13n), concentration difference was statistically

Table 1 Changes of various indicators in patients of after 72h ACI pathogenesis ($\bar{X} \pm S$)

Indicators	Control Group (20n)	Reserch Group 72h (20n)
IGF-1	326.5±72.2	239.17±65.3 [▲]
LP (a)	190.1±45.8	481.07±98 [▲]
HCY	12.3±3.4	19.97±5.3 [▲]
CRP	1.1±0.9	9.87±2.1 ^{▲▲}
UA	317.7±50.9	367.32±86.8
TCH	4.9±0.6	5.88±1 [▲]
TG	1.3±0.5	2.43±2 ^{▲▲}
HDL-C	1.3±0.3	1.09±0.1 [▲]
LDL-C	2.5±0.6	3.26±1 [▲]

significant ($P < 0.05$); HCY (20.83 ± 5.3 and 12.73 ± 4.2) concentration difference was statistically significant ($P < 0.05$), and the conventional treatment group concentration was recovered, but had no statistical significance ($P > .05$). CRP both before and after treatment, the concentration change was statistically significant for butylphthalide treating group (8.28 ± 2.1 mg/L and 2.48 ± 1.2 mg/L), and the conventional treatment group (11.08 ± 2.4 mg/L and 4.32 ± 1.4 mg/L). LP(a) both before and after the treatment, the concentration did not change significantly, with no statistical significance ($P > .05$).

Table 2 Analysis of correlation between IGF-1 and ACI risk factors

	IGF-1		CRP		HCY		LP(a)		HDL		TG	
	72h		72h		72h		72h		72h		72h	
	r	t	r	t	r	t	r	t	r	t	r	t
IGF-1	-	-	-0.021	0.101	-0.542	2.956	0.026	0.121	-0.417	2.105♦	0.46	2.374♦
CRP	-0.021	0.101	-	-	-0.159	0.736	0.123	0.568	0.158	0.733	0.015	0.069
HCY	-0.542	2.956	-0.159	0.736	-	-	-0.015	0.067	0.062	0.285	-0.227	1.068
LP(a)	0.026	0.121	0.123	0.568	-0.015	0.067	-	-	-0.098	0.451	-0.047	0.216
TCH	0.189	0.883	0.264	1.254	-0.143	0.662	0.345	1.684	0.228	1.072	0.188	0.877
HDL	-0.417	2.105♦	0.158	0.733	0.062	0.285	-0.098	0.451	-	-	-0.267	1.27
LDL	-0.137	0.632	0.195	0.911	-0.165	0.767	0.381	1.887♦	0.451	2.313♦	-0.446	2.284
UA	-0.134	0.620	0.043	0.197	-0.243	1.148	0.202	0.945	-0.089	0.409	-0.074	0.34

Table 3. Changes of indicators of application of ACI in patients with butylphthalide group and control group ($\bar{X} \pm s$)

Indicators	control group		butylphthalide group	
	Before treatment, 72h	After treatment, 13day	Before treatment, 72h	After treatment, 13day
IGF-1	249.97±80.1	256.48±78.4	227.73±75.3	306.40±73.2*
LP (a)	525.39±98.2	545.50±86.3	423.44±92.5	406.68±82.7
HCY	19.32±5.1	16.57±4.0	20.83±5.3	12.73±4.2*
CRP	11.08±2.4	4.32±1.4**	8.28±2.1	2.48±1.2**
UA	378.88±86.8	398.82±74.0	352.30±82.4	340.25±72.1
THC	6.30±1.2	5.66±1.2	5.32±1.1	4.54±1.2*
TG	2.78±2.1	2.59±2.1	2.97±2.2	2.85±2.3
HDL-C	1.04±0.2	1.26±0.3	1.07±0.1	1.46±0.3*
LDL-C	3.55±1.2	3.26±1.0	2.88±1.2	2.22±1.1

CONCLUSION: In ACI patients, treatment with butylphthalide capsule led to increased IGF-1 level and decreased CRP level. While butylphthalide treatment did not significantly affect many conventional cardiovascular risk factors, it appeared to have anti-inflammatory effect which serve as a molecular mechanism for its impact on IGF-1 level.

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Expression of TREM2 and CARD9 in a Mouse Model of Alzheimer's Disease

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OBJECTIVE: Recent studies show that inflammation plays an important role in mechanisms of Alzheimer's disease (AD). Triggering receptor expressed on myeloid cells 2 (TREM2), which is a receptor on the surface of microglial membrane, and is closely tied to AD. Caspase recruitment domain containing protein 9 (CARD9) is an adaptor protein, and one study suggests that further TREM signal requires CARD9 in the dendritic cells. This study aimed to evaluate the relationship between them and AD.

METHODS: Bilateral injection of amyloid β peptide (A β) in hippocampus was used to build AD model, and intraperitoneal

injection of IL-4 was used to activate TREM2 signaling pathway. Morris water maze test detected cognitive function of mice. Western blotting (WB) was detected CARD9, TREM4 level in different groups.

RESULTS: WB shows that the expression level of TREM2 in IL-4 treatment group (IL-4 + AD) was significantly higher than that in AD group (3.21 ± 0.05 vs 1.16 ± 0.04 ; $P < .05$), while the expression level of CARD9 in IL-4 treatment group (IL-4 + AD) was significantly higher than that in AD group (2.82 ± 0.10 vs 1.07 ± 0.03 ; $P < .05$). Water maze test shows that the escape latency of IL-4 treatment group (IL-4 + AD) was significantly shorter than that in AD group (37.71 ± 20.22 vs 45.20 ± 22.27 ; $P < .05$).

CONCLUSION: Expression of TREM2 and CARD9. TREM2 was associated with cognitive function in a mouse model of AD. These molecules may become new potential drug targets in the treatment of AD.

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Proteomics Study of Human Clara-like Lung Cancer Cell and Alveolar Type II Epithelial-like Cancer Cell

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BACKGROUND: Both clara-like lung cancer cell and alveolar and type II epithelial-like lung cancer cell are common type of primary lung adenocarcinoma, however these two kinds of cancer cells are indistinguishable under light microscope, and can only be identified through ultrastructure of secretory granules similar to clara cell or plate lamellar bodies similar to alveolar type II epithelial cell under electron microscope. The electron microscope detection is expensive, complicated, and the work cycle is long, therefore clinical lung cancer cell typing is quite difficult. This study aimed to search for the tumor markers of clara-like lung cancer cell and alveolar type II epithelial-like cancer cell by proteomics methods, facilitating clinicians and researchers to identify these two cell types with easier and cheaper methods such as immunohistochemistry.

METHODS: Human clara-like lung cancer cell line NCI-H358 and alveolar type II epithelial-like lung cancer cell line A549 were cultured, total proteins were extracted, and separated with 2D-DIGE (two-dimensional fluorescence difference gel electrophoresis), protein expression were compared and analyzed with analysis software DeCyder. Protein dots that differentially expressed 5 times between two groups were digested from preparation gel, identified by mass spectrometry, and verified by western blot. The verified proteins were detected in surgical lung cancer specimens from 32 patients by immunohistochemistry, and the results were compared with transmission electron microscope.

RESULTS: Proteomic analysis identified 38 different protein spots, 8 of which showed more than 5-fold difference in their expression between the two cell lines. Five proteins were identified by mass spectrometry, including UCHL1 (ubiquitin carboxyl-terminal hydrolase isozyme L1, highly expressed in A549 cells), ERO1L, CK19, CK8 and PRDX2 (cytoskeletal 19, cytoskeletal 8, peroxiredoxin 2, highly expressed in NCI-H358 cells). The expressions of ERO1L in NCI-H358 cells were significantly higher than that in A549 cells through western blot verification ($P = .02$), but differences of UCHL1 level in A549 cells and NCI-H358 cells were not significant. The positive rate of ERO1L in surgical lung cancer specimens was 72.4%. According to the results of immunohistochemistry, 5 ERO1L positive lung adenocarcinoma tissues, 5

ERO1L total negative lung adenocarcinoma tissues, 2 ERO1L positive squamous cell carcinoma tissues, 1 ERO1L total negative squamous cell carcinoma tissue, 1 ERO1L positive large cell carcinoma and 1 ERO1L negative small cell carcinoma were chosen for electron microscope observation, and 4 of 5 ERO1L-positive cases of lung adenocarcinoma showed pulmonary clara-like cancer cells, and none of ERO1L-negative tissues showed clara-like lung cancer cells under electron microscope.

CONCLUSIONS: ERO1L appeared to a strong candidate to become a tumor marker for identification of human clara-like lung cancer cell subtype in lung adenocarcinoma, and ideal markers for identification of alveolar type II epithelial-like lung cancer cells have not been identified.

P130

The Association between Insomnia Symptoms and Mortality: A Prospective Study of US Men

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BACKGROUND: Insomnia is the most common sleep disorder. This present study aimed to determine whether insomnia is associated with mortality risk.

METHODS: A total of 23,447 US men had insomnia symptoms during 6 years of follow-up, and people with a cancer diagnosis and people with missing values for insomnia questions were excluded from this study. Deaths were identified from state vital statistics records, the National Death Index, reports given by the families, and the postal system.

RESULTS: The ratios of total mortality were 1.25% for difficulty of initiating sleep, 1.24% for non-restorative sleep, 1.09% for difficulty maintaining sleep and 1.04% for early-morning awareness. Men with difficulty of initiating sleep and non-restorative sleep had 55% and 32% increased risk of CVD mortality.

CONCLUSIONS: Some insomnia symptoms were associated with mortality risk. Especially men with difficulty of initiating sleep and non-restorative sleep have an increased risk of total mortality and CVD mortality.

P131

Health Status, Quality of Life and Chronic Diseases of Centenarians in Zhongxiang, Hubei Province, China

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OBJECTIVE: This study aimed to investigate the health status, quality of life and chronic diseases of centenarians and to discuss factors affecting their longevity in Zhongxiang, Hubei.

METHODS: Several visits were made to Zhongxiang, Hubei, covering seven randomly selected towns and 28 centenarians from January 2013 to August 2014. Questionnaires include Mini Mental State Examination (MMSE), Geriatric Depression Scale (GDS), Activities of Daily Living (ADL), nutritional status and quality of life scores. Physical examinations were also performed and the results were analyzed and evaluated.

RESULTS: Among the 28 centenarians aged 100–113 with average age of 103.03, seven were male, and twenty-one female. For MMSE score, 11 individuals had normal cognitive function, seventeen various degrees of cognitive dysfunction, and no one had dementia. For GDS score, 22 individuals were normal, six had mild depression, and no one had severe depression. For ADL scores, all 28 centenarians had various degrees of decline. The minimum score

was 36 and the maximum score was 87 with average level of 73. For nutritional status score, 28.6% of centenarians were normal while 71.4% had potential malnutrition and malnutrition. For quality of life, the overall score for quality of life was 76.18 ± 8.38 . For major chronic diseases, 75% had hyperlipidemia, 74.2% malnutrition, 71% anemia, 46% high blood pressure, 42% hypocalcemia, 39% hyponatremia, and 21.4% renal insufficiency. Blood cells were reduced by 21.4%, and hyperuricemia and cataract were reduced by 18% respectively followed by chronic bronchitis and hypoalbuminemia by 14% each, and coronary heart disease were reduced by 7%. Various degrees of deafness, decreased vision and missing teeth were common.

CONCLUSIONS: Centenarians have a good overall quality of life, with an average of 1–6 chronic diseases per person. However, nutritional status of the centenarians should be improved.

P132

Comparison of Health Services Accessibility between Urban and Rural among the Elderly over 65 Years Old in Chongqing, China

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BACKGROUND: At present, there is a large gap between elder health service and the actual demand. How to use limited resources to improve the life quality of old people and to provide health care for them becomes a problem. Health services accessibility was associated with the people's health, especially among the elderly. This study aimed to explore the health service accessibility (Space accessibility, Economic accessibility) among the elderly.

METHODS: During July 2014, 402 eligible the elderly over 65 years old in a district in Chongqing were recruited for the multi-stage stratified random sampling survey. The mixed questionnaire was adopted to investigate the elderly aged followed by interview study.

RESULTS: The study comprised 198 (49.3%) urban respondents and 204 (50.7%) rural respondents. 19.70% of urban respondents and 42.65% of rural respondents indicated that it was not convenient to see a doctor. Rural respondents' health insurance coverage was 95.59%, and the urban respondents' medical insurance coverage was 96.46%. Logistic regression analysis found that factors affect the low accessibility of "seeing a doctor" are "expensive medical charges (OR = 2.026, 95% CI [1.172–3.505])", "poor medical personnel technology (OR = 2.236, 95% CI [1.290–3.876])", "limited categories of medicines (OR = 3.936, 95% CI [2.265–6.839])", "living in the rural areas (OR = 3.223, 95% CI [1.786–5.814])" and "poor medical care personnel service (OR = 3.196, 1.607–6.358)".

CONCLUSION: Rural health service accessibility was relatively poorer than that of urban among the elderly in Chongqing, western China. Future policies should focus on these factors to improve the accessibility of health service, and narrow the gap between urban and rural areas.

P133

Depleted Aldehyde Dehydrogenase1A1 Reverses Cisplatin Resistance of Non-small Cell Lung Cancer Cells A549/DDP

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OBJECTIVE: Cisplatin is the first-line chemotherapy for the treatment of non-small cell lung cancer (NSCLC). However, its

resistance has been a major obstacle in the management of NSCLC. Aldehyde dehydrogenase1A1 (ALDH1A1) over-expression has been observed in a variety of cancers, including lung cancer. The purpose of this study was to investigate the effect of ALDH1A1 expression on cisplatin resistance and to explore potential mechanism.

METHODS: To measure its potential role in cisplatin resistance, a short hairpin RNA (sh RNA) was used to knockdown ALDH1A1 expression. The cell viability of A549/DDP cells was detected by MTT, and phosphorylation level of AKT was detected by western blot assay.

RESULTS: ALDH1A1 was over-expressed in cisplatin resistant cells A549/DDP compared with its parental cells A549. The ALDH1A1 depletion significantly decreased A549/DDP proliferation (0.89 ± 0.06 vs 0.69 ± 0.02 , con vs sh, $P < .05$) and reduced cisplatin resistance (71.92 ± 2.56 vs 47.10 ± 1.99 , con vs sh, $P < .05$). Besides, PI3K/Akt pathway was activated in A549/DDP, and ALDH1A1 knockdown reduced phosphorylation level of AKT. Moreover, the combination of ALDH1A1-sh RNA and PI3K/Akt pathway inhibitor LY294002 markedly inhibited cell viability (0.69 ± 0.02 vs 0.35 ± 0.00 , sh vs sh+LY, $P < .001$) and increased sensitivity of resistant cells to cisplatin (38.19 ± 1.23 vs 47.10 ± 1.99 , sh+LY vs sh, $P < .05$).

CONCLUSION: ALDH1A1 depletion could reverse cisplatin resistance in human lung cancer cell line A549/DDP, and may act as a potential target for the treatment of lung cancers resistant to cisplatin.

P134

The Role of HIF-1 α in Hippocampal Neurons Injury Induced by 1.5%-21% Intermittent Hypoxia

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BACKGROUND: Obstructive sleep apnea hypopnea syndrome (OSAHS), a common sleep and breathing disorder, is often intertwined with cognitive dysfunction. Cognitive dysfunction is associated with hippocampal neuron injury induced by intermittent hypoxia (IH). However, the precise mechanisms of IH-induced hippocampal neurons injury are not fully understood. HIF-1 is obligate heterodimer consisting of an O₂-labile α subunit and a stable β subunit, which is the primary transcriptional response to hypoxic stress and plays an important role in the hypoxia related diseases. This study aimed to investigate the role of HIF-1 α in hippocampal neuron injury induced by 1.5–21% intermittent hypoxia through the inhibitor of HIF-1 (YC-1) pretreatment.

METHODS: The primary hippocampal neuronal cultures were prepared and maintained by using neonatal SD rats. The hippocampal neurons, which had been cultured to the seventh day, were further divided into four groups: normal control group, IH group, IH+YC-1 group and YC-1 group. Hippocampal neurons apoptosis were detected by terminal deoxynucleotidyl transferase-mediated dUTP-nick end-labeling. Apoptosis associated protein cleaved-caspase3 and HIF-1 α expressions were detected by western blotting.

RESULTS: IH exposure caused an increased expression of HIF-1 α and up-regulation of apoptosis in hippocampal neurons compared with that in the normal control group, demonstrated by western blotting and TUNEL respectively ($P < .05$). Furthermore, compared with the IH group, the inhibitor of HIF-1 (YC-1) pretreatment significantly decreased IH-mediated hippocampus apoptosis and accompanied with the down-expression of HIF-1 α ($P < .05$).

CONCLUSION: These results demonstrate a critical role of HIF-1 in the regulation of hippocampal neurons apoptosis induced by 1.5–21% intermittent hypoxia. Inhibitor of HIF-1 (YC-1) has a protective effect against IH-induced hippocampal neurons apoptosis.

P135

Study of Preparation Technology of Hypoglycemic Granule of *Cyclocarya paliurus*

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OBJECTIVE: This study aimed to investigate preparation technology of glucose-lowering *Cyclocarya paliurus* and determination of total flavonoid content.

METHODS: Extraction *Cyclocarya paliurus* of glucose-lowering active ingredient was extracted and wet granule way to make *Cyclocarya paliurus* glucose-lowering particles was adopted. Preparation of different proportion of the accessories and condensate of proportion was groped, and the ultraviolet spectrophotometry was used to determine the content of total flavonoids in preliminary research.

RESULTS: According to the different proportion of the materials for exploration, the best proportion with accessories for *Oriental otellia alismoides* was determined as follows: dextrin starch: lactose is 1:1:2, preparation for the grain size, the percent of qualified was 97.27%. Rutin reference substance in 10–50 $\mu\text{g/mL}$ range in a linear relationship, and the linear equation were $\text{Abs} = 8.8866\text{pc} + 0.00147$ ($R^2 = 0.9997$, $n = 5$), $\text{RSD} = 3.4656\%$.

CONCLUSION: Preparation of *Cyclocarya paliurus* glucose-lowering particles was simple. The determination method of total flavonoid content was simple and accurate, and can be used for the quality control of *Cyclocarya paliurus* granules.

P136

Artery Calcification, Osteoporosis, and Serum Adiponectin Levels in Chinese Elderly

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OBJECTIVE: This study aimed to investigate the relationship between serum adiponectin level and thoracic aortic calcification (TAC) in 333 Chinese elderly over the age of 65.

METHODS: This is a cross-sectional study in subjects living in Changsha city and its district area. Serum total adiponectin level was measured by ELISA. TAC was evaluated by posterior-anterior chest radiographs. The diagnosis of osteoporosis was established using dual energy X-ray absorptiometry according to the World Health Organization criteria.

RESULTS: Results showed that the mean serum adiponectin level was $6.3 \pm 3.7 \mu\text{g/mL}$, and TAC was detected in 58% of subjects ($n = 192$), and comorbidities of TAC and osteoporosis were observed in 33.6% ($n = 112$) subjects. The percentage of TAC or comorbidities of TAC and osteoporosis was higher in subjects with low adiponectin tertile level than that in patients with intermediate or high adiponectin tertile level. Multivariable logistic regression analysis showed that each standard deviation of lower adiponectin level was associated with an odds ratio (OR) for increased TAC prevalence (OR = 1.36; 95% CI 1.17–1.85) after adjusted for CVD risk factors. Further adjustment for bone mineral density showed a persisted significance (OR = 1.49; 95% CI 1.28–1.91).

CONCLUSION: Serum adiponectin level was associated with TAC and lower serum adiponectin level may be a useful indicator of artery calcification in elderly.

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A Case Report: An Older Patient with Upper Gastrointestinal Bleeding Treated with Acid Suppression Developed Clostridium Difficile Colitis with Frequent Recurrence

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THE CASE: Upper Gastrointestinal (UGI) bleeding treated with acid suppression increase the risk of Clostridium Difficile Colitis (CDC). An 85 year-old Chinese male, with hypertension and Type2 DM, had colon cancer resection and colostomy that caused prolonged hospitalization for 6 months. During peri-and postoperative period, he had several episodes of infections that required antibiotics and stress-induced UGI bleeding treatments. We treated his UGI bleeding by intravenous omeprazole and subsequently changed to oral preparation 40 mg/day. One month later, he suddenly had watery diarrhea with hypotension and stool test result confirmed as CDC that treated with oral Vancomycin for 10 days. His symptoms and stool test improved during medication period but omeprazole was continued at that time as the fecal occult blood test (FOBT) still positive. The second CDC episode occurred 3 days after Vancomycin discontinuation and he had oral Vancomycin plus Metronidazole for the next 14 days also sucralfate to replace omeprazole. He regained improvement but the third CDC episode recurred 2 weeks after the latest treatment discontinuation and oral Vancomycin with metronidazole were administered for 3 weeks. His symptoms and repeated stool test had subsided before discontinuation of treatment; but his routine FOBT constantly revealed positive and managed by intravenous Ranitidine 150 mg twice daily. For the last 1 month of his hospitalization, we noticed no recurrence of CDC and UGI-bleeding.

DISCUSSION: Acid suppressor aim to treat or prevent UGI bleeding increase the risk of CDC incidence in elderly that consider higher risk in Proton Pump Inhibitor and mucosa protector than Antihistamine-2 agent. Further research is required on each acid suppression agents since they might have different risk on CDC incidence and recurrence.

P138

Effect of Different Intermittent Hypoxia on Hippocampal Neuron Autophagy

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PURPOSE: Cognitive dysfunction is one of the common complications in elderly patients with obstructive sleep apnea hypopnea syndrome (OSAHS). Intermittent hypoxia (IH), a major pathological component of OSAHS, can induce hippocampal neuronal injury. It is likely that IH-induced hippocampal neuronal injury may be a major mechanism for the cognitive dysfunction. Autophagy is an important pathophysiological mechanism for maintaining intracellular homeostasis. In our study, we aimed to investigate the intermittent hypoxia-induced autophagy in hippocampal neurons.

METHODS: The primary hippocampal neuronal cultures were prepared and maintained by using neonatal SD rats. The hippocampal neurons, which had been cultured to the seventh day, were further divided into four groups with different duration of intermittent hypoxia: 2 h, 4 h, 8 h, and 12 h. We established two different conditions of intermittent hypoxia cell models. Cultured hippocampal neuronal were either exposed to IH (Episodic cycles of 5% O₂, 5% CO₂, and balance N₂ for 90s and 21% O₂, 5% CO₂, balance N₂ for 30s) or to IH (Episodic cycles of 1.5% O₂, 5% CO₂, and balance N₂ for 5 min and 21% O₂, 5% CO₂, balance N₂ for 10 min). The normal control group was exposed to normoxia (21% O₂) continuously. Western blot were used to examine the protein

levels of microtubule-associated protein light chain 3(LC3) after IH.

RESULTS: In first intermittent hypoxia model (5% O₂), IH exposure (8 h and 12 h) significant up-regulated the expression of LC3II compared to normal control group ($P < .05$). However, there was no significant difference of LC3II expression between normal control group and IH 2 h and 4 h groups. In second intermittent hypoxia model (1.5% O₂), IH exposure significant up-regulated the level of LC3II compared to normal control group ($P < .05$). Furthermore, with the extension of IH duration, the LC3II expression was increased gradually.

CONCLUSION: Intermittent hypoxia induced an up-regulation of autophagy activation. The level of autophagy activation was dependent upon the severity of intermittent hypoxia.

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Effects of Electrolyte and Acid-base Imbalance to a Group of Elderly Patients Hospitalized in a Single Center

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OBJECTIVE: The aim of this study was to analyze the situation of elderly patients confined in a single-center who eventually died because of electrolyte and acid-base imbalance.

METHODS: A total of 12,097 elderly (≥ 65 years old) individuals were confined in Tianjin Cancer Hospital on May 1, 2012 to April 30, 2013. Among these, 500 patients died. Thorough analysis and investigation of the electrolyte, acid-base imbalance, and other related factors were conducted.

RESULTS: It was noted that 1,169 patients suffered from electrolyte and acid-base imbalance. 311 had hyponatremia, 172 suffered from hypernatremia, 229 had hypokalemia whereas 105 had hyperkalemia. 237 and 115 patients had acidosis and alkalosis, respectively. Analysis showed that gender, age, medical conditions such as heart failure, high blood pressure, lung infection and renal dysfunction were some of the risk factors that contribute to the occurrence of electrolyte and acid-base imbalance ($P < .05$).

Logistic regression analysis showed that gender, age, underlying medical conditions such as heart failure, high blood pressure, lung infection, renal dysfunction are risk factors to electrolyte and acid-base balance disorders which may later lead to death.

Table 1 Annual case of hospitalization and death caused by electrolyte and acid-base imbalance in elderly patients with gender and age distribution

Age	65-75 years old				76-85 years old				>85 years old			
Distribution	Elderly patients hospitalized annually		Death		Elderly patients hospitalized annually		Death		Elderly patients hospitalized annually		Death	
Sex	M	F	M	F	M	F	M	F	M	F	M	F
# of Individuals	2553	2327	86	47	2154	1796	147	105	633	317	57	58

Table 2 Annual cases of hospitalization and death caused by electrolyte and acid-base imbalance in elderly patients with gender and age distribution

Type of electrolyte imbalance	Annual hospital admission in elderly patients (person)	Incidence (%)	Eventual death in patients upon admission (person)	Incidence (%)	Mortality (person)	Incidence (%)
Hyponatremia	1170	31.2	183	35.4	116	17.8
Hypernatremia	650	17.3	56	10.8	118	18.1
Hypokalemia	780	20.8	94	18.6	135	20.7
Hyperkalemia	200	5.3	27	5.4	78	11.9
Acidosis	1120	29.9	103	19.9	134	20.6
Alkalosis	850	22.7	54	9.9	71	10.9
Total	3750	100	517	100	652	100

Mortality among patients hospitalized due to electrolyte and acid-based imbalance.

Table 3 Mortality among patients hospitalized due to electrolyte and acid-based imbalance.

Type of acid and electrolyte imbalance		65-75 years old		76-85 years old		>85 years old	
		Admission	Mortality	Admission	Mortality	Admission	Mortality
Hyponatremia	Mild	49	47	53	36	6	16
	Moderate	4	6	15	3	6	2
	Severe	19	3	26	1	5	2
Hypokalemia	Mild	7	29	26	44	5	21
	Moderate	5	4	10	2	9	6
	Severe	13	14	10	9	9	6
Hypernatremia	Mild	17	59	27	40	12	19
	Moderate	15	14	8	51	4	13
	Severe	55	43	36	67	12	24
Acidosis	Mild	10	15	25	35	19	21
	Severe	10	15	25	35	19	21
Total Person		194	234	236	288	87	130

Table 4 Factors related to electrolyte and acid-base imbalance

	Regression Coefficients	WalsX2	P	OR	OR of 95% C.I.	
					Lower	Upper
Sex	-0.314	5.661	0.017	0.730	0.564	0.946
Age	0.017	3.494	0.032	1.018	0.999	1.036
Days hospitalized	0.007	1.706	0.191	1.007	0.996	1.018
Heart Failure	0.891	20.851	0.000	2.437	1.663	3.572
Hypertension	0.331	6.163	0.013	1.392	1.072	1.807
Diabetes	0.077	0.274	0.600	1.080	0.081	1.440
Stroke	0.185	0.596	0.440	1.203	0.753	1.923
Lung infection	1.025	52.244	0.000	2.786	2.110	3.678
Renal insufficiency	0.611	10.473	0.001	1.842	1.272	2.667
Urinary tract infections	-0.138	0.224	0.636	0.871	0.490	1.546
Gastrointestinal bleeding	0.815	3.014	0.083	2.259	0.900	5.668
Diarrhea	0.827	1.970	0.160	2.286	0.720	7.257
Tumor	-0.441	7.375	0.057	0.643	0.468	0.885

CONCLUSION: Severe and life-threatening electrolytes and acid-base imbalance were associated with various risk factors such as age, sex, and other pre-existing medical conditions like heart failure, high blood pressure, lung infection, and renal insufficiency, etc.

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Application of Brachial-ankle Pulse Wave Velocity in Arteriosclerosis Detection in the Elderly Population

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OBJECTIVE: This study aimed to analyze the characteristics of brachial-ankle pulse wave velocity (baPWV) between different genders and to investigate its influencing factors in elderly population.

METHODS: A total of 927 cases of elderly residents were randomly selected from 19 villages in Shengze area, Jiangsu Province. The indexes such as age, gender, height, weight, waist circumference, hip circumference and blood pressure were collected through standardized questionnaire and physical examination. The biochemical indicators were measured while baPWV was tested by automatic arteriosclerosis measurement system.

RESULTS: The differences in age, systolic blood pressure, heart rate, bilirubin, creatinine, uric acid, triglycerides and baPWV were statistically significant between male and female ($P < .05$). BaPWV was raised with the increase of age. Overall, baPWV of female was faster than that of male ($P < .01$). Blood pressure ($r = 0.36-0.585$, $P < .001$), age ($r = 0.261-0.299$, $P < .001$) and heart rate ($r = 0.298-0.327$, $P < .001$) were strongly correlated with baPWV, while uric acid ($r = 0.121-0.211$, $P < .01$) had weak correlation with baPWV. The association between TG ($r = 0.088-0.111$, $P < .05$) and baPWV was extremely low. Multi-variable linear stepwise regression analysis showed that systolic blood pressure, heart rate and age were the influencing factors of baPWV, which were influenced greatest by systolic blood pressure ($\beta > 0.5$, $P < .001$).

CONCLUSION: baPWV increased with age. Monitoring baPWV in elderly patients, particularly females, might be important.

P141

Mechanisms of Cognitive Dysfunction in Elderly Aggravated by Ischemia and Hypoxia

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OBJECTIVE: Atherosclerosis, chronic obstructive pulmonary disease (COPD), obstructive sleep apnea-hypopnea syndrome (OSAHS) and other ischemic diseases in elderly patients are common age-related diseases. Ischemia and hypoxia are one of the characteristic pathological changes in these diseases that can cause or aggravate cognitive dysfunction. This study aimed to investigate potential mechanisms of cognitive impairment caused by ischemia and hypoxia.

RESULTS: Neuronal cells are extremely sensitive to hypoxia. A long period of ischemia and hypoxia in elderly patients with vascular diseases can reduce metabolic activity of brain tissue and cause cerebral hypoperfusion and varying degrees of neural damages, such as white matter sparse, atrophy of the hippocampus and bilateral temporal lobe, which can eventually cause cognitive dysfunction. The possible mechanisms of these pathological changes mainly express in the following aspects: 1. Hypoxia can enhance the levels of oxygen radicals and inflammation cytokines in neuronal cells. Oxygen radicals can directly oxidize DNA, proteins and other biological molecules. Massive release of inflammatory cytokines such as tumor necrosis factor (TNF- α), interleukin (IL-1 β) can lead to neural damages such as apoptosis; 2. Hypoxia can induce mitochondrial damage and the dysfunction of mitochondrial autophagy such as removing damaged proteins and transporting them to lysosome; 3. Receptor for advanced glycation endproducts (RAGE) in blood-brain barrier can promote β -amyloid deposits and hyperphosphorylation of Tau protein, which can lead to neurodegenerative diseases; 4. Hypoxia-inducible factor (HIF-1 α) can lead to cellular metabolic disorders through regulating mitochondrial movement regulator (HUMMR) on mitochondrial membranes; 5. The dysfunction of L-type calcium channel protein (LCC) on the cellular membranes can induce a series of cascade reactions, leading to neuronal excitotoxicity and neuronal conductive disorders, which can lead to apoptosis.

CONCLUSION: The mechanisms of ischemia and hypoxia-induced neurocognitive dysfunction in elderly are very complex, which mainly include neural damages through oxygen radicals and inflammation cytokines, mitochondrial and autophagic dysfunction, β -amyloid deposits and hyperphosphorylation of Tau protein, calcium channel protein functional disorders. Further studies of these molecular mechanisms may provide a theoretical basis in prevention and treatment of aged vascular disease associated with cognitive dysfunction.

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Influence of Th17/Treg Imbalance on Heart Remodeling of SHR with Hyperhomocysteinemia

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OBJECTIVE: This study aimed to explore the influence of Th17/Treg imbalance on the heart remodeling of spontaneously hypertensive rat (SHR) with hyperhomocysteinemia.

METHODS: 7-week-old male SHR were divided into three groups: control group (group C) which were fed with perfect compound feed without methionine, methionine group (group M) which were fed with perfect compound feed added 2% methionine, and treatment group (group T) fed as group M in the first 8 weeks, and then as the fodder of group C, but added with vitamin B₆ 12 mg·kg⁻¹·d⁻¹, vitamin B₁₂ 0.09 mg·kg⁻¹·d⁻¹ and folate 4 mg·kg⁻¹·d⁻¹. The weight, blood pressure and echocardiography were checked every 4 weeks, and the proportion of Th17 cells and Tregs were analyzed by flow cytometry and related factors in blood were detected by ELISA.

RESULTS: Plasma concentration of homocysteine was increased gradually in group M, which reached the peak on week 16 (31.28 ± 3.230) and umol/L on week 8 vs (29.23 ± 2.533) umol/L on week 12 vs (48.22 ± 3.855) umol/L on week 16, $P < .01$. There was no significant difference of plasma homocysteine among every time in group C. While the group T manifested a high concentration of plasma homocysteine on week 8, then decreased to normal until week 16 ($P < .01$). The plasma homocysteine level was different among three the groups on week 12 and 16 ($P < .01$). The weight of SHR in group M and T was going up slowly compared to that of group C from week 4 ($P < .01$). There was no difference about blood pressure in the three groups. Echocardiography displayed an increase tendency of left ventricular and diastolic dimension (LVEDD) and left ventricular mass index (LVMI) in group M and group T than that in group C before week 8. On week 12, the LVEDD was the largest in group M, and the smallest in group C, while LVMI maintained the same trend on week 16 ($P < .05$). There was no difference of left ventricular ejection (LVEF) in the three groups at any time point. Compared with that of group C, SHR in group M and group T had a higher plasma IL-6 and IL-17A level on week 8, which kept increasing after week 8 in group M, while SHR decreased in group T. The plasma concentration of TGF- β and IL-10 appeared higher in group M and group T than in group C when fed until week 12, especially obvious in group T. By flow cytometry, the proportion of CD4⁺IL17⁺T cells ((6.46 ± 0.821) % in group C vs (10.50 ± 1.177) % in group M vs (7.58 ± 0.647) % that in group T, $P < .01$) and CD4⁺CD25⁺FOXP₃⁺T cells ((3.46 ± 0.279) % of group T more than (3.08 ± 0.340) % of group M, and the latter was more than (2.92 ± 0.277) % of group C) were disparate among all groups on week 12 ($P < .01$), and the trend continued until week 16. There was a correlation between the Th17/Treg ratio and plasma homocysteine level.

CONCLUSION: Th17 and Treg may be important in heart remodeling of SHR with hyperhomocysteinemia.

P143

Lifestyle Factors Associated with Type 2 Diabetic Kidney Diseases: A Systematic Review/Meta-Analysis

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OBJECTIVE: This study aimed to systematically investigate the independent lifestyle risk factors associated with diabetic kidney disease (DKD) in patients with type 2 diabetes mellitus.

METHODS: Medline, EMBASE and four kinds of Chinese databases were searched (January 1960- December 2012). Papers were independently selected, data were extracted and the quality of included studies was assessed. We calculated the combined relative risk (RR) or odds ratio (OR) and 95% CI of correlation factors, or the mean and standard deviation of the related factors. RevMan 5.2 was used for meta-analysis.

RESULTS: Eight studies involving a total of 9540 patients with type 2 diabetes were included. Of the 8 included studies, there were 6 cross-sectional studies. The remaining 2 studies were cohort study and case-control study with poor to moderate quality of methodology. The results showed that lifestyle factors with beneficial effects on DKD include moderate work/exercise [adjusted OR (95% CI): 0.24 (0.08, 0.71)], taking deep color vegetables (≥1 time/day) 0.24 (0.06, 0.92), taking fruits (1 time/week~1 time/day vs <1 time/week) 0.30 (0.14, 0.66) and taking fruits (≥1 time/day vs <1 time/week) 0.28 (0.13, 0.60).

CONCLUSION: While high quality research is lacking, current evidence suggest beneficial effects of certain life style actors on DKD including moderate exercise/work and taking deep colored vegetables and increased fruit intake. Prospective cohort studies with high quality are needed.

P144

Inhibitory Effect of Paclitaxel Liposome on the Growth of A549 Tumor Cells of Human Lung Cancer

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OBJECTIVE: This study aimed to investigate inhibitory effects and pharmacokinetics of arginine glycine aspartic acid (RGD) as well as microRNA-34a co-modified paclitaxel loaded nano liposomes (RGD miLPs-34a/PTX) on growth of tumor sphere of A549 lung cancer cells.

METHODS: The stability of pharmacological and pharmacokinetic properties of RGD miLPs-34a/PTX was evaluated by fluorescence spectrophotometric analysis. The tumor sphere model with the A549 lung cancer stem cells was developed and the distribution of RGD miLPs-34a/PTX in the A549 lung tumor sphere was investigated by near-infrared cell imaging. The inhibitory effect of RGD miLPs-34a/PTX on the A549 lung cancer was detected by measuring the cancer stem cell growth in the tumor sphere.

RESULTS: RGD miLPs-34a/PTX has excellent stability in 50% of the medium with serum. Pharmacokinetic analysis has shown that the elimination rate of micro-RNA-34a was faster than those of miLPs-34a/PTX and RGD miLPs-34a/PTX in the tumor of A549 lung cancer stem cells. The difference was statistically significant ($P < .05$). Stem cell imaging experiments showed that fluorescence intensity of RGD miLPs-34a/PTX in the tumor was significantly higher than that of miLPs-34a/PTX. Cancer stem cell growth inhibition study showed that the inhibitory effect of RGD miLPs-34a/PTX

on the growth of tumor spheres was significantly stronger than that in the other liposome groups, with the difference statistically significant ($P < .01$). The groups with drugs had statistically significant inhibitory effects on the cancer stem cells than those in the control saline group ($P < .01$).

CONCLUSION: RGD co-modified paclitaxel loaded micro-RNA-34a nano liposomes had strong inhibitory effect and could serve as a promising therapeutic drug delivery system for the A549 lung cancer stem cells.

P145

Related Factors on Decreased Ability of Daily Life in Elderly Patients with Chronic Kidney Disease

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OBJECTIVE: This study aimed to investigate the factors associated with decreased ability of daily life in elderly patients with chronic kidney disease.

METHODS: A cross-sectional study was conducted between Sept. 1, 2013 and June 31, 2014 in West China Hospital of Sichuan University and Kangfu Kidney Disease Hospital in Chengdu. All patients were older than 60 years with chronic kidney disease. They received a comprehensive geriatric assessment. The contents of the survey include: ①. general survey: which included social-demographic characteristics, lifestyle, the primary diseases and if having dialysis; ② laboratory parameters in the last 3 months which included blood routine, biochemical indexes and urine indicators; ③ the comprehensive geriatric assessment indexes which included activities of daily living, step state and balance, falls, walking aids, hearing, vision, constipation, urinary incontinence, drug use status, comorbidity, cognitive function, emotional status, nutritional status, and social support. Statistical analyses were performed with SPSS software, version 17.0.

RESULTS: A total of 542 patients were included in this study, of whom 332 were male and 210 female. The mean age was 76.12 ± 8.86 years, with a median age of 76 years (60–101 years old). Univariate analysis showed that the difference of age, marital status, serum albumin, risk of malnutrition, malnutrition, cognitive function, depression score, comorbidity index, subjective support and the use of social support was statistically significant ($P < .05$). Multiple logistic regression analysis showed that independent factors associated with the ADL impairment in the elderly CKD patients included age [(70–79 years old OR = 1.811, 95% CI 1.006–3.258), ≥ 80 years old OR = 5.416, 95% CI 2.990–9.812], marital status (OR = 1.904, 95% CI 1.150–3.152), depression (OR = 1.327, 95% CI 1.219–1.444), risk of malnutrition (OR = 2.813, 95% CI 1.702–4.651) and malnutrition (OR = 5.132, 95% CI 2.508–10.501).

CONCLUSION: The factors associated with decreased ability of daily life in the elderly patients with chronic kidney disease include age, being widowed, depression, and poor nutritional status.

P146

Application of Different Dosage of Oxycodone Hydrochloride Injection Combined with Propofol on Elderly Patients Undergoing Therapeutic ERCP

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OBJECTIVE: This study aimed to compare the effects of different dosage of oxycodone hydrochloride injection with propofol on elderly patients undergoing therapeutic ERCP.

METHODS: Ninety elderly patients (60–75 years) were scheduled for therapeutic ERCP, and the patients were allocated into three groups (Group O1, O2, O3) according to the dose of oxycodone. There were 30 cases in each group. Each group received 0.05 mg/kg, 0.1 mg/kg, 0.15 mg/kg and propofol 2 mg/kg after 2 minutes. Basic vital signs and BIS value were recorded at the following time points: entering the operating room (T0), after induction (T1), entering the endoscope (T2), taking the stone (T3), exiting endoscope (T4), the patient awake (T5) and the duration of surgery, recovery time, patients cooperation, and adverse effect (including twisted body and respiratory depression).

RESULTS: There was no significant difference in general condition such as weight, age, duration of anesthesia, surgery time, et al ($P > .05$). Compared with the former two groups, MAP, SpO₂ was decreased obviously in group O3 ($P < .05$), and compared with O2 group, HR in T3 was obviously increased in group O2, and HR in T3–T4 was obviously reduced in group O3 ($P < .05$). The incidence of twisted body was greater in group O1, and the incidence of respiratory depression and nausea and vomiting were greater in group O3 ($P < .05$).

CONCLUSION: The addition of 0.1 mg/kg oxycodone to propofol increased anesthetic efficacy and reduced incidence of respiratory depression and other adverse effects on elderly patients undergoing therapeutic ERCP.

P147

MicroRNA-200 Family and Prognostic Value in Various Carcinomas: a Meta-Analysis

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BACKGROUND: MicroRNAs regulate gene expression and also play a significant role as prognostic biomarkers of various cancers. Recently, some studies have shown that miR-200 families act as novel biomarkers for the prediction of cancer outcomes. This meta-analysis was designed to review current evidence supporting the associations between miR-200 families and the prognosis of patients with various cancers.

METHODS: Eligible published databases including PubMed, Embase and Chinese National Knowledge Infrastructure (CNKI) databases were searched for articles until December 12, 2014. We performed a meta-analysis by calculating pooled hazard ratios (HR) and 95% confidence intervals (CI). Data were extracted from studies comparing overall survival (OS), progression-free survival (PFS) or recurrence-free survival (RFS). A bivariate random-effects model was adopted, and meta-regression and subgroup analyses were performed. In addition, an analysis of publication bias was conducted.

RESULTS: For OS, the pooled HR was 1.54 (95% CI 1.01 to 2.33), which showed that high miR-200 family was clearly related to poor survival in various carcinomas, but no significant association was found in PFS or RFS (HR = 0.92, 95% CI 0.67 to 1.26 and HR = 0.65, 95% CI 0.29 to 1.48, respectively). Subgroup analysis indicated that up-regulated miR-200 family was linked to poor OS in Asians (HR = 2.19, 95% CI 1.27 to 3.78) but not in Caucasians (HR = 0.94, 95% CI 0.46 to 1.91). Similarly, high miR-200 expression could not clearly predict the relationship with PFS and RFS in both Asians and Caucasians. For cancer type, high miR-200 also predicted poor OS among lung cancer patients (HR = 3.09, 95% CI 1.75 to 5.46), while it was not revealed in others. Besides, only elevated miR-200c of the miR-200 family indicated a significant poor OS (HR = 2.25, 95% CI 1.39 to 3.64).

CONCLUSIONS: Our findings suggest that miR-200 family might be a potentially useful biomarker for predicting cancer prognosis, especially for lung cancer in Asians.

P148

Gene Expression Profile of the Hippocampus in A β Induced Alzheimer's Disorder Model Rats

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BACKGROUND: Beta amyloid (A β) is believed to be a key pathological hallmark of Alzheimer disease (AD) and the identification of the involved biological process or pathway is considered to be a primary way to elucidate the etiology and pathogenesis of this disease. This study aimed to evaluate gene expression profile of the hippocampus in A β -induced AD rat model.

METHODS: In present work, injection of abnormal amyloid-A β into the hippocampus of rats was applied to set up the experimental A β -induced AD model rats. The memory impairment of the AD model rats was validated by Morris water maze tests.

RESULTS: Our results demonstrated that the learning and memory of rats in the AD group were decreased and the deposition of A β in the rats' brain led to neuronal cells damage. The cell density and morphology were declined. Gene microarray was used to scan the gene expression level, and gene expression levels of 17 genes were significantly alerted (at least more than 2-fold), which included 7 up regulated and 10 down regulated genes respectively. Compared to the control group, a large number of enzymes, hormones involved in catalytic activity (like hydrolase, glycosidase, endopeptidase and et.al) and neuropeptides and those of a few of genes involved in metal homeostasis, signal transduction and neurotransmission were significantly modified in the model group.

CONCLUSION: The results suggested the involvement of specific genes expressed by the hippocampus in the pathogenesis of AD. They provided a new insight into the regulation of gene expression level in Alzheimer disease and its progression.

P149

Effect of Early Intravenous Administration of Panax Notoginseng on Elderly with Acute Lacunar Infarction

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OBJECTIVE: This study aimed to investigate the effect of early intravenous administration of Panax notoginseng on elderly with acute lacunar infarction.

METHODS: A total of 64 elderly-patients were recruited, who were diagnosed in 24 hours as acute lacunar infarction for the first time. Thirty-one of the patients treated with Panax notoginseng were divided into treatment group and thirty-three of the patients without Panax notoginseng therapy were divided into control group. Both groups received similar treatments based on acute stroke management guideline such as antiplatelet, lipid-lowering agent, and physical rehabilitation. The treatment group received Panax notoginseng 750 mg intravenously once daily for 4 weeks, while the control group received normal saline. We assessed NIHSS, ADL, MMSE, MoCA scores and physical rehabilitation compliance on both groups before and after the treatment of Panax notoginseng.

RESULTS: Compared to the control group, the treatment group had significant improvement in ADL score (7.1 ± 8.5 vs

2.0 ± 7.1 , $P < .05$) and MoCA (3.4 ± 2.5 vs 1.6 ± 1.1 , $P < .05$), NIHSS score (1.0 ± 1.1 vs 0.3 ± 1.0 , $P < .01$), as well as higher physical rehabilitation compliance (3.1 ± 0.7 vs 2.4 ± 1.0 , $P < .05$).

CONCLUSION: Treatment with Panax notoginseng improved daily self-care activities, cognitive function, and higher compliance on physical rehabilitation for elderly patients with acute lacunar infarction.

P150

CYP2C19 Poor Metabolizer is Related to Silent Myocardial Ischemia in Chinese Elderly Women after PCI

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BACKGROUND: Polymorphisms of CYP2C19 have been associated with low responsiveness to clopidogrel, which is verified to be a risk factor for ischemic cardiac events. However, the contribution of these genotypes to platelet reactivity in asymptomatic patients after percutaneous coronary intervention has not been studied yet. This study aimed to evaluate the contribution of these genotypes to platelet reactivity in asymptomatic elderly patients after percutaneous coronary intervention.

METHODS: A total of 302 Chinese patients were classified into 2 groups after 1 month of receiving successful percutaneous coronary intervention (PCI): Silent myocardial ischemia (SMI) group (183) and non-SMI group (119). CYP2C19 genotype was tested by Polymerase Chain Reaction-Restriction Fragment Length Polymorphism (PCR-RFLP). Multivariate analysis and linear and logistic regression models were used to assess the clinical information and blood results.

RESULTS: Patient genotype distribution was in Hardy-Weinberg equilibrium. The univariate analysis showed that the CYP2C19 gain-of-function alleles did not affect asymptomatic events in both groups. ($\chi^2 = 16.376$, $P > .05$). After adjusting for common risk factors to all patients, logistic regression analysis indicated that the odds ratios (OR) of CYP2C19 loss-of-function alleles (vs gain-of-function alleles) in elderly women with SMI was 3.946, 95% CI: [1.317, 3.628], $P = .016$, with the reference category of non-SMI group.

CONCLUSION: CYP2C19 loss-of-function genotype was associated with silent myocardial ischemia in Chinese Elderly women who underwent initial revascularization.

P151

Impact of Orthostatic Blood Pressure Change on Geriatric Syndromes in Elders with Hypertension

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OBJECTIVE: The aim of this study was to investigate the relationship between orthostatic change in blood pressure and geriatric syndromes in hypertensive elders referred to comprehensive geriatric assessment (CGA).

METHODS: In a population-based cross-sectional study, a total of 176 hypertensive patients aged 65 years and older who underwent CGA were recruited. Supine and standing blood pressures (BP) were measured, and then the differences between supine and upright BP and pulse pressure (PP) were calculated. Orthostatic hypotension (OH) was defined as either a decline of

at least 20 mmHg in systolic BP and/or at least 10 mmHg in diastolic BP within 3 minutes of standing position. CGA included activities of daily living (ADL), cognitive functioning, depression symptoms, nutritional status, fall risk, comorbidity and polypharmacy.

RESULTS: Thirty-six patients (20.5%) were diagnosed as OH. Compared with non-OH, patients with OH had larger pulse pressure difference in the third minute after orthostatic change ($t = -2.396$, $P = .021$), much worse in BP control ($\chi^2 = 16.294$, $P < .001$), lower ADL score ($t = 2.693$, $P = .011$), higher score of the Tinetti Performance Oriented Mobility Assessment (POMA) ($t = 3.121$, $P = .003$). Elderly patients who had larger PP difference within 3 minutes after changing posture suffered from worse in BP control, higher percentage of polypharmacy and more depression symptoms (all $P < .05$).

CONCLUSION: Orthostatic hypotension might be an important sign for the presence of geriatric syndromes in older patients with hypertension.

P152

Circulating Levels of Chemerin in Patients with Congestive Heart Failure

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BACKGROUND: Chemerin is an adipocytokine that regulates insulin resistance and inflammation, linked to congestive heart failure (CHF). The aim of this study was to investigate whether circulating chemerin levels increase in CHF patients and the relationships with clinical biochemical parameters of CHF severity.

METHODS: Plasma chemerin concentrations were quantified with ELISA for 52 CHF patients with cardiovascular disease and normal cardiac function (NHF), and 26 healthy controls (NCD).

RESULTS: Chemerin was significantly elevated in CHF patients (220.0 ± 9.7 ng/ml) compared to NHF (165.6 ± 8.0 ng/ml) and NCD (147.2 ± 11.3 ng/ml) patients ($P < .001$ for both). NYHA class IV CHF patients had higher chemerin levels (294.8 ± 23.0 ng/ml, $n = 9$) compared to class III (221.6 ± 14.2 ng/ml, $n = 17$) and class II (193.0 ± 12.4 ng/ml, $n = 26$) CHF patients ($P < .01$ for both). Chemerin positively correlated with homeostasis model assessment of insulin resistance (HOMA-IR) and body mass index (BMI) in both CHF and NHF groups and with TNF- α ($\beta = 0.376$, $P = .002$) in the CHF group after adjustment for age and sex.

CONCLUSION: Plasma chemerin levels were elevated in CHF. They were independently associated with markers of insulin resistance and obesity in all patients with inflammation in CHF patients.

P153

Evaluation of Teach-back Method for Improving Health Literacy of Senior Citizens in Nursing Homes

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OBJECTIVE: This study aimed to explore the effect of Teach-back method for the health literacy of senior citizens in nursing homes of Urumqi.

METHODS: Ten out of twenty-seven registered nursing homes were randomly extracted according to their size, and the 10 nursing homes were randomly divided into intervention group and control group. Strict accordance with the inclusion criteria, 126 patients in the experimental group and 134 patients in the control group of each nursing home were selected. The experimental group took Teach-back and peer education methods, and the control group took the peer education method, and the duration was 6 months.

RESULTS: Before the intervention, the difference between the total score and each dimension score (health knowledge, health belief, health behavior and health skills) of the health literacy of the two groups was not statistically significant ($P > .05$). After the intervention, the health literacy total score of the experimental groups was (110.10 ± 17.683), up from (47.57 ± 19.567), and the health literacy total score of the control groups was (74.96 ± 27.162), up from (49.54 ± 12.380). The 4 dimension scores of the two groups were significantly higher than before, and the difference was statistically significant ($P < .05$). Through comparing the difference between the two groups, general health literacy score and each dimension scores in the experimental group were significantly higher than those in the control group, and the difference was statistically significant ($P < .05$).

CONCLUSION: Teach-back method improved the health literacy level of senior citizens through a short-term education intervention.

P154

Dyslipidemia in Type 2 Diabetic Rats

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AIM: The objective of the study was to evaluate and compare blood fat levels between normal control rats and type 2 diabetic rats.

METHODS: Forty male Sprague-Dawley rats, 8 weeks of age, weighing 180–200 g, were randomly divided into 2 groups: control group (NC group), diabetes mellitus group (DM group). The rats from NC fed with regular chow for 2 weeks received vehicle. The rats in diabetes mellitus groups were fed with high-diet for 2 weeks, and then streptozotocin (STZ) was injected intraperitoneally in a dose of 35 mg/kg. After at least 4 weeks, all rats were fasted overnight and taken tail blood samples for blood fat levels.

RESULTS: After more than 6 weeks, the weight of the normal control group was 398.28 ± 61.00 g, however, the weight of rats from the diabetes mellitus group was 318.60 ± 79.08 g. The total cholesterol level was 0.38 ± 0.09 mmol/L in the control group vs 0.57 ± 0.21 mmol/L in the diabetes mellitus group. The triglyceride level of the control group was 1.55 ± 0.43 mmol/L, and that of diabetes mellitus group was 1.43 ± 0.18 mmol/L. The level of low density lipoprotein cholesterol in control group was 0.39 ± 0.14 mmol/L, and that of diabetes mellitus group was 0.36 ± 0.10 mmol/L.

CONCLUSION: Compared to the control rats, diabetic rats had higher triglyceride levels, but similar levels of total cholesterol and low density lipoprotein.

P155

Characteristics of Thyroid Nodules and Thyroid Function and Their Correlation with Atherosclerosis and Cardiac Function in the Elderly

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OBJECTIVE: This study aimed to investigate the prevalence and characteristics of thyroid nodules and thyroid dysfunction in elderly as well as their relationships with atherosclerosis and cardiac function.

METHODS: A total of 839 inpatients were randomly selected from the First Hospital of Lanzhou University. According to the inclusive criteria, thyroid ultrasonography and thyroid function test results from 442 patients and other clinical data were analyzed.

RESULTS: 1) The prevalence of thyroid nodule with aging ($\chi^2 = 40.84$, $P = .000$) reached 68.6%. It was negatively correlated with academic degree ($\chi^2 = 14.44$, $P = .002$). The age (OR = 3.734, $P = .000$), gender difference (OR = 3.246, $P = .006$) and BMI (OR = 1.371, $P = .045$) were independent risk factors for thyroid nodules. The prevalence of hypothyroidism was 46.6% with the age ($\chi^2 = 25.51$, $P = .000$). However detection of hyperthyroidism was 1.4%, and was unrelated with aging ($\chi^2 = 4.12$, $P = .390$) in the elderly. Thyroid nodules increased atherosclerosis risk ($r = 0.190$, $P = .000$), and was unrelated with cardiac function ($r = 0.030$, $P = .530$). Thyroid dysfunction would increase the risks ($r = 0.650$, $P = .000$) of atherosclerosis and diastolic dysfunction ($r = 0.099$, $P = .037$).

CONCLUSIONS: In the elderly, the prevalence of thyroid nodules and hypothyroidism was high and increased with the age. Thyroid dysfunction also increased risk for atherosclerosis.

P156

Demand and Utilization Analysis of Community Health Services for the Elderly in Zhejiang Province

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OBJECTIVE: This study aimed to evaluate the needs and utilization status of community health services for the elderly

METHODS: A stratified multi-stage sampling and six district/county areas were extracted based on the economic and social development in Zhejiang Province. A self-designed questionnaire survey was conducted for the elderly in sampling areas.

RESULTS: The number of senile people suffering from at least one chronic disease is 734 (86.0%). Age, smoking, alcohol consumption were significantly different between the prevalence of chronic diseases ($P < .05$). Full awareness of the elderly health management services was only 22.2%, and regular physical examination was the most wanted management services of the older people. The utilization of preventive health services was 85.2% and was influenced by many factors. Community health services satisfaction was 64.5%.

CONCLUSIONS: The needs and utilization of community health service were great and balanced, but low awareness of related policies to support the community health service should be emphasized and addressed.

P157

Application of Noninvasive Positive Pressure Ventilation in Patients with Recurrent Ischemic Stroke

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OBJECTIVE: The aim of this study was to investigate the effect of Noninvasive Positive Pressure Ventilation (NPPV) for the treatment of recurrent ischemic stroke.

METHODS: A case-control study of 76 elderly patients who were admitted to the geriatric department was conducted. The patients in the control group were treated with conventional drugs.

The exposure group was treated with conventional drug and NPPV therapy for 5 courses. Every course lasted for 4 weeks, twice in daytime for an hour each time and once during the night for continuous therapy. The interval between two courses was 2 weeks. Study endpoint was termination of all five courses. The recurrence rate of ischemic stroke, improvement of Scandinavian Stroke Scale (SSS) score, blood gas analysis, Vascular Endothelial Growth Factor (VEGF), soluble Inter Cellular Adhesion Molecule-1 (sICAM-1), and soluble Vascular endothelial Cell Adhesion Molecule-1 (sVCAM-1) were used as assessment indexes.

RESULTS: Compared with control group, the exposure group had significantly lower recurrence rate of ischemic stroke (Control group: 23.68%, experimental group: 2.63%). The baseline of SSS score was improved (before and after treatment difference: the control group: 0.05 ± 5.18 , the experimental group: 4.72 ± 4.51). The level of PaO₂ was increased (before and after treatment difference: the control group: -9.61 ± 15.45 , the experimental group: 2.29 ± 16.84) and the level of PaCO₂ (before and after treatment difference: the control group: -2.07 ± 6.72 , the experimental group: 5.92 ± 5.31), sICAM-1 (before and after treatment difference: the control group: 41.57 ± 83.20 , the experimental group: 99.34 ± 92.19) and sVCAM-1 (before and after treatment difference: the control group: 48.19 ± 92.53 , the experimental group: 48.19 ± 92.53) were all decreased.

CONCLUSION: NPPV appeared to be a useful treatment modality for patients with stroke.

P158

Serum Levels and Clinical Significance of Hcy, FA, VitB₁₂ and hs-CRP in Elderly Hypertensive Patients in Plateau Tibetan and Han Nationality

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OBJECTIVE: The study was to evaluate the clinical significance of Homocysteine (Hcy), Folate acid (FA), Vitamin B₁₂ (VitB₁₂) and high-sensitivity C-reactive protein (hs-CRP) in Tibetan and Han nationality patients with hypertension in plateau area.

METHODS: Two elevation regions, Xining (altitude, 2260 m), and Yushu (altitude, 3800 m) were chosen. According to serum Hcy level, patients were divided into hypertension group and healthy control group. Each group was divided into two subgroups, Tibetan and Han nationality. All subjects were divided into four groups, with 30 cases in each. Enzyme immunoassay method was used to detect the serum levels of Hcy, FA and vitamin B₁₂ in each group. Automatic biochemistry analyzer was used to inspect the hs-CRP serum level.

RESULTS: A total of 120 patients were included in this study. Compared with control group for the same altitude, serum Hcy and hs-CRP levels in the hypertension group were significantly higher ($T = 6.543$, $T = 8.670$, $P < .01$), while FA level was significantly decreased ($T = -9.675$, $P < .01$). VitB₁₂ had no statistically significant difference. For the different altitude, serum Hcy and hs-CRP levels for the hypertension group in Yushu region were higher than that of Xining hypertension group ($T = 6.745$, $T = 5.332$, $P < .05$), while for FA and VitB₁₂, there was no statistically significant difference. With the same altitude of Tibetan and Han, the serum levels of Hcy in Tibetan people were significantly higher ($T = -7.675$, $P < .01$), and those of FA and VitB₁₂ were significantly lower ($T = 7.440$, $T = 8.432$, $P < .05$).

CONCLUSIONS: Plateau living with low oxygen environment affected the levels of serum Hcy and hs-CRP.

P159

Study on the Incidence and Effectiveness of Falls Intervention Programs in Community-dwelling Older People in Beijing

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OBJECTIVE: This study aimed to understand the epidemiological characteristics of falls and to evaluate the effectiveness of fall-prevention programs in older adults living in an urban community in Beijing.

METHODS: A cross-sectional study as baseline survey was conducted in the Longtan community of Dongcheng District in Beijing in 2005. People aged 60 or above were recruited using stratified cluster sampling method. Intervention strategies including giving lectures and distributing free literature on falls-related knowledge, and making personalized programs and health education on falls prevention were provided to the interviewees regularly every year after the time of baseline survey. A follow-up survey was conducted in the same community in 2009. Information related to falls was collected with a standardized structured questionnaire by face-to-face interview.

RESULTS: The incidence of falls among 1,512 interviewees was 18.0% (272) during the past 12 months at baseline, while 84 fell with fall rates of 17.8% among 472 interviewees in the follow-up survey in 2009. The standardized incidence rate of falls was 11.6% after the intervention, which was decreased compared to 18.0% before the intervention ($\chi^2 = 7.330$, $P = .007$). After the intervention, the proportions of people obtaining falls-related knowledge, never fearing of falls and often fearing of falls, and people with normal gait and with negative emotion were 30.7%, 60.0%, 15.7%, 68.9% and 37.5% respectively, which were higher than those before the intervention (20.6%, 52.1%, 12.5%, 22.5%, 23.5%; $\chi^2 = 12.803$, 204.770, 13.868, 21.765, all $P < .05$); however, the percentage of older adults often taking medicine was 35.0% before intervention, which was decreased to 25.6% after intervention ($\chi^2 = 9.712$, $P = .002$).

CONCLUSIONS: The incidence of falls among elderly urban community-dwellers in Beijing was high, and targeted multiple interventions were effective in reducing the incidence of falls.

P160

Incidence and Cause of Falls in a Community Dwelling Older Population in Beijing

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OBJECTIVE: This study aimed to investigate the incidence of falls, and to explore the causes of falls in community-dwelling elderly people in Beijing.

METHODS: A cross-sectional study was conducted in a community in Beijing, and 1,512 people aged 60 years or over were recruited using stratified cluster sampling method. Information about any falls was collected with a standardized structured questionnaire by face-to-face interview. Multilevel model analysis was used to explore the reasons for falls in the elderly.

RESULTS: The fall incidence rate among 1,512 interviewees was 18.0% (272) during the past year, and the total frequency of falls was 379 times. A total of 180 female and 92 male older adults fell with the incidence of falls which was higher among women than among men (20.1% vs 14.9%, $\chi^2 = 7.45$, $P = .006$). The rates of fall incidence for groups of 60~, 65~, 70~, 75~, 80~, 85~95 years old were 14.5% (37/256), 15.4% (74/481), 19.5% (74/380), 20.2% (49/

242), 23.1 (24/104) and 28.6 (14/49), respectively, and increased with ageing ($\chi^2_{\text{for trend}} = 10.37$, $P = .001$). Among the 379 times of falls, 300 times (79.2%) were related to health problems, 291 times (76.8%) were related to behavioral and psychological factors, and 284 times (74.9%) were related to environmental factors. Poor balance was the most common cause of falls as to health problems, followed by leg weakness, and attention problems and uneven ground were the most common causes of falls as to behavioral and psychological factors and environmental factors, respectively. Multilevel model analysis showed that for self-reasons, the main influence factor was location of falls; but for external reasons, the main influence factors were season and time of day of the falls.

CONCLUSIONS: The incidence of falls among elderly urban community dwellers in Beijing was high. Falls in the elderly resulted from both intrinsic and extrinsic factors including biological factors, behavioral and psychological factors, social factors and environmental factors, which suggests intervention strategies should be targeted at various factors related to falls.

P161

Clinical Research on Pathogen Distribution and Drug Resistance in Acute Exacerbation of Chronic Obstructive Pulmonary Disease in Elderly Patients

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OBJECTIVE: This study aimed to analyze sputum culture and drug sensitive test of inpatients and to investigate pathogen species and drug resistance.

METHODS: We collected sputum samples from 426 older inpatients (≥ 60 years old) and 74 non-older inpatients (40–59 years old) with acute exacerbation chronic obstructive pulmonary disease (AECOPD) in our hospital during Jan. 2012 and Dec. 2012. Bacteria and fungus were cultured and antibiotic sensitivity was tested for any positive cultures.

RESULTS: Positive rate of sputum culture was 30.52% in 426 older inpatients, of which single infection accounted for 80%, and mixed infection accounted for 20%. A total of 164 pathogens were found: *G⁻* plays a major role, 96 stains (58.54%) followed by fungus in the second, 63 stains (38.41%). *G⁺* was rare with only 5 stains (3.05%). Top three *G⁻* in sputum culture were as follows: 25 *Acinetobacter baumannii* (15.24%), 24 *Klebsiella pneumonia* (14.63%), and 22 *Pseudomonas aeruginosa* (13.41%). Positive rate of sputum culture was rare in 74 patients with only 11 cases (14.86%), which was statistically different compared with the elderly group. *Acinetobacter baumannii* infection which was more in age 70 or older elderly group and in the patients with basic disease was statistically significant. The rate of pathogen detection in smokers was higher than that of no-smokers, *Pseudomonas aeruginosa* infection rate was higher in smokers than that of no-smokers, and it had statistical significance. Half of *Acinetobacter baumannii* were resistant to antibiotic agents.

CONCLUSION: Top three pathogens in sputum culture in older inpatients with AECOPD in our hospital were *Acinetobacter baumannii*, *Klebsiella pneumonia*, *Pseudomonas aeruginosa*, and many of them are resistant to antimicrobial agents. The rate of pathogen detection in older inpatients is higher than that of patients for 40–59 years old in our hospital. The resistance rate of *acinetobacter baumannii* to antimicrobial agents reached to 50%.

P162

Selection of Treatment Facility and its Influencing Factors in Elderly Patients with Chronic Diseases in a Community in Changsha

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OBJECTIVE: This study aimed to investigate preferences of treatment institution and its influencing factors for elderly patients with chronic diseases in a community in Changsha.

METHODS: By using systematic sampling, 400 elderly patients with chronic diseases were included for the investigation by questionnaires from June to August 2012. The influencing factors were analyzed by binary logistic regression model or multinomial logistic regression model on the basis of the numbers of response variables.

RESULTS: For mild conditions with self-consciousness, more than half of the elderly patients with chronic diseases (206 cases, 51.5%) preferred community health service centers including private clinics (67 cases, 16.8%), municipal hospitals (92 cases, 23.0%), top first class hospitals (35 cases, 8.8%). In serious conditions in self-consciousness, patients preferred top first class hospitals (238 cases, 59.5%) and municipal hospitals (162 cases, 40.5%), no one selected the private clinics or community health service centers (72–443.240, $P < .001$). The analysis of impact factors for the first treatment institution selection in patients with serious conditions in self-consciousness showed that compared with patients selecting the municipal hospital, the following personnel preferred to choose top first class hospitals: patients from employment staff (OR = 32.38) and retirees (OR = 15.17), patients with monthly income more than 5,000 yuan (OR = 134.28), rural cooperative medical patients (OR = 661.42), patients with high confidence on community health service centers (OR = 4.23), patients with BP and serum glucose determination at a frequency of once a month to every 3 months (OR = 83.93), patients unknowing the telephone numbers of community health service centers (OR = 3.02), patients living far away from community health service centers (OR = 15.70), and patients with little information about community health service centers (OR = 1.62). The analysis of impact factors for the first treatment institution selection in patients with mild conditions in self-consciousness showed that compared with patients selecting the private hospitals, the following personnel preferred to choose community health service centers: patients with primary education level (OR = 0.15), retired persons (OR = 0.14), patients with higher health attention (OR = 0.26), while persons with less health attention (OR = 10.65) and patients living far away from community health service center (OR = 5.22) were more willing to choose the private hospitals.

CONCLUSIONS: Patient preferences for treatment facilities were affected mainly by the perceived severity. Age, education level, employment status, income, health insurance type, trust and understanding of community health service centers, and the emphasis on health and the distance from community health services centers were also influential factors.

P163

The Association of Biological Molecules with Frailty in Older Adults

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BACKGROUND: Frailty is a common and important geriatric syndrome characterized by declines in physiological reserve and function of multiorgan systems, leading to increased vulnerability for adverse health outcomes. Emerging evidence demonstrates the value of biological molecules in the circulation as predictor of frailty in older adults. This study aimed to review and discuss the

recent advances of biological molecules and to explain or predict of frailty, so as to present the current state of the epidemiological evidence by different experimental designs.

METHODS: PubMed-MEDLINE and Wanfang Med Online were searched from 2002 to 2014. The search was restricted to English-language articles using the following groups of MeSH headings and key words: frail elderly, frail, frailty, aged, older adults, biological markers, inflammation, hormone, coagulation, oxidative stress, gene, mechanism, and pathogenesis. Experimental design includes cross-sectional and perspective or pilot study in the community, hospital and laboratory settings.

RESULTS: Higher levels of IL-6 and CRP were associated with frailty after adjusting for age, gender, BMI, smoking, comorbidity, medication and lifestyle confounders. The frail subjects had significantly lower serum testosterone and 25 (OH) D levels, higher adiponectin and coagulation factors (D-dimer, fibrinogen, VIII). Components of the IGF-I system may be independent predictors of frailty. Furthermore, frailty status is associated with blunted cortisol reactivity, higher oxidative damage (8-OHdG, GSSG, MDA, carbonyl proteins) and ApoE genes. Lower serum sirtuins and higher glycoprotein may represent new potential and sensitive markers. The association of frailty with the molecules was studied using multinomial or binary logistic regression in most of researches. Finally, the possible reasons of some conflict results and its influencing factors were identified.

CONCLUSIONS: The biological molecules in the circulation could predict cumulative decline of homeostasis while there is currently no consensus on the association of biological molecules with frailty. Further prospective data and new predictive models by statistical methods would enable the nature of this relationship to be explored. This review highlights the need to improve knowledge about the molecular markers of frailty in older adults.

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A Treatment Study of Danhong Injection (Herbal TCM Product from Salviae Miltiorrhizae and Flos Carthami Tinctori) on the Elderly Unstable Angina Patients

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OBJECTIVE: The aim of this study was to investigate the treatment effect of Danhong Injection (DHI) in the elderly patients with unstable angina pectoris as well as its potential mechanisms.

METHODS: A total of 155 patients with unstable angina pectoris were randomly classified into study group (80 cases) and control group (75 cases). Patients in the two groups were all administered with routine medications. Additionally, DHI 40 ml was administered intravenously to the patients (1/day for 14 days) in the study group while Prostaglandin E₁ 10 µg was administered intravenously to the patients (1/day for 14 days) in the control group. Platelet aggregation rate, plasma biochemical and endothelial indicators were examined for all patients before and after the treatment.

RESULTS: There was significant difference in the inhibition rate of ADP and AA induced platelet aggregations in each group after treatment (Pre-treatment AA, ADP-induced PAG and post-treatment in study group: 40.6 ± 16.5 , 49.6 ± 14.0 and 13.2 ± 10.3 , 36.9 ± 11.3 ; pre-treatment AA, ADP-induced PAG and post-treatment in control group: 42.8 ± 15.6 , 52.8 ± 8.6 and 15.1 ± 8.3 , 44.1 ± 12.3). The difference of ADP induced platelet aggregation rate was statistically significant between the two groups ($P < 0.01$) but AA induced platelet aggregation rate was not after treatment. The plasma circulatory endothelial cells (CECs) in the treatment group were significantly less than the control group after treatment ($P < 0.01$) (pre-treatment and post-treatment CECs in study group: 8.69 ± 1.24 and 2.69 ± 0.35 ; pre-treatment and post-treatment CECs in control: 8.40 ± 1.82 and 4.72 ± 0.59). The plasma level of

NO was elevated (pre-treatment and post-treatment NO in study group: 73.09 ± 15.48 and 94.13 ± 14.40 ; pre-treatment and post-treatment NO in control: 71.18 ± 14.92 and 90.20 ± 16.24) and endothelin (ET) decreased compared with pre-treatment in both two groups ($P < 0.01$) (pre-treatment and post-treatment ET in study group: 81.55 ± 1.19 and 47.70 ± 1.30 ; pre-treatment and post-treatment ET in control: 80.94 ± 2.62 and 46.78 ± 2.00) and the difference between the two groups was not statistically significant. No abnormal changes in blood routine examination, liver or renal functions were seen in all patients.

CONCLUSIONS: DHI was proved to be an effective and safe agent for the treatment of patients with unstable angina pectoris.

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Association between Androgen Receptor CAG Repeat Polymorphism and the Metabolic Syndrome in a Cohort of Old Chinese Men

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BACKGROUND: Elderly men experience both a dramatic decline of androgen levels and an increased incidence of metabolic syndrome (MS). Since most androgen effects are mediated through androgen receptor (AR), and CAG repeat polymorphism in exon 1 of AR gene modulates AR activity, whether CAG repeat plays a significant role in this clustering of metabolic disorders needs to be examined.

METHODS: Four hundred and sixty-one elderly men, with a median age of 72 (60–96) years old, were enrolled in this study. Baseline body weight, height, waist and hip circumferences and blood pressure were measured. Fasting blood samples were drawn and leucocyte DNAs were extracted for the determination of AR CAG repeat length. The subjects were divided into: (1) subnormal total testosterone (TT) group ($n = 296$) or normal TT group ($n = 165$) according to TT levels (≤ 11.0 or > 11.0 nmol/L); (2) short, middle or long CAG group according to CAG repeat length (< 21 , $21-23$, > 23). The incidence of men with MS was compared among different groups.

RESULTS: MS was more common in elderly men with hypogonadism (72.1% vs 23.8%, $P = .023$) and short CAG (53.5% vs 19.1% in middle CAG group, $P = .031$; and vs 31.2% in long CAG one, $P = .043$). Men with more than one component of MS was more presented with a shorter CAG repeat length than those with only one component [(17.2 ± 6.8) vs (25.1 ± 10.6) , $P = .021$]. When considered as a continuous variable, however, AR CAG repeat length was not associated with fasting glucose, glycosylated hemoglobin level and waist-to-hip ratio (all $P > .05$), but positively associated with triglyceride ($r = 0.44$, $P = .038$).

CONCLUSIONS: Short CAG repeat (less than 21) was associated with increased risk of MS. However, the association was not linear.

P166

Propranolol on the Treatment of Hemangioma via Blocking Beta Receptors of Mast Cells

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OBJECTIVES: While propranolol is known as an effective treatment for infantile hemangioma, little is known about the

mechanism of its action. We hypothesized that propranolol may weaken mast cells function in the process of angiogenesis of hemangioma.

METHODS: To test this hypothesis, the effect of propranolol on beta-receptors of HUVECs and HMC-1 were assessed, and propranolol on blood vessels formation and inflammation of HUVECs were detected. Then pre-treated HMC-1 with propranolol, the supernatant secretion was collected by HMC-1 and the supernatant was added to HUVECs.

RESULTS: The results revealed down-regulated angiogenesis formation, which was conformity with the lower inflammatory medium. In addition, HMC-1 supernatant was divided into 3 groups: control, with $10 \mu\text{M}$ and $50 \mu\text{M}$ to HUVECs to explore whether the mechanism propranolol of treating IH was associated with β -receptors and inflammatory infiltration of mast cells. And the β -receptors agonist was added in the system to observe the proliferation and apoptosis ability of HUVECs.

CONCLUSIONS: Propranolol could block beta receptors of mast cells and reduce the release of inflammatory mediators via down-regulating NF- κ B pathway activity protein p65, which in turn may facilitate prevention of angiogenesis formation of hemangioma via down-regulating ERK pathway protein expression.

P167

Impact of Renal Insufficiency on All-Cause Death, Cardiovascular Death and Thromboembolism Events in Non-valvular Atrial Fibrillation Patients Aged over 45 Years

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OBJECTIVE: This study aimed to investigate whether renal insufficiency was an independent risk factor for thromboembolism events, all-cause death and cardiovascular death in patients with non-valvular atrial fibrillation aged over 45 years.

METHODS: A total of 825 patients with non-valvular atrial fibrillation aged over 45 years were selected from cases in the PLA General Hospital from July 2008 to June 2011 in this retrospective study. The baseline general conditions, diseases, laboratory examinations, echocardiography parameters and adverse events during the observation period were recorded. The patients were divided into two groups according to the baseline glomerular filtration rate. Logistic analysis and multivariate Cox regression analyses were adopted to calculate the adjusted hazard ratio (HR) of renal insufficiency for clinical adverse events.

RESULTS: The prevalence of renal insufficiency ($\text{eGFR} < 60 \text{ ml/min/1.73 m}^2$) was 21.09% in the 825 non-valvular atrial fibrillation patients without anticoagulation therapy. Over 75 years, female, diabetes mellitus, heart failure and left ventricular hypertrophy were independent risk factors of chronic renal insufficiency. During a total of 2279.58 man years' follow-up (mean 33.16 ± 22.24 months), 209 patients died of all-cause death and 61 patients died of cardiovascular death. A total of 139 patients had thromboembolism events (stroke or TIA or peripheral arterial thromboembolism) and 133 patients had cardiovascular events (acute myocardial infarction and/or acute heart failure). The all-cause mortality, cardiovascular mortality and rates of thromboembolism events and cardiovascular events were higher in patients with chronic renal insufficiency than those in the control group ($P < .05$). After adjust for CHA₂DS₂VASc score, renal insufficiency was an independent predictor of the all-cause death (HR: 1.794, 95% CI: 1.330–2.419, $P < .001$), cardiovascular death (HR: 2.865, 95% CI: 1.706–4.811, $P < .001$), thromboembolism events (HR: 1.636, 95% CI: 1.121–2.388, $P = .011$) and cardiovascular events (HR: 2.371, 95% CI: 1.651–3.404, $P < .001$). In addition, left

atrial enlargement (LAVI > 40 ml/m²) was an independent risk factor of all-cause death, thromboembolism events and cardiovascular events, but not for cardiovascular death.

CONCLUSIONS: After adjusting for CHADS₂ score or CHA₂DS₂VASc score, chronic renal insufficiency was an independent risk factor for death, cardiovascular death, thromboembolism event and cardiovascular event in non-valvular atrial fibrillation patients without anticoagulation therapy. Left atrial enlargement was independently associated with all-cause death, thromboembolism event and cardiovascular event.

P168

Prognostic Values of Beta-2 Microglobulin and Ambulatory Blood Pressure Parameters for Risks of Cardiovascular Events in Chinese Elderly Patients with Isolated Systolic Hypertension

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OBJECTIVE: The aim of this study was to investigate the relationships of serum B2M concentration and ambulatory blood pressure parameters with risks of cardiovascular events in Chinese elderly patients with Isolated Systolic Hypertension (ISH) who did not have severe renal insufficiency (serum creatinine > 3.0 mg/dl).

METHODS: Renal function parameters such as serum B2M concentration, serum cystatin C (CysC) concentration and estimating glomerular filtration rate (eGFR), ambulatory blood pressure parameters, and echocardiographic data were evaluated in 460 patients with ISH aged >60 years in this observational, retrospective study. The study population was stratified by the median serum B2M concentration at baseline. The Cox proportional hazard model was adopted to calculate adjusted hazard ratios of risk factors for cardiovascular events and mortality.

RESULTS: During a median follow-up period of 3.25 years, 43 patients died of non-cardiovascular causes and 20 patients died of cardiovascular causes. A total of 87 patients had stroke/TIA, 27 patients had acute myocardial infarction/acute heart failure, and 28 patients had intracranial hemorrhage/major bleeding. Multivariate analysis showed that higher serum B2M concentration was an independent predictor of increased risk of all-cause mortality, ischemic stroke and/or TIA, acute myocardial infarction and/or acute heart failure, and major bleeding (HR: 3.17, 95% CI: 1.65–6.08; HR: 1.62, 95% CI: 1.05–2.48; HR: 5.43, 95% CI: 1.98–14.88; and HR: 3.46, 95% CI: 1.13–10.56, respectively). In addition, extremely low mean daytime DBP (<60 mmHg) was an independent risk factor of cardiovascular death (HR: 5.71, 95% CI: 1.89–17.20). No association was found between measures of BP variability and clinical outcomes.

CONCLUSIONS: Higher serum B2M concentration was independently associated with the risk of cardiovascular events and all-cause death in elderly ISH patients with creatinine ≤3.0 mg/dl. Excessive low DBP was also an independent risk factor of cardiovascular death.

P169

Experimental Study on Anti-atherosclerosis Effect of 2 New ADP-Receptor Antagonists: Prasugrel and Ticagrelor

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OBJECTIVE: Endothelial dysfunction and inflammation are important factors in the development of atherosclerosis. Animal

studies proved that clopidogrel could inhibit the process of atherosclerosis. This study aimed to focus on 2 new ADP-receptor antagonists, prasugrel and ticagrelor, and to explore their anti-atherosclerosis effects and potential mechanisms.

METHODS: Rabbit arterial intimal injury model was established by balloon dilatation accompanied with high-fat diet. Rabbits were divided into 3 groups and administered clopidogrel, prasugrel or ticagrelor by gavage respectively. Platelet aggregation rate was measured by light turbidimetric. The level of tissue inflammatory factors was measured by western blot. The positive rate of macrophages and smooth muscle cells in arterial plaque sites was measured by immunohistochemical analysis. Circulating endothelial cell counts were also measured.

RESULTS: The platelet aggregation rate in the 3 groups, clopidogrel, prasugrel and ticagrelor group, was all reduced (28%, 24%, 8% vs 60%) ($P > .05$); the level of CRP (17.3, 14.8, 16.1 vs 29.5 mg/L), vascular adhesion molecule -1 (8.1, 6.4, 7.2 vs 13.9 ng/ml) and monocyte chemoattractant protein -1 (13.2, 12.5, 11.9 vs 26.7 ng/L) in the 3 groups were all decreased after intervention ($P < .05$). Western blot showed that the expression of aortic vascular adhesion molecule-1 and monocyte chemoattractant protein-1 in the 3 groups were also decreased. Circulating endothelial cells, the positive rate of macrophages and smooth muscle cells in arterial plaques were all reduced ($P < .05$).

CONCLUSION: The new ADP-receptor antagonists, prasugrel and ticagrelor had potent anti-atherosclerosis effect. Likely through improving endothelial functions and inhibiting inflammatory reactions.

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The Effect of Simvastatin on Atherosclerotic Calcification

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OBJECTIVE: This study aimed to investigate the effect of simvastatin on atherosclerotic calcification and its potential mechanism.

METHODS: Twenty-four male ApoE^{-/-} mice on a C57BL/6J genetic background and 12 male C57BL/6J mice were selected when they were 8-week-old. 24 male ApoE^{-/-} mice were randomly divided into model group and simvastatin group (n = 12). 12 male C57BL/6J mice were regarded as control group. All mice received adaptive feeding in the animal center for 2 weeks after which they were fed by a high-fat diet instead. Nine weeks later, animals were treated with intragastric administration as follows: control group and model group received 0.2 ml PBS buffer per day for 8 weeks; simvastatin group received simvastatin (20 mg/kg dissolved in 0.2 ml PBS buffer for intragastric administration) every day for 8 weeks. The body weight of mice was recorded every day during the whole experiment. They were sacrificed and their aorta and aortic sinus were separated for further use. The aorta of mice was stained with oil red O and aortic sinus paraffin sections were processed with HE, Von kossa, TUNEL and immunohistochemical staining to observe plaque size, atherosclerotic calcification, apoptosis and expression of related proteins.

RESULTS: Body weight of the three groups increased gradually over time ($P > .05$). Serum lipid parameters of TG, TC, LDL-C, and HDL-C in the model group were higher, compared with those in the control group and simvastatin group ($P < .01$). Percentage of aortic plaque area of model group and simvastatin group were (54.50 ± 15.41)% and (33.69 ± 9.72)%, simvastatin group reduced ($P < .05$). And the mean plaque area of aortic sinus of simvastatin group was less than that of model group ($P < .05$). Percentage of calcification area of simvastatin group (2.33 ± 0.73)% was lower than that of model group (10.87 ± 2.41)% ($P < .05$).

Apoptosis rate of control group was $(30.90 \pm 1.75)\%$, model group $(66.43 \pm 4.05)\%$ and simvastatin group $(47.01 \pm 5.94)\%$, respectively. Apoptosis rate of model group was higher than that of control group ($P < .01$). And it was reduced in simvastatin group compared with the model group ($P < .05$). Protein expressions in aortic sinus of the three groups of GRP78, CHOP and Caspase12 were observed in varying degrees. The highest expression levels of the three proteins were observed in model group, which were significantly decreased in simvastatin group.

CONCLUSION: Simvastatin reduced endoplasmic reticulum stress-mediated apoptosis involved atherosclerotic calcification.

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Efficacy and Safety of Combination Antiplatelet Therapy in Very Old Patients with Acute Coronary Syndrome

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BACKGROUND: Persistent platelet activation may contribute to cardiovascular events in very old patients. Chronic use of platelet inhibitors could improve morbidity, mortality and quality of life in this population. Although aspirin and clopidogrel are widely used in patients with acute coronary syndrome (ACS), the benefit-risk of combination of aspirin and clopidogrel in very old patients with ACS is still a controversial issue.

OBJECTIVE: The aim of this study was to determine whether clopidogrel with aspirin (C+A) will produce more potent platelet inhibition vs aspirin or clopidogrel alone in very old patients with ACS.

METHODS: A total of 340 patients (>75 years old) with ACS were randomly assigned to 3 treatment groups: (1) 100 mg aspirin daily, (2) 75 mg clopidogrel daily, (3) 100 mg aspirin + 75 mg clopidogrel daily. Platelet tests were performed after 4 weeks after randomization. Details of any serious adverse events and major adverse cardiac events (MACE) were also recorded by ECG, echo, CT or MRI and laboratory examinations.

RESULTS: There were no deaths. Combination therapy resulted in significant greater inhibition than either inhibitor used alone ($P < .05$), and aspirin or clopidogrel resistance was significantly improved ($P < .05$). Accordingly, MACE decreased significantly in combination therapy group, whereas there was no significant difference in serious adverse events of the three treatment groups.

CONCLUSION: This study demonstrated synergistic antiplatelet effects of combined aspirin and clopidogrel therapy, and these findings could have significant impact on future design of antiplatelet strategies for very old patients with ACS.

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Influence of Sex Hormone Changes on the Risks for Thrombotic and Atherosclerotic Diseases

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OBJECTIVE: Some evidence suggests that decreased androgen and imbalanced sex hormones might play an important role in the pathogenesis of cardiovascular disease. This study was to evaluate the influence of sex hormone changes on the risks for thrombotic and atherosclerotic diseases.

METHODS: First, 28 male New Zealand white rabbits were randomized to castration groups and sham operation group after being fed with cholesterol rich diet. The castration groups were then randomized to 3 groups: controlled group, hypotestosteronemia group and physiotestosteronemia group. From the 2nd week to

10th week after castration, the latter two castration groups were injected testosterone undecanoate (TU) 3 mg/kg, 6 mg/kg intramuscularly once every 2 weeks respectively. Sex hormones and coagulation indicators were measured at baseline, 2nd week and 10th week after castration. Secondly, 247 Patients aged over 65 were enrolled in the study, including castrated prostatic cancer patients (groupI), non-castrated prostatic cancer patients (groupII) and non-prostatic cancer patients (group III). Sex hormones, coagulation indicators, platelet function and blood biochemical indicators were measured for all patients. Details of any major adverse cardiac events (MACE) were also recorded during follow up.

RESULTS: In the animal study, at 2nd week and 10th week after castration, total testosterone were respectively 0.80 ± 0.2 nmol/L and 0.71 ± 0.1 nmol/L in castrated rabbits as well as 0.77 ± 0.24 nmol/L and 4.7 ± 2.1 nmol/L in hypotestosteronemia rabbits. Sex hormones were imbalanced significantly in castrated rabbits and hypotestosteronemia rabbits. Accordingly, the risks for thrombotic and atherosclerotic diseases increased significantly in castrated rabbits and hypotestosteronemia rabbits but it was not in physiotestosteronemia rabbits. Secondly, baseline information of the three patient groups was similar. Testosterone were 0.99 ± 1.54 ng/ml, 3.99 ± 1.87 ng/ml and 3.98 ± 1.83 ng/ml respectively in group I, groupII and group III. Sex hormones were imbalanced significantly in castrated prostatic cancer patients. Accordingly, the risks for thrombotic and atherosclerotic diseases increased significantly in castrated prostatic cancer patients. The incidences of MACE were 39.3%, 21.3% and 15.6% respectively in group I, groupII and group III. The MACE was also increased significantly in castrated prostatic cancer patients.

CONCLUSION: Imbalanced sex hormones appeared to be a significant risk factor for thrombotic and atherosclerotic diseases.

P173

A Comparative Study in the Very Old Patients Treated with Different Antiplatelet Agents

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OBJECTIVE: The aim of this study was to investigate and compare antiplatelet effects and side-effects of combination therapy with different antiplatelet agents in the very old patients (≥ 75 years).

METHODS: A total of 263 patients aged 75 years old or over with cardiovascular diseases were included in the study. Patients were randomly divided into seven groups: aspirin 100 mg, aspirin 100 mg plus marzulene-s 0.67 g, plavix 75 mg, aspirin 100 mg plus plavix 75 mg, talcom 50 mg, plavix 75 mg plus PPIs, and the placebo group. The treatment lasted for 16 weeks. Before and at the end of the study, gastroscopy was performed to detect gastrointestinal bleeding. Blood specimens were collected in all eligible patients to test blood routine values, prothrombin time (PT), prothrombin activity (PA), activated partial thromboplastin time (APTT), fibrinogen (FIB), and platelet aggregation rate (PAG).

RESULTS: Platelet aggregation rate decreased significantly in all treatment groups compared with placebo group ($P < .05$) after treatment. Although there was an inclination that the platelet aggregation rate in aspirin plus plavix group was lower than that in aspirin or plavix alone group, the difference was not significant among them. The difference of platelet aggregation rate was not significant among PPIs added group and non-PPIs groups. The difference of the gastric bleeding occurrence was not significant among all the groups, and also not among marzulene-s added group and non-marzulene-s groups. Aspirin or Plavix resistance was similar to that in literature.

CONCLUSIONS: Aspirin, clopidogrel or combination of other antiplatelet agents suppressed platelet aggregation well in the very old patients, and all of them could be used safely in this group. PPIs did not decrease the inhibitory effect of platelet aggregation compared with the non-PPIs groups.

P174

The Effects of Voglibose on Type 2 Diabetes in the Elderly

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OBJECTIVE: This study aimed to evaluate the effects of Voglibose on type 2 diabetes in the elderly.

METHODS: Voglibose was given to 20 old patients in accordance with WHO diagnostic standard of type 2 diabetes for 12 weeks. Fasting blood glucose, 2-hour post-meal blood glucose, blood hemoglobin A1C, body weight, side effects and the frequency of hypoglycemia were analyzed before and after the treatment.

RESULTS: After 12 week treatment of Voglibose, levels of FBG, PBG and HbA1C were all decreased significantly than those of baselines. The fasting blood glucose was (8.68 ± 1.07) mmol/L and (7.20 ± 0.60) mmol/L before and after treatment of Voglibose respectively ($P < .01$). The 2-hour post-meal blood glucose was (11.65 ± 1.71) mmol/L and (8.80 ± 0.69) mmol/L before and after treatment of Voglibose respectively ($P < .01$). The blood hemoglobin A1C was (7.58 ± 0.66)% and (6.83 ± 0.53)%, respectively ($P < .01$). The level of serum insulin, TG, Tc and Cr had no change before and after the treatment. Hypoglycemia, weight gain, side effects events were not frequent in the Voglibose group.

CONCLUSION: Voglibose had good effects on Type 2 Diabetes in the elderly in terms of blood glucose control with little side effects.

P175

L-arginine Ameliorates the Progression of Autoimmune Myocarditis

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OBJECTIVE: Nitric oxide (nitric, oxide, NO) plays a dual role. On the one hand, it can inhibit inflammatory process under physiological conditions, and on the other hand, a large amount of NO can be involved in inflammation in autoimmune myocarditis. This study aimed to investigate the effects of N-nitro-L-arginine methyl ester (L-NAME), a kind of inducible nitric oxide synthase (iNOS) inhibitor, in the treatment of Balb/C mice with experimental autoimmune myocarditis (EAM) and to discuss the therapeutic mitochondrial mechanism induced by apoptosis.

METHODS: Sixty male 4- to 5- week old Balb/C mice were divided into normal control group, model control group and experimental group randomly ($n = 20$). Model control group and experimental group were created into EAM by injection of porcine cardiac myosin subcutaneously in double groin and axilla and intraperitoneal injection of pertussis toxin on day 0 and 7 respectively. Model control group was intraperitoneally administered 5 mg/ (kg·day) of physiological saline after injective myosin and pertussis toxin. Experimental group was intraperitoneally given 5 mg/kg/day of L-NAME on day 1–21. At the end of intervention, mice were euthanatized and hearts were harvested on day 21. The inflammatory score, fibrosis score, protein expression levels of Cas-

pase-3, Caspase-8 and Caspase-9, serum nitric oxide (NO) level, iNOS, iNOS mRNA, Caspase-3, Caspase-8 and Caspase-9 mRNA, cardiac ROS production rate and mitochondrial membrane potential were measured. Meanwhile, heart weight/body weight was calculated (HW/BW).

RESULTS: Inflammatory score, cardiac intestinal fibrosis score, cardiac apoptotic index, protein expression levels of Caspase-3, Caspase-8 and Caspase-9, HW/BW, level of NO and activity of iNOS, expression levels of iNOS mRNA, Caspase-3, Caspase-8 and Caspase-9 protein were all significantly higher in model control group and experimental group than in normal control group ($P < .01$), and the levels in model control group was higher than those in experimental group ($P < .01$). HW/BW was only a little elevation in model control group compared with the experimental group.

CONCLUSION: L-NAME protected cardiac myocyte likely via suppressing the activity of iNOS and further decreased production of NO in EAM. The mechanism may be related to inhibiting the apoptosis of cardiac myocyte mediated by caspase family and protecting the mitochondrial function.

P176

Relationship between Polymorphisms in p38beta MAPK and Pre-diabetes in Individuals of Han Chinese Descent

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OBJECTIVE: The present study aimed to investigate the relationship between p38beta MAPK gene polymorphisms and pre-diabetes in individuals of Han Chinese descent.

METHODS: A total of 1,466 non-consanguineous Han Chinese individuals were enrolled in the study. Mass spectrometry was performed to detect three haplotype-tagging single nucleotide polymorphisms (SNPs) of p38beta MAPK. Logistic regression analysis was performed to compare the difference in risk of pre-diabetes and to evaluate the interaction between genotypes and risk factors.

RESULTS: Subjects carrying allele A at rs2072878 possessed a significantly increased risk of pre-diabetes. The subjects with genotype AA had higher pre-diabetic risk than those with genotype AG or GG. After age stratification, the results showed that SNP rs2072878 was closely associated with the risk of pre-diabetes in subjects aged ≥ 60 years, and the risk of pre-diabetes in the subjects with genotype AA was increased by 1.429-folds when compared to those with genotype AG or GG. In pre-diabetes, homozygotes of AA at SNP rs2072878 had lower fasting plasma insulin levels, decreased 30-min and 120-min insulin levels after 75-g OGTT compared to the non-carriers.

CONCLUSION: In Han Chinese subjects, genetic variants in p38beta MAPK were associated with increased risk for pre-diabetes as well as plasma insulin levels after OGTT.

P177

Ankle Brachial Index and Pulse Wave Velocity is Associated with Subclinical Kidney Damage in Community Based Han Chinese

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OBJECTIVE: Our study aimed to explore whether the Ankle Brachial index (ABI) and brachial-ankle Pulse Wave Velocity

(baPWV) are associated with subclinical kidney damage in community based Han Chinese.

METHODS: A total of 2,127 subjects (860 men and 1267 women) aged 60 years and over were recruited in Beijing. Subclinical kidney damage was defined as urinary albumin-to-creatinine ratio (UACR) ≥ 30 mg/g. BaPWV was divided by quartile. Logistic regression was used to determine odd ratio (OR) and 95% confidence intervals (CI) of ABI and baPWV with subclinical kidney damage.

RESULTS: ABI was associated with UACR in multivariate-adjusted model (OR 0.88, 95% CI 0.80–0.97 by every 0.1 unit increasing of ABI), especially in hypertension (OR 0.82, 95% CI 0.73–0.93) and diabetes group (OR 0.84, 95% CI 0.69–0.98). BaPWV groups were also significantly associated with UACR, and odds ratio of having subclinical kidney damage for baPWV quartile II, III, and IV were 1.07 (0.67–1.69), 1.15 (1.07–1.32), and 1.23 (1.04–1.46) in multivariate-adjusted model. For hypertension and diabetes patients, only baPWV quartile IV group had higher odds ratio.

CONCLUSION: In Chinese community-based elderly Han population, ABI and baPWV were associated with subclinical kidney damage after adjusting for other risk factors. The association of ABI with subclinical kidney damage was stronger in patients with hypertension and/or diabetes.

P178

Correlative Analysis of Aspirin Resistance in the Very Old Patients with Coronary Artery Disease

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OBJECTIVE: This study aimed to evaluate incidence and correlative factors of aspirin resistance in the elderly patients with coronary artery disease.

METHODS: A total of 515 patients aged 60 or over were included in the study. Patients were divided into three groups according to their age and possible fact of having CAD: very old patients with CAD (age ≥ 75 , n = 285), very old patients without CAD (age ≥ 75 , n = 56), common old patients with CAD (60 $\leq$ age ≤ 74 , n = 174). All of the patients had taken aspirin at least for 2 weeks or longer. After admission, blood specimens were collected in all eligible patients to test blood routine, biochemical makers, platelet aggregation rate (PAG) and coagulating indexes. The incidence of aspirin resistance and its related factors were compared firstly. After that, the patients diagnosed as AR were divided into two groups: aspirin alone group, aspirin and clopidogrel combination group. PAGs were analyzed comparatively between them.

RESULTS: The incidence of aspirin resistance in the very old patients with CAD was statistically higher than that of the other two groups. In very old patients with CAD group, the incidence of aspirin resistance was increased with aging. The increased PLT counts, levels of FBG, LDL-C, and CRP, accompanied with diabetes mellitus were related factors of AR in very old patient with CAD group, however, in common old patients with CAD group, only increased PLT counts, CRP levels and accompanied with diabetes mellitus were related factors with AR. The patients with AR could obtain better effects after combination treatment with aspirin plus clopidogrel without increasing of adverse reactions.

CONCLUSIONS: Aspirin resistance is a common phenomenon in different aged groups. Its incidence increases with age, particularly in old patients with CAD. The related factors of AR are not the same in different aged groups.

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Cardiac Expression of Sirt1 in Aged Spontaneous Hypertensive Rats and its Role in Cardioprotection of Rosuvastatin

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OBJECTIVE: This study aimed to explore the role of Sirt1 intracellular signal pathway in the protective effect of rosuvastatin on the antioxidative damage of cardiomyocyte of aged SHR via testing the cardiac Sirt1 expression in vivo and in vitro.

METHODS: Male 20-week-old SHR were given vehicle or 40 mg/kg/d rosuvastatin, 10 mg/kg/d amlodipine, and both for 10 weeks. Age-matched male WKY was set for control. Noninvasive rat tail blood pressure was obtained and echocardiography was performed to obtain left ventricular (LV) interventricular septal and posterior wall thickness (IVST and PWT), LV end-diastolic dimension (LVEDD), end-systolic dimension (LVESD) and the LV ejection fraction (LVEF). Early and late transmitral velocity (E and A wave), the septal mitral annulus systolic and early diastolic velocity (Sm and Em) was also assessed. Invasive left ventricular hemodynamic measurement was performed. Heart rate, mean arterial pressure, LV end-diastolic pressure (EDP), maximal slope of systolic pressure increment (+dP/dt_{max}) and diastolic pressure decrement (-dP/dt_{max}), time constant of LV pressure decay (τ) were all calculated. Hearts were harvested and masson trichrome staining was used to evaluate of fibrosis deposition. Plasma and cardiac malondialdehyde (MDA) was measured to determine the oxidative stress, and terminal deoxynucleotidyl transferase-mediated dUTP nick-end labelling (TUNEL) assay was performed to assess the apoptosis. Expression of protein Sirt1/FoxO1/Caspase3/Cleaved Caspase3/Bcl-2/Bax was determined by Western Blotting. Primary neonatal rat ventricular myocytes were employed to investigate the impact of rosuvastatin on express of Sirt1/FoxO1 by the treatment of rosuvastatin in various concentration or time period. Lentiviral vectors encoding short hairpin RNA (shRNA) targeting Sirt1 was designed to knockout the gene of Sirt1. Reactive oxygen species (ROS), and MDA were measured and MTT was tested. Cell survival was assessed by flow cytometric analysis of Annexin V and propidium iodide (PI) staining and TUNEL staining. RT-PCR and western blotting was performed to determine the expression of Sirt1/FoxO1 and Caspase3/Cleaved Caspase3/Bcl-2/Bax.

RESULTS: Blood pressure decreased after treatment of amlodipine alone or combined with rosuvastatin. Interstitial fibrosis deposition increased in group of SHR control and decreased in amlodipine and rosuvastatin group. Coadministration of both medications gives rise to an even more significant decrease in interstitial fibrosis deposition. There was a significant increase in IVST and PWT and there was no alteration in LVEDD, LVEF, E/A and Sm in SHR groups. E/Em rose significantly in SHR groups, among which groups with medication enjoyed a lower E/Em than SHR control. There was no difference in +dP/dt_{max} between each group. -dP/dt_{max}, LVEDP and τ significantly increased in SHR control and reversed significantly in amlodipine and rosuvastatin group. Coadministration group shared the same level of these values with WKY control. SHR control represented increased plasma and cardiac MDA concentration and a higher apoptosis index, which was ameliorated in each medication and even to the same level of WKY control in coadministration group. Cardiac expression of Sirt1/FoxO1/Bcl-2 decreased and Cleaved Caspase3/Bax increased profoundly in SHR, while the situation was reversed in rosuvastatin and coadministration groups. Compared with control, cardiomyocyte of H₂O₂ group showed higher level of ROS and MDA, lower cell viability by MTT, and higher rate of apoptosis by TUNEL and flow cytometric analysis. These damages were alleviated significantly in rosuvastatin group and knockout of Sirt1 gene diminished the protective effect of rosuvastatin. Downregulation of Sirt1/FoxO1/Bcl-2 and upregulation of Cleaved Caspase3/Bax were

observed in cardiomyocyte of H₂O₂ group and rosuvastatin treatment reversed them. However, knockout of Sirt1 gene weakened the effects significantly.

CONCLUSION: Aged SHR are known to have LV hypertrophy, myocardial fibrosis, LV diastolic dysfunction, as well as increased oxidative stress and downregulation of cardiac Sirt1 expression. Results from this study showed that rosuvastatin could reverse these alterations and possibly exert the cardiac protective effects via Sirt1/FoxO1 intracellular signal pathway.

P180

Establishing a Predictive Model of Clopidogrel Resistance in Elderly Chinese Patients with Coronary Heart Disease

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OBJECTIVE: This study aimed to evaluate the possible risk factors of clopidogrel resistance in elderly patients with coronary heart disease and to establish a risk prediction model and risk score for identifying clopidogrel resistance.

METHODS: Patients with coronary artery disease took the therapy of combination of aspirin and clopidogrel. After 5 days, platelet aggregation was measured using light transmission aggregation with AA 0.5 mmol/L or adenosine diphosphate 10 μ mol/L as agonist. Clopidogrel resistance was defined as the value of platelet aggregation $\geq 43\%$. 120 post PCI patients were validation sample. We identified potential risk factors, conducted univariate analysis and multi-factor logistic regression to determine the independent risk factors and set up a risk model. The calibration and discrimination of the model were assessed by the Hosmer-Lemeshow (H-L) test and the area under the receiver operating characteristic (ROC) curve, respectively. We finally produced a risk score according to the coefficient β and rank of variables in the logistic regression model.

RESULTS: A total of 565 patients were analyzed (76.0 $\pm$ 11 years, 74.9% males). The prevalence of clopidogrel resistance was 38.58%. Glycemia (OR = 1.014; 95% CI: 1.004–1.023), history of AMI (OR = 2.323; 95% CI: 1.103–4.892) and body mass index (OR = 1.057; 95% CI: 1.017–1.099) were independently associated with clopidogrel resistance. P values of H-L test were all higher than 0.05, and the areas under the ROC curve were all larger than 0.70. The risk score was successfully established: glycemia >7.0 mmol/L:1 score, history of AMI: 1 score, Body mass index: 20–25 kg/m²: 2 score; >25 kg/m²:4 score.

CONCLUSIONS: The prevalence of clopidogrel resistance was high in elderly patients many of whom may take combination of dual antiplatelet therapy including clopidogrel. Glycemia, body mass index, and history of AMI were independently associated with clopidogrel resistance. Establishing the appropriate forecasting model is important for identifying the high risk patients with clopidogrel resistance and decreasing cardiovascular events.

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The Agreement between Different Platelet Assays to Identify Clopidogrel Resistance in Elderly Patients with Coronary Artery Disease

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OBJECTIVE: Definition of clopidogrel resistance remains controversial. Different platelet tests have been linked with recurrent ischemic events and proposed for daily practice. This study

aimed to compare the agreement between three methods to detect platelet function for diagnosing clopidogrel resistance in elderly Chinese patients with coronary artery disease (CAD).

METHODS: We prospectively investigated the agreement of platelet tests to isolate clopidogrel resistant in elderly CAD patients (age: 60–75 years) receiving 75 mg clopidogrel maintenance. After 7 days, platelet function was assessed with light transmission aggregation (LTA, clopidogrel resistance if $>43\%$), thrombelastography (TEG, clopidogrel resistance if $<30\%$) and Verify Now Point-of-care assay (VN, clopidogrel resistance if PRU >240).

RESULTS: Seventy consecutive patients were included. The rates of clopidogrel resistance were respectively: 50% (n = 35) with the LTA, 25% (n = 18) with the TEG and 32% (n = 23) with the VN. We observed significant correlation between different platelet tests assessing clopidogrel response: $r = 0.6$ ($P < .0001$) for LTA and TEG, $r = 0.693$ ($P < .0001$) for LTA and VN and $r = 0.564$ ($P < .0001$) for TEG and VN. However, using the most common thresholds, the agreement between the difference tests was poor: 0.29 for LTA and TEG, 0.51 for LTA and VN and 0.3 for TEG and VN.

CONCLUSION: This study showed that assessment of platelet function inhibition by clopidogrel is highly test-specific. Indeed, our results demonstrated a poor agreement between different platelet assays and suggested that identification of clopidogrel resistant is test-dependent.

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Postprandial Hypotension: Clinical Characteristics in the Very Old Chinese Male Patients

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OBJECTIVE: The aim of this study was to assess the characteristics of postprandial hypotension (PPH) in the very old Chinese male patients.

METHODS: A total of 349 male patients were recruited in this study. They were divided into two groups according to their different ages. Group 1 was the general old group consisting of 163 cases (aged between 65–80), and group 2 was the very old patients group of 186 cases (aged over 80). The changes of blood pressure after meal were assessed by ambulatory blood pressure monitoring every 15 minutes. The baseline clinical data were also recorded.

RESULTS: The prevalence of PPH was 59.3%, and the prevalence of PPH after breakfast (33.8%) and lunch (32.1%) were higher than that after supper (20.9%), $P < .05$. The prevalence of PPH in group 2 was significantly higher than that of group 1 (63.4% vs 54.6%), $P < .05$. Patients with hypertension had a higher incidence of PPH than those without hypertension (59.7% vs 23.1%). Hypertension and age were significant risk factors for PPH (OR = 2.188, 95% CI: 1.134–4.223, $P = .02$; OR = 1.86, 95% CI: 1.112–3.11, $P = .018$). On the contrary, acarbose was a protective factor for PPH (OR = 0.4, 95% CI: 0.189–0.847, $P = .017$). The characteristic of the actual decrease in blood pressure of PPH was 20–40 mmHg and the maximum was 90 mmHg. PPH usually occurred at 30–60 minutes after a meal and lasted 30–120 minutes.

CONCLUSIONS: The result of this study indicated that the prevalence of PPH in the very old Chinese male patients was significantly higher than those in the general male patient population. And the decline of blood pressure in the former group was much greater. This phenomenon was worth prevention and intervention.

P183

Lipid-lowering Efficacy and Safety of Policosanol on Cardiovascular Diseases in the Very Old Patients

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OBJECTIVE: The aim of the study was to comparatively observe policosanol's lipid-lowering efficacy and safety in the very old patients with cardiovascular diseases.

METHODS: Two hundred and ninety-four elderly patients with dyslipidemia were included. All patients were randomly divided into four groups: Policosanol 20 mg group (group A, n = 64); Policosanol 10 mg group (group B, n = 72); atorvastatin 20 mg group (group C, n = 91); policosanol 10 mg + atorvastatin 20 mg group (group D, n = 62). The following indexes were observed at 0 day, 12th week, 24th week and 52th week: TC, TG, LDL-C, HDL-C, Lp (a), ApoA1, ApoB, ALT, AST, CK, and Mb.

RESULTS: From 12th week, LDL-C and TC levels started to decrease in each group, and LDL-C and TC levels decreased more significantly with time. At 52th week, the changes of LDL-C and TC levels were most significant in group D among four groups. LDL-C and TC levels in group A were lower than those in group B at 52th week. There was no significant change of HDL-C level in each group after treatment. There was no significant change of TG level in group A and B after treatment. However, TG level in group C and D at 52th week was statistically lower than baseline but there was no significant difference of TG level between group C and D. Lp (a), apo B and apo B/AI were significantly lower than baseline in each group after treatment. Lp (a) level decreasing was most significant in group C and D. Apo B level lowering was most significant in group D. Decreasing of Apo B/AI in group C and D was more significant than in group A and B but there was no statistical change of apo B/AI in group A and B at 52th week. No severe liver damage and rhabdomyolysis occurred in policosanol group A and B during the whole study. Four patients suffered from liver damage in group C and three patients suffered from liver damage in group D but the liver function gradually resumed to baseline after discontinuing the medication. One patient presented significant gastrointestinal symptoms in group C and the symptoms also alleviated with atorvastatin discontinued.

CONCLUSIONS: Policosanol could effectively modulate blood lipid levels mainly by decreasing TC and LDL-C levels but it had little effect on HDL-C and TG levels. The effects were increased with increasing of dosage of policosanol and the combination with routine dose of atorvastatin could enhance the lipid-lowering effects. 10 mg, 20 mg policosanol and the combination of little dose of Policosanol with routine dose of atorvastatin were safe and well tolerated in the elderly patients.

P184

Efficacy and Safety of Rivaroxaban in the Prevention and Treatment for Thrombo-embolic Disease in the Very Old Patients

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OBJECTIVE: The aim of this study was to evaluate the effects and safety of rivaroxaban in the prevention and treatment of thromboembolic disease in the very old patients by comparison with warfarin.

METHODS: All enrolled patients were followed up for 12 months. During that period, the hemoglobin, platelets, coagulation, liver and kidney function were monitored at baseline,

1 month, 3 months, 6 months and 12 months respectively after treatment. The primary efficacy endpoints included all-cause mortality, recurrent or new venous thrombosis, pulmonary embolism, cerebrovascular and cardiovascular events. The primary safety endpoints included major and clinically relevant non-major bleeding, liver and kidney function worsening. In the first part, 82 venous thrombosis patients treated in our hospital from March 2012 to March 2014 were enrolled with an average age of 81.5 ± 5.6 years old. The enrolled patients were randomly divided into rivaroxaban group with 39 patients (30 males and 9 females) and warfarin group with 43 patients (34 males and 9 females). The rivaroxaban group was given low molecular weight heparin and oral rivaroxaban 10 mg, 1/d for the first 3 days, and then oral rivaroxaban 10 mg, 1/d was continued to be given alone. The warfarin group was given individual dose of warfarin and low molecular weight heparin therapy for the first several days until International Normalized Ratio (INR) got to 1.6 to 2.5 (the average INR 2.04 ± 0.42). Oral warfarin was continued to be given and INR was monitored to keep INR stabilize at 1.6 to 2.5. In the second part, 90 patients from March 2012 to March 2014 at high risk of venous thrombosis in our hospital were enrolled, with an average age of 82.4 ± 5.5 years old. All enrolled patients were also randomly divided into rivaroxaban group with 44 patients (34 males and 10 females) and warfarin group with 46 patients (38 males and 8 females). The rivaroxaban group was given oral rivaroxaban 10 mg, 1/d. The warfarin group was given titrated dose of warfarin and INR monitored constantly to keep INR stabilize at 1.6 to 2.5.

RESULTS: In the first part, thrombi all disappeared in the rivaroxaban and warfarin groups. There were no recurrent or new venous thrombosis, pulmonary embolism and deaths in the two groups. At 1st, 3rd, 6th and 12th month in warfarin group, the average INR was 1.98 ± 0.44 , 2.04 ± 0.53 , 2.03 ± 0.49 and 2.01 ± 0.46 respectively. During follow up, the differences of efficacy and safety were not statistically significant between the two groups. Hemoglobin, platelet, coagulation, liver and kidney function had no significant changes both in rivaroxaban group and warfarin group during follow up. In the second part, there were also no deaths among the patients at high risk of thrombosis. At 1st, 3rd, 6th and 12th month in warfarin group, the average INR was 2.02 ± 0.46 , 1.98 ± 0.54 , 2.03 ± 0.56 and 2.04 ± 0.46 . During follow up, the differences of efficacy and safety were not statistically significant between the two groups. Hemoglobin, platelet, coagulation, liver and kidney function had no significant changes both in rivaroxaban group and warfarin group during follow up.

CONCLUSIONS: Oral rivaroxaban appeared to be an effective way for the treatment and prevention for thrombo-embolic diseases in the very old patients. 10 mg daily of rivaroxaban was safe.

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The Multi Effects of Policosanol on Cardiovascular Diseases in the Very Old Patients

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OBJECTIVE: The aim of the study was to evaluate policosanol's multi effects beyond lipid lowering in the very old patients with cardiovascular diseases.

METHODS: A total of 294 elderly patients with dyslipidemia were included. All patients were randomly divided into four groups: Policosanol 20 mg group (group A, n = 64); Policosanol 10 mg group (group B, n = 72); atorvastatin 20 mg group (group C, n = 91); policosanol 10 mg + atorvastatin 20 mg group (group D, n = 62). Medications were taken orally before going to bed at night once daily in each group. The following indexes were observed at 0d, 12th week, 24th week and 52th week: platelet

counts, platelet aggregation rates, CECs, and carotid artery intima-media thickness.

RESULTS: ADP induced platelet aggregation rate was decreased significantly in group A than that of baseline ($48.79 \pm 20.29\%$ vs $40.37 \pm 23.56\%$) but AA induced platelet aggregation rate was not. Both AA and ADP induced platelet aggregation rates had no significant changes in other groups. CECs, hs-CRP and Hcy levels were all decreased significantly at 52th week after treatment in each group. There was no statistical change of IMT in each group after treatment.

CONCLUSIONS: Besides lipid modulating effects, policosanol had other effects such as improving endothelial functions and preventing progression of atherosclerosis.

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Mitochondrial tRNAThr Gene Mutation in Maternally Inherited Hypertension and the Regulatory Mechanism in Adiponectin Pathway

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BACKGROUND: Development of essential hypertension (EH) in the offspring has significant correlation with maternal EH. In addition, mtDNA mutations are related with elevated blood pressure, but the underlying mechanism is unclear. Based on a study with small sample size and two pedigrees maternal family members, we tried to investigate potential mechanisms of how EH characterized with maternal inheritance was caused by mitochondrial mutations.

METHODS: A total of 115 pedigrees were enrolled in this study. The subjects answered questionnaires and received full mitochondrial genome sequencing analysis. According to the incidence of hypertension in three-generations of Chinese family, they were divided into maternally-inherited EH pedigrees (group A, n = 17), non-maternally-inherited EH pedigrees (group B, n = 65), and normal control pedigrees (group C, n = 33). In order to clarify the relationship between mtDNA C15910T and maternally inherited EH, all maternal family members' whole mitochondrial genome carrying mtDNA C15910T mutation from Zhuozhou and Inner Mongolia pedigrees were analyzed, and their blood biochemical indices were compared. To further reveal the mechanism through which mtDNA C15910T caused maternally inherited EH, peripheral blood mononuclear cells (PBMCs) from Zhuozhou pedigrees were cultured in vitro. They were divided into four groups, hypertension matrilineal (n = 6, EH+Mu), non-hypertension matrilineal (n = 6, Mu), hypertension without mutation (n = 6, EH), and normal controls (n = 6, con).

RESULTS: The results indicated that mitochondrial tRNA gene mutation was significantly correlated with EH. Mitochondrial tRNA gene mutation occurred more frequently in patients with EH. It was found that the overall incidence rate of EH was 59.3% in Zhuozhou and Inner Mongolia pedigrees maternal family members, and 90% in males, was significantly higher than in the general population in China (33.5%). The age at onset of EH in members carrying mtDNA C15910T was earlier than in gender- and age-matched groups. The mtDNA C15910T, which was extraordinarily conserved from mollusc to human mitochondria, was located at D-loop of tRNA^{Thr}. The serum sodium and chloride levels were increased in members carrying mtDNA C15910T, while more significantly in maternal members accompanied by EH. In vitro, we

found that PBMCs carrying mtDNA C15910T were characterized by more viability and proliferation. The increased ATP production resulted in raised intracellular ROS. The mitochondrial dysfunction resulted in reduced APN levels secreted from adipose tissue, causing hypoadiponectinemia. Hypoadiponectinemia further promoted cell proliferation, which in turn produced more ROS. This vicious cycle promoted the occurrence of EH with maternally inherited mtDNA C15910T. The APN, AdipoR1, and PGC-1 α were reduced, but ERR α significantly upregulated in C15910T mutation group.

CONCLUSIONS: mtDNA C15910T induced maternally inherited EH, either directly, and/or by elevating serum sodium and chloride levels. APN pathways may play important roles in the pathogenesis of EH. Other potential mechanisms could include changing APN, AdipoR1, PGC-1 α or APN, ERR α signal pathway leading to high blood pressure.

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Association among Sex Hormone Levels, Abnormal Metabolisms and Vascular Diseases in Elderly Chinese Men

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OBJECTIVE: The aim of this study was to investigate the metabolic and gonadal risk factors of vascular diseases in elderly Chinese males.

METHODS: After screening, 337 subjects aged 60–90 were selected to be enrolled into this single-center, cross-sectional study, and their sex hormone levels and metabolic parameters were assessed. The nonparametric K-sample test was used for inter-group comparisons. Linear and logistic regression analyses were utilized to compare the sex hormone levels between patients with and without abnormal metabolisms, vascular diseases. Odds ratios (OR) in cross-table analysis and exp (B) in logistic regression analysis were used to evaluate the vascular risks of dependent factors. R² of logistic regression equation was used to estimate the goodness-of-fit of vascular diseases logistic regression models. Receiver Operating Characteristic (ROC) analysis was used to find tangent points.

RESULTS: 1. Obesity and overweight, hyperglycemia and dyslipidemia were the most important abnormal metabolism that was related to decreased androgen indices. Including SHBG in the stepwise regression was increased about 30% of the explanation effect, and all metabolic indices were excluded. 2. Hypertension increased the risk of cardiovascular diseases (CAVD) in elderly men for approximately 3-fold. The number of metabolic diseases was also correlated with incremental risks of CAVD; presence of one abnormality approximately increased the risk approximately 62%. But no significant relationship could be found between sex hormones and CAVD. 3. Cerebrovascular disease (CEVD) development was closely associated with both metabolic syndrome and sex hormone levels, and their explanation effects of single action and combined action were 13.2%, 12.55 and 28.5%. C-peptide might be the underlying mechanism of the metabolic syndrome's effect on CEVD. In regression with all metabolic continuous variables included, the testosterone/estradiol (T/E2) ratio had significant relationships with CEVD risk ($P < .05$). C-peptide = 2.43 U/L and FE2 = 0.66 were the tangent points in ROC analyses.

CONCLUSIONS: Age and abnormal metabolism were independently important factors associated with the sex hormone levels in elderly Chinese men, which were all mediated by SHBG. Metabolic diseases and sex hormones play different roles in the development of CAVD and CEVD.

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Regulatory Effect of Anxa1 on Proliferation of MIN6 Cells

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BACKGROUND/AIM: Anxa1 (Annexin 1, also known as lipocortin-1), is a member of calcium-dependent phospholipid-binding proteins with diverse functions. Previous studies suggest that anxa1 protein is involved in change of cell proliferation in different tissues and anxa1 protein increased in rat's islets with aging. However, it is unclear that how Anxa1 affects proliferation of pancreatic beta cells. The aim of this study was to investigate regulatory effect of anxa1 on proliferation of MIN6 cells.

METHODS: In this study, after upregulation of Anxa1 expression by plasmid transfection and downregulation of Anxa1 expression by small interfering RNA (siAnxa1) in MIN6 beta cell line, CCK8 assay was used to determine the degree of cell proliferation and cell cycle analysis was performed by using flow cytometry (FCM). Subsequently, the expression of cell cycle-related proteins, such as Cyclin D1, Cyclin E and CDK2 proteins were also detected by western blot. The relationship between the PI3K/Akt/mTOR signaling pathway and Anxa1 was also evaluated.

RESULTS: Overexpression of Anxa1 in MIN6 cells increased the extent of cell proliferation. Knockdown of Anxa1 expression using siRNA decreased the degree of cell proliferation. Overexpression of Anxa1 in MIN6 cells induced a significant increase in the percentage of cells in the S and G (2)/M phase, as demonstrated by flow cytometry. In contrast, the percentage of cells in the S and G (2)/M phase was decreased in cells when Anxa1 was inhibited, compared with that of control cells. Expression of Anxa1 protein in MIN6 cells was positively related with Cyclin D1, Cyclin E and CDK2 proteins and positively regulated the PI3K/Akt/mTOR signaling pathway.

CONCLUSIONS: This study observed a positive correlation between Anxa1 protein and proliferation of MIN6 cells. Anxa1 protein might affect proliferation of MIN6 cells by regulation of Cyclin D1, Cyclin E and CDK2 protein and the PI3K/Akt/mTOR signaling pathway.

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Relationship between Pulse Pressure and Atherosclerotic Renal Artery Stenosis in Middle-aged and Elderly Patients with Hypertension

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OBJECTIVE: The aim of the present study was to analyze the relationship between the pulse pressure (PP) and atherosclerotic renal artery stenosis (ARAS) in middle-aged and elderly patients with hypertension, and to evaluate the clinical significance of PP.

METHODS: We retrospectively analyzed the data of 257 patients with hypertension undergoing renal arteriography at General Hospital of PLA between 2009 and 2014. Their mean age was 64.09 ± 7.81 years (42–85 years). According the results of arteriography, they were divided into two groups: the one with ARAS (n = 145), and the other with non-ARAS (n = 112). Their PP and multinomial clinical indexes with ARAS were statistically analyzed.

RESULTS: There was no statistical difference between the two groups in PP and other clinical indexes ($P > .05$). Single factor correlation and Logistic analysis showed that age and PP and plasma creatinine levels (Cr) and new onset of hypertension after 50 years of age were strongly positively related to ARAS. The age and PP and Cr were a team of continuous variable. Especially, PP was an independent factor.

CONCLUSION: PP was an independent risk factor of ARAS in middle-aged and elderly patients with hypertension. PP could serve as an early predictive indicator and a new therapeutic target for ARAS.

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Study on Risk Scoring System of Atherosclerotic Renal Artery Stenosis

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OBJECTIVE: The aim of this study was to set up a risk scoring system of Atherosclerotic Renal Artery Stenosis (ARAS) in atherosclerotic patients.

METHODS: Based on the previous research of prediction factors with ARAS, the regression coefficients were used to set up the prediction rule. Further, the prediction rule was transferred to risk scoring system and put back to test its sensitivity and specificity by ROC in 260 patients with renal artery angiography.

RESULTS: The risk factors for ARAS include age, serum creatinine, pulse pressure and newly onset of hypertension after 50 years old. Age, pulse pressure and serum creatinine were used as continuous variables and newly onset of hypertension as dichotomized variable. According to our prediction system, the score of 260 patients was from 7 to 15.5. With the score increasing, the probability of ARAS increased too. The sensitivity was 73.9% and specificity was 72.1%.

CONCLUSION: Our study established a prediction model that may help to identify ARAS patients or high risk for developing ARAS from the general atherosclerotic population.

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Association between Physical Inactivity and Mortality in Octogenarian Patients with Acute Coronary Syndrome

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BACKGROUND: Whether physical inactivity is responsible for all-cause mortality increase in octogenarian patients with acute coronary syndrome (ACS) remains unclear. In this study, we aimed to investigate association between physical inactivity and long-term all-cause mortality in octogenarian ACS patients.

METHODS: In this study, we included a total of 353 hospitalized patients, aged ≥80 years, with ACS during the period of 5-year follow-up. Association between physical inactivity and all-cause mortality was analysed by multivariable Cox aggression.

RESULTS: Patients with physical inactivity tend to have lower survival rates (21.2% vs 56.5%, $P < .001$) and higher mortality rates (78.8% vs 43.5%, $P < .001$), and patients with physical inactivity had a worse long-term outcome as compared with those with physical activity (chi-square 27.52, log rank $P < .001$). Physical inactivity still was an independent predictor for long-term all-cause mortality independent of confounders including age, prior heart failure, stroke, ejection fraction, beta blocker, clopidogrel and PCI (HR: 2.35, 95% CI: 1.26–4.37, $P = .007$).

CONCLUSIONS: Our study demonstrates that physical inactivity, an independent risk factor, is closely related to the long-term all-cause mortality in octogenarian ACS patients, and enhancing physical activity is equally important for these population.

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The Effect of Overexpression of Sarco/endoplasmic Reticulum Ca^{2+} -ATPase (SERCA2a) on Endoplasmic Reticulum Stress-induced NF- κ B Activation in Neonatal Rat CardiomyocytesZ. G. Qu,^{1,2,*} X. C. Lu,^{1,*} X. Y. Li,¹ X. H. Liu,³ F. F. Xu,³ Y. Qu⁴.

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OBJECTIVE: Overexpression of SERCA2a attenuates unfolded protein response (UPR), and decreases the tissue and systemic inflammation markers in heart failure animal models. This study aimed to investigate the effect of overexpression of SERCA2a on endoplasmic reticulum stress (ERS) induced NF- κ B activation in neonatal rat cardiomyocytes.

METHODS: Neonatal SD rat cardiomyocytes were infected with adenoviral vectors carrying SERCA2a (MOI = 2 pfu/cell) or Enhanced Green Fluorescent Protein (EGFP) gene (as control). 48 hours after the gene transfer, the cardiomyocytes were treated with tunicamycin (TM, 10 μ g/ml) to induce ERS for 24 hours.

RESULTS: The results showed that the overexpression of SERCA2a mitigated TM induced elevations of cytosolic Ca^{2+} levels ($P < .01$), and decreased the intracellular reactive oxygen species (ROS) productions ($P < .05$), compared with the TM group. Compared with the TM+Ad-EGFP group, the overexpression of SERCA2a decreased the protein levels of phosphorylated inositol-requiring kinase 1 (Phospho-IRE1) ($P < .01$), C/EBP-homologous protein (CHOP) ($P > .05$), cleaved Caspase 12 ($P < .01$), and cleaved Caspase 3 ($P < .05$), along with the mRNA transcripts for spliced X-box binding protein 1 (XBP1 (s)) ($P < .05$). Compared with the TM group, the overexpression of SERCA2a decreased NF- κ B DNA binding activity ($P < .01$), the mRNA transcripts for IL-1 β ($P < .05$), and IL-1 β concentration in the culture medium ($P < .05$) significantly. However, the overexpression of SERCA2a induced more NF- κ B p65 nuclear translocation, which indicated the occurrence of endoplasmic reticulum overload response (EOR). The protein levels of TNF receptor-associated factor 2 (TRAF2) before treatment with TM were decreased significantly by the overexpression of SERCA2a.

CONCLUSIONS: Overexpression of SERCA2a attenuated ER stress-induced upregulation of NF- κ B DNA binding activity and decreased IL-1 β production in neonatal rat cardiomyocytes, through calcium homeostasis maintenance, less intracellular ROS production and attenuation of the IRE1 signaling pathway activation.

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Clinical Application of a New Miniaturized Echocardiographic Scanner in Elderly Patients with Cardiovascular DiseasesL. Zhang,¹ G. Zhi,² W. Zhang,² H. P. Hui,¹ Q. Chen,¹ H. B. Liu¹.

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BACKGROUND: Modern ultrasonic miniaturization has evolved in the creation of a pocket-sized echocardiographic scanner which can provide black-white anatomic and color-flow images in real time. The aim of this study was to assess the clinical application of a new miniaturized echocardiographic scanner (MINI) for cardiovascular diseases in comparison with standard echo machine (STANDARD) in elderly patients.

METHODS: One hundred and sixty consecutive elderly patients (age ≥ 65) with cardiovascular diseases underwent a sequential echocardiographic assessment by MINI (by Vscan, GE) and STANDARD (Sequoia512, Siemens) exam by 2 echocardiographic exports, and the scanning time was recorded. Segmental left ventricle (LV) endocardial border delineation was scored (4 = great, 3 = good, 2 = medium, 1 = poor) to describe image quality. Atrial and ventricle dimensions, left ventricle wall thickness, regional and global LV function, grades of valve disease, and pericardial effusion were compared between MINI and STANDARD exam to assess the diagnostic accuracy.

RESULTS: There were 310 abnormalities found by MINI exam in all one hundred and sixty consecutive cardiologic patients and the average scanning time was 176 ± 31 sec. The mean image quality score was 2.3 ± 1.0 with MINI and 2.5 ± 1.0 with STANDARD exam ($P > .05$). Moderate and severe heart enlargement, LV systolic dysfunction, valve regurgitation and valve stenosis, pericardial effusion was not missed by MINI exam. In the study, 38 cases were diagnosed as regional wall motion abnormality by MINI exam and the diagnostic sensitivity and specificity of myocardial infarction by MINI exam was 92% and 98%. The consistency of MINI and STANDARD exam to assess LV global ejection fraction was good (Kappa = 0.75, $P < .001$).

CONCLUSIONS: The new miniaturized echocardiographic scanner demonstrates the following characteristics: being small and light, shortening scanning time, having high image quality and diagnostic accuracy. It is feasible in routine clinical practice, especially where doctors require a fast and simple exam to know the basic information of cardiac structure and function, such as at outpatient clinics, in conduit room, in coronary care unit, and other setting.

P194

Xuezhikang Reduces Arterial Stiffness in Patients with Essential Hypertension, Independent of Lipid Lowering Effects

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OBJECTIVE: Increased arterial stiffness occurs early in hypertension and is considered to play an important role in cardiovascular events. Xuezhikang, an extract of Cholestin, contains statin-like components. It is hypothesized that treatment with Xuezhikang can reduce arterial stiffness in patients with essential hypertension.

METHODS: One hundred essential hypertension patients were prospectively enrolled to Xuezhikang group (1,200 mg/day) and placebo group. Physical examination, lipid profiles, high sensitivity C reactive protein (hs-CRP), matrix metalloproteinases-9 (MMP-9), arterial stiffness parameters, including β , Ep, AC, AI and PWV β that were measured by Echo Tracking technique were obtained at baseline and the patients were given 6 months' therapy in all subjects.

RESULTS: Ninety cases with completed data were included at the end of the study. The Xuezhikang group (n = 46; age, 58.4 ± 10.0 years) and placebo group (n = 44; age, 57.1 ± 10.4 years) were well balanced with respect to clinical characteristics, use of medications, lipid profiles and arterial stiffness parameters. Xuezhikang therapy significantly reduced the levels of β , Ep and PWV β from baseline (8.4 ± 3.1 vs 6.8 ± 2.1 , $P = .007$; 122.8 ± 43.9 vs 100.7 ± 33.2 , $P = .009$; 6.7 ± 1.2 vs 6.1 ± 1.0 , $P = .013$), but placebo group didn't. There were significant reductions in total cholesterol by and LDL cholesterol in Xuezhikang group (5.5 ± 0.7 vs 5.1 ± 0.6 , $P = .013$; 3.4 ± 0.6 vs 2.9 ± 0.5 , $P = .001$), but not in placebo group. Xuezhikang therapy also significantly reduced hs-CRP and MMP-9 from baseline [2.1 (0.4-10.0) vs 1.4 (0.3-4.1), $P = .020$; 17.2 ± 2.4 vs 12.7 ± 3.8 , $P < .001$], but placebo group didn't. Of interest, the percent change of PWV β in

Xuezhikang group was significantly correlated with the percent change of hs-CRP or MMP-9 ($r = 0.144$, $P = .043$; $r = 0.278$, $P = .030$ respectively), but not with lipid profiles.

CONCLUSIONS: Six-month treatment with Xuezhikang reduced arterial stiffness, independent of lipid lowering effects in hypertensive patients. Xuezhikang appeared to have some statins' pleiotropic effects, such as anti-inflammatory and protective against extracellular matrix degeneration that might play a role in improving arterial stiffness in patients with essential hypertension.

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Assessment of Risk Factors for Cognitive Impairment

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BACKGROUND: Evidence has demonstrated that many risk factors contribute to cognitive impairment in the elderly individuals. However, it still needs to identify the risk factors and potential protective factors of cognitive impairment.

OBJECTIVE: This study aimed to evaluate the risk factors and potential protective factors for cognitive impairment in elderly individuals living in the community in Shanghai city.

METHODS: We included 160 patients with cognitive impairment and 520 normal elders who were all from database of 1,068 Shanghai community-dwelling subjects in China's national study on the evaluation, early recognition, and treatment of psychological problems in the elderly, supported by the Ministry of Science and Technology, National Pillar Program, between January 2010 and December 2013. All subjects were screened for general condition, living style, mood and sleep quality, past medical history, and family history. Binary logistic regression was used to investigate the risk factors and potential protective factors of cognitive impairment among age, gender, educational level, professional nature, living place, living style, mood, sleep quality, dieting, past medical history such as hypertension, heart disease and other physical diseases, and family history such as dementia and depression.

RESULTS: Cognitive impairment onset was affected by patients who were higher in age, doing physical labor, lack of exercises, and who were suffering from hypertension or other physical diseases. Living in rural areas, with non-vegetarian, having too much or too little sleep, diabetes increased marginally significant for cognitive impairment. High schooling and involvement in physical activities could lower the risk of cognitive impairment. Besides, we also found that patients with history of surgery had lower risk of cognitive impairment.

CONCLUSIONS: Age, professional nature and physical activity were the most important factors to influence cognitive function in this sample.

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Comparative Study on Cognitive Function and Neuroimage between Amnesic and Vascular Mild Cognitive Impairment

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OBJECTIVE: In order to assess if the cognitive presentation and grey and white matter change differ between amnesic and vascular MCI, two populations of MCI underwent Neuropsychological Test Battery and voxel-based morphometry.

METHODS: All participants were from the database of 1,068 Shanghai community-dwelling subjects in China's national study

on the evaluation, early recognition, and treatment of psychological problems in the elderly, supported by the Ministry of Science and Technology, National Pillar Program, between January 2010 and December 2013. In Part one, 44 patients with aMCI, 44 patients with vMCI and 44 NC subjects were selected and assessed by the Neuropsychological Test Battery. The Neuropsychological Test Battery comprises the following cognitive tests of various cognitive domains: the Digit Span (forward and backward span length), Auditory Verbal Learning Test, Associative Learning Test, Visual Recognition Function Test, Verbal Fluency Test, Wechsler's Mapping, and Wechsler's Block Diagram. In Part two, 35 patients with aMCI, 10 patients with vMCI and 35 NC subjects underwent 3D magnetic resonance imaging. GM atrophy and WM damage and correlations between cognitive measures and VBM were performed by using optimized voxel-based morphometry.

RESULTS: In Part one, all of cognitive tests in various cognitive domains in the Neuropsychological Test Battery showed significant difference between patients and normal control subjects ($P < .05$). However, there was not any significant difference between patients with aMCI and vMCI assessed by the Neuropsychological Test Battery ($P > .05$). Wilks' Lambda discriminant analysis showed that cognitive measures of MMSE, MoCA, NTB, CDR, HIS and ADL achieved a classification accuracy of 87.7% for distinguishing aMCI (81.1%), vMCI (90.0%) and NC (92.3%) (Wilks' Lambda = 0.549, $\chi^2 = 51.643$, $P = .027$). In Part two, compared with control subjects, patients with aMCI exhibited significant grey matter volume reductions mainly in bilateral temporal lobe and limbic system (left > right). In addition, white matter volume reductions were found predominately in left temporal lobe and hippocampus. Conversely, patients with vMCI had grey matter volume reductions in right frontal lobe, right basal ganglia, and bilateral limbic system (left > right). White matter damaged in bilateral frontal, temporal, occipital, limbic and right parietal lobe and right corpus callosum ($P < .001$, voxel size > 100 mm³). Patients with aMCI had a larger grey matter volume loss significantly in bilateral limbic lobe, left frontal and temporal lobe, right parietal and occipital lobe compared with patients with vMCI. Patients with vMCI had worse white matter volume loss significantly in the left frontal, limbic lobe and basal ganglia compared with patients with aMCI ($P < .01$, voxel size > 100 mm³). Both the MMSE and the MoCA showed a similar relationship with grey and white matter degeneration in patients with aMCI, reflecting greater cognitive impairment with decreased grey matter in the left temporal lobe and hippocampus and with white matter lesion in frontal, temporal, occipital, limbic lobe and cerebellum. Both the MMSE and the MoCA showed a similar relationship with grey matter degeneration in patients with vMCI, reflecting greater cognitive impairment with decreased grey matter in the left thalamus and hippocampus. For patients with vMCI, white matter loss of the right thalamus, sub-lobar and left corpus callosum was correlated with the MMSE, and the bilateral limbic lobe, cerebellum, and the right thalamus and the right temporal lobe was correlated with the MoCA ($P < .001$, voxel size > 1000 mm³).

CONCLUSION: These findings suggest different patterns of cognitive change in healthy older adults and MCI patients.

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Changes and the Significance of Th17/Treg Immune Imbalance in Secondary Systemic Infection in Patients with Severe Acute Pancreatitis

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OBJECTIVE: Th17/Treg immune balance is beneficial to stable state of immune system. We investigated the changes and

significance of Th17/Treg immune imbalance in secondary systemic infection in patients with severe acute pancreatitis.

METHODS: Twenty-one patients with severe acute pancreatitis and secondary systemic infection (infection group), and 25 patients with severe acute pancreatitis alone (non-infection group) were selected, and 20 healthy cases underwent annual health check-up (control group) in this study. The expression levels of Th17/Treg cells and related cytokines were compared between the groups.

RESULTS: There were significant differences in mortality rate and duration of ICU stay between infection group and non-infection group [23.8% vs 4.0%, (11.3 ± 3.4) d vs (7.5 ± 2.8) d, $\chi^2 = 3.949$, $t = 2.890$, $P = .047$ and $.045$]. The percentages of Th17 cell and Treg cell, Th17/Treg ratio, mRNA expressions of IL-6, IL-17, IL-23, TGF- β and orphan receptor - γ t were higher in infection and non-infection groups than those in control group [(26.4 ± 1.2)%, (12.8 ± 0.9)% vs (3.1 ± 0.8) %; (6.7 ± 1.6)%, (4.2 ± 1.3)% vs (1.3 ± 0.4)%; (4.3 ± 1.0)%, (3.2 ± 1.1)% vs (2.4 ± 0.9)%; (7.1 ± 0.8) ng/L, (5.3 ± 0.7) ng/L vs (0.2 ± 0.1) ng/L; (22.9 ± 2.4) ng/L, (15.6 ± 2.8) ng/L vs (10.3 ± 1.5) ng/L; (15.7 ± 2.1) ng/L, (10.2 ± 1.5) ng/L vs (8.3 ± 1.4) ng/L; (23.6 ± 2.2) ng/L, (16.3 ± 1.7) ng/L vs (11.6 ± 1.1) ng/L; (0.052 ± 0.014), (0.035 ± 0.010) vs (0.004 ± 0.001); $F = 15.761$, 55.745, 9.437, 102.788, 21.038, 16.239, 36.957, 23.924, respectively, $P = .555$, .000, .014, .000, .002, .004, .000, .000]. The mRNA expressions of IL-10 and Foxp3-T were lower in infection and non-infection groups than those in control group [(6.4 ± 1.1) ng/L, (10.5 ± 2.1) ng/L vs (15.4 ± 2.0) ng/L; (0.005 ± 0.001), (0.020 ± 0.007) vs (0.032 ± 0.009), $F = 18.995$ and 20.608, $P = .003$ and $.002$].

CONCLUSIONS: Secondary infection could aggravate the Th17/Treg immune imbalance in patients with severe acute pancreatitis and extend the ICU length of stay.

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C1q/ Tumor Necrosis Factor-related Protein 9 Decreases Pro-inflammatory Cytokines Expression in Macrophages by Inhibiting Nuclear Factor- κ B Translocation

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BACKGROUND: Chronic inflammation is involved in atherosclerosis, and macrophage cells play important role in this process. The aim of the study was to investigate the anti-inflammatory effect of C1q/ tumor necrosis factor (TNF) -related protein 9 (CTRP9) in RAW264.7 mouse macrophage cells treated with oxidized low density lipoprotein (oxLDL), and to explore its mechanism.

METHODS: RAW264.7 mouse macrophage cells were divided into three groups: the control group, the oxLDL group (treated with oxLDL) and the gCTRP9 + oxLDL group (pretreated with recombinant globular domain of CTRP9 and stimulated by oxLDL). Foam cells were detected by oil red O staining. Western blot was used to detect the expressions of pro-inflammatory cytokines tumor necrosis factor- α (TNF- α) and monocyte chemoattractant protein 1 (MCP-1). In addition, the expression levels of NF- κ B p65 in cytoplasm and nucleus proteins extraction were both determined.

RESULTS: Relative levels of MCP-1 and NF- κ B were increased in the oxLDL group as compared with those of the control group (1.66 ± 0.09 vs 1.03 ± 0.10, 0.52 ± 0.11 vs 1.03 ± 0.06, both $P < .05$). The expression levels of TNF- α and MCP-1 were decreased in gCTRP9 + oxLDL group as compared with those of the oxLDL group (both $P < .05$). The expression level of NF- κ B p65 in nucleus 2 and 8 h after treatment was lower in the

gCTRP9 + oxLDL group than that of the oxLDL group (1.03 ± 0.06 vs 0.17 ± 0.10, 0.31 ± 0.03, both $P < .05$).

CONCLUSIONS: oxLDL induced the expressions of inflammatory cytokines of TNF- α and MCP-1 in macrophages. gCTRP9 pretreatment reduced oxLDL-induced pro-inflammatory effect. Nuclear factor- κ B translocation may be involved in this process, suggesting that gCTRP9 may play a protective role in anti-inflammatory and anti-atherosclerosis.

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Value of Methionine Loading Test in Mild Vascular Cognitive Impairment

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BACKGROUND: Homocystine (Hcy) is an independent risk factor for ischemic vascular disease. Methionine loading test (MLT) is a required method to find out latent higher Hcy. This study aimed to investigate the value of MLT in the mild vascular cognitive impairment (VCI) after acute cerebral infarction.

METHODS: Fasting plasma homocystine (Hcy) level and homocystine level after MLT were measured by high-performance liquid chromatography methods. We chose 240 patients with normal level of fasting plasma Hcy (normal group), 159 patients with normal level of Hcy after MLT, 81 patients with hyperhomocysteinemia after MLT (hyperhomocysteinemia group), and 112 patients with fasting hyperhomocysteinemia (fasting hyperhomocysteinemia group) in this study. The Montreal Cognitive Assessment (MoCA) and Mini-Mental State Examination (MMSE) were conducted in normal, hyperhomocysteinemia and fasting hyperhomocysteinemia groups on admission, at 7, 14, 30 days after treatment.

RESULTS: Logistic regression analysis showed that the increased level of Hcy might be an independent risk factor for VCI [OR: 1.285, 95% CI: 1.038-1.265, $P < .05$]. The scores of MMSE and MoCA were lower in patients with fasting hyperhomocysteinemia and patients with hyperhomocysteinemia after MLT than in patients with normal fasting plasma Hcy at 7, 14 and 30 days after treatment ($P < .01$ or $.05$), while the scores had no significant differences among the three groups on admission ($P > .05$). There were no significant differences in MMSE and MoCA scores between patients with fasting hyperhomocysteinemia and patients with hyperhomocysteinemia after MLT on admission, 7, 14 and 30 days after treatment ($P > .05$).

CONCLUSIONS: Hcy may be an independent risk factor for VCI. MLT may help identify such dormant vascular risk factor for VCI, which offers a valuable detection method for early intervention and prevention in the clinical medicine.

P200

Comparison of Clinical and Pathological Features of Idiopathic Membranous Nephropathy in Elderly and Non-elderly Patients

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BACKGROUND: Idiopathic membranous nephropathy is one of the most common types of Nephrotic syndrome. In this study, we explored the differences of clinical and pathological characteristics of renal biopsy in elder and non-elder patients with idiopathic membranous nephropathy.

METHODS: The clinical and pathological data of idiopathic membranous nephropathy diagnosed by renal biopsy in our hospital in the past recent 10 years were retrospectively analyzed and compared.

RESULTS: The proportion of elderly idiopathic membranous nephropathy patients accounted for 26.29% and less than 45 years of age accounted for 25.26%. The proportion of them is similar. The occurrence of hypertension, levels of blood serum creatinine, microscopic hematuria and urinary N-acetyl- β -D-glucosaminidase were higher, while urinary osmotic pressure, glomerular filtration rate and blood albumin were lower in elderly group than in non-elderly group (all $P < .05$). The proportion of glomerulosclerosis, tubulointerstitial lesions and vascular lesions were higher in elderly group. The positive rate of C1q of the elderly group was lower than that of the non-elderly, while IgG4 positive percentage and intensity of elderly group were higher than that of non-elderly (all $P < .05$).

CONCLUSIONS: Proportion of non-elderly patients with idiopathic membranous nephropathy is increasing. Elderly patients are more susceptible to hypertension, hematuria and renal function lesion.

P201

Clinicopathological Features and Treatment in Male Breast Cancer

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J. F. Ma and Y. Qiao contributed equally to this work as co-first authors.

BACKGROUND: As increasing prevalence and minimal awareness of male breast cancer, we investigated the clinicopathological features, diagnosis and treatment experience, in order to improve the recognition of male breast cancer and to prepare for the study of standardized treatment of this condition.

METHODS: Data of epidemiological characteristics, clinical and pathological parameters, treatment and outcome from 43 male breast cancer (MBC) patients were collected and analyzed at Peking University Cancer Hospital and Beijing Hospital from January 2009 to December 2014.

RESULTS: Of these patients, 42 (97.6%) cases presented with located tumors, 11 (24.5%) cases were on pathological stage III or stage IV, and 40 (93%) cases were with hormone receptor positive breast cancer and 3 (6.9%) cases were with human epidermal growth factor receptors (HER)-2 positive breast cancer. Patients were stratified according to more or less than 70 years of age, and there were no significant differences in clinicopathological features between the two year groups. But 3 cases with HER2 over-expressed were all less than 70 years old. 38 (88.4%) patients underwent surgical treatment, among which 32 HR-positive patients (74.4%) received tamoxifen as adjuvant endocrine therapy. The median follow-up periods was 31 months (6.1–55.4 months), and 7 (16.3%) patients developed local recurrence or distant metastasis.

CONCLUSIONS: Male breast cancer was often diagnosed at a later stage with inferior outcome. The majority of male breast cancer was found to be HR positive, hence hormonal therapy could be effective.

P202

Analysis of Highly Cited Papers Published in Chinese Core Journals of Neurological Diseases from 2005-2013

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BACKGROUND: Neurological conditions account for a significant proportion of elderly diseases. In this study, we analyzed highly cited papers published in Chinese core journals of neurological diseases to provide proper research direction and important

theoretical guidance of thesis writing and to provide better clinical services.

METHODS: The highly cited papers (cited by the top 20 papers) published in 6 important journals of neurological diseases since 2005–2013 were retrieved by Chinese citation database and ranked by cited frequency. The distribution of column and time, original articles and their content features were analyzed.

RESULTS: Among 6 journals of neurological diseases, the rate of citation frequency ≥ 5 was the most in Chinese Journal of Neurosurgery (23.9%); zero citation paper was the least in Chinese Journal of Neuromedicine (14.5%). The cited frequency was largely different among the journals as well as the same journal. The highly cited papers were mainly published in 2005–2008. Original articles accounted for the largest proportion in 4 journals, followed by review, while guideline, editorial or expert forum accounted for some proportion in particular journal. Of the highly cited papers, clinical papers were more than basic researches, and most of clinical papers were cross-sectional studies. Experimental subject of most basic study was animal, and single center study was a bigger contribution to the highly cited paper. The research contents majored in common and critical diseases such as prevalence mechanism and diagnosis or treatment of cognition disorders, cerebrovascular diseases, neurodegenerative diseases and mental diseases. The content was different as for particular emphasis on each journal.

CONCLUSIONS: The influence of journal is largely dependent on a few highly cited papers. More attention should be paid to hot issues, the topic of solving practical problems in clinical practice, development of guideline and consensus, prospective multicenter study cooperation in attempt to the promotion of neurological study quality.

P203

Effects of Suppressive Therapy of Thyroid-stimulating Hormone on Emotion Status in Elderly Patients

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BACKGROUND: Thyroid function is closely related to cognition and emotion. This study aimed to evaluate effects of suppressive therapy of thyroid-stimulating hormone (TSH) on emotion status in elderly patients undergoing resection of differentiated thyroid cancer (DTC).

METHODS: Serum levels of free T_3 (FT3), free T_4 (FT4) and TSH were measured in elderly patients. 165 patients were enrolled, in whom 74 cases with DTC underwent complete resection and TSH suppressive therapy [completely (CSG, $n = 33$) and partially (PSG, $n = 41$)], 43 cases with nodular goiter without TSH suppressive therapy (WSG) underwent bilateral subtotal thyroidectomy and 48 cases had normal thyroid function (normal control group, NCG). Hamilton Anxiety Rating Scale (HAMA) and Hamilton Depression Rating Scale (HAMD) were used to evaluate patients' emotion status.

RESULTS: There were significant differences in scores of HAMA between CSG and other three groups [PSG ($P < .05$), WSG ($P < .01$) and NCG ($P < .01$)]. However, there was no significant difference in HAMD scores among the four groups. Serum levels of TSH and FT4 were associated with the total scores of HAMA in CSG (both $P < .05$).

CONCLUSIONS: The elderly patients with suppressive therapy had more severe anxiety symptoms than those in the other groups, and their anxiety symptoms were correlated with thyroid dysfunction.

P204**The Effect of Cognitive Intervention on Language Function and Quality of Life in Elderly Patients with Post-stroke Aphasia**

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BACKGROUND: Post-stroke aphasia is usually accompanied by cognitive disorder, which affects overall rehabilitation. We investigated the effect of cognitive intervention on language function and quality of life (QOL) in elderly patients with post-stroke aphasia.

METHODS: Fifty-five elderly patients (aged ≥ 60 years) with post-stroke aphasia were randomly divided into the control group ($n = 27$) and the experimental group ($n = 28$). The control group received comprehensive treatment including speech-language therapy (SLT), while the experimental group was subjected to additional cognitive intervention. All patients were enrolled in an eight-week rehabilitation program. The mini-mental state examination (MMSE) was used to assess cognitive function, and a device for diagnosis and treatment of language disorders, ZM2.1, was used to evaluate language function, and the 36-item short-form health survey (SF-36) (Chinese version) was used to assess quality of life (QOL), before and after treatment.

RESULTS: After treatment, MMSE scores, language function scores in all categories and QOL scores in all dimensions were improved in both groups ($P < .05$ for both). Furthermore, after treatment, there were significant differences between the two groups in MMSE scores of orientation, recall, attention and calculation, and in scores of overall language ability and the individual categories ($P < .05$ for all). Scores of the experimental group vs the control group for the individual categories of language function were as follows: simple instructions (76.6 ± 14.1 vs 67.4 ± 19.3), complex instructions (66.1 ± 12.8 vs 58.2 ± 14.9), yes or no (72.5 ± 12.1 vs 63.0 ± 14.1), naming (55.0 ± 19.5 vs 43.3 ± 22.2), simple comprehension (67.5 ± 21.4 vs 55.6 ± 20.6), complex comprehension (44.8 ± 17.0 vs 35.0 ± 18.9), listening and reading words (65.4 ± 16.7 vs 53.7 ± 19.3), calculation (39.3 ± 25.8 vs 25.9 ± 19.2), memory (36.4 ± 18.7 vs 26.3 ± 17.8), matching (75.9 ± 18.6 vs 65.3 ± 17.1), simple general knowledge (68.3 ± 18.2 vs 58.0 ± 19.5), complex general knowledge (58.7 ± 17.4 vs 50.0 ± 13.3), orientation (70.7 ± 19.6 vs 60.5 ± 17.2) and comparison (59.9 ± 14.6 vs 50.2 ± 17.5) ($P < .05$ for all). There were significant differences between the two groups in emotional function (66.7 ± 18.2 vs 53.1 ± 21.2), general health (67.2 ± 12.6 vs 60.7 ± 9.8), mental health (71.0 ± 5.6 vs 63.1 ± 4.2), social functioning (64.7 ± 9.0 vs 59.3 ± 10.2) and vitality (55.4 ± 14.8 vs 46.9 ± 15.6) ($P < .05$ for all).

CONCLUSIONS: Cognitive intervention combined with SLT can contribute to the recovery of language function and improvement of QOL in elderly patients with post-stroke aphasia.

P205**Risk Factors for Bladder Tumor Recurrence and its Prognosis in Patients with Primary Upper Tract Urothelial Carcinomas after Radical Nephroureterectomy**

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BACKGROUND: It is controversy whether bladder tumor recurrence affects patient's survival in primary upper tract urothelial carcinomas after radical nephroureterectomy. This study aimed to analyze risk factors for bladder tumor recurrence in primary upper tract urothelial carcinomas after radical nephroureterectomy,

and to evaluate the relationship between bladder tumor recurrence and subsequent oncological outcomes.

METHODS: The data were retrospectively analyzed in 115 patients of primary upper tract urothelial carcinomas. The expressions of tumor markers Ki-67 and P53 protein were detected by immunohistochemistry. Combined with traditional clinicopathological factors and molecular markers, we analyzed the risk factors of bladder tumor recurrence and studied the role of bladder tumor recurrence on subsequent oncological outcomes.

RESULTS: The incidence of bladder cancer was 11.3% (13/115) in patients with the primary upper tract urothelial carcinomas after radical nephroureterectomy, and the mean follow-up time was 48.50 months (7–130 months). Multivariate survival analysis revealed that more than 65 years ($P = .040$), ureteral tumors ($P = .008$) and the lower expression of Ki-67 ($P = .041$) were independent predictors for bladder cancer. The mean progression free survival time was (92.1 ± 13.3) and (112.8 ± 4.3) months in bladder tumor group and non-bladder tumor group. There was no significant difference between those two groups ($P > .05$). The mean disease specific survival time was (113.5 ± 4.3) and (116.0 ± 5.0) months. There were no significant difference between those two groups ($P > .05$).

CONCLUSIONS: Advanced age, lower ureteral tumors and low expression of Ki-67 were independent predictors for bladder tumor recurrence in primary upper tract urothelial carcinomas after radical nephroureterectomy. There was no relationship between bladder tumor recurrence and oncological outcomes. However, close follow-up and rigorous treatment are essential for patients with high risk factors.

P206**Prospective Study of Muscle Mass and Strength in Elderly Patients with Inguinal Hernia**

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BACKGROUND: Muscle mass declines with age. This study aimed to evaluate muscle mass in elderly patients with inguinal hernia was evaluated.

METHODS: A total of 30 male hospitalized patients aged 70–90 years with inguinal hernia were involved in the study group, and 10 male healthy volunteers aged 40–50 years were considered as the control group. Human body measurement was conducted. Body composition was detected by bioelectrical impedance analysis and dual energy X-ray scanning. Stomach and thigh muscles were scanned by CT and the muscle area was calculated by using special software.

RESULTS: Grip strength index and calf circumference were less in the study group than in the control group. Total body muscle mass measured by bioelectrical impedance analysis was the same as that measured by dual energy X-ray scanning, and it was reduced in the study group as compared with the control group. The limb skeletal muscle index in study group was 7.53 ± 0.78 , sarcopenia type I (male $<7.01 \text{ kg/m}^2$) was 22.7%, and sarcopenia type II (male $<6.08 \text{ kg/m}^2$) was 0. The mass of stomach and thigh muscles was less in study group than in the control group by CT scanning, but there was no difference in vertical spinal muscle mass between both groups. L3 skeletal muscle index was 70.0% (male <52.4).

CONCLUSIONS: Muscle mass and strength further declined in elderly patients with inguinal hernia than healthy older adults, and the incidence of sarcopenia was significantly higher in the former than in the latter.

P207

Correlation of Post-stroke Depression with Stroke Deficit Scale and Serum Brain-derived Neurotrophic Factor Levels in the Elderly

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BACKGROUND: Post-stroke depression (PSD) is one of common complications after stroke. Studies showed that brain-derived neurotrophic factor is involved in depression onset. This study aimed to explore the association of post-stroke depression (PSD) with the scale of stroke deficit and serum brain-derived neurotrophic factor levels in the elderly.

METHODS: A total of 150 elderly stroke patients from the department of neurology of our hospital were recruited from August 2010 to September 2013. Patients were assessed by the Hamilton depression rating scale (HAMD), the National Institutes of Health Stroke Scale (NIHSS), and the Activities of Daily Living Scale (ADL). In addition, serum BDNF levels were determined by ELISA.

RESULTS: Post-stroke depression occurred in 42.0% of the 150 elderly patients suffered (63/150). There were significant differences in neurological function impairment and ADL scores between the PSD group and the non-PSD group [(14.72 ± 4.06) vs (5.51 ± 1.93) and (40.63 ± 17.06) vs (79.66 ± 29.31), $t = 18.49$ and 9.47 , respectively, $P < .0001$ for both). HAMD and NIHSS scores showed positive correlations with the degree of depression ($r = 0.998$ and 0.962 , $P = .0012$ and 0.0375 , respectively). ADL scores and serum BDNF levels decreased as the degree of depression increased, and were negatively correlated with the degree of depression ($r = -0.985$ and -0.995 , $P = .0149$ and 0.0049 , respectively).

CONCLUSIONS: The degree of post-stroke depression was positively correlated with the scale of neurological dysfunction and negatively correlated with activities of daily living scale and serum BDNF levels in the elderly, emphasizing the need for early rehabilitation and treatment to improve microcirculation and brain metabolism leading to improved neurological function.

P208

Effect of Nebulized Corticosteroids on Long-term Poorly Controlled Asthma in Elderly Patients

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OBJECTIVE: Treatment of asthma in the elderly remains inadequate. We evaluated the efficacy and safety of long-term nebulized corticosteroids and inhaled bronchodilators therapy on asthma in long-term poor controlled elderly patients.

METHODS: Prospective, randomized control study was conducted in 63 elderly asthma patients, who received inhaled corticosteroids or combined with other long-term medication treatment for asthma control for more than 3 months but were still in moderate to severe persistent asthma state. They were randomly divided into nebulizer treatment group ($n = 31$) and dry powder inhalation group ($n = 32$), and they were treated correspondingly with Budesonide inhalation suspension 1 mg and Salbutamol 2.5 mg Bid or Budesonide/Formoterol powder 320/9 µg inhalation Bid. Patients in the two groups were evaluated for the differences in lung function, acute exacerbations of asthma, asthma control test (ACT) and adverse effects after 12-week treatment.

RESULTS: The improvement in morning peak expiratory flow (PEF) and ACT score were better in the test group than in the control group [(29.2 ± 14.4) ml vs (15.8 ± 13.5) ml, (4.8 ± 2.2) vs (3.0 ± 2.7), $t = 3.715$ and 2.897 , $P = .000$ and 0.005], but there were

no statistical differences in average daily use of relievers and severe acute exacerbations between the test and control groups ($t = -1.512$, $P = .136$; $\chi^2 = 2.238$, $P = .135$). The local adverse effects caused by inhaled corticosteroids had no significant difference between the two groups.

CONCLUSIONS: Compared with powder inhalation, nebulizer inhalation administration of corticosteroids and bronchodilators could improve control of asthma symptoms and lung function in elderly patients with serious asthma condition. This 12-week continuous treatment also demonstrated the safety profile with minimal side effects.

P209

Stilbene Li Strong Heart Capsule in the Treatment of Chronic Heart Failure in the Elderly: A Clinical Study

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OBJECTIVE: This study aimed to evaluate therapeutic effect of stilbene Li strong heart capsule on quality of life, pro-BNP and left ventricular ejection fraction in elderly patients with chronic heart failure.

METHODS: From March 2013 to March 2015, a total of 60 cases of patients with chronic heart failure months (the American Heart Association Standard, Guideline for Clinical Trials of New Patent Chinese Medicines in the treatment of heart failure in study of the diagnostic standards) were recruited from geriatric ward and outpatient department of Inner Mongolia Medical University and were randomly divided into group of conventional treatment with western medicine and group of 30 cases by the use of stilbene Li strong heart capsule. The control group was given conventional western medicine for chronic heart failure (angiotensin converting enzyme inhibitors (ACEI), angiotensin II receptor blockers (ARB), Beta-Blockers and aldosterone antagonist, with or without use of digoxin, loop diuretics and vasodilators), and treatment group was given stilbene Li strong heart capsule, 4 capsules each time and 3 times a day, and course of treatment for the two groups was 3 months. The following indicators of quality of life questionnaire, Minnesota heart failure, TCM syndrome integral, Lee's scale pro-BNP level, left ventricular ejection fraction (LVEF) were observed before and after treatment.

RESULTS: Compared with the conventional treatment, patients with left ventricular ejection fraction increased. The plasma NT-proBNP went down significantly and Lee's ratings dropped significantly.

CONCLUSION: On the basis of conventional treatment, application of stilbene Li strong heart capsule may improve cardiac function, athletic stamina as well as ventricular remodeling.

P210

Improvement in Quality Life of Patients with Alzheimer's Disease and Depression by Nursing Intervention Measures

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OBJECTIVE: This study aimed to explore new nursing intervention measures' impact on quality of life in Alzheimer's disease patients with depression.

METHODS: Forty patients with Alzheimer's disease and depression were select from the geriatric neurology ward according to the study criteria as study objects, and randomly divided into basic nursing care group (20 cases) and nursing intervention measure group (20 cases) with 1 year intervention. Basic nursing care

group included daily routine, sleeping, medication and safety measures, while nursing intervention measure group had psychiatric rehabilitation nursing besides the routine measures. Cornell Scale for Depression in Dementia (CSDD) and Mini-Mental State Examination (MMSE) were performed in all patients for the assessment at pre-intervention and post-intervention.

RESULTS: Compared to the situation before implementing nursing interventions, patients' perception and depression in dementia of the two groups were significantly improved, and compared with basic nursing care measures, the improvement of clinical symptoms was more evident with the new nursing intervention measures.

CONCLUSION: These new nursing intervention measures had beneficial effect on patients with AD and depression.

P211

Effects of Fluoxetine on White Matter Changes in a Rat Model of Depression

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OBJECTIVE: This study aimed to investigate the effects of fluoxetine on white matter changes in a rat model of depression.

METHODS: Sprague-Dawley rats were divided into control group, CUS (chronic unpredictable stress, CUS) group and FLX (fluoxetine, FLX) group. After four week chronic unpredictable stress stimulation, the control group and the CUS group rats were treated with saline, and the FLX group rats were treated with fluoxetine at dose of 5 mg/kg/day. The sucrose preference was measured weekly. The white matter volume, the total length and total volume of the myelinated fibers in white matter, the total volume of the myelin sheaths in the white matter were calculated with the stereological methods.

RESULTS: The sucrose preference of the CUS group was significantly lower than that of the control group and FLX group ($P < .05$, $P < .05$, respectively). According to the stereological results, the white matter 2 volume of CUS rats was significantly decreased by 21.14% compared with the control group ($P < .05$), and decreased by 12.54% compared with FLX group ($P > .05$). The total volume, total length and mean diameter of the myelinated fibers, the volume of the myelin sheaths and the volume of the axons in the white matter of the CUS group were significantly decreased when compared to those of the control group ($P < .05$), whereas no significant differences in these parameters of the stereological results were found between the CUS groups and FLX group ($P > .05$).

CONCLUSIONS: Fluoxetine had no effects on the white matter and myelinated fibers in the white matter of CUS rats.

P212

Dynamic Changes of AGEs and RAGE in the Accelerated Progression of Age-related Intrarenal Small Arterial Stiffness in Hypertensive Rats

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OBJECTIVE: This study aimed to dynamically investigate the morphological change of age-related intrarenal small arterial

(IRSA) in spontaneously hypertensive rat (SHR) and Wistar-Kyoto (WKY) rats respectively and to explore the dynamic changes of advanced glycation end-products (AGEs) and receptor for AGEs (RAGE) in the progression of age-related IRSA stiffness accelerated by hypertension.

METHODS: SHR and WKY rats were respectively randomized into 4, 12, 24, 48 and 72-week-old group ($n = 16$). Minimal renal vascular resistance (minRVR) was detected at each group. Renal arcuate arteries (RAA) and interlobular arteries (RILA) were analyzed by EVG, HE, Sirius-red staining, α -smooth muscle actin (α -SMA) immunofluorescent technique and proliferating cell nuclear antigen (PCNA), AGEs, RAGE immunohistochemical staining, and various morphological indexes and medial cell proliferation index (PI) were measured. The concentration of serum AGEs was assessed by ELISA.

RESULTS: The minRVR and the morphological indexes of medial nuclear area, medial collagen area and medial thickness of RAA and RILA had shown an age-dependent increase since 24w in SHR and 72w in WKY respectively, accompanied by the increased expression of RAGE in media. The expressions of AGEs in serum and both IRSA media were also significantly elevated at 72w WKY consistently, while the levels of AGEs began to increase from 48w in SHR. Interestingly, the collagen area and the expression of RAGE in RILA media at 12w SHR had begun to be much higher than that in 4w SHR group ($P < .05$).

CONCLUSION: Hypertension accelerated the progression of age-related intrarenal small arterial stiffness. RILA stiffness occurred earlier than age-related RAA stiffness. The potential mechanisms likely involve up-regulated RAGE in IRSA media, while up-regulated AGEs in serum and IRSA media may also be involved in the late stage.

P213

Intravenous Zoledronic Acid Versus Oral Alendronate in the Treatment of Osteoporosis in Older Adults

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OBJECTIVE: This study aimed to evaluate and compare intravenous zoledronic acid and oral alendronate for the treatment of osteoporosis in the elderly.

METHOD: The study was designed to compare a once yearly intravenous dose of zoledronic acid 5 mg with a once weekly oral dose of alendronate 70 mg with respect to markers of serum β -C-terminal telopeptide of type I collagen (β -CTX), procollagen type I C-terminal propeptide (PICP), N-terminal midfragment of osteocalcin (NMID) and bone mineral density in 90 older patients with osteoporosis in China.

RESULTS: The increasing rate of lumbar spine and femoral neck BMD in zoledronic acid group were 12.85%, 13.51% respectively, while in alendronate group they were 11.92%, 12.73% respectively. There were no significant differences ($P > .05$). The declining rates of serum β -CTX, PICP and NMID in zoledronic acid group were 17.73%, 54.75%, 47.35%. While in alendronate group they were 19.22%, 53.65% and 46.8%. There were no significant differences ($P > .05$). There was no serious adverse reaction in the two groups, and the drug cost was very close. Because of the poor compliance of oral alendronate, intravenous zoledronic acid had an advantage.

CONCLUSION: Treatment with intravenous zoledronic acid appeared to be better than oral alendronate in the treatment of osteoporosis.

P214

The Impact of KLF2/RelA Imbalance on Neutrophil Apoptosis of Neutrophilic Asthma Guinea Pig Model

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OBJECTIVE: This study aimed to explore the role of KLF2 and RelA in neutrophil apoptosis of neutrophilic asthma.

METHODS: Forty adult male guinea pigs were randomly divided into three groups: control group (group A), neutrophilic asthma group (group B) and neutrophilic asthma treatment group (group C). The total number of cell and differential cell counts in BALF was detected. Flow cytometry was used to detect apoptosis rate of neutrophils, and RT-PCR was used to detect the expression of KLF2 and RelA mRNA in inflammatory cells of bronchoalveolar lavage fluid and blood neutrophils. Western blot was used to detect the protein expression levels of KLF2, RelA in neutrophils of blood.

RESULTS: The total number of cells and proportion of neutrophil of BALF in group B were significantly higher than that of group A and group C ($P < .05$). The neutrophil apoptosis rate in group B was significantly lower than that of group A ($P < .05$), and group C was higher than group B ($P < .05$) but still lower than group A. The proportion of KLF2/RelA in BALF and blood neutrophils of group B in RNA and protein levels was decreased than that of group A and group C ($P < .05$). Neutrophils apoptosis rate was positively correlated with the KLF2/RELA protein ratio ($r = 0.8826$, $P < .001$).

CONCLUSION: Neutrophil apoptosis rate in neutrophilic asthma guinea pigs was reduced. The imbalance of KLF2/RelA may play an important role in the reduction of neutrophil apoptosis in neutrophilic asthma.

P215

Serum Ferritin Levels Are Associated with Carotid Atherosclerosis in Chinese Postmenopausal Women: The Shanghai Changfeng Study

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OBJECTIVE: Postmenopausal women are at increased risk of cardiovascular disease. Increased serum ferritin level may be involved in the pathogenesis. The aim of the present study was to investigate the relationship of ferritin and carotid atherosclerosis in postmenopausal women.

METHODS: A total of 1,178 postmenopausal women (mean age, 60.8 years) were enrolled in this study. A standard interview, anthropometric measurements and laboratory analyses were performed for each participant. Bilateral CIMTs (carotid intima-media thickness) were measured using ultrasonography, and the presence of carotid plaques was assessed. Serum ferritin was measured using electrochemiluminescence immunoassay.

RESULTS: The results showed that serum ferritin was 181.9 ± 65.8 ng/mL in the postmenopausal women. Multivariate linear stepwise regression analysis demonstrated that age (standardized $\beta = 0.233$, $P < .001$), ALT (alanine transaminase) (standardized $\beta = 0.194$, $P < .001$), logHOMA-IR (homeostasis model assessment index for insulin resistance) (standardized $\beta = 0.181$, $P < .001$), TG (triglyceride) (standardized $\beta = 0.083$, $P = .003$), Hb (hemoglobin) (standardized $\beta = 0.080$, $P = .004$) and PPG (2 h glucose levels following a 75-g oral glucose challenge) (standardized $\beta = 0.079$, $P = .004$) were independently associated with serum ferritin. Compared with subjects with ferritin in the first quartile, subjects with ferritin in the fourth quartile had greater CIMTs and higher prevalence of carotid plaque. After adjusting for conventional CVD risk factors, Hb, WBC (white blood cell), logUACR and liver function, postmenopausal women with ferritin in the fourth quartile had a 1.587-fold increased risk of carotid plaques relative to those in the lowest quartile.

CONCLUSION: These results suggest serum ferritin was independently and positively associated with carotid atherosclerosis in postmenopausal women and that ferritin may be implicated in atherosclerosis.

P216

The Effect of High Plasma Homocysteine on Cardiac Remodeling in Spontaneously Hypertensive Rats and on Folic and Vitamin B₁₂

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OBJECTIVE: This study aimed to explore the effect of high plasma homocysteine (Hcy) on cardiac remodeling in hypertensive rat and the protective effect of folic and vitamin B₁₂.

METHODS: Seven-week-old spontaneously hypertensive rats (SHR) were divided into four groups according to different diets. At the end of 20 week period, plasma Hcy levels and blood pressure were measured, and cardiac structure and function were detected using echocardiography and invasive hemodynamic method. Picro-sirius red (PSR) staining was used to check the fibrosis of paraffin embedding ventricle section.

RESULTS: The plasma Hcy level was increased in group M, while there was no statistical difference among the other three groups. Rats in group M had the largest heart weight, and systolic blood pressure was lower in group M compared with group C. LVEDP was increased and $-dp/dt_{max}$ was decreased in group M as compared to group C, T and M+C. The latter three groups had the similar parameters of LVEDP and $-dp/dt_{max}$. Systolic function assessed by LVFS and $+dp/dt_{max}$ was decreased most in group M, and there were no statistical significance about systolic function between T and C groups, but rats in M + C had poorer systolic function than rats in group C ($P < .05$). Perivascular fibrosis increased significantly in group M than that in the others, which was reduced towards group C values in group T, however, values were significantly greater in the M+C group than in group C. Changes in myocardium fibrosis were similar to those observed in perivascular fibrosis except those in group T, which showed a slightly improvement of collagen deposition as compared to group M + C.

CONCLUSIONS: High level of Hcy accelerates systolic and diastolic dysfunction through accumulation of fibrosis in myocardium and vascular wall which led to adverse cardiac remodeling. Folic and vitamin B₁₂ had a protective effect on heart function by reversing adverse changes from Hcy.

P217**Effects of Aspirin on Proliferation, Differentiation, Mineralization of MC3T3 Cells and Its potential Mechanism**

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OBJECTIVE: This study aimed to explore the proliferation, differentiation and mineralization of MC3T3 cells in different concentration of aspirin and its possible mechanisms.

METHODS: MTT experiments were conducted to test the proliferation of MC3T3 cells in different concentration of aspirin (0, 0.25, 0.5, 1, 1.5, 2, 5, 10 mmol/L). Alkaline phosphatase staining, quantitative analysis and alizarin red staining were used to show the differentiation and mineralization of MC3T3 cells in different concentrations of aspirin. Western Blot and Real-time PCR were used to explore the possible mechanism of aspirin on MC3T3 cells.

RESULTS: The results of Thiazolyl blue tetrazolium bromide (MTT) experiments showed that the Optical density (OD) values of low concentration aspirin (0.25, 0.5, 1, 1.5, 2 mmol/L) groups had no difference with the control group ($P > .05$); while the high concentration groups were lower ($P < .01$). The result of ALP staining, quantitative analysis and alizarin red staining showed that the group of 1.5 mmol/L aspirin had much more positive particles than other experimental groups. The results of Western Blot and Real-time PCR showed that the content of osteoprotegerin (OPG) protein, Col1a protein and OPGmRNA in 1.5 mmol/L Aspirin group were much more than the control group ($P < .01$), while the Receptor activator of NF- κ B ligand (RANKL) protein and RANKLmRNA were lower ($P < .01$).

CONCLUSIONS: While low concentrations of aspirin had no effect on proliferation of MC3T3 cells, high dose aspirin suppressed cell proliferation. However, aspirin at 1.5 mmol/L had stronger osteogenic effects and mineralization than other dose groups and controls. Low dose aspirin could up-regulate OPG and down-regulate RANKL to prevent osteoporosis through OPG/RANKL/RANK signaling pathways.

P218**Relationships between Serum High-sensitivity C-Reactive Protein, Lipid Parameters and Age-related Body-composition Changes in the Elderly**

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OBJECTIVE: Aging is associated with progressive loss of lean body mass and increase of fat mass. In this study, we examined the association of serum high-sensitivity C-reactive protein (hs-CRP)) and serum lipid measures (total cholesterol (TC), triglyceride (TG), low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), Lipoprotein (a) (LP(a)), apolipoprotein A1 (ApoA1), apolipoprotein B (ApoB)) with age-related body-composition changes (loss of muscle and increase of fat mass) in the elderly.

METHODS: We conducted this study using 2013–2014 data from hospitalized Chinese people of 65 years or older. Total body fat percentage and appendicular skeletal muscle mass were assessed with a dual-energy X-ray absorptiometry scan. The participants were classified into four groups based on both total body fat percentage and appendicular skeletal muscle mass.

RESULTS: The analysis included data of 336 participants (mean age 71.63 years; 171 males and 165 females). Subjects with muscle wasting had higher hs-CRP level, compared with obese or no obese groups. Furthermore, in the muscle wasting with obesity group, the hsCRP level was the highest. In the obese or no obese group, the Lipoprotein (a) level of the subjects with muscle wasting

was significantly higher than those with no muscle wasting. Correlation analysis showed that ASM/height² was negatively correlated with hsCRP. On the contrary, total body fat percentage was positively correlated with hsCRP. Logistic regression analysis showed that Muscle wasting was positively associated with hsCRP and LP (a), while obesity was positively correlated with hsCRP.

CONCLUSIONS: Our study showed that age-related muscle wasting appeared to be significantly associated with hsCRP no matter the subjects had obesity or not. But there was no obvious correlation between lipid parameters and age-related body-composition changes.

P219**Effect of Large Doses of L-Glutamine on Newly Diagnosed Type 2 Diabetes Patients above 65**

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OBJECTIVE: This study aimed to comprehensively evaluate the effect of large doses of L-Glutamine (L-Gln) on newly diagnosed type 2 diabetic patients above 65.

METHODS: Forty-four patients with newly diagnosed type 2 diabetes were treated in geriatric section of Yunnan First People's Hospital. They were divided into treatment group (n = 22) and control group (n = 22). The patients in control group were only treated by Metformin Hydrochloride (0.25 g each time, three times a day), and the patients in treatment group were given the treatment of L-Gln (4.02 g/d) and Metformin Hydrochloride (0.25 g each time, three times a day). Before and after the treatment for 24 weeks, 15 mL of blood was collected from each patient. Automatic biochemical analyzer was used to determine Liver function (AST, ALT, TBA and AKP), kidney function (BUN, CREA, UA, and Ccr), heart function (LDH, CK, CK-MB, a-HBDH), Lipids (TC, TG, HDL-C and LDL-C) and blood glucose. A kit was used to detect serum insulin, serum homocysteine (Hcy), antioxidant capacity (GSH, CAT, SOD and NO) and inflammatory factors (IL-1 α , IL-1 β , IL-6 and TNF- α), and to analyze the difference before and after treatment.

RESULTS: Before treatment, two groups had no differences in blood pressure, weight, BMI medication history, biochemical parameters, etc. After treatment for 3 months, blood sugar, serum insulin, CHOL, TG, HDL-C, LDL-C, AST, ALT, AKP, GSH, CAT, SOD, NO, r-GT, LDH, CK, CK-MB, a-HBDH, TNF- α , IL-1 α , IL-1 β , IL-6, and Hcy in both of the two groups were improved compared to the results before treatment. After treatment, the levels of LDH, CK, CK-MB, a-HBDH and Hcy in the treatment group were lower compared with those in the patients of the control group. However, GSH, CAT, SOD and NO were higher in the treatment group than those in the control group. Moreover, with regards to BUN, CREA, UA, Ccr, CHOL, TG, HDL-C, LDL-C, AST, ALT, AKP, TNF- α , IL-1 α , IL-1 β , blood sugar, serum insulin and IL-6, there was no difference between the two groups.

CONCLUSIONS: L-Gln treatment appeared to reduce Hcy levels, improve antioxidant capacity, and restore cardiac function in patients with type 2 diabetes.

P220**Association between Resting Heart Rate at Admission and Cardiac Function in Elderly Patients with Non-ST Elevation Myocardial Infarction**

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OBJECTIVE: This study aimed to investigate the relationship between resting heart rate (RHR) at admission and heart function in elderly patients, with non-ST elevation myocardial infarction (NSTEMI).

METHODS: A total of 90 elderly NSTEMI patients were enrolled in this study. 46 of them were male and 42 female. The average age was (68.15 ± 5.38) years old. All elderly NSTEMI patients were tested with resting electrocardiogram and divided into three groups according to the levels of resting heart rate. RHR1 group: RHR <70 bpm (30 cases), RHR2 group: $70 \text{ bpm} \leq \text{RHR} < 90$ bpm (28 cases), RHR3 group: RHR ≥ 90 bpm (32 cases). Arrhythmia, heart failure, cardiac death, shock and infarction and other major adverse cardiac event were determined in each patient during hospitalization. The variability of above mentioned index was contrasted among the three groups and the relations between every index and RH were analyzed.

RESULTS: General information, age, male patient ratio, smoking rate and hypertension duration were similar among the three groups (all $P > .05$). Compared with RHR1 group, RHR3 group showed higher levels of FPG, total cholesterol (TC), high density lipoprotein (HDL-C), low density lipoprotein (LDL-C) and total glycerin (TG) ($P < .01$), and RHR2 group showed higher levels of FPG, TC, HDL-C and LDL-C ($P < .01$), and levels of TG ($P < .05$) was higher in RHR2 group. Compared with RHR2 group, the levels of FPG, TC and LDL-C were significantly increased in the RHR3 group ($P < .01$), while the levels of TG and HDL-C only increased ($P < .05$). Compared with RHR1 group, the levels of LVDD were significantly increased in the RHR2 and RHR3 group, and the levels of LVEF were significantly decreased in the RHR2 and RHR3 group. RHR3 group was more clearly than RHR2 group ($P < .05$). Compared with RHR2 group, the levels of LVDD were significantly increased in the RHR3 group, and the levels of LVEF were significantly decreased in the RHR3 group ($P < .05$). With the level of resting heart rate rising, arrhythmia, heart failure, cardiac shock, death, infarction had a trend of rising. Although every index was similar among the three groups ($P > .05$), they had rising trend. By partial correlation analysis, the levels of FPG, LDL-C, TC, TG were positively correlated to RHR ($r = 0.399, 0.256, 0.248, 0.323$ respectively, all $P < .01$). However, the levels of HDL-C were negatively correlated to RHR ($r = -0.293, P < .01$). The levels of LVDD were positively correlated to RHR ($r = 0.509, P < .01$), and the levels of LVEF were positively correlated to RHR ($r = 0.255, P < .05$).

CONCLUSION: RHR was closely correlated with LVDD and LVEF in elderly patients with NSTEMI. Elevated RHR at admission predicted end organ damage, arrhythmia, heart failure, cardiac shock, and death.

P221**The Relationship between Red Blood Cell Volume Distribution Width and the Degree of the Coronary Artery Lesions and Short-term Outcomes in Patients with Acute ST-segment Elevation Myocardial Infarction**

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OBJECTIVE: This study aimed to investigate the relationship of red blood cell volume distribution width (RDW) with extent of the coronary artery lesions and short-term outcomes in

patients with acute ST segment elevation myocardial infarction (STEMI).

METHODS: A total of 228 STEMI patients from the cardiovascular department of the first affiliated hospital of Liaoning Medical University and from Emergency Intensive Care Unit (EICU) of Beijing Anzhen hospital were enrolled in this study (male 172, aged 26–78; female, 56, aged 50–74). Blood routine test and RDW were measured after being admitted, and vessels lesion and Gensini score were determined according to coronary angiography (CAG) $\pm$ percutaneous coronary intervention (PCI), and then cardiovascular events (MACE) were observed within one month. According to baseline RDW quartile points, 228 patients with STEMI were divided into four groups: $\leq 12.6\%$ group 1 (54 cases, 12.7–13.0%); group 2 (56 cases, 13.1–13.4%); group 3 (48 cases, $\geq 13.5\%$); and group 4 (70 cases). Coronary angiographic findings and prognosis were analyzed. Bi-variance correlation was evaluated by Spearman correlation coefficient to explore the relation of RDW, the extent of coronary lesion and cardiac events.

RESULTS: Differences in White blood cell counts ($P = .023$), serum creatinine ($P = .040$) and CRP ($P = .000$), blood triglycerides ($P = .024$), LVEF ($P = .000$), BNP ($P = .009$), cTnI ($P = .022$) among study groups of patients were statistically significant. Changes of RDW in patients with STEMI was positively correlated with Gensini score ($r = 0.198, P = .035$). As RDW values increased gradually, major cardiovascular events also rose. Logistic regression analysis showed that RDW ($OR = 2.351, 95\% \text{ CI: } 1.156-4.781, P = .018$), age ($OR = 0.926, 95\% \text{ CI: } 0.859-0.997, P = .042$), Gensini score ($OR = 1.031, 95\% \text{ CI: } 1.006-1.058, P = .017$), serum creatinine ($OR = 0.962, 95\% \text{ CI: } 0.925-1.000, P = .013$), CRP ($OR = 1.151, 95\% \text{ CI: } 1.048-1.265, P = .003$) and cTnI ($OR = 0.887, 95\% \text{ CI: } 0.808-0.974, P = .012$) are independent risk factors for MACE within one month after STEMI.

CONCLUSIONS: RDW had significant correlation with Gensini score in patients with STEMI patients. Elevated RDW appeared to be a predictive factor for cardiovascular events for these patients.

P222**Relationship between Body Mass Index and Ambulatory Blood Pressure Parameters in Elderly Essential Hypertensive Patients**

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OBJECTIVE: The aim of this study was to explore the relationship between body mass index and 24-hour ambulatory blood pressure monitoring parameters in elderly patients with essential hypertension.

METHODS: A total of 119 patients (male 68 cases, female 51 cases) with essential hypertension (EH) were selected and chosen from our hospital between June 2013 and July 2014. They were 60 to 80 years old, with the mean age (68.18 ± 5.88) . The patients were divided into three groups: normal weight (NW Group): $18.5 \text{ kg/m}^2 \leq \text{BMI} < 24 \text{ kg/m}^2$; overweight (OW group): $24 \text{ kg/m}^2 \leq \text{BMI} < 28 \text{ kg/m}^2$, and obese (OB group): $\text{BMI} \geq 28 \text{ kg/m}^2$, and all the subjects received 24-h ambulatory blood pressure monitoring (Ambulatory Blood Pressure Monitoring, ABPM). The impact of different BMI between ABPM on various parameters and their correlation was analyzed.

RESULTS: No significant difference was observed between the three study groups in age, sex, smoking history, drinking history, total cholesterol (TC), triglyceride (TG), high density lipoprotein-C (HDL-C), low-density lipoprotein-C (LDL-C), and fasting blood glucose (FBG) ($P > .05$). All the indexes of the three groups were significantly increased ($P < .01$): 24-h average systolic (24 h SBP), 24-hour average diastolic blood pressure (24 h DBP), the mean daytime systolic blood pressure (dSBP), daytime average

diastolic blood pressure (dDBP), nighttime mean systolic pressure (nSBP), nighttime average diastolic (nDBP), nighttime systolic blood pressure load (NSBPL), and nighttime diastolic blood pressure load (nDBPL). Compared with the NW group, nighttime SBP load in the OW group increased, and the difference was statistically significant ($P < .05$). OB group Nighttime systolic blood pressure in the OB group had significantly higher load, and the difference was statistically significant ($P < .01$). By comparison with NW group, there were changes in daytime, nighttime average systolic and diastolic blood pressure were higher, and difference was statistically significant ($P < .01$). Compared with the OW group, nighttime diastolic blood pressure, diastolic blood pressure at night standard deviation significantly increased in OB group, and the difference was statistically significant ($P < .05$). BMI and 24 h SBP ($r = 0.397$), 24hDBP ($r = 0.397$), dSBP ($r = 0.425$), dDBP ($r = 0.453$), nSBP ($r = 0.388$), nDBP ($r = 0.572$), nighttime SBP load ($r = 0.176$), night DBP load ($r = 0.404$), daytime DBP changes in standard deviation ($r = 0.182$), standard deviation change in nighttime SBP ($r = 0.163$), and DBP night ($r = 0.183$) was positively correlated with changes in standard deviation ($P < .05$).

CONCLUSION: In elderly patients with essential hypertension, body mass index had significant associations with mean systolic blood pressure, 24 h average diastolic blood pressure, daytime systolic and diastolic blood pressure, night time systolic and diastolic blood pressure, nighttime systolic blood pressure load, and blood pressure variability.

P223

Correlation between Brachial-ankle Pulse Wave Velocity, Ankle-brachial Index and the Severity of Coronary Lesions

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BACKGROUND: Brachial-ankle pulse wave velocity (baPWV) is a unique measure of systemic arterial stiffness that is measured by brachial and tibial arterial wave analyses. Ankle brachial index (ABI) is the ratio of ankle vs arm systolic blood pressure which initially proposed as a diagnostic tool for peripheral artery disease. Both of them are independently associated with total mortality and cardiovascular events in prospective studies. The aim of this study was therefore to investigate whether ABI and baPWV correlate more closely with coronary atherosclerosis lesion.

METHODS: A total of 251 subjects (177 males and 74 females) were selected from the patients who were suspected of having coronary artery disease. In all patients (aged 60.52 ± 7.95 years), baPWV, ABI and other baseline characteristics were measured before the coronary angiography. Coronary artery Gensini scoring systems was calculated to quantify the severity of coronary artery atherosclerosis. Based on Gensini scores, patients were assigned into four groups: group with normal coronary artery (Score 0–1), group with mild lesions (Score 2–20), group with moderate lesions (Score 20–40), and group with severe lesions (Score >40).

RESULTS: Between the groups with normal coronary arteries ($n = 55$), mild lesions ($n = 100$), moderate lesions ($n = 68$), and severe lesions ($n = 28$), there were no significant differences in term of gender, age, body mass index, smoking rates, diabetes prevalence and blood pressure ($P > .05$). Biochemical test results including triglycerides, total cholesterol and HDL cholesterol were also similar ($P > .05$). The LDL cholesterol was higher in the groups with moderate lesions (3.63 ± 0.38 mmol/L) and severe lesions (3.73 ± 0.28 mmol/L) than that in the normal coronary artery (3.03 ± 0.76 mmol/L) and mild lesions (3.22 ± 0.28 mmol/L). When baPWV was compared between the groups with normal

coronary arteries (1470.90 ± 20.42 cm/s), mild lesions (1603.41 ± 19.84 cm/s), moderate lesions (1908.32 ± 30.24 cm/s), and severe lesions (2209.23 ± 36.04 cm/s), the differences were statistically significant between any two groups ($P < .05$). When ABI was compared, the group of normal coronary arteries (1.31 ± 0.30) was much higher than the other groups (1.10 ± 0.47 , 0.90 ± 0.27 , 0.86 ± 0.09 in the groups with mild, moderate, and severe lesions respectively). Apparently the ABI in severe lesions group was lower than any other groups. When correlations between Gensini scores and baPWV, ABI, LDL-C and HDL-C were analyzed using Spearman rank correlation analysis, the results were consistent with that of variance analysis. The Gensini scores were directly correlating with baPWV and LDL-C ($P < .05$, correlation coefficients (r) were 0.466, 0.598, respectively), whereas the Gensini scores were inversely correlated with ABI and HDL-C ($P < .05$, correlation coefficient (r) were -0.910 and -0.150 , respectively).

CONCLUSIONS: Our results showed that the degree of coronary atherosclerosis lesions directly correlated with baPWV and inversely correlated with ABI. With the progression of coronary atherosclerosis lesions, the level of PWV was gradually increasing whereas the level of ABI was gradually decreasing. baPWV and ABI could be used for early screening and assessment of coronary atherosclerosis lesions at subclinical stages.

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Relationships between Sarcopenia and Inflammatory Factors and Impact of Vitamin D3 Treatment

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OBJECTIVE: This study aimed to explore whether there is association between elevated inflammatory cytokines IL-6, TNF- α and hs-CRP and sarcopenia, to observe the changes in the levels of these inflammatory factors and to find out the possibility of improvement in muscular functions under the intervention with active Vitamin D₃ (calcitriol), so as to provide theoretical evidence for the safety and efficacy of Vitamin D₃ in treating sarcopenia.

METHODS: Patients aged ≥ 60 years were selected from the Department of Preventive Medicine for Veterans in our hospital. Based on Appendicular Skeletal Muscle Index (ASMI), gait speed, and grip strength, 60 patients were assigned to the normal muscular volume group and the treatment group, who received examinations: including hepatic function, renal function, blood calcium, blood phosphor, blood glucose, IL-6, TNF- α and hs-CRP. At 12 weeks and 24 weeks, interventional treatment with calcitriol was given to the patients in the treatment group, and the changes in inflammatory cytokines, ASMI, gait speed and grip strength were compared.

RESULTS: Levels of IL-6 [(682.8 ± 129.8) vs (35.6 ± 14.7) pg/ml] and TNF- α [(62.8 ± 17.4) vs (8.7 ± 4.5) pg/ml] were higher in the treatment group than that in the control group, with statistically significant differences between the two groups ($P < .01$); the levels of IL-6 (412.7 ± 128.6) pg/ml and TNF- α (42.5 ± 16.5) pg/ml at 12 weeks of calcitriol intervention, and the levels of IL-6 (289.5 ± 100.6) pg/ml and TNF- α (20.7 ± 11.5) pg/ml at 24 weeks were decreased, and the decreases were more significant at 24 weeks than at 12 weeks with statistically significant differences ($P < .05$). By 24 weeks, there had been improvements in ASMI [(males: 4.62 ± 0.95 , females: 3.02 ± 0.43) vs (males: 4.92 ± 0.82 , females: 3.25 ± 0.58) kg/m²], gait speed [(0.5 ± 0.1) vs (0.7 ± 0.1) m/s] and grip strength [(males: 20.5 ± 2.6 , females: 15.5 ± 1.5) vs (males: 21.5 ± 1.5 , females: 16.2 ± 1.5) kg]; however, the differences were not statistically significant. There were no statistically significant differences between the two groups in the changes in hepatic function, renal function, blood calcium, blood phosphor, or blood glucose.

CONCLUSION: Elevated IL-6 and TNF- α was associated with sarcopenia. Calcitriol intervention was effective for lowering IL-6 and TNF- α level and might be of significance for improving the patients' muscular functions, and the calcitriol dose 0.5 μ g q.d. was considered safe.

P225

The Relationship between Leukoaraiosis and Cognitive Function

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OBJECTIVE: We aimed to investigate the relationship between the performance of periventricular leukoaraiosis (LA) in MRI and cognitive function in elderly patients with hypertension in Shanghai city.

METHODS: Ninety elderly veterans with hypertension in Shanghai city were evaluated one by one to assess the prevalence of cognitive function impairment using both Mini-mental state examination (MMSE) and Montreal Cognitive Assessment (MoCA). They were then divided into 3 groups. Group A: people with normotension and normal cognitive function; Group B: people with hypertension and normal cognitive function, whereas people with hypertension and cognitive function impairment were divided into Group C. All subjects in the 3 groups were paired one by one, according to the following factors: age, gender, nationality, place of residence, years of education, type II diabetes, hyperlipidemia, and Parkinson's disease. Head MRI were operated in selected patients. Two experienced radiologists scored the periventricular white matter lesion in separated workstations independently. The LA score in the three groups were recorded followed by statistical comparison.

RESULTS: The results showed that there were significant differences among the three groups ($\chi^2 = -34.745$, $P < .05$). Further pairwise comparison suggested that there were significant differences between Group A and Group B ($\chi^2 = -10.506$, $P = .033$). The significant difference between Group B and Group C was ($\chi^2 = -10.029$, $P = .040$), and between Group A and Group C was ($\chi^2 = -25.112$, $P < .001$). An affinity correlation between white matter degeneration and cognitive dysfunction with hypertension was observed. LA score increased significantly in hypertensive patients and cognitive impairment.

CONCLUSIONS: As a result, LA score can be used for early diagnosis of hypertensive patients with cognitive dysfunction.

P226

Etiologic and Clinical Analysis of 229 Cases of Invasive Pulmonary Fungal Disease

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OBJECTIVE: By reviewing the epidemiology of invasive pulmonary fungal disease (including clinical diagnosis) in our hospital in recent 5 years, this study aimed to evaluate the underlying etiology, clinical features, treatment and outcomes of invasive pulmonary fungal disease.

METHODS: Patients of 18 years old and older with fungal disease were identified from Xiangya Hospital of Central South University between 2009 and 2014. The information collected from the patients included the demographic data, diagnostic methods, pathogenic bacteria, underlying conditions, clinical features, treatment and outcomes. Descriptive method was used for statistical analysis.

RESULTS: A total of 229 cases of invasive pulmonary fungal disease were retrieved. There were 152 (66.4%) males and 77 (33.6%) females, ranging from 22 to 93 years of age. There were

96 elderly patients, and 133 adult patients. Hospital acquired infection accounted for 30.6%. Pathogenic bacteria were as follows: candida (57.7%), aspergillus (22.7%), cryptococcus (5.7%), mucor (3.1%), histoplasmosis (1.3%), nocardia (0.4%), penicillium (0.4%), and uncertain pathogen (8.7%). Clinical diagnosis of 173 cases of pathogenic bacteria was as follows: candida (76.3%), aspergillus (17.3%), mucor (1.7%), and uncertain pathogen (4.7%). Diagnosis of 56 cases of pathogenic bacteria was as follows: aspergillus (39.3%), cryptococcus (23.2%), mucor (7.1%), histoplasmosis (5.4%), nocardia (1.8%), penicillium (1.8%), and uncertain pathogen (21.4%). The basic top five diseases of the invasive pulmonary candidiasis were as follows: hypertension (31.1%), COPD (30.3%), pulmonary tuberculosis (28.8%), coronary heart disease (22.7%) and diabetes (17.4%). The top three risk factors of the invasive pulmonary disease were as follows: long-term broad-spectrum antibiotics (37.9%), catheter (22.7%), and long-term application of hormones (15.9%). The basic top five diseases of the invasive pulmonary aspergillosis were as follows: pulmonary tuberculosis (40.4%), hypertension (26.9%), COPD (19.2%), bronchial asthma (15.4%), bronchiectasis (15.4%) and lung cancer (15.4%). The top three risk factors of the invasive pulmonary were as follows: long-term broad-spectrum antibiotics (19.2%), hypoproteinemia (9.6%), and long-term application of hormones (7.7%). The common clinical manifestations of invasive pulmonary fungal disease were cough (89.5%), expectoration (80.3%), dyspnea (43.7%), fever (30.1%), hemoptysis (13.1%), and chest pain (11.8%). But all had no specificity. Imaging findings of invasive pulmonary fungal disease had obvious diversity. Main findings were pneumonia (71.3%); mass type (9.6%), nodular type (8.7%), hollow (8.7%), and aspergillus ball (1.7%). This research showed that effective rate (cure rate and improvement rate) of drug treatment for invasive pulmonary candidiasis was 79.7%, and mortality rate was 7.8%. Effective rate of drug treatment for invasive pulmonary aspergillosis was 71.2%, and mortality was 8.8%.

CONCLUSIONS: Common pathogen of invasive pulmonary fungal disease included candida, aspergillus, cryptococcus and mucor. The common basic diseases of invasive pulmonary candidiasis and invasive pulmonary aspergillosis were hypertension, COPD, tuberculosis. The common risk factors were long-term broad-spectrum antibiotics, catheter, and long-term application of hormone. The common clinical manifestations of invasive pulmonary fungal disease were cough, expectoration and dyspnea, and main imaging features were pneumonia type.

P227

Meta-analysis: Probiotics for the Prevention and Treatment of Antibiotic-associated Diarrhea in Elderly Patients

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OBJECTIVE: Antibiotic-associated diarrhea (AAD) is a common complication of antibiotic therapy. Of particular importance is AAD which may result in severe illness, death and high healthcare costs in elderly patients. The prevention and treatment effect of probiotics for AAD have been recognized, however, they remain controversial. Here, we evaluated the efficacy of probiotics for the prevention and treatment of AAD in elderly patients.

METHODS: By searching PubMed, Elsevier and Wanfang Databases, we selected all the reports on randomized controlled trials (RCTs) comparing probiotics, including *Bifidobacterium*, *Bacillus subtilis*, *Lactobacillus acidophilus* and *Enterococcus*, to placebo or blank control in elderly patients who needed antibiotic treatment for meta-analysis. Statistical analysis was performed according to Cochrane guidelines by using Stata version 12.0 software.

RESULTS: Twelve reports involving a total of 4,236 participants who met the inclusion criteria. Mean age of patients was

78 ± 8 years. Compared with the non-probiotics groups, probiotics significantly prevented AAD (RR 0.81, 95% CI: 0.69–0.96). In addition, combined treatment with several kinds of probiotics had a better effect than alone-to-treat for elderly AAD patients (RR 1.28, 95% CI: 1.14–1.45).

CONCLUSION: Supplementation with probiotics may be effective in preventing AAD for elderly patients. Furthermore, the effect of combined treatment is better than alone. However, it is difficult to draw definitive conclusion with current reported data. Future clinical trials should be focused on the appropriate time of probiotics supplementation.

P228

Association between TLR4 and Treg in *Helicobacter Pylori* Colonization and Inflammation in Mouse

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BACKGROUND: The World Health Organization has categorized *H. pylori* as a class I carcinogen/definite human carcinogen. This study aimed to investigate immune pathogenic mechanism of *Helicobacter pylori* infection via TLR signaling and Treg cells in mice.

METHODS: We examined the relationship between TLR signaling and Treg cells and their effects on *H. pylori* colonization and inflammation. After blockage of either TLR4 or CD25, each mouse was orally administered *H. pylori*. Twelve weeks after the last inoculation, mice were sacrificed and serum, saliva, and gastric mucosa were collected.

RESULTS: The number of *H. pylori* colonies and the expression of MyD88 and NF- κ B p65 in the non-blocked group were significantly lower than that in the TLR4 blocked group ($P < .05$), and were significantly higher than that of the CD25 blocked group ($P < .05$). Conversely, the inflammatory grades and the expression of Foxp3 in the non-blocked group were significantly higher than those of the TLR4 blocked group ($P < .05$), and were significantly lower than those of the CD25 blocked group ($P < .05$). The levels of Th1 and Th17 cytokines in the gastric mucosa of mice in the non-blocked group were significantly higher than in the TLR4 blocked group ($P < .001-.05$), but were significantly lower than in the CD25 blocked group ($P < .001-.05$).

CONCLUSION: The balance between TLR signaling and Treg cells might be important for limiting *H. pylori* colonization and suppressing inflammatory response in infected mice.

P229

Clinical Analysis of Osteoporosis in Elderly Patients with Hypertension

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OBJECTIVE: This study aimed to evaluate factors that influence the osteoporosis in elderly patients with hypertension.

METHODS: According to the inclusion and exclusion criteria, aged ≥ 60 years, 380 cases (185 males and 195 females) of elderly patients with hypertension, from January 2013 to June 2014 in our department, were recruited in this study. All patients were tested for bone mineral density, and according to the bone mineral density, the data were divided into bone mass normal group, the bone loss and osteoporosis group. The general clinical data and the main risk factors were compared in different genders. At the same time, the data were used for statistical analysis.

RESULTS: (1) Compared with the osteoporosis group, body mass index (BMI) in normal bone mass and reduced bone mass group was higher while HDL-C and Hcy were lower, and the difference was statistically significant ($P < .05$); Compared with reduced bone mass group, Hcy was lower, and the difference was statistically significant ($P < .05$). (2) The comparison of first to fourth lumbar BMD in different gender, BMD was lower in women than men, and the difference was statistically significant ($P < .01$). Men normal bone mass accounted for 55.05% and bone loss accounted for 31.89%. OP accounted for 14.05% and women normal bone mass accounted for 15.38%. The detection rate of OP in female was higher than that in male. In 60–69 age group, the 70–79 age group and ≥ 80 age group, the difference of detection rate of osteoporosis was statistically significant ($\chi^2 = 9.62$, $P = .047$). The detection rate of OP increased with age. (3) The difference of SBP variability, DBP variability and dynamic pulse pressure were statistically significant ($P < .05$). Compared with osteoporosis group, SBP variability, DBP variability and dynamic pulse pressure value in normal bone mass and reduced bone mass group were reduced.

CONCLUSIONS: In elderly patients with hypertension, high BP variation, advanced age, and high Hcy were risk factors for OP, with high BP variation being the strongest. High BMI might be protective against OP in this patient population.

P230

Effects of Comprehensive Geriatric Assessment Intervention on Chinese Han Older Patients with Multiple Chronic Comorbidities

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BACKGROUND: From the beginning of the 21st century, older adults have become the fastest growing segment of the population, and aging has brought serious challenges to China. A large number of the elderly suffered from multiple chronic comorbidities, which has raised specific treatment challenges. Comprehensive geriatric assessment (CGA) as a well-established instrument was widely implemented to improve the care of the elderly.

OBJECTIVE: The aim of this study was to investigate the effects of CGA intervention for older patients with multiple chronic comorbidities.

METHODS: One hundred patients (age ≥ 65) with multiple geriatric conditions admitted in our hospital from October 2014 to June 2015 were recruited in our study and randomly assigned into CGA ($n = 50$) and control ($n = 50$) group. Geriatrics in control group received conventional therapy, while patients in CGA group were given CGA intervention and consultation therapy as well as conventional therapy. All patients received clinical health assessment before and after treatment. The assessment scales included instrumental activities of daily living (IADL), activities of daily living (ADL), Hamilton anxiety (HAMA), mini-mental state examination (MMSE), self-rating depression scale (SDS), arsons insomnia scale (AIS), mini nutrition assessment (MNA), Tinetti balance scale (TBS), and Tinetti gait scale (TGS). These scores were assayed and compared between the two groups.

RESULTS: Major chronic conditions observed in this study included hypermedication (96%), cerebral infarction sequela (93%), chronic bronchitis (81%), cardiomyopathy (73%), and others. The scores of SDS, AIS, and HAMA were significantly reduced after treatment and 3 months after treatment as compared with those before treatment in both CGA and control group ($P < .01$). The scores of IADL, ADL, MMSE, MNA and Tinetti scale were significantly higher after treatment relative to that before treatment in two groups ($P < .01$). After treatment and 3 months after treatment, the scores of SDS, AIS, and HAMA in CGA group were lower than that in control group ($P < .01$), while the scores of the others were higher relative to control group ($P < .01$).

CONCLUSIONS: Most of older patients were perplexed by multiple chronic comorbidities. The CGA intervention was shown to underpin the effective improvement of the later-life quality in elderly patients with multiple chronic comorbidities.

P231

Association between Hemoglobin A1c and Frailty in Elderly Community-dwelling Individuals

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BACKGROUND: Frailty, an early stage of disability, has proved to be a predictive factor of variety of poor outcomes in old populations. Studies have shown that higher Hemoglobin A1c (HbA1c) level was accompanied with high prevalence of frailty among old community-dwelling people. But the relationship between HbA1c level and prevalence of frailty still remains unknown.

METHODS: In this cross-sectional study, 20 communities were selected from total 60 communities which were covered by Pinyi Community Health Service Center by cluster sampling method and residents aged 65 years and over participated in this survey. Frailty was measured by "FRAIL" scale. HbA1c was surveyed by "A1CNow+" (German, Bayer). Demographic characteristics, medical history and drug use were collected by self-reporting. Multivariate logistic regression model was used to analyze the relationship of HbA1c level and frailty.

RESULTS: The prevalence of pre-frailty and frailty in the whole group (n = 668) was 60.63% and 6.74% respectively. Prevalence of frailty status in different HbA1c level was described as following: yze t (n = 517, pre-frail: frail = 61.51%:6.19%), 6.0–7.0% (n = 99, pre-frail: frail = 66.67%: 7.07%), 7.0–8.0% (n = 37, pre-frail: frail = 81.08%: 5.41%), ed as (n = 15, pre-frail: frail = 53.33%: 26.67%). Reference to normal HbA1c level (bA1c I, HbA1c levels of 7.0–8.0% and .0% le was associated with higher prevalence of pre-frailty ($P = .020$, OR = 3.151, 95% CI = 1.200–8.271) and frailty ($P = .014$, OR = 6.958, 95% CI = 1.486–32.587) respectively. After adjusting BMI, MNA-SF score, medical insurance and chronic diseases, high levels of HbA1c still have association with frailty status significantly.

CONCLUSION: Prevalence of pre-frailty was higher than that of frailty. Higher HbA1c levels were associated with pre-frailty or frailty status among community-living old persons.

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Energy Expenditure Increased While Fat Utilization Decreased during Exercise in Obese Male Adults

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OBJECTIVE: Increased physical activity is often recommended for obesity treatment, but the substrate utilization during exercise is different between individuals at the same exercise intensity. The aim of this study was to investigate the best intensity, which elicits maximal fat oxidation, determined using a cycle-ergometer during exercise in obese adults.

METHODS: The sample included 119 middle aged (52 ± 9 years old) males. A total of 32 were normal-weight; 65 overweight; and 22 obese. Energy consumption, respiratory exchange ratio and respective contribution (%) of fat and carbohydrate to total energy expenditure were determined using indirect calorimetry. A cycle-ergometer based protocol, including five workloads (10W, 20W, 30W, 40W and 50W), was used to describe fat oxidation kinetics and to determine the best intensity eliciting maximal fat oxidation (Fatmax).

RESULTS: With the increase of exercise intensity, energy expenditure (kcal/min) increased markedly ($r = 0.593$, $P < .001$) in total subjects. Further, the relationship between the energy expenditure and the body mass index (BMI) was analyzed during different workload exercise. It was found that with the increase of BMI, the energy expenditure increased at mild intensity (10W and 20W) ($r = 0.387$, $P < .001$; $r = 0.279$, $P = .003$), while the fat utilization significantly decreased ($r = -0.287$, $P = .018$; $r = -0.203$, $P = .034$). And it showed no relationship between the energy expenditure and the BMI at moderate intensity (30W, 40W and 50W). Furthermore, with the increase of BMI, the intensity of Fatmax also increased ($r = 0.373$, $P = .002$).

CONCLUSIONS: These results highlight that although energy expenditure increased, fat utilization decreased during exercise in obese male adults. Therefore, obese people need more intensive exercise to utilize body fat and lose weight.

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Beneficial Effect of Forest Bathing for Elderly Patients with Stable Chronic Obstructive Pulmonary Disease

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OBJECTIVE: The aim of this study was to investigate the effect of forest bathing as a natural therapy for elderly patients with stable chronic obstructive pulmonary disease.

METHODS: Eighteen elderly patients with stable chronic obstructive pulmonary disease were randomly divided into two groups. One group was sent to a broad-leaved evergreen forest to experience a 4-day trip, and the other group was sent to a city area in Hangzhou as control. Peripheral blood was collected and proportions of natural killer (NK), NKT-like and CD8⁺ T cells as well as perforin and granzyme-B expression in these cell populations were analyzed by flow cytometry. Pro-inflammatory cytokines and stress hormones were detected as well. Besides, profile of mood states (POMS) evaluation was used to assess the mood state change of subjects.

RESULTS: No significant change of indicators in the city group was observed after the experiment. In forest group, the proportion of CD8 T cells and NKT-like cells expressing perforin decreased significantly after experiment both compared to their baseline (CD8 T cells 3.02% vs 22.65%, $P < .001$; NKT-like cells 6.84% vs 36.22%, $P < .001$) and compared to city group (CD8 T cells 3.02% vs 10.90%, $P < .01$; NKT-like cells 6.84% vs 16.87%, $P < .05$). Similarly, the proportion of NK cells expressing perforin also decreased compared to baseline (54.32% vs 93.28%, $P < .001$). Subjects exposed to forest environment also showed a significant reduction of pro-inflammatory cytokines and stress hormones. Meanwhile, POMS evaluation showed that the scores in the negative subscales were decreased after forest bathing trip.

CONCLUSION: Taken together, our results indicated that forest bathing had beneficial effect on COPD patients by reducing the inflammation and stress level and thus improve their health condition.

P234**Relationship between Body Composition and Resting Energy Expenditure in Old People**

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OBJECTIVE: Aging is associated with changes in basal Energy Expenditure (BEE) and body composition. The aim of this study was to study the association between REE and body composition in old adults.

METHODS: Ninety-two old adults (age range: 66–89 years; men/women: 53/39) were included. The BMI range was 18.5–38.2 kg/m². REE and nonprotein respiratory quotient (npRQ) were detected by indirect calorimetry (Italy, COSMED). Body composition was determined by DEXA, and abdominal adiposity was determined by magnetic resonance imaging 3.0.

RESULTS: Increasing weight and body mass index (BMI) were associated with increasing BEE ($P < .05$), and was not correlated with BEE/kg both in man and women. While BEE increased with fat mass (FM) ($P < .05$) and FM percent total weight (FM %) ($P < .05$), BEE/kg decreased with the increment of FM ($P < .01$) and FM% ($P < .05$). In contrast, BEE/kg increased with the elevation of free fat mass (FFM) ($P < .01$) and FFM percent total weight ($P < .05$). The negative correlation between REE and trunk fat ($P < .01$) was more obvious compared with the fat of extremities ($P < .05$). And abdominal subcutaneous and visceral adipose tissue were both inversely correlated with BEE ($P < .05$), and there were no significant differences between subcutaneous and visceral adipose tissue. NpRQ was unrelated with weight, fat mass and free fat mass.

CONCLUSION: In elderly people, FM and FFM influenced BEE/kg inversely. Regard to body fat, trunk fat had more impact on BEE/kg than extremity fat. These findings require confirmation in prospective studies.

P235**Macrophage Migration Inhibitory Factor Levels in Obstructive Sleep Apnea Hypopnea Syndrom Patients and its Injury to Endothelial Cells**

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OBJECTIVE: This study aimed to investigate the expression of macrophage migration inhibitory factor (MIF) in obstructive sleep apnea hypopnea syndrom (OSAHS) patients and its injury to endothelial cells.

METHODS: Study participants ($n = 71$) were divided into control group ($n = 20$), mild OSAHS group ($n = 19$), moderate OSAHS group ($n = 15$), and the severe OSAHS group ($n = 17$). Fasting peripheral blood samples (about 4 ml) were collected in the morning and plasma macrophage migration inhibitory factor (MIF) levels were measured using standard ELISA assay. Peripheral blood mononuclear cell (PBMC) in control group and severe OSAHS patients were separated, and then PBMC was co-cultured with umbilical vein endothelial cells (HUVEC) 72 h. The apoptosis of HUVEC was detected by flow cytometry, and endothelial cell function index in the supernatant fluid such as ET - 1, NO, sICAM-1 and IL-6 was detected with ELISA method.

RESULTS: Plasma MIF levels in control group and mild, moderate and severe OSAHS patients were (25.65 ± 8.01), (27.55 ± 9.15), (31.42 ± 13.93), and (38.66 ± 14.66) ng/ml, respectively. The severe OSAHS group had increased MIF levels than the control group ($P < .01$), and by comparison with moderate, mild OSAHS group and the control group, there were no statistically significant differences ($P > .05$). The correlation between MIF and OSAHS clinical indicator: MIF with Apnea hypoventilation

index (AHI) ($r = 0.365$, $P = .008$) and Oxygen index reduction (ODI) ($r = 0.308$, $P = .308$) were positively correlated, but MIF and lowest blood oxygen ($r = 0.323$, $P = .323$) showed a negative correlation. Endothelial cell apoptosis rate in control group and severe OSAHS group was respectively (2.94 ± 1.018) %, (8.23 ± 3.013) %, and there was statistically significant difference between two groups, $P < 0.001$. ET - 1 test results in control group and severe OSAHS group was respectively (6.71 ± 5.52), (9.88 ± 4.79) pg/ml, and there was no statistically significant difference between two groups, $P = .141$. sICAM-1 test results in control group and severe OSAHS group was respectively (10.895 ± 7.641), (20.277 ± 6.786) ng/ml, with statistically significant difference between the two groups, $P = .014$. NO test results in control group and severe OSAHS group was respectively (35.174 ± 15.688), (25.085 ± 5.151) $\mu\text{mol/L}$, and there was no statistically significant difference between two groups, $P = .067$. IL-6 test results in control group and severe OSAHS group was respectively (219.557 ± 41.692), (436.331 ± 177.581) $\mu\text{mol/L}$, and there was statistically significant difference between two groups, $P < .001$.

CONCLUSION: MIF levels increased with the severity of disease and were significantly elevated in severe OSAHS patients. It may be involved in the development of OSAHS, and is closely related to the degree of disease severity and hypoxia. Apoptosis of endothelial cells in severe OSAHS patients was also higher than that of control group, and MIF might participate in the progress of injury.

P236**Characteristics of Vascular Access in the Elderly Patients on Maintenance Hemodialysis**

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OBJECTIVE: This study aimed to evaluate long-term patency and prevention of complications of vascular access in the elderly patients on maintenance hemodialysis (MHD).

METHODS: MHD patients were divided into two groups according to the age: the elderly group (≥ 65 years old, $n = 116$), and the young group (< 65 years old, $n = 130$). The types of vascular access and the characteristics of construction and use of the vascular access were analyzed.

RESULTS: The proportion of catheter was higher in the elderly group (44% vs 10%, $P < .01$), and women tend to use catheter regardless of the age. The rate of emergency dialysis was higher in the elderly patients (19% vs 5.4%, $P < .01$). The prevalence of diabetes, cardiovascular disease and peripheral vascular disease was significantly higher in the elderly, otherwise, the ratio of fistula and catheter was similar in the elderly patients combined with diabetes or peripheral vascular disease (50.7% vs 49.3%, $P > .05$). The levels of BMI, hemoglobin and albumin were significantly lower in the elderly catheter group ($P < .01$). In the elderly groups, fistula was established in elbow in 5 cases, significantly higher than that of the young group (8.3% vs 0.88%, $P < .001$). The primary patency of fistula was less than the young group, but the cumulative patency was similar after interventions between the two groups. The proportion of PTA or surgical treatment was higher than the young group ($P < .01$). The prevalence of disabilities of fistula mature was higher in the elderly (10% vs 3.5%, $P < .01$). The rate of the cuffed catheter was higher in the elderly group. There were no significant differences between the primary catheter patency and the cumulative catheter patency in the two groups. The incidence of catheter-related infections was increased in the elderly ($P < .05$).

CONCLUSIONS: Vascular catheters were used more frequently in the elderly patients. The proportion of catheter-related

infections was higher in the elderly patients than that in the young, but the patency of catheter was similar in both groups. A-V fistula in the elbow was constructed more in the elderly patient. In the elderly, the patency and the maturation of the fistula were inferior to the young, and patency became similar to that in the young after active interventions. This study suggests that age is an adverse factor and there are distinctions in the selection, construction and utilization of vascular access in the elderly MHD patients.

P237

NAADP Mediates Calcium Ion Concentration via Two Pore Channels in Pulmonary Artery Smooth Muscle Cells

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BACKGROUND: Pulmonary hypertension (PH) is associated with profound vascular remodeling and alterations in Ca^{2+} homeostasis in pulmonary arterial smooth muscle cells (PASMCs). Previous studies show that Nicotinic acid adenine dinucleotide phosphate (NAADP) is the most potent Ca^{2+} -mobilizing messenger that releases Ca^{2+} mediated by the two-pore channels (TPCs) TPC1 and TPC2 from endolysosomal organelles in rat PASMCs. We hypothesized that this signaling pathway also expressed in human PASMCs and was involved in the control of numerous physiological functions.

METHODS: We examined the expression of TPC1 and TPC2 mRNA in intralobar PASMCs with Quantitative Real-time RT-PCR, and with Fluo3 fluorescence probe measurement of free Ca^{2+} concentration in the cytoplasm.

RESULTS: RT-PCR showed that TPC1 and TPC2 were expressed in rat and human PASMCs, and TPC1 was the major subtype. Application of membrane-permeant NAADP acetoxymethyl ester to PASMCs elicited a concentration dependent increase in global $[\text{Ca}^{2+}]_i$ which was independent of extracellular Ca^{2+} and blocked by the NAADP antagonist Ned-19 or the vacuolar H^{+} -ATPase inhibitor bafilomycin A1.

CONCLUSIONS: These observations suggest that NAADP mediates complex Ca^{2+} interactions via TPCs between endolysosome and sarcoplasmic reticulum to regulate vascular reactivity and other cellular functions in both rat and human PASMCs. It indicates a new direction for further studies on the pathological roles of TPCs and NAADP in the vascular system.

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Quality Control Circle Activities on the Elderly Continuous Blood Purification Treatment

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AIMS: The aim of this study was to investigate the effect of quality control circle (QCC) activities on the improvement of the elderly continuous blood purification (CBP) treatment.

METHODS: A total of 49 elderly severe and acute patients from June 2013 to June 2014 in the first affiliated hospital of Nanjing Medical University were retrospectively analyzed. The subjects were divided into two groups: the control group with 27 cases, and the experimental group with 22 cases.

RESULTS: After QCC activities, compared with the control group, the waiting time (from receiving the alarm to the beginning

of CBP) was shortened from 55.6 ± 7.5 min to 24.8 ± 3.9 min ($P < .01$), and APACHE II scores were reduced from 26.6 ± 4.0 to 22.0 ± 3.5 ($P < .01$), and blood pressure and heart rate were improved. Mortality rate was lowered (37% vs 22.7%, $P < .01$).

CONCLUSION: QCC activities could improve the process and quality of the elderly CBP treatment, so as to save the time for emergency, to stabilize hemodynamics, and to reduce mortality.

P239

Factors Contributing to Hyperuricemia in Elderly Patients with Type 2 Diabetes

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BACKGROUND: This study aimed to evaluate factors contributing to hyperuricemia among Chinese patients who were over age 60 with type 2 diabetes.

METHODS: A total of 454 patients diagnosed with type 2 diabetes aged over 60 were studied. Data of physical examination, resting blood pressure, fasting plasma glucose (FPG), blood urea nitrogen (BUN), serum creatinine (Scr), serum uric acid (SUA), total cholesterol (TC), triglyceride (TG), high density lipoprotein cholesterol (HDL-C), low density lipoprotein cholesterol (LDL-C), glycosylated hemoglobin A1c (HbA1c) and 24 hours urinary albumin excretion (24 h-UAE) were collected. The prevalence of hyperuricemia and the risk factors associated with hyperuricemia were investigated.

RESULTS: In this study, hyperuricemia occurred in 18.1% (82/454) patients. Age, body mass index (BMI), serum triglyceride and 24 h urinary albumin excretion (24 h-UAE) were significantly higher and diabetic course, estimated glomerular filtration rate (eGFR) were significantly lower in hyperuricemia patients than those non-hyperuricemia subjects. Hyperuricemia was positively correlated with age, BMI, serum triglyceride and 24 h-UAE ($r = 0.115$, $r = 0.159$, $r = 0.179$, $r = 0.168$, respectively, all $P < .001$) and negatively correlated with diabetic course and eGFR ($r = -0.136$, $r = -0.274$, respectively, both $P < .001$). Hyperuricemia was more common in males than in females. Multiple stepwise regression analysis revealed that gender, BMI, serum triglyceride, 24 h-UAE and eGFR were independent risk factors for hyperuricemia.

CONCLUSIONS: Hyperuricemia is common in elderly patients with type 2 diabetes and gender, BMI, serum triglyceride, 24 h-UAE and eGFR are independent risk factors for hyperuricemia.

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Analysis of Factors Affecting Recurrence of Benign Paroxysmal Positional Vertigo in Elderly Patients

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BACKGROUND: The aim of this study was to analyze factors for the recurrence of benign paroxysmal positional vertigo (BPPV) in the elderly.

METHODS: A total of 59 elderly patients with BPPV who received diagnosis and treatment at our hospital, and experienced recurrence during a 24-month follow-up were selected as the recurrent group and 59 elderly patients without BPPV recurrence after treatment were randomly selected as the non-recurrent group. Factors, including age, gender, semicircular canal involvement, precipitating factors, time of recurrence, magnetic resonance imaging of

the head, hypertension, diabetes mellitus, hyperlipidemia, posterior circulation ischemia, and obstructive sleep apnea-hypopnea syndrome (OSAHS), were compared between the two groups, and their relevance to BPPV recurrence was analyzed using binary logistic regression.

RESULTS: The peak time of recurrence was 6–9 months after initial recovery. The mean age was (72.4 ± 7.8) years in the recurrent group and (70.8 ± 8.6) years in the non-recurrent group. There were 15 (25.4%) males and 44 (74.6%) females in each group, and the composition had no significant differences between the two groups ($t = 3.797$, $P = .052$; $\chi^2 = 0.000$, $P = .095$). Posterior semicircular canal involvement was common in both groups at the onset of BPPV and there was no significant difference ($\chi^2 = 7.918$, $P = .095$). In the recurrent group, recurrence affected the same semicircular canal in 31 patients (52.5%), a different semicircular canal on the same side in 16 patients (27.1%), and semicircular canals on the opposite side or both sides in 12 patients (20.3%). Possible precipitating factors for BPPV recurrence included insomnia, psychological factors, fatigue, migraine, upper respiratory infection and cough, with the first two showing the strongest association ($\chi^2 = 4.933$, $P = .026$). The combination of hyperlipidemia and posterior circulation ischemia increased the risk for BPPV recurrence ($OR = 5.822$, 95% CI : 2.317–14.629, $P = .000$; $OR = 1.078$, 95% CI : 1.044–8.267, $P = .041$). The incidence of lacunar infarction was significantly higher in the recurrent group than in the non-recurrent group ($\chi^2 = 5.446$, $P = .020$), and the infarcts were more often found in the frontal lobe ($\chi^2 = 4.259$, $P = .039$) and the parietal lobe ($\chi^2 = 8.859$, $P = .003$), which indicated that arteriolar sclerosis or occlusion might be a cause for BPPV recurrence. Patients generally visited the neurology or emergency medicine department at the initial episode and the first recurrence, but were usually referred to the otorhinolaryngology department without much delay in later recurrent episodes.

CONCLUSIONS: BPPV recurrence can be reduced by instituting social and psychological intervention and by mitigating precipitating factors. It is of great significance to control blood lipid levels effectively, to improve posterior circulation ischemia and to take targeted interventions in order to prevent BPPV recurrence in the elderly.

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Correlation between TNF- α -238 Gene Polymorphism and Hepatitis B Virus Reactivation in Patients with Malignant Tumors after Chemotherapy

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BACKGROUND: The aim of this study was to investigate the correlation between the single nucleotide polymorphism (SNP) of tumor necrosis factor (TNF- α) gene promoter-238 and hepatitis B virus (HBV) reactivation in patients with malignant tumors after chemotherapy.

METHODS: Polymerase chain reaction-restriction fragment length polymorphism assay (PCR-RFLP) was used to detect the SNP at TNF- α -238 site among 100 malignant tumor patients with HBV infection. HBV-DNA levels in patients were detected before and after chemotherapy. HBV reactivation was defined as that HBV-DNA level which was 10-fold greater than the increase compared with that before chemotherapy or 1×10^9 copy/ml higher.

RESULTS: The quantification of HBV-DNA was higher after chemotherapy than before chemotherapy [(3.02 ± 0.68) log copies/ml vs (2.49 ± 0.23) log copies/ml, $t = -7.383$, $P = .000$]. Among the patients with malignant tumor and HBV infection, the genotype frequency of G/A was higher in HBV reactivation group than in non-reactivation group after chemotherapy [27.3% (6/22) vs 3.8% (3/78), $\chi^2 = 11.499$, $P = .001$].

CONCLUSIONS: HBV reactivation is associated with TNF- α -238 gene polymorphism in malignant tumor patients with HBV infection after chemotherapy.

P242

Clinical Characteristics of Elderly Patients with Ulcerative Colitis

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BACKGROUND: This study aimed to analyze the clinical characteristics of elderly patients with ulcerative colitis (UC).

METHODS: A total of 166 cases with diagnosed UC were recruited. The clinical characteristics were compared between the elderly group ($n = 35$, ≥ 60 years) and the young group ($n = 131$, 20–59 years).

RESULTS: The number of male patients was more than that of female patients in both groups. The incidence of abdominal pain in the elderly group was significantly lower than that in the young group ($P < .05$). The number of severe patients in the older group was significantly lower than that of young group. Severe UC was 27.5% in the young versus 11.4% in the elderly ($P < .05$). The main disease location in the elderly was total colitis (31.4%), while the ratio was lower compared with that in the young group. However, the proctitis in the elderly was more common than in the young group. The extraintestinal manifestations and the ratio of receiving immune inhibitor and operation were no significant differences between the two groups.

CONCLUSIONS: Compared with the young group, there are a lower incidence of abdominal pain, a higher number of severe patients, a lower incidence of total colitis, and a higher incidence of proctitis in the elderly group.

P243

Application of the Streamlined Liner of the Pharynx Airway and the ProSeal Laryngeal Mask Airway in Elderly Patients Undergoing Laparoscopic Cholecystectomy

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BACKGROUND: This study aimed to compare the efficacy of the streamlined liner of the pharynx airway (SLIPA) and the proSeal laryngeal mask airway (PLMA) in airway management in elderly patients undergoing laparoscopic cholecystectomy.

METHODS: One hundred and twenty patients aged 60–75 years with American Society of Anesthesiologists (ASA) grade I or II undergoing laparoscopic cholecystectomy were randomly divided into three groups: the endotracheal intubation group (Group T, $n = 40$), the SLIPA group (Group S, $n = 40$) and the PLMA group (Group P, $n = 40$). Endotracheal intubation and laryngeal mask airway insertion were conducted after induction of anesthesia. All the patients were ventilated with intermittent positive pressure ventilation. Mean arterial pressure and heart rate were recorded before induction (T0), successful insertion (T1), ten minutes after pneumoperitoneum (T2), successful extubation (T3), and ten minutes after extubation (T4). The rate of successful placement at first attempt, time taken for placement, airway sealing pressure, and side effects were recorded. The incidences of post-extubation cough, backflow, aspiration, laryngospasm, bleeding, gastric distension and sore throat within 2 days after intubation were also recorded.

RESULTS: The successful rates of insertion were 92.5%, 92.5%, 95%, respectively, at first attempt and 100% at second attempt in the three groups ($\chi^2 = 0.268$, $P < .05$). Changes in emodynamic parameters were significantly different between Group T and

Group S or P after induction ($t = 4.076$, $P < .05$). Time taken for placement was shorter in Group S than in Groups T and P, which meant that the placement of SLIPA was easier than that of endotracheal intubation or PLMA ($t = 43.561$, $P < .05$). The airway sealing pressure was higher in Group P than in Group S, but had no statistically significant difference between the two groups ($t = 0.363$, $P > .05$). There was no significant difference in post-extubation complications, including backflow, aspiration, laryngospasm and gastric distension ($t = 0.321$, $P > .05$), among the three groups, but incidences of post-extubation cough and pharyngalgia were higher in Group T than in Group S and Group P ($\chi^2 = 26.674$, 10.568 , $P < .05$).

CONCLUSIONS: SLIPA and PLMA can both provide adequate ventilation during operation, with few complications. SLIPA placement was easier, while PLMA had good airway sealing and thus was more suitable for elderly patients undergoing laparoscopic cholecystectomy.

P244

Analysis of Cardiovascular Risk Factors Associated with Abdominal Obesity in Middle-aged to Elderly Women in Guangxi Nanning

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Supported By: This work was supported by a grant from National Natural Science Foundation of China (30972709, 81061120527).

BACKGROUND: This study aimed to investigate the cardiovascular risk factors associated with abdominal obesity in middle-aged to elderly women in South China.

METHODS: A total of 1,303 abdominal obesity females aged from 40 to 85 years old in Nanning, Guangxi province, were selected and screened via cross-sectional study. Medical history and physical examination, blood pressure, blood biochemistry analysis for all subjects were recorded. The waist circumference over 80 cm in female was considered as abdominal obesity. Principal component analysis was used to analyze the major cardiovascular risk factors associated with abdominal obesity.

RESULTS: The principal component analysis showed that the first, second, third and fourth principal component of the characteristic values were greater, and their cumulative rate was 65.43%. The first principal component was blood lipid profiles including TCH and LDL, and the variance was 21.80%. The second principal component was blood pressure (BP) including systolic BP and diastolic BP, and the variance was 18.44%. The third principal component was blood glucose especially fasting blood glucose, the variance was 13.51%. The fourth principal component was uric acid (UA) only, and the variance was 11.69%.

CONCLUSIONS: Dyslipidemia is the most important factor associated with female abdominal obesity in middle-aged to elderly in South China. Elevated blood pressure, blood glucose and uric acid also influence female abdominal obesity.

P245

Analysis of Risk Factors for Venous Thromboembolism in Elderly Patients with Lung Cancer

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BACKGROUND: This study aimed to investigate the risk factors for venous thromboembolism in elderly patients with lung cancer.

METHODS: A retrospective analysis was carried out in 869 elderly patients who were treated in our hospital from Mar. 2010 to Mar. 2014. And the venous thromboembolism and its related risk factors in elderly lung cancer patients were analyzed.

RESULTS: Thirty-five cases (4.35%) were complicated with venous thromboembolism. Multivariate logistic regression analysis showed that adenocarcinoma, basic diseases, and D-dipolymer ≥ 300 ng/ml belonged to the independent risk factors for the complication of venous thromboembolism in elderly patients with lung cancer ($OR = 2.839$, 1.586 and 10.514 , respectively, $P = .007$, $.022$ and $.000$).

CONCLUSIONS: The risk factors for the complication of venous thromboembolism should be monitored in the treatment of elderly patients with lung cancer. Early anticoagulation therapy should be performed to improve clinical effect and to reduce the incidence of complications.

P246

Medication Use for Secondary Prevention in Elderly Patients with Acute Myocardial Infarction

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BACKGROUND: This study aimed to investigate medication use for secondary prevention medications in elderly patients with acute ST-segment elevation myocardial infarction (STEMI) at hospital and after discharge.

METHODS: In this retrospective study, a total of 180 elderly survivors (113 male, 67 female) with STEMI were enrolled from August 2008 to July 2010 in Luhe Hospital. All the patients were followed up in clinic or by telephone from November 2010 to January 2011. The average follow-up period was 12.2 ± 5.1 (6–24) months. The influencing factors on the secondary prevention medications were analyzed.

RESULTS: Compared with the medication taken in hospital, a significant decline was observed in the use of aspirin (91.7% vs 98.3%), clopidogrel (90.6% vs 100%), β -blockers (66.7% vs 78.3%), ACEI/ARB (58.9% vs 77.2%) and lipid-lowering drugs (81.1% vs 95.6%) (all $P < .05$) during the period of follow up. The main influencing factors on medication application after discharge were the patients' poor compliance and doctors' failure of prescription in accordance with the guidelines.

CONCLUSIONS: There is a wide gap between guidelines and real world medication usage for secondary prevention in elderly patients with STEMI after discharge.

P247

Effects of JiaweiDitan Decoction on Oxidative Stress and Inflammatory Response in Elderly Patients with Obstructive Sleep Apnea Hypopnea Syndrome

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BACKGROUND: This study aimed to investigate the effects of JiaweiDitan Decoction on oxidative stress and inflammatory reaction in elderly patients with obstructive sleep apnea hypopnea syndrome (OSAHS).

METHODS: A total of 60 OSAHS patients with deficiency of phlegm and blood stasis mutual junction were enrolled, and randomly divided into treatment group ($n = 30$, receiving JiaweiDitan Decoction+ conventional treatment for 12 weeks) and control group ($n = 30$, receiving conventional treatment for 12 weeks). The

clinical symptoms integral, Epworth Sleepiness Scale (ESS), apnea hypopnea index (AHI), minimum oxygen saturation (SaO₂), superoxide dismutase (SOD), malondialdehyde (MDA), tumor necrosis factor (TNF)- α interleukin (IL)-6, and C-reactive protein (CRP) were observed before and after treatment.

RESULTS: The clinical symptoms integral, AHI value, minimum SaO₂, ESS score, levels of SOD, MDA, TNF- α , IL-6 and CRP were significantly improved in treatment group after treatment as compared with pretreatment [(11.65 $\pm$ 3.82) points vs (14.32 $\pm$ 4.25) points, (21.18 $\pm$ 12.37) times/h vs (29.16 $\pm$ 13.58) times/h, (83.24 $\pm$ 7.42)% vs (76.92 $\pm$ 11.91)%, (7.12 $\pm$ 4.84) points vs (10.01 $\pm$ 4.16) points, (99.24 $\pm$ 13.15)103 $\times$ U/L vs (87.511 $\pm$ 14.82) 103 $\times$ U/L, (8.56 $\pm$ 3.23) μ mol/l vs (11.25 $\pm$ 3.41) μ mol/l, (50.63 $\pm$ 10.57) ng/l vs (58.92 $\pm$ 11.65) ng/l, (78.12 $\pm$ 15.92) ng/l vs (89.13 $\pm$ 16.54) ng/l, (9.93 $\pm$ 5.25) mg/l vs (13.59 $\pm$ 4.92) mg/l, t = 2.559, 2.379, 2.467, 2.480, 3.243, 3.137, 2.887, 2.651, 2.786, respectively, all P < .05]. All the above indicators after treatment had significant differences between the treatment group and the control group [(11.65 $\pm$ 3.82) points vs (13.89 $\pm$ 4.45) points, (21.18 $\pm$ 12.37) times/h vs (28.03 $\pm$ 13.12) times/h, (83.24 $\pm$ 7.42)% vs (78.26 $\pm$ 10.15)%, (7.12 $\pm$ 4.84) points vs (9.56 $\pm$ 7.12) points, (99.24 $\pm$ 13.15)103 $\times$ U/L vs (90.13 $\pm$ 13.56) 103 $\times$ U/L, (8.56 $\pm$ 3.23) μ mol/l vs (10.86 $\pm$ 3.65) μ mol/l, (50.63 $\pm$ 10.57) ng/l vs (56.52 $\pm$ 11.04) ng/l, (78.12 $\pm$ 15.92) ng/l vs (87.81 $\pm$ 15.61) ng/l, (9.93 $\pm$ 5.25) mg/l vs (12.97 $\pm$ 5.03) mg/l, t = 2.092, 2.018, 2.169, 2.009, 2.642, 2.585, 2.111, 2.380, 2.290, respectively, all P < .05].

CONCLUSIONS: JiaweiDitan Decoction could improve clinical symptoms and suppressed oxidative stress and inflammatory reaction in elderly patients with OSAHS.

P248

Relationship between Hypertension and Cognitive Function in Community-Dwelling Elderly of Beijing

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BACKGROUND: This study aimed to investigate the relationship between hypertension and cognitive function in community-based elderly people aged 60 and over in Beijing.

METHODS: A total of 264 patients with hypertension and 271 cases with normal blood pressure were included in this study. The Montreal Cognitive Assessment (MoCA) and Mini-Mental Status Examination (MMSE) were used to determine cognitive change.

RESULTS: MMSE scores of total score showed no significant difference (27.98 $\pm$ 2.68 vs 28.37 $\pm$ 2.81, t = 1.634, P = .103). The subscores of memory were significantly decreased in hypertensive group (2.27 $\pm$ 0.84 vs 2.46 $\pm$ 0.76, t = 2.747, P = .006). The MoCA scores of total score, subscores of visuospatial and executive, language, delayed recall were significantly decreased in hypertensive group (24.61 $\pm$ 4.55 vs 25.61 $\pm$ 4.44, t = 2.572, P = .01; 3.78 $\pm$ 1.37 vs 4.04 $\pm$ 1.22, t = 2.355, P = .019; 2.22 $\pm$ 0.80 vs 2.39 $\pm$ 0.74, t = 2.698, P = .007; 2.39 $\pm$ 1.60 vs 2.66 $\pm$ 1.60, t = 1.978, P = .048).

CONCLUSIONS: Hypertension has adverse impact on cognitive function of elderly patients. MoCA is sensitive in clinical assessment of cognitive impairment in Chinese elders living in Beijing.

P249

Effect of Continuous Blood Purification for the Treatment of Cardiorenal Syndrome Type I

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BACKGROUND: The aim of this study was to evaluate the effect of continuous blood purification (CBP) on cardiorenal syndrome (CRS) type I.

METHODS: Clinical data of 42 patients with CRS type I at our hospital were collected from January 2012 to June 2014. We observed and compared changes in mean arterial pressure (MAP), heart rate, respiration rate, acute physiology and chronic health evaluation (APACHE) II score, and urinary volume before and 5 days after CBP. Meanwhile, levels of serum creatinine (Scr), cysteine proteinase inhibitor Cystatin C (CysC), serum creatinine (cTn) and B-type natriuretic peptide (BNP) were monitored. In addition, dynamic changes in cardiac index (CI), intrathoracic blood volume index (ITBI), global end-diastolic volume index (GEDI), central venous pressure (CVP), and extravascular lung water index (ELVWI) were monitored using the pulse induced contour cardiac output plus monitoring system (PiCCO plus), and changes in left ventricular ejection fraction (LVEF) before and 5 days after CBP was measured by color Doppler ultrasound.

RESULTS: There was no significant difference in MAP in patients with CRS type I before and 5 days after CBP (t = 1.7798, P = .0791). Tachycardia and tachypnea were improved, while urine volume was increased and the APACHE II score was decreased significantly, 5 days after CBP (t = 9.7507, 22.4307, 15.6590 and 4.6221, respectively, P < .05 for all). Plasma levels of Scr, CysC, cTn and BNP after treatment were lower than those before treatment [(126.75 $\pm$ 68.27) μ mol/L vs (413.63 $\pm$ 126.12) μ mol/L, (1.12 $\pm$ 0.76) mg/ml vs (4.14 $\pm$ 1.12) mg/ml, (2.63 $\pm$ 0.41) ng/ml vs (3.52 $\pm$ 0.72) ng/ml, (807.62 $\pm$ 427.71) pg/ml vs (3300.26 $\pm$ 567.62) pg/ml, respectively, P < .05 for all]. Myocardial contractility (CI and LVEF), cardiac preload (ITBI, GEDI, CVP) and lung related parameters (ELVWI) were also significantly improved after CBP (t = 7.1649, 31.4519, 5.2859, 5.9448, 12.9955, 4.6722, respectively, P < .05 for all).

CONCLUSIONS: CBP can reduce clinical symptoms of CRS type I and improve cardiac and renal function. It may serve as an important auxiliary measure for the treatment of patients with cardiorenal syndrome type I.

P250

Clinical Value of Electrocardiographic Poor R-wave Progression

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BACKGROUND: Poor R-wave progression (PRWP) is a common terminology of electrocardiogram (ECG) interpretation, but its value in the diagnosis of anterior myocardial infarction (MI) remains disputed. This study was to evaluate the value of PRWP in the diagnosis of anterior MI.

METHODS: Resting electrocardiograms (ECGs) of consecutive patients who underwent adenosine triphosphate (ATP) stress single photon emission computed tomography (SPECT) myocardial perfusion imaging (MPI) from 2000 to 2007 in Beijing Hospital were analyzed retrospectively. Four standard PRWP criteria namely Marquette, Zema, Warner and Depace as well as 10 new criteria from A to J were used to interpret ECGs. The detection rates of anterior MI between PRWP positive group and negative group were compared by using the χ^2 test.

RESULTS: Out of 1,003 subjects, 811 met the inclusion criteria. The detection rate of anterior MI in PRWP positive group (17.8–42.1%) was significantly higher than that in PRWP negative group (5.4–6.5%, all $P < .05$). But the sensitivity and positive predictive value of these criteria were relatively low. It was found that most patients with anterior MI had PRWP and ST-T changes at the same time, and patients without anterior MI only presented with PRWP and without ST-T changes.

CONCLUSIONS: ECG PRWP is common, especially in patients with anterior MI, and we should suspect anterior MI when ECG presents PRWP in combination with ST elevation and or T wave inversion in precordial leads.

P251

Value of Ischemia Modified Albumin in Elderly Patients with Acute Coronary Syndrome

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BACKGROUND: This study aimed to investigate the diagnostic value of ischemia modified albumin (IMA) in elderly patients with acute coronary syndrome (ACS).

METHODS: A total of 153 elderly patients suspected for ACS were included in this study as case group. Blood samples were obtained for the measurement of IMA, CK-MB and cTnT in all patients at emergency department and 12 hours (12 h) after. Routine check of 18-lead electrocardiogram (ECG), echocardiography, chest X-ray, abdominal ultrasound, coronary angiography were performed in all cases. Eighty healthy people were selected as the control group. The sensitivity and specificity of IMA, cTnT, ECG and the combination of the three indicators for early diagnosis of ischemia in elderly patients with ACS were separately evaluated. The value of IMA in stratification of elderly patients with ACS was also evaluated.

RESULTS: Of the 153 cases, 95 were diagnosed ACS [including 36 cases with acute ST elevation myocardial infarction (STEMI), 48 with acute non-ST elevation acute myocardial infarction (NSTEMI) and 11 with unstable angina (UA)], and 86 were found cTnT positive at 12 h. The sensitivity of IMA for cTnT positive at 12 h and the diagnosis of ACS were [95.3% (82/86) and 96.8% (92/95), respectively], and negative predictive value were [92.7% (51/55) and 94.5% (52/55), respectively]. The sensitivities and negative predictive values of cTnT [26.7% (23/86), 25.3% (24/95) and 50.8% (65/128), 44.5% (57/128)] and ECG [40.7% (35/86), 41.1% (39/95) and 52.3% (56/107), 47.7% (51/107)] at admission were poor. The sensitivity of combination of the three indicators was [98.8% (85/86) and 97.9% (93/95), respectively] and negative predictive value was [98.1% (50/51) and 96.1% (49/51), respectively]. IMA in ACS group was significantly higher than in non-ACS group ($P < .05$).

CONCLUSIONS: IMA can be used for early diagnosis and risk stratification in elderly patients with ACS.

P252

Analysis of Influencing Factors on Survival in Elderly Patients with Chronic Heart Failure

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BACKGROUND: This study aimed to evaluate influencing factors on survival in elderly patients with chronic heart failure.

METHODS: A total of 298 patients with chronic heart failure were enrolled in this study and 20 exposure factors were analyzed by univariate and multivariate COX model.

RESULTS: Univariate analysis showed that age, multiple etiologies, NYHA classification, NT-proBNP, LVEF, ACEI and beta-blocker using have significant influence on prognosis of patients with heart failure (RR was 3.007, 2.354, 2.283, 4.385, 1.381, 0.721 and 0.506, respectively, and 95% CI 1.589–5.675, 1.178–4.704, 1.534–4.617, 1.900–10.121, 1.573–3.212, 0.562–0.837 and 0.283–0.905, respectively, all $P < .05$). Multivariate analysis showed that age, multiple etiologies, NT-proBNP, beta blockers using have significant independent influence on survival of patients with heart failure (RR was 2.683, 2.953, 2.469 and 0.628 respectively, with 95% CI 1.344–5.355, 1.456–5.987, 1.032–5.912 and 0.463–0.958, respectively, all $P < .05$).

CONCLUSIONS: Age, underlying cause, NT-proBNP, and beta blockers usage were independent influencing factors of survival in elderly patients with chronic heart failure.

P253

Analysis of Cognitive Dysfunction in Elderly Patients with Chronic Heart Failure

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BACKGROUND: The aim of this study was to investigate cognitive dysfunction in elderly patients with chronic heart failure (CHF).

METHODS: A total of 127 patients aged ≥ 60 years were enrolled in this study and were divided into 3 groups: CHF group ($n = 53$, with CHF), non-CHF group ($n = 53$, with cardiovascular disease but without CHF) and control group ($n = 21$). Cognitive level of all patients was evaluated by using Montreal cognitive assessment scale.

RESULTS: There were 77.4% of the patients in CHF group, 53.7% in non-CHF group and 38.1% in control group who had cognitive impairment ($P < .05$). Cognitive score was significantly lower in CHF group than in other 2 groups (both $P < .05$). Cognitive dysfunction was deteriorated with the increase of NYHA classification or decrease of left ventricular ejection fraction.

CONCLUSIONS: Chronic heart failure may be one of the important factors aggravating cognitive impairment in the elderly.

P254

Detection of the Relationship between Ambulatory Blood Pressure and Blood Pressure Variability and Ankle Brachial Index in Elderly Men with Essential Hypertension

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BACKGROUND: This study aimed to investigate the relationship between ambulatory blood pressure and blood pressure variability and ankle brachial index (ABI) in elderly men with essential hypertension.

METHODS: Elder patients with essential hypertensive ($n = 160$) were divided into two groups according to the ABI. The ambulatory blood pressure monitoring was performed in all the patients.

RESULTS: Mean daytime pulse pressure [(62.4 $\pm$ 13.8) mmHg vs (53.0 $\pm$ 13.0) mmHg] and mean night pulse pressure [(67.3 $\pm$ 17.0) mmHg vs (55.4 $\pm$ 20.0) mmHg] and mean 24 h pulse pressure [(63.0 $\pm$ 16.0) mmHg vs (52.9 $\pm$ 13.6) mmHg] ($P < .05$), maximal night systolic blood pressure (SBP) [(146.5 $\pm$ 17.4) mmHg vs (135.5 $\pm$ 17.1) mmHg] and night SBP standard deviation [(12.4 $\pm$ 4.0) mmHg vs (10.1 $\pm$ 4.2) mmHg] ($P < .05$) of low ABI patients were all significantly higher than those of normal ABI patients.

CONCLUSIONS: Maximal night systolic blood pressure, night SBP standard deviation, mean night pulse pressure, mean daytime pulse pressure, mean 24 hour-pulse pressure may be risk factors for low ABI in elderly men with essential hypertension.

P255

Association between Plasma Amino-terminal Pro-B-type Natriuretic Peptide and Cardiac Structure, Function in Geriatric Inpatients

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BACKGROUND: This study aimed to evaluate the relationship between amino-terminal pro-B-type natriuretic peptide (NT-proBNP) and cardiac structure and function in geriatric inpatients and to assess the influences of age, pulmonary infection, hemoglobin, renal function on plasma NT-proBNP.

METHODS: A total of 148 consecutive inpatients were included in this cross-sectional study. According to ACC/AHA guidelines for heart failure, the patients were divided into normal control group, cardiac abnormality group and heart failure group. Plasma NT-proBNP level was measured by enzyme linked immunosorbent assay (ELISA). Cardiac structure and function were assessed by two-dimensional and Doppler transthoracic echocardiography.

RESULTS: Compared to normal control group, plasma NT-proBNP level was elevated in both cardiac abnormality group and heart failure group, but the difference was statistical significant only between heart failure group and normal control group ($P = .005$). Plasma NT-proBNP level was elevated with the increase of number of cardiac abnormalities parameters. Left atrial enlargement ($r = 0.251$, $P = .002$), valve diseases ($r = 0.242$, $P = .003$) and atrial fibrillation ($r = 0.260$, $P = .001$) were positively and ejection fraction ($r = -0.191$, $P = .020$) was negatively correlated with Plasma NT-proBNP level. Pulmonary infection ($r = 0.200$, $P = .015$) and coronary heart disease ($r = 0.188$, $P = .022$) were positively and hemoglobin ($r = -0.186$, $P = .024$) was negatively correlated with NT-proBNP level.

CONCLUSIONS: Compared to normal control group, plasma NT-proBNP level is elevated in both cardiac abnormality group and heart failure group. Atrial fibrillation, left atrial enlargement, valvular disease, left ventricular ejection fraction, pulmonary infection, hemoglobin and coronary heart disease are all correlated with plasma NT-proBNP level.

P256

The Role of Plasma Amino-terminal Brain Natriuretic Peptide Precursor and Left Ventricular Ejection Fraction in Evaluation of Chronic Congestive Heart Failure in the Elderly

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This work was supported by a grant from the Scientific Research Projects of Public Health Bureau in 2011 [Public Health Office of Bureau (2011)].

BACKGROUND: The aim of this study was to investigate the relationship between plasma amino-terminal brain natriuretic peptide precursor (NT-proBNP), left ventricular ejection fraction (LVEF) and New York Heart Association (NYHA) classification in elderly patients with chronic congestive heart failure.

METHODS: From January 2006 to October 2011, 308 patients (aged more than 60 years) with chronic congestive heart failure

were enrolled and divided into systolic heart failure group (SHF group, LVEF $<50\%$, $n = 210$) and diastolic heart failure group (DHF group, LVEF $>50\%$, $n = 98$). One hundred and fifty patients without structural heart disease were selected and hospitalized during the same period as control group. Plasma NT-proBNP was measured by electrochemistry luminescence immunoassay. Color Doppler echocardiography was performed to evaluate left ventricular structure, systolic function (LVEF) and diastolic function (Ea/Aa).

RESULTS: Plasma NT-proBNP level was positively correlated with NYHA classification ($r = 0.862$, $P < .01$). LVEF was negatively correlated with NYHA grading ($r = -0.614$, $P < .05$). And plasma level of NT-proBNP was not correlated with LVEF ($r = -0.136$, $P = .09$). Plasma NT-proBNP was negatively correlated with LVEF ($r = -0.735$, $P = .008$) in SHF group, and positively correlated with Ea/Aa ratio in DHF group ($r = 0.614$, $P = .009$).

CONCLUSIONS: Both plasma NT-proBNP levels and LVEF can reflect the cardiac function in elderly patients with chronic heart failure. LVEF combined with plasma NT-proBNP may assist in reaching the diagnose of diastolic heart failure in the elderly.

P257

Relationships between Serum Resistin and Blood Lipids Levels in Old Male Patients

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BACKGROUND: This study aimed to investigate the correlation between serum resistin and blood lipids in old male patients.

METHODS: A total of 220 male patients over 80 years old were selected and divided into hyperlipidemia group ($n = 80$) and normal lipid group ($n = 140$) according to their blood lipids level. The serum resistin concentration was measured by ELISA.

RESULTS: Serum resistin level was significantly higher in hyperlipidemia group than in normal lipid group [(7.16 ± 0.77) ng/ml vs (6.21 ± 0.63) ng/ml, $P < .05$]. Serum resistin level was negatively correlated with TC, LDL-C, HDL-C ($\gamma = -0.231, -0.254, -0.243$, all $P < .01$), and positively correlated with TG ($\gamma = 0.127$, $P < .01$). Resistin was independently correlated with HDL-C after controlling the age and medical history ($P = .017$).

CONCLUSIONS: Serum resistin levels were higher in old male patients with hyperlipidemia and may be related to the disorder of blood lipid.

P258

Preventive Effect of Lung Recruitment Maneuver on Pulmonary Complications in Elderly Patients after Off-pump Coronary Artery Bypass Surgery

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BACKGROUND: The aim of this study was to evaluate the role of lung recruitment maneuver in preventing pulmonary complications for elderly patients after off-pump coronary artery bypass surgery.

METHODS: A total of 106 patients aging above 60 years who underwent primary off pump coronary artery bypass grafting from Jan. 2012 to Dec. 2012 were enrolled in this research, including 82 males, 24 females, ranging from 60 to 70 years old (mean 66 ± 4) in age. The patients were randomly divided into mechanical

ventilation group (control) and lung recruitment maneuver group (treatment group). The control group received routine mechanical ventilation therapy while the treatment group received twice lung recruitment maneuver respectively after back in ICU 1 h and 2 h except the routine management. The oxygen index (OI), ventilation time and the complications such as hypoxemia, pulmonary atelectasis and pulmonary infection were observed and compared between the two groups. The hypoxemia was defined as $OI < 200$ and the delay of extubation was defined as ventilation time ≥ 24 h.

RESULTS: No patients died in the hospital. The treatment group was superior to control group in OI at 3 h after in ICU and the moment weaning from respirator (272 ± 45 vs 251 ± 48 and 287 ± 32 vs 262 ± 41 , both $P < .01$). The treatment group was also superior to control group in the morbidity of hypoxemia (16 cases, 30.2% vs 2 cases, 3.8%, $P = .0003$), pulmonary infection (4 cases, 7.5% vs 0 cases, 0%, $P = .041$), delay of extubation (9 cases, 17% vs 2 cases, 3.8%, $P = .026$), noninvasive mechanical ventilation after weaning from respirator (12 cases, 22.6% vs 3 cases, 5.7%, $P = .012$) and pulmonary atelectasis and pulmonary infections. There were no patients who ceased LRM because of unstable hemodynamics.

CONCLUSIONS: LRM in early period of post-operation could improve oxygenation in elderly patients after OPCABG, shorten ventilation time, and reduce the morbidity of hypoxemia.

P259

Ischemic Stroke and Bleeding Risk in Patients with Non-valvular Atrial Fibrillation Treated with Warfarin

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BACKGROUND: The aim of this study was to evaluate the ischemic stroke and bleeding risk in patients with non-valvular atrial fibrillation treated by different dosage of warfarin.

METHODS: One hundred and seventy seven non-valvular atrial fibrillation patients, older than 65 years old, were divided into high risk bleeding group ($HAS-BLED \geq 3$, $n = 103$), and low bleeding risk group ($HAS-BLED < 3$, $n = 74$). Patients in the high-risk group were treated with low intensity of anticoagulation, with international normalized ratio (INR) between 1.5 and 2.5. And the low-risk group was given moderate-intensity anticoagulation with INR in 2.0–3.0. The major/minor bleeding and ischemic stroke risk between the two groups were observed and recorded.

RESULTS: The median follow-up time of the 177 patients was 16.9 (6.3–30.9) months. Eight cases of minor bleeding events occurred during follow-up period in high-risk group, with the incidence rate of 5.5% (person* year). Five cases were in low-risk group, with the incidence rate of 4.7% (person* year). No statistical difference of minor bleeding risk was observed between the two groups ($HR = 1.88$, $P = .15$). No case of serious bleeding events during follow-up period was found. 2 ischemic stroke events were observed in high-risk group (1.4%) and 1 in low-risk group (0.9%) respectively, which demonstrated that there was no statistical difference of ischemic stroke risk between the two groups ($HR = 0.89$, $P = .94$).

CONCLUSION: The risk of ischemic stroke and bleeding in patients with non-valvular atrial fibrillation treated by warfarin can be significantly reduced by different anticoagulation intensity according to HAS-BLED score.

P260

Changes in the Number and Function of Circulating Blood Endothelial Progenitor Cells in the Elderly Patients with Acute Myocardial Infarction and its Effects on Cardiac Function

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BACKGROUND: This study aimed to explore the mechanism and its clinical significance of the effects of endothelial progenitor cells (EPCs) in the elderly patients with acute myocardial infarction (AMI) on infarcted heart repair through comparison of the number and function of EPCs between different aged patients with AMI and to analyze variation of Akt and SDF-1 expression.

METHODS: Arterial blood samples (8–10 ml) were extracted from elderly patients with AMI ($n = 26$) and the young and middle-aged AMI patients ($n = 24$). Circulating blood endothelial progenitor cells (EPCs) was evaluated by flow cytometry analysis (FACS). The expressions of Akt and SDF-1 in EPCs were determined by enzyme-linked immunosorbent assay (ELISA). MI was induced in the inbred Lewis rats by left anterior descending artery ligation. $1 \times 10^7/200$ ml of EPCs from the elderly MI patients or the young and middle-aged AMI patients or equal volume of saline were immediately injected into the MI rats through the tail veins. At 28 days of post-operation, cardiac function was measured by echocardiography. The number of EPCs was observed in the infarcted heart under a fluorescence microscope. The vascular density was examined by immunofluorescence (IF).

RESULTS: FACS revealed that the number of circulating blood EPCs in the elderly AMI patients was significantly smaller than in the young and middle-aged AMI patients ($47.23\% \pm 14.92\%$ vs $89.76\% \pm 7.27\%$, $P < .001$). ELISA showed that the expression levels of Akt and SDF-1 were notably lower in the elderly AMI patients than in the young and middle-aged AMI patients (Akt: $19.04 \pm 6.41\%$ vs $43.96 \pm 15.91\%$; SDF-1: $25.81 \pm 6.32\%$ vs $64.04 \pm 16.35\%$, both $P < .001$). After these cells were transplanted into the infarcted rats, fluorescence microscopic examination showed that the number of the engrafted elderly EPCs homing into the infarcted hearts was significantly less than that of the engrafted young and middle-aged EPCs [$(3.69 \pm 1.97) \text{ mm}^2$ vs $(12.01 \pm 5.44) \text{ mm}^2$, $P < .001$]. IF showed that the vascular density was evidently smaller in the rats receiving the elderly EPCs therapy than in the rats receiving the young and middle-aged EPCs therapy ($42 \pm 9/\text{mm}^2$ vs $96 \pm 15/\text{mm}^2$, $P < .001$). Echocardiography showed that the cardiac function indexes (LVEF and LVFS) were significantly lower in the rats receiving the elderly EPCs therapy than in the rats receiving the young and middle-aged EPCs therapy (LVEF: $58.1 \pm 5.0\%$ vs $73.8 \pm 7.9\%$; LVFS: $35.4 \pm 3.8\%$ vs $59.0 \pm 7.6\%$, $P < .001$).

CONCLUSIONS: The number and function of circulating endothelial progenitor cells in elderly patients with MI were lower and impaired, respectively compared with those in young and middle-aged, which may be associated with impairment in Akt-SDF-1 signaling pathway.

P261**Gender Differences of In-hospital and Follow-up Outcomes in Elderly Patients Undergoing Percutaneous Coronary Intervention**

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BACKGROUND: This study aimed to evaluate gender differences of the short-term and 1-year outcomes in elderly patients undergoing Percutaneous Coronary Intervention (PCI).

METHODS: A total of 1,105 elderly patients undergoing PCI were consecutively recruited. Among them, 739 were males, 366 were females. The clinical features, procedure outcomes and clinical prognosis were compared between the two groups.

RESULTS: The prevalence of smoking (20.3% vs 5.7%, $\chi^2 = 39.670$, $P < .001$), prior myocardial infarction (27.9% vs 19.9%, $\chi^2 = 8.156$, $P < .001$) as well as the proportion of abnormal serum creatine (20.2% vs 10.9%, $\chi^2 = 14.718$, $P < .001$) were significantly higher in men than in women. The prevalence of hypertension (73.2% vs 65.8%, $\chi^2 = 6.284$, $P < .01$) as well as the proportion of anemia (20.5% vs 15.8%, $\chi^2 = 3.702$, $P = .054$) was significantly higher in women than in men. The rates of intravascular ultrasound (5.8% vs 1.9%, $\chi^2 = 8.645$, $P < .001$) and radial access (79.6% vs 73.2%, $\chi^2 = 5.642$, $P = .02$) were significantly higher in men than in women. The rate of in-hospital NACE was significantly higher in women than in man (7.7% vs 3.9%, $\chi^2 = 6.946$, $P < .001$), mainly driven by the higher rate of access site-related major bleeding (3.0% vs 0.5%, $\chi^2 = 11.100$, $P < .01$). No significant difference was detected of 1 year MACE between men and women (6.2% vs 7.9%, $\chi^2 = 1.117$, $P = .29$), and the rates of death, myocardial infarction and target vessel revascularization were also similar between men and women (all $P > .05$).

CONCLUSIONS: The in-hospital NACE was significantly higher in elderly women compared with elderly men who had undergone PCI, which was mainly driven by the difference of access site-related major bleeding. Proper management of the access site bleeding is an effective way in improving PCI quality in elderly women.

P262**Epidemiologic Features of Sarcopenia and its Related Indexes in Elderly Population in Beijing**

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BACKGROUND: This study aimed to investigate the epidemiologic features of sarcopenia and its related indexes in elderly population in Beijing.

METHODS: A cross-sectional study was conducted by interviewing the elderly population in Beijing with the standard questionnaire including demographic information, history of diseases, lifestyle, etc. Body muscle mass was measured by using a bioelectrical impedance analyzer (BIA). Physical function was evaluated by the subjects' 6-m gait speed (GS), and muscle strength was determined by handgrip strength (HS).

RESULTS: Muscle mass declined with aging in the elderly ($F = 6.77$, $P = .0013$), and the elderly handless muscle mass in the urban than in the rural ($t = 4.32$, $P = .038$). HS and GS significantly declined with aging ($F = 7.62$ and 15.79 , $P = .0005$ and 0.000), especially in females and the rural elderly. Logistic regression showed

that aging and female were associated with the high risk of HS decline (OR = 1.13, 1.92, both $P < .05$), while the urban elderly had lower risk of HS decline (OR = 0.10, $P < .05$). Aging, high body mass index, irregular lifestyle were positively related to GS decline (OR = 1.12, 1.25 and 2.77 respectively, all $P < .05$). The risk of GS decline was lower in the urban elderly having physical activity and hobbies (OR = 0.08, 0.27 and 0.11 respectively, all $P < .05$).

CONCLUSIONS: Sarcopenia-related indexes are significantly associated with age and have population-specific difference. Positive and healthy lifestyle contribute to maintain muscle function in the elderly.

P263**Accuracy of Symptom Recall of Functional Constipation in Elderly Patients**

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BACKGROUND: This study aimed to evaluate accuracy of symptom recall of functional constipation (FC) and its potential influencing factors in elderly patients.

METHODS: All enrolled patients were ≥ 60 years old, and met Rome III FC diagnostic criteria. At the beginning of study, all subjects reported the duration of constipation, frequency and severity of 6 constipated symptoms during the past 3 months (recall symptoms); then they took diary for 2 weeks (diary symptoms). Symptom score was evaluated by the independent investigators according to the unified and standard forms. The consistency of recalled symptoms and diary symptoms, and the potential influencing factors for the accuracy of recalled symptoms were analyzed.

RESULTS: A total of 115 patients were enrolled in this study, with average age of (70.1 ± 6.5) years old. The overall scores of recall symptoms and diary symptoms were (27.6 ± 7.4) and (26.1 ± 7.6) respectively. The overall score of recalled symptoms had a good consistency with the overall score of diary symptoms [equation of linear regression intercept 0.437 (95%CI: -1.917–2.790), regression coefficient 0.930 (95% CI: 0.848–1.013)], and the good consistency were found in the score of each constipation symptom (all $\kappa > 0.75$). Gender, age, labor intensity and economic situation influenced the accuracy of symptom recall in FC patients (all $P < .05$), but education level, duration and severity of constipation, subtype of FC were not the influencing factors for the accuracy of symptom recall (all $P > .05$).

CONCLUSIONS: The accuracy of symptom recall is high in elderly patients with FC, and gender, age, labor intensity and economic situation might influence the recall accuracy of individual constipation symptoms.

P264**Efficacy of Green Tea Extract on Weight Control in Metabolic Syndrome: a Meta-analysis**

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BACKGROUND: This study aimed to systematically review efficacy of green tea extract (GTE) on weight control in metabolic syndrome (MS).

METHODS: Related randomized controlled trials (RCTs) about GTE and MS were searched and identified from PubMed,

Scopus, PsycINFO, EMBase, the VIP database, CNKI, and Wanfang Resource Database. Valid data were extracted according to strict inclusion and exclusion criteria. The RevMan 5.2.2 software was used to carry out the meta-analysis.

RESULTS: A total of 10 trials involving 626 patients were included. Meta-analysis showed that: (1) compared with the placebo group, body weight (SMD = -0.19, 95% CI -0.34 to -0.03, $P = .02$) and waist circumference (SMD = -0.19, 95% CI -0.36 to -0.02, $P = .03$) were significantly decreased in the GTE group; 2) no significant difference was found between the two groups in BMI (SMD = -0.14, 95% CI -0.31 to 0.03, $P = .10$), hip circumference (SMD = 0.15, 95% CI -0.11 to 0.42, $P = .26$) or waist-to-hip ratio (WMD = -0.02, 95% CI -0.05 to 0.01, $P = .19$).

CONCLUSIONS: Green tea extract can significantly reduce body weight and waist circumference in patients with metabolic syndrome.

P265

Application of Ultrasonography in Differential Diagnosis of Benign and Malignant Thyroid Nodules in the Elderly

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BACKGROUND: The aim of this study was to investigate clinical application of ultrasonography in differential diagnosis of benign and malignant thyroid nodules in the elderly.

METHODS: Ultrasonography data of 117 elderly patients with thyroid nodules confirmed by surgery and pathology were analyzed retrospectively from Jan. 2013 to Feb. 2015. Their sonogram features were summarized.

RESULTS: Among 117 cases, there were 85 cases (72.6%) with benign thyroid nodules and 32 cases (27.3%) with malignant thyroid tumors according to pathological diagnosis. The accordance rate of ultrasonography for diagnosing the benign and malignant thyroid nodules was 88.03%, and only 14 cases were misdiagnosed.

CONCLUSIONS: Ultrasonography was useful in differential diagnosis of benign and malignant thyroid nodules in the elderly.

P266

Application of Extracorporeal Cardiac Shock Wave Therapy in Treating Coronary Heart Disease in the Elderly

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BACKGROUND: The aim of this study was to evaluate efficacy and safety of cardiac shock wave therapy (CSWT) for coronary heart disease in elderly patients.

METHODS: Eleven patients with refractory angina pectoris were enrolled. The ischemic area was determined by the 99mTechnetium-MIBI single-photon emission computed tomography (SPECT) and coronary angiography, and treated with CSWT. The CSWT was performed during treatment of 3 months, with 9 times totally. The clinical evaluations included the clinical symptoms, Canadian Cardiovascular Society (CCS) class scores, New York Heart Association class (NYHA), Seattle angina questionnaire (SAQ), 6-min walking distance and the use of dosage of nitroglycerin, left ventricular end diastolic diameter (LVEDD) and left ventricular ejection fraction (LVEF). The amelioration of ischemic myocardial was analyzed by SPECT. The safety of CSWT was evaluated by changes in clinical symptoms, ECG monitoring, blood pressure and oxygen saturation, blood levels of creatine kinase, creatine kinase-myocardial band isoenzymes (CK-MB), troponin T (TnT), alanine transfer enzyme (ALT),

creatinine (CR), brain natriuretic peptide and high sensitive C reactive protein (hs CRP).

RESULTS: The CCS class scores, NYHA, 6-min walking distance and the dosage of nitroglycerin were significantly improved at 4 and 12 months after treatment as compared with pretreatment. The steady state of angina pectoris and the frequency score in SAQ were significantly improved. The 21 cardio-vessel segments were treated, and the improvement rate of resting myocardial perfusion was 6.2% (6/13). The effective rate was 38.5% (5/13), and the obvious effective rate was 7.8% (1/13) after treatment. The improvement rate of loaded myocardial perfusion was 57.1% (12/21), the effective rate was 47.6% (10/21), and the obvious effective rate after treatment was 9.5% (2/21). There were no significant changes in levels of CK, CK-MB, TNT, ALT, Cr, BNP and hs CRP, heart rate, systolic blood pressure, diastolic blood pressure and oxygen saturation after treatment as compared with pretreatment.

CONCLUSIONS: CSWT is a safe and effective treatment for coronary heart disease in the elderly, and its therapeutic effect could maintain for at least one year.

P267

Association between Serum Triiodothyronine Level and Coronary Artery Disease in Elderly Patients with Low Triiodothyronine Syndrome

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BACKGROUND: This study aimed to investigate the relationship between serum triiodothyronine (T3) levels and coronary artery lesions in elderly patients with coronary heart disease combined with three low T3 syndromes.

METHODS: A total of 986 cases of low T3 syndrome in elderly patients with coronary artery disease were defined as group A (low T3 group), and 1,074 cases with normal free triiodothyronine (FT3), free thyroxine (FT4), and thyroid stimulating hormone (TSH) in elderly patients of coronary heart disease were assigned to group B (control group). The levels of FT3, FT4, and TSH in two groups were determined. The level of coronary artery disease was evaluated by lesion count and Gensini score according to the results of coronary arteriography analysis.

RESULTS: Group A had higher age than the group B ($P < .05$). The levels of total cholesterol and LDL-C were higher in group A than in group B ($P < .05$). Serum FT3 values, in group A was lower than that of group B [(1.0 ± 0.6) pmol/L vs (4.1 ± 0.5) pmol/L, $P < .05$], serum FT4 and TSH were no significant different between the two groups. Group A's total gensini score was higher than group B [(43.5 ± 18.3) vs (24.5 ± 10.1) , $P < .05$]. Vessel disease incidence was higher in the low T3 group than that of control group ($P < .05$). In the low T3 group, severe coronary artery stenosis (arbitrary 1 of 3 major vascular stenosis in excess of 75%) and complete occlusion (arbitrary 1 of 3 major vascular occlusions) were higher compared with group B ($P < .05$). Multivariate logistic regression analysis showed that age and low FT3 were related to number and degree of coronary artery lesions.

CONCLUSIONS: The lesions of coronary heart disease in the elderly patients with low T3 syndrome were mainly multivessel and two-vessel involving a wide range and higher severity.

P268**Analysis of Causes of Death in Elderly Patients Undergoing Maintenance Hemodialysis**

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BACKGROUND: The aim of this study was to analyze the causes of death in elderly patients undergoing maintenance hemodialysis (MHD).

METHODS: A total of 105 patients aged ≥ 60 years undergoing MHD treatment for more than 3 months in 2010 were included with complete basic registration information in this retrospective cohort study. The deadline of the follow-up was 30 Jun. 2013.

RESULTS: Among 105 patients, the median age was 69 years and the median dialysis duration was 27 months. 33 patients died during the follow-up period. The main causes for the death were cardiovascular disease (11 cases, 33.3%), infection (7 cases, 21.2%) and cerebrovascular disease (5 cases, 15.2%). Age and the proportion of patients with cardiovascular diseases and peripheral vascular calcification score were higher, and the proportion of hypertension patients was lower in death group than in survival group. After adjusting for age, body mass Index, kt/v, and calcification score, COX regression analysis indicated that complications with cardiovascular disease (HR = 2.143, 95% CI: 1.010–4.613, $P = .049$), without hypertension (HR = 0.293, 95% CI: 0.097–0.889, $P = .03$), and with lower serum level of albumin (HR = 0.838, 95% CI: 0.748–0.904, $P = .002$) were the main risk factors for mortality in elderly patients on MHD.

CONCLUSIONS: The main causes for death in elderly MHD patients were cardiovascular disease, infection and cerebrovascular disease. Malnutrition and comorbid cardiovascular disease might be the main risk factors for mortality in this elderly patient population.

P269**Clinical Effect of Minimally Invasive Transforaminal Lumbar Interbody Fusion Combined with Unilateral Pedicle Fixation on Elderly Lumbar Degenerative Diseases**

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BACKGROUND: The aim of this study was to analyze the clinical effect of minimally invasive transforaminal lumbar interbody fusion (MI-TLIF) in combination with unilateral pedicle fixation on lumbar degenerative diseases in the elderly.

METHODS: Between June 2011 and June 2013, 51 elderly patients with lumbar degenerative diseases who underwent MI-TLIF in combination with unilateral pedicle fixation, including 39 males and 12 females, aged from 60 to 72 years (average 65 years old), were included in this study. All the 51 cases had mono-segmental lesion, including 11 cases with the lesion in L3, 4, 22 cases with the lesion in L4, 5, and 18 cases with the lesion in L5–S1. The operation time, blood loss, postoperative drainage and complications were recorded. Postoperative outpatient follow-ups were conducted once every 2–3 months for all cases, and JOA scoring criteria (29 points) of lower back pain were adopted to evaluate the clinical effects. Thin-layer helical CT scanning reconstruction was applied to evaluate the fusion rate of bone grafting. X-ray film of lumbar vertebra AP & LAT was employed to evaluate the intervertebral space height, the internal fixation and the position of cage.

RESULTS: The operation time ranged between 50–90 min with a mean value of 65 min; blood loss ranged between 80–180 ml

(average 110 ml); and postoperative drainage volume ranged between 20–70 ml (average 40 ml). Healing phase I was achieved in 49 cases, while post-operative local cutaneous necrosis around the incision site occurred in 2 cases and the wound healed well after wound dressing. Dural laceration without nerve root injury occurred in 2 cases. The follow-up in 51 patients ranged between 18–24 months (average 20 months). In terms of JOA scoring, the preoperative score was (9.7 ± 1.6) , the post-operative score was (21.4 ± 1.4) 7 d after operation and the score was (25.4 ± 1.2) in the final follow-up. During the final follow-up, 35 cases were evaluated as excellent, 11 cases as good and 5 cases as moderate. The excellent and good rate was 90.2%, and the fusion rate of bone grafting was 88.2%. The intervertebral space height ranged from (9.2 ± 1.5) mm before operation to (11.2 ± 1.3) mm 7 d after operation, to (11.0 ± 1.2) mm in the final follow-up. There were significant differences in intervertebral space height before and after operation (both $P < .05$), and no difference was found between 7d after operation and the final follow-up ($P > .05$). Looseness, fracture or cage displacement was not observed.

CONCLUSIONS: The clinical effect of MI-TLIF in combination with unilateral pedicle fixation on lumbar degenerative diseases in the elderly is satisfactory. It has the advantages of high fusion rate, less trauma, quicker recovery and is cost-effective and convenient. However, the indications for operation should be restricted and the long-term curative effect remains to be evaluated in further clinical studies.

P270**Present Situation on Blood Pressure Control Rate and Antihypertensive Drug Use in the Elderly Coronary Heart Disease Patients with Diabetes Mellitus and Hypertension**

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OBJECTIVE: This study aimed to investigate the present situation on blood pressure control rate and antihypertensive drug use in elderly patients with coronary heart disease (CHD), diabetes mellitus (DM) and hypertension (HP).

METHODS: In this multicenter, non-intervention and cross-sectional survey, the elderly coronary heart disease patients with diabetes mellitus and hypertension aged 60 years or over were recruited from 165 hospitals in 21 provinces or cities across China from April to July 2011. Antihypertensive therapy, blood pressure control rate in different target value, ways of antihypertensive drugs and standard treatment recommended by guideline were investigated in the study.

RESULTS: A total of 1,379 elderly patients with coronary heart disease, DM and HP were selected from 7,962 cases of elderly patients with CHD (accounting for 17.3%). (1) the control rate of blood pressure was 17.1% for antihypertensive target value $<130/80$ mmHg; the success rate for achieving the target value $<140/80$ mmHg was only 27.5%; the control rate was 39.6% for the target value $<140/90$ mmHg; the success rate for achieving the target value $<150/90$ mmHg was 51.7%; the control rate of the elderly patient aged 70 years or over gradually increased with increasing of age; success rate of elderly patient older than 85 years was the highest, whereas success rate of elderly patient aged 71–71 years old was the lowest; success rate in male patients was close to that of female patients, and success rate for men was slightly higher than those in women; (2) Of the 1,379 subjects, 1,317 effective interviewed cases received antihypertensive therapy (effective interviewed rate was 97.8%); the mostly used drugs were angiotensin receptor blockers (ARB)/angiotensin converting enzyme inhibitors (ACEI) utilization rate (usage rate was 76.8%), followed by calcium channel blockers (CCB) (utilization rate was 65.5%), beta-

blockers (usage rate was 44.6%), thiazide diuretics (usage rate was 26.3%) respectively; (3) combination of two drugs was the most common way in antihypertensive medication (accounting for 41.2%); other ways were three drugs combination (accounting for 28.9%), method of single drug (23.9%); CCB were the common drugs in the way of single drug (accounting for 8.8%), and CCB combined with ARB were the most common way in combination of two drugs (11.7%); CCB combined with ARB and β -blockers were frequently used in combination of three drugs (9.2%); (4) 987 cases of patients received standard treatment recommended by guideline (accounting for 76.6%); the rate of standard usage in combination of two drugs was 71.9%, and the rate of standard usage in combination of three drugs was 66.1%.

CONCLUSION: ARB and ACEI are the major drugs in antihypertensive medication while diuretic drugs are now rarely used. Combined medication is the main method of antihypertensive therapy where a substantial part of irregular medication not in conformity with the relevant recommend guidelines is found.

P271

Present Situation on Blood Sugar Control and Related Influencing Factors, Hypoglycemic Drug Use in the Elderly Coronary Heart Disease Patients with Diabetes Mellitus and Hypertension

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OBJECTIVE: The aim of this study was to investigate the present situation on blood sugar control and related influencing factors, hypoglycemic drug use in the elderly coronary heart disease (CHD) patients with diabetes mellitus (DM) and hypertension (HP).

METHODS: In this multicenter, non-intervention and cross-sectional survey, the elderly CHD patients aged 60 years or over were recruited from 165 hospitals in 21 provinces or cities across China from April to July 2011. The elderly CHD patients with DM and HP were selected from the survey as subjects to be investigated for blood sugar control, its influencing factors, and the usage of hypoglycemic drugs.

RESULTS: A total of 1,018 CHD patients with DM and HP whose blood glucose was well controlled were screened out from

7,962 cases, accounting for 12.8%. Proportion of patients with $HbA_{1C} < 6.5\%$ and $< 7.0\%$ was 23.5% (239/1018) and 51.2% (521/1018), respectively. Blood sugar success rate of patients with blood pressure up to standard was higher than that of patients with blood pressure not up to standard ($P < .05$), and blood sugar success rate decreased with the loose of blood pressure control standard, however, blood sugar success rates among different blood pressure control standards were found no statistical differences ($P > .05$). Single factor analysis showed that age, BMI, alcohol, dyslipidemia, stroke or TIA, the types of CHD, and angina that occurred in nearly two months were associated with $HbA_{1C} < 7.0\%$ ($P < .05$). The overall treatment rate was 94.1% (958/1008) in the survey, and α -glycosidase inhibitor was the most commonly prescribed hypoglycemic drugs (usage rate was 51.3%), followed by metformin (46.7%) and insulin (46.7%). Combination medication was the main way of therapy (63.9%); treatment of oral antidiabetic drug (OAD) only accounted for 53.0%; insulin (INS) only accounted for 13.9% of insulin (INS), combination of INS and OAD accounted for 31.2%; blood sugar success rate of INS was 47.5 ($HbA_{1C} < 7.0\%$); OAD only was 59.5% and combination of INS and OAD was 34.3%; blood sugar success rate of single drug was 60.2%; combination medication was 42.8%; multivariable Logistic regression analysis showed that the SBP and SDP, diabetes duration, hypoglycemic way were positively correlated with unsuccessful glucose control ($HbA_{1C} \geq 7\%$); stroke or TIA was negatively correlated with unsuccessful glucose control ($HbA_{1C} \geq 7\%$); SBP and SDP, BMI, hypoglycemic way were positively correlated with unsuccessful glucose control ($HbA_{1C} \geq 6.5\%$).

CONCLUSION: The overall rate of hypoglycemic therapy is high in elderly CHD patients with DM and HP, but overall blood sugar success rate is low, and the rate of patients with blood pressure up to standard was higher than that of patients with blood pressure not up to standard, however, blood pressure does not have to maintain a minimum standard, and the standard of below 150/90 mmHg was appropriate. There is much difference in success rates of such patients because of a variety of treatment, and a lot of problems with drugs usage were found. Therefore, measures should be taken to strengthen the management of such cases, and especially positive control of blood pressure is made to improve blood sugar success rate.